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PRINCIPLES OF MICRO ECONOMICS

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Dr. Aruna Kumar Dash

LURNEXA PUBLICATION

PRINCIPLES OF MICROECONOMICS FOR BUSINESS AND MANAGEMENT

By
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Hyderabad

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*I dedicate this book to my beloved parents,
whose love and sacrifices shaped my life*

*To my wife, for her constant support, and
endless encouragement*

*To my daughter 'Pari', my source of joy
and inspiration*

Authors Profile



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I hope this textbook will be a useful resource to students, teachers, researchers and lifelong learners in the field of Microeconomics for Business and Management. I hope this book will help not only to improve academic understanding, but also to help society by increasing economic understanding and informed decision-making.

Finally, I dedicate this work to our families and parents for their constant love, patience, and encouragement.

Thank you all for making my dream come true.

Aruna Kumar Dash

TABLE OF CONTENTS

Unit-1: Introduction to Micro Economics	1-25
What do You Mean by Economics?	1
What Economics Studies?	1
Do you Think Resources are Limited?	4
Branches of Economics	8
Nature of Microeconomics	10
Scope of Micro Economics	10
Is Economics a Required Course in MBA and Engineering Program?	13
Managerial Economics	14
Steps Need to Follow Making a Business Decision More Effectively	15
Production Possibilities Curve	16
Can PPC Will Shift in Short Run?	19
Opportunity Cost	21
Features of Capitalist Economy	22
Features of Command Economy	22
Features of Mixed Economy	23
Review Questions	25
 Unit 2: Demand and Supply Analysis	 26-74
Demand	26
What is Law of Demand?	27
Demand Schedule	28
Demand Curve	28
Market Demand Curve	29
Change in Quantity Demanded	30
Change in Demand or Shift in Demand	30

Exceptions of the Law of Demand	37
Supply	38
Law of Supply	39
Supply Schedule	40
Supply Curve	40
Market Supply Curve	41
Change in Quantity Supply	41
Change in Supply	42
Market Equilibrium	46
Numerical Example	48
Effects of Government Intervention on Price Control	53
Price Ceiling	54
Price Floor	55
Minimum Wage Law	55
Why do we Study Elasticity	56
Types of Elasticity of Demand	57
The Price Elasticity of Demand	57
Price Elasticity of Demand Interpretation	58
Degree of Price Elasticity of Demand	59
Elastic Demand (Relatively Elastic in Demand)	60
Relatively Inelastic in Demand or Inelastic Demand	60
Unitary Elastic Demand	61
Perfectly Elastic in Demand	62
Perfectly Inelastic Demand	63
Price Elasticity of Demand in a Straight-Line Demand Curve	63
Methods of Measuring Elasticity of Demand	64
Percentage Method	64
Point Method (Geometric Method)	65

ARC Method	65
The Determinants of Price Elasticity of Demand	65
Cross Price Elasticity of Demand	66
Interpretation of Cross Price Elasticity of Demand	68
Income Elasticity of Demand	68
Interpretation of Income Elasticity	68
Promotional/Advertisement Elasticity of Demand	69
Elasticity of Supply	69
Total Revenue and the Price Elasticity of Demand	72
Review Questions	74

Unit 3: Consumer Behaviour **76-104**

Consumer Behaviour	76
The Concept of Utility	77
Marshallian Cardinal Utility Analysis	78
Diminishing Marginal Utility	79
Total Utility and Marginal Utility	80
Theoretical Relationship between TU and MU	81
Consumers Equilibrium through Cardinal Approach	81
Consumer Equilibrium in One Commodity Context	82
Equi-Marginal Utility	82
Criticisms of Cardinal Utility Analysis	84
Consumer Surplus	84
Ordinal Utility Approach	85
Indifference Curve	85
Indifference Map	86
Indifference Schedule	86
Properties of Indifference Curve	87

Critical Thinking Question: Why Can't an Indifference Curve be Horizontal	88
Vertical or Upward Slopping Curve?	89
Slope of the Indifference Curve	90
Principle of Diminishing MRS	90
Budget Line	91
Slope of the Budget Line	93
Effects of Changes in Prices on Budget Line	93
Effects of Changes in Income on Budget Line	93
Differences between Indifference Curve and Budget Line	94
Consumer Equilibrium	94
Price Consumption Curve	96
Income Consumption Curve	99
Normal Good vs. Inferior Good	100
Engel Curve	100
Engel Curve for Normal Goods	101
Engel Curve for Inferior Goods	102
Price Effect Is the Combination of Income and Substitution Effect	103
Review Questions	105

Unit-4: Theory of Production **107-158**

Production Function	107
Short Run Production Function	109
Total Product (TP), Marginal Product (MP), and Average Product	109
The Law of Variable Proportion	110
Increasing Return to Factor	113
Diminishing Return to Factor	114

Negative Return to Factor	114
Long Run Production Function	115
ISO Quant	115
ISO Quant Schedule	115
ISO Quant Curve	116
ISO Quant vs. Indifference Curve	117
Properties of ISO Quant	117
Why Can't an ISO Quant Slopes Horizontal or Vertical and Upward Slopping	118
Diminishing Marginal Rate of Technical Substitution (MRTS)	119
Why Can't ISO Quant Concave to the Origin and Downward Slopping	120
Why Can't Two ISO Quant Curve Cannot Intersect with Each Other	121
ISO Quant Map	121
ISO-Cost Line	121
Producers Equilibrium or Least Cost Combination of Factors	122
Expansion Path	123
Returns to Scale	124
Increasing Returns to Scale (IRS)	124
Constant Returns to Scale (CRS)	126
Decreasing Returns to Scale (DRS)	126
Application of Returns to Scale	127
Cost Theory	129
Fixed Cost	130
Variable Cost	130
Social Cost	131
Sunk Cost	131
Explicit Costs	132

Implicit Costs	132
Accounting Costs	132
Operating Cost	132
Total Cost	133
Average Cost	133
Average Fixed Cost	134
Average Variable Cost	135
Average Total Cost	136
Marginal Cost	137
Relationship between Average Cost and Marginal Cost	138
Relationship between AC, MC and AVC	139
Short Run vs. Long Run Cost	140
Short Run Total Cost	141
Relationship between Short Run Production Function and Short Run Cost	141
Relationship between Average Cost and Marginal Cost	144
Long Run Total Cost Vs. Short Run Total Cost	144
Long Run Cost Curves	144
Why LAC Is U-Shaped in General?	145
Learning by Doing (Learning Curve)	148
Long Run Average and Marginal Cost	148
LAC is Envelope of Short Run Average Cost Curve	149
Return to Factor Vs. Return to Scale	151
Return to Factor	151
Return to Scale	151
Return to Scale Vs. Economies of Scale	151
Internal and External Economies of Scale	152
Internal Economies of Scale	152
Advantage of Internal Economies of Scale	152

External Economies of Scale	154
Advantages of External Economies of Scale	154
Diseconomies of Scale	154
Internal Diseconomies of Scale	154
External Diseconomies of Scale	155
Economies of Scope	155
Economies of Scale vs. Economies of Scope	155
Break-Even Point	155
Marginal and Total Revenue	156
Numerical Questions and Its Solutions	156
Review Questions	158

Unit-5: Market Structures **160-213**

Market Structures	160
Perfect Competition	161
Price and Output Determination under Perfect Competition	162
Equilibrium (or Profit Maximizing) Conditions	165
Critical Thinking Questions	170
Short-Run Price and Output Determination of a Perfectly Competitive Firm (Profit Case)	172
Price and Output Determination of a Perfectly Competitive Firm in the Short Run (Normal Profit or Break-Even Case)	172
Short-Run Price and Output Determination of a Perfectly Competitive Firm (Loss Case)	172
Critical Thinking Questions	172
Shut-Down Point	172
Long-Run Price and Output Determination in a Perfectly Competitive Market	173
Allocative Efficiency	174
Monopoly	176

Why Monopolies Arise?	177
Perfectly Competitive Firm versus Monopoly	178
Demand Curve of a Perfectly Competitive Firm and a Monopoly	178
A Monopoly's Total, Average, and Marginal Revenue	179
Demand Curve and Marginal Revenue Curves for a Monopoly	179
Profit Maximization or Equilibrium of a Monopoly Company/Firm in Short Run	180
Short Run Price and Output Determination under Monopoly (Profit Case)	181
Short Run Price and Output Determination under Monopoly (Loss Case)	182
Short Run Price and Output Determination under Monopoly (Break Even Point)	182
Price and Output Determination under Monopoly in the Long Run	182
Dead Weight Loss	183
Producer and Consumer Surplus	184
Price Discrimination	185
Degrees of Price Discrimination	186
Equilibrium if Discriminating Monopolist	187
Monopolistic Competition	189
Characteristics of Monopolistic Competition	190
Monopolistic Competition vs. Perfectly Competitive Market	191
Profit Maximization or Equilibrium of a Monopolistic Firm	192
Price and Output Determination under Monopolistic Competition in the Short Run	193
Price and Output Determination under Monopolistic Firm in the Short Run (Loss)	194
Price and Output Determination under Monopolistic Firm in the Short Run (Break Even Points)	195

Price and Output Determination under Monopolistic Firm in the Long Run	195
Oligopoly	196
Non Pricing Strategy	197
Features or Characteristics of Oligopoly Market Structure	197
Sources or Causes of Oligopoly	198
Collusive Vs. Non-Collusive Oligopoly	199
Pricing Strategy Followed in an Oligopoly Market	199
Cournot's Duopoly Model	200
Limitations of Cournot Model	202
Price Leadership Model	203
Price Leadership by a Low-Cost Firm	203
Barometric Price Leadership	204
Price Leadership of a Dominant Firm	204
Cartel	205
The Centralized Cartel	206
Market Sharing Cartel	207
Cartel Success Story	207
Cartel Failure Story	207
Conditions for Cartel Success	207
Kinked Demand Curve	208
Criticism of Kinked Demand Curve	211
Pricing in Practice	211
Review Questions	213
Unit-6: Factor Pricing	214-252
Factor Pricing	215
Rent	215

Determination of Land Rent	216
Wages	217
Interest	218
Nominal Interest Rate	218
Real Interest Rate	218
Saving	219
Investment	219
Investment Demand Curve	219
Classical Theory of Interest Rate Determination	220
Keynesian Liquidity Preference Theory of Interest	220
Keynesian Theory of the Demand for Money	221
Transaction Demand for Money	221
Precautionary Demand for Money	222
Speculative Demand for Money	222
Supply of Money	223
The Keynesian Theory of Determination of Interest Rate	223
Liquidity Trap	225
Characteristics of a Liquidity Trap	225
Profit	226
Schumpeter's Innovations Theory of Profits	227
The Marginal Productivity Theory of Distribution	227
Clark's Version of Marginal Productivity Theory	228
Externality	231
Positive Externality	232
Negative Externality	233
Critical Thinking Question	234
Private Goods Vs Public Goods	234
Asymmetric Information	236

Adverse Selection	236
Moral Hazard	237
Market Failure	238
Causes of Market Failure	239
Government Failure	240
Reasons for Government Failure	241
Role of Government in Promoting Welfare	241
Pareto Optimality	246
Consumption Efficiency	246
Production Efficiency	248
Product Mix Efficiency: Optimum Mix of Commodities	250
Review Questions	252
Index	254-260

UNIT-1

INTRODUCTION TO MICRO ECONOMICS

"A dream is not that which you see while sleeping; it is something that does not let you sleep."

— A. P. J. Abdul Kalam

What do you mean by Economics?

Economics is a social science which studies how individuals, firms, and governments make choices to satisfy their unlimited wants with a given resource. Economics differs from pure science. It keeps on changing and it is very dynamic compared to pure science. For example, H₂O. Water is formed from the combination of two hydrogen atoms and one oxygen atom. Combining two hydrogen atoms with one oxygen atom consistently produces water, regardless of the time period. Similarly, Newton's 3rd Law which states that for every action, there is an equal and opposite reaction. It may be said that pure science is relatively static as compared to economics. You might wonder, "Why should we study economics?" In 2000, there was one survey of top-level positions in companies, I mean the CEO and COO, and it found that people in top-level positions are either economists or have engineering backgrounds. Economics is the study of how individual, firm, and governments make choices to satisfy their unlimited wants with a given resource. In simpler terms, economics is all about decision-making.

What Economics Studies?

Economics studies how individuals, firms, and governments make decisions to satisfy their unlimited wants with given resources. An individual is defined as a person, and you are regarded as such. A firm is a company or business organization that is treated as a production unit. It converts inputs into outputs. A firm produces goods and services for individuals. An individual is the consumer unit, and a firm is the production unit. A government is the organization through which political authority is exercised over a society.



As an individual, you need to decide whether to visit Manali to see snowfall or to buy a car.
As an individual, you make many decisions in your life. For instance,

- ❖ Whether you should go for an MBA or MFC.
- ❖ Whether you should be a risk lover or risk averter
- ❖ Whether you should work hard or follow a shortcut

Each decision an individual makes has a small or a big consequence.



A firm is a production unit that makes multiple decisions. For instance,

- ❖ **Strategic decisions** (mergers and acquisitions, entering a new market)
- ❖ **Operational decisions** (inventory and supply chain management)
- ❖ **Financial decisions** (e.g., investment, financing, dividend policies)
- ❖ **Marketing decisions** (e.g., Product, price, place, and promotion)
- ❖ **Human resource decisions** (e.g., hiring, firing, training, compensation)
- ❖ **As in a household**, each decision a firm makes has a small or large consequence.



A government takes multiple decisions to ensure the smooth functioning of the economy. For instance,

- ❖ **Economic Decisions:** The government has a limited budget. The government may decide whether to spend more money on defense expenditure or to spend more money on education
 - ❖ **Taxation and Spending:** How much tax the government will impose and how much the government will spend
 - ❖ **Trade Policy** (tariffs, quotas, trade agreements)
 - ❖ **Social Policy Decisions** such as Education policy, Healthcare funding, and Welfare programs and subsidies
 - ❖ **Environmental Decisions**, such as policies related to mitigating the adverse impact of climate change and pollution control
 - ❖ **Defense and Foreign Policy:** To safeguard an economy, the government takes decisions on Military spending and National security measures
- As with households and firms, each decision a government makes has a small or large consequence.

Economics aim is to satisfy their unlimited wants with given resources



Have you seen on TV that many people want to participate in “Kaun Banega Crorepati”? Why? The answer is very simple

- (i) For more visibility.
- (ii) To earn more money.

The question arises why individuals, and firms want to earn more money. They want to satisfy their unlimited wants.

Do you think human wants are limited? Of course not. When one want is satisfied, another want comes to its place.

We can say that human wants are unlimited. For example, as a student, you might be happy with your Bike. Once you become the Chief Executive Officer (CEO), you may not prefer a bike; you may prefer a BMW. Again, after some years, you may buy a higher-end car by paying 10 crores.

Do you think Resources are Limited?

Resources are limited from the household, firm, and government perspectives.

An individual's point of view, resources are your money and time

A firm point of view, resources are land, labour, capital, and raw materials.

From the government's point of view, resources are natural resources, money, etc

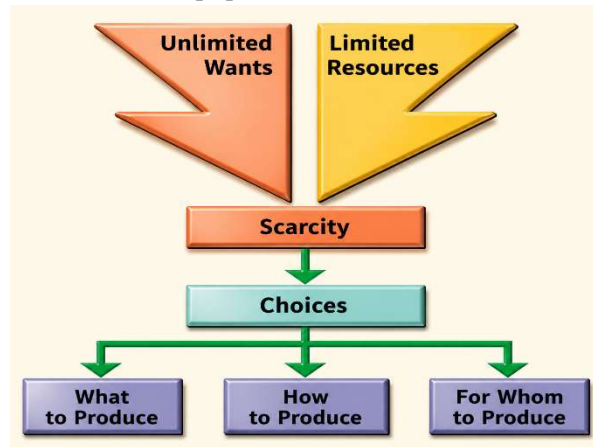
Resources are limited, and wants are unlimited. Scarce resources are utilized by individuals, firms, or governments to produce goods and services to satisfy the wants. As resources are scarce, so are goods and services. Scarcity occurs when the demand for goods surpasses the supply.

There is a choice because there is scarcity. Consider a farmer with one acre of land. The farmer cannot produce sugarcane, rice, vegetables simultaneously on one acre of land. The farmer will decide which crops to produce and which not to. Because resources are limited, a farmer must make decisions about which crops to grow, the quantity to produce, and who will buy the product.

Choice refers to the decision-making process individuals, firms, and governments take in their day-to-day lives. Economics revolves around making choices, which arise from scarcity. Without scarcity, choices would not exist, and consequently, there would be no economics. From morning to night, individuals continuously make decisions influenced by this scarcity. On the one hand, we have unlimited wants; on the other hand, we have limited resources, which leads to scarcity. Producers must decide which goods to produce, the quantity, and for whom to produce in times of scarcity.

All the countries in the world irrespective of developed or developing countries in the world faces three fundamental problems of an economy

- (a) What to produce-means which goods to produce
- (b) How to produce-means the techniques used in the production process
- (c) For whom to produce- whether you are producing for high income, middle income or low income or the entire population.



Goods

Goods are classified into three categories:

- (i) Consumer durables
- (ii) Consumer non-durables.
- (iii) Services.

Goods are called tangible items. We refer to goods as physical items. Goods include physical items such as pens, chairs, books, and laptops. You can realize it, feel it, and touch it.

Consumer Goods

Individuals use Consumer Goods for personal use to satisfy their wants. *Examples:* Food, Clothing, Smartphone, medicines, etc.

Consumer Durable Goods

Durable goods last a long time and can be used repeatedly. *Examples:* Cars, furniture, AC, laptop etc.

Consumer Non-Durable Goods

Non-Durable Goods are consumed quickly or have a short life span. *Examples:* Milk, oil, paper, soap, FMCG Products

Services

Services are known as intangibles. You can only realize it. For example, services provided by professors, doctors, scientists, tourism, IT, and banking. At the end of the class, students will determine whether it was effective. Similarly, X is unwell and consulted a doctor. The doctor gave medicine for 5 days. After 5 days, X will realize whether the doctor's services were effective. Note: you can't store services, whereas you can store goods.

Capital Goods

Capital goods are those goods which are used by a firm to produce other goods. *Examples:* Machinery, tools, buildings, equipment, tractors, trucks, etc

Public Goods

Public goods are goods provided by the government for societal welfare purposes. These goods are characterized as non-excludable, which means access to them cannot be restricted to specific individuals. For instance: roads, defence, fire station, police station, parks, Government hospitals. Similarly, a taxpayer can't say, "I am paying tax." Hence, I should use the road and you should not, as you are not paying tax.

Public goods are non-rival, which means one person's use of the good does not lessen its availability to others. Here, two individuals are not rivals, and no one can be excluded from consumption. For instance, street lighting: One person using the light does not reduce the light available to others. Similarly, national defense: One citizen's protection does not diminish that of other citizens.



Private Goods

Private goods differ from public goods as they are not provided by the government for public use. Private goods are characterized as excludable. It means that access to a good or service can be restricted to those who do not make payment. For instance, a movie ticket is considered as excludable as only individuals who purchase it can watch the movie.

Private goods are characterized as rival as well. It means that one individual's usage of a good reduces its availability for others. For instance, if you eat a pizza, another person cannot eat the same pizza



Rival goods are limited in quantity; when one person consumes them, their availability decreases for others. Private Goods are usually sold in markets. *Examples:* Your laptop, property, books, Watch, house etc.

Free Goods

Nature gives free goods for the survival of living beings. Free goods are unlimited in supply. No cost is involved in the production process. All individuals can use free goods without paying. The cost of production is zero for a free good. Examples are sunlight, free air, rainwater, Ocean water, etc



Economic Goods

Goods which are economically useful can be said to be economic goods, which are basically the antithesis of free goods. Payment is done when these services are accessed

There is a cost incurred in the production of an economic good. There is scarcity of supply of these goods. There are limited amounts of these goods. In the production of economic goods, the producers utilize factors of production like land, labour, capital, and entrepreneurship, and pay factor payments like rent, wages, interest, and profit.



Branches of Economics

Microeconomics is the branch of economics which examines how individual entities, such as customers, businesses, and markets, behave and make decisions. For example, if someone asks an individual about their income, then we are talking about microeconomics. Similarly, if someone asks an individual how many people they have employed in their company, then we are talking about microeconomics.

One area of economics is microeconomics. Microeconomics examines how people, households, and firms make decisions and allocate limited scarce resources. It examines how people make decisions, what influences those decisions, and how those decisions impact supply, demand, and prices in markets.

Microeconomics examines decision-making processes of individuals, households, and firm regarding how much to produce, how much to buy and which product to sell. In microeconomics, emphasis is placed on a single market.

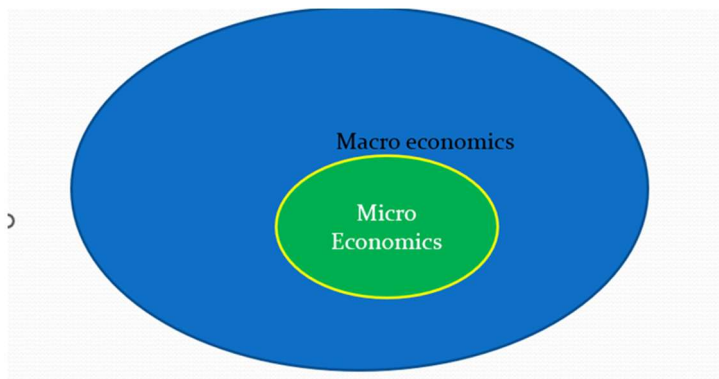
For instance, automobile industry. How increase in crude oil price will change the demand and supply of automobile industry. Economics is divided into the study of how individual markets operate and the study of how the economy as a whole operates.

- ❖ Microeconomics is the study of a particular individual (Can be a household, firm, or a particular market)
- ❖ Microeconomics focuses on specific components of the economy rather than analyzing the economy as a whole.

- ❖ Microeconomics focuses on consumer and producer behavior, along with the theories of demand, supply, and the firm.
- ❖ Microeconomics examines the behavior of individual consumers and producers, focusing on how they achieve equilibrium.
- ❖ Microeconomics deals with individual output, individual income, and individual product price.
- ❖ In micro economics we study the individual aspects such as:
 - (i) A firm
 - (ii) Personal income
 - (iii) Personal savings
 - (iv) A firm's investment
 - (v) A consumer
 - (vi) A producer

Macroeconomics

Macroeconomics is the branch of economics that studies how an economy functions as a whole. Policy makers are more concerned about macroeconomic variable compared to micro economic variables. Macroeconomics is more dynamic and broader as compared to micro economics.



- ❖ It studies income, employment, and output by analyzing the overall functioning of the economy.
- ❖ It examines the behavior of all the units combined, not just one.
- ❖ Macro Economics deals with national output/GDP, not individual output; aggregate income, not individual income; and the general price level of an economy, not individual prices.

Key Macro Economics Indicators are GDP, Employment, Unemployment, Inflation, Exchange rate, Interest rate, and money supply

Note: Macroeconomics encompasses the broader aspects of economic theory, while microeconomics focuses on specific individual and business decisions within that framework.

Nature of Microeconomics

The nature of microeconomics is defined by its key features.

- ❖ **Individual Analysis:** Micro economics studies about an individual; it can be a producer or consumer. It doesn't study economy as a whole. For instance, how many people are working in your company? How a company decides the price of the product they produce. All these are studied in micro economics.
- ❖ **Price Determination:** It analyzes the determination of prices for goods and services based on demand and supply factors. Increased demand for goods and services raises prices, while decreased demand lowers them.
- ❖ **Resource Allocation:** Resources are physical resources and human resources. Physical resources are land, factories, machinery and equipment, raw materials, and natural resources. Human resources include teachers, doctors, engineers, managers, factory laborers etc. All the resources are scarce irrespective of a consumer, producer and a country point of view. Microeconomics examines the utilization of limited scarce resources by individuals and firms in a best way.
- ❖ **Decision-Making:** Microeconomics provides individuals and firm the ability to decide on pricing for production and consumption. Microeconomics helps individuals decide what to buy and how much to buy. Individuals and firms have limited income. They must choose goods and services that satisfy them the most. It helps a firm in choosing what, how, and how much to produce. For instance, if a product is in great demand, a firm can choose to increase production.
- ❖ **Partial Equilibrium:** It examines a single market at a time, assuming other factors remain constant. For instance, when we say stock market has no relationship with the commodity market, then we are talking about partial equilibrium. Partial equilibrium is opposite to general equilibrium where simultaneous equilibrium of price and quantity across all the markets. If we will say stock market, bond market, commodity markets and foreign exchange markets are related with each other, then we are talking about general equilibrium.

Scope of Micro Economics

The scope of microeconomics is extensive, encompassing a variety of economic activities. Some of the key areas that microeconomics covers include:

- ❖ **Consumer Behavior:** Microeconomics is a field of study focused on the behavior of individual consumers. It examines how consumers make decisions regarding their purchasing choices, including the quantity of goods or services they buy, the prices they are willing to pay, and the satisfaction they will get from it. Consumer behavior plays an important role as it affects demand and supply.

When consumer prefer a product might be due to change their taste and preference or increase their income, then demand for these goods increases. With increase in demand for these goods producers will produce more these goods to enhance their profit. On the contrary, if consumers don't like a product they won't buy that product. As a producer they won't produce that product because it will increase their inventory and company will incur losses. For instance, few years ago LML Vespa was dominating the two-wheeler segment because people like it.

Now due to change in taste and preference people, they don't like it. If two-wheeler manufacturing company will produce LML Vespa now it will reduce company profitability and inventory of the company will increase. Price of a product depends upon the forces of demand and supply.

The demand for goods depends on the behaviour of consumers, whereas the supply is dependent upon the costs and behaviour of producers. An increase in price causes quantity demanded to fall, *ceteris paribus*. Both of them are in inverse relation to each other. An increase in price results in an increase in quantity supplied. In case of increase in prices, the consumers will be dissatisfied, whereas the producers will be satisfied.

❖ **Production and Costs:**

Microeconomics examines firm behavior regarding production decisions, including what to produce, the quantity, and pricing strategies. It examines how firm use scarce resources like land, labour, capital, and technology to efficiently produce goods and services.

Microeconomics examines how firms use technology and determine production costs when producing goods and services. Firms seek to optimize production by minimizing costs while maximizing output. Microeconomics provides insights into how firms make production decisions to enhance profitability.

❖ **Market Structure:** market structures include perfect competition, monopoly, monopolistic competition, and oligopoly. In market structure, perfect competition is one extreme and the monopoly is on the other extreme. Monopolistic competition and oligopoly in between.

Perfect Competition is a situation where many numbers of buyers and sellers exist and none of them can affect the market price. Monopoly is the single firm that dominates the whole market with its product, which has no close substitutes.

A monopoly consists of one seller with multiple buyers. The monopoly business enterprise can influence the price of its product, but the firms operating under perfect competition are price takers. There exists another form of market organization known as monopolistic competition.

Oligopoly competition refers to the market structure where there are few firms producing homogeneous or differentiated products. Microeconomics discusses the impact of market structures on firm and consumer behavior, as well as their effects on prices and quantities in the market. Microeconomics examines the impact of competition, consumer demand, and production costs on pricing decisions and profit margins.

- ❖ **Theory of Factor Pricing:** Factor pricing studies how the prices of factors of production are determined. The factors of production include land, labour, capital, and entrepreneur. Land has alternative uses. Land is rewarded as rent. Labour is rewarded as wages. Labour can be skilled and unskilled labour. Capitalist used in the production of goods and services. It includes machinery, tools, buildings, equipment, and technology that help in producing other goods.

Capital is rewarded as interest. Entrepreneur is the decision makers whose aim is to increase the value of the firm. Entrepreneur combines factors of productions and takes decision which product to produce, how much to produce and which technology to be used. Entrepreneur is rewarded as profit. Microeconomics explains how demand and supply for these factors influence their earnings in the economy.

- ❖ **Optimum Allocation of Resources:** Microeconomics focuses on the allocation of resources within the economy. It tells how efficiently resources are distributed among consumers and producers. Micro economics analyze how resources are allocated efficiently in different sectors of the economy. Resources are limited for consumers, producers and government point of view. Human wants are unlimited but resources are limited. Hence, optimum allocation of resources plays an important role.
- ❖ **Public Policy:** Public policy falls within the scope of microeconomics as it analyzes how government decisions affect individual economic units and markets. Microeconomics studies how governmental policies such as taxes, subsidies, and regulations affect how businesses and consumers behave in the economy. Public policy in microeconomics examines how government actions impact price, demand, supply, production, consumption, and resource allocation. Policies like taxation, subsidies, minimum wages, price controls, and market regulations are examined for their impact on consumers and producers. For instance, microeconomics, examines how a tax on alcohol affects the market's pricing, consumer demand, and producer supply.
- ❖ **Welfare Economics:** Welfare Economics is a branch of microeconomics that analyzes the impact of economic activities on individual and societal well-being. Welfare economics analyzes resource allocation efficiency and assesses whether the distribution of goods and services enhances social welfare. Welfare economics examines important issues such as income distribution, market efficiency, consumer

satisfaction, and the role of government intervention in maximizing social benefit. Welfare economics is a crucial aspect of microeconomics, aiming to enhance economic welfare for both individuals and society as a whole.

- ❖ **Microeconomics is the Foundation of Macroeconomics:** Microeconomics serves as the foundation of macroeconomics by explaining how the actions of individual consumers, households, workers, and firms contribute to overall economic outcomes. Micro foundations of macroeconomics examine how individual choices related to consumption, saving, investment, production, and employment impact larger macroeconomic factors like national income, inflation, unemployment, and economic growth. For instance, when consumers cut spending, aggregate demand decreases, impacting overall economic growth. Thus, microeconomics provides the foundation for understanding
- ❖ **Macroeconomic Behavior.** Understanding microeconomic policy is essential for formulating effective macroeconomic policy.

Relationship between Microeconomics and Business Management

The nexus between microeconomics and business management is important as microeconomics assists managers in making effective business decisions. Understanding consumer behavior, demand, supply, pricing, production, cost analysis, and market competition is crucial for successful business management. Business managers apply microeconomic principles to determine production decisions, pricing strategies, and profit maximization while minimizing costs. Understanding customer demand helps managers in planning production and marketing strategies. Microeconomics serves as a foundation for business management by enhancing decision-making, planning, and resource utilization within organizations.

Is economics a required Course in MBA and Engineering Program?

There was one survey in the US in 2000. It was found that 90% of people in higher-level positions in a company (CEO, COO) are either economists or have Engineering backgrounds.

Yes, knowing economics very well will help you in following aspects.

- ❖ **Understanding Market Dynamics:** Economics explains how markets operate — how supply and demand influence prices, and how consumer, producer, and government behavior influences markets. This helps managers make smarter pricing, marketing, and production decisions.
- ❖ **Resource Allocation & Efficiency:** Economics teaches how to allocate scarce resources efficiently, which is crucial for optimizing budgets, managing costs, and maximizing profits in any business.

- ❖ **Decision**-to do a better cost-benefit analysis of a firm
- ❖ **Macroeconomic Awareness:** Students will get exposure to how economic indicators like GDP, inflation, consumption, investment, and unemployment affect business environments, investment opportunities, and risks
- ❖ **Policy Impact:** Knowing how the government's fiscal policy and the nation's central bank's monetary policy impact the business environment
- ❖ **Analytical & Critical Thinking Skills:** Economic models and theories develop logical thinking and problem-solving skills that are transferable across all business functions. Why does the performance of the economy fall, and what will be its consequences on producers, consumers, and the governments Why does inflation arise, and how does it affect the firm's profitability and households' livelihood.

Managerial Economics

Managerial economics is the application of economic theory and decision science (tools) by a manager to increase the effectiveness of decision-making. The demand and supply theory, production and cost theory, pricing theory, and other micro and macroeconomic theories are related to economic theory. Analytical tools like linear programming, statistical and econometric methods, etc. The use of statistics and mathematics is known as econometrics.

What decisions does a manager make?

Strategic Decisions

- Setting goals and objectives of an organization
- Allocating resources in the production and distribution process
- Entering new markets or launching new products
- Formulating policies and business strategies for an organization

Operational Decisions

- Managing inventory levels
- Approving budgets and expenditures
- Ensuring compliance with regulations

Financial Decisions

- Setting departmental budgets
- Approving expenses
- Takes Cost-cutting measures

Other Decisions

- Decisions related to product, Price, Place and promotion
- Decision related to business ethics
- Decision related to cost-benefit analysis while hiring and firing employee

Steps Need to Follow Making a Business Decision More Effectively

- 1) Identify the Problem- Clearly understand the problem that requires attention
- 2) Collect necessary information- Gather useful and relevant data related to the situation
- 3) Evaluate the available evidence- Examine the data and think about potential results.
- 4) Select best options- Compare alternatives and choose the most suitable solution.
- 5) Implement the decision- Implement the selected plan.
- 6) Review your decision- To evaluate the decision's effectiveness and identify areas for improvement.

Let's assume you are an HR manager at a Fortune 500 company. In a particular year (i.e., 2007), many people left your company. The Chief Executive Officer (CEO) called the HR manager and told them to look into it.

The problem is now identified:

“Why are many people leaving your organization?”

Before taking any action, gather relevant information. HR has started collecting information from candidates who are quitting the company through exit interviews and from managers across departments. Here, different people will give different information. As a manager, you need to weigh the evidence (i.e., whether it is correct).

For example, what are the most likely reasons people left your company? Managers need to choose among the alternatives based on the collected information and implement them. After implementation, the Manager role is not over.

The most important thing is that the manager reviews their decisions. Without reviewing the decision, it is difficult to say whether the managerial decision-making process is more effective. After reviewing the decision, some errors can be rectified, and the decision will provide an effective outcome.

Production Possibilities Frontier or Production Possibilities Curve

The production possibilities frontier demonstrates the various combinations of two goods that an economy produces. This assumption is based on the use of all resources efficiently.

The total amount of goods that can be produced using the given level of resources is represented through the PPF.

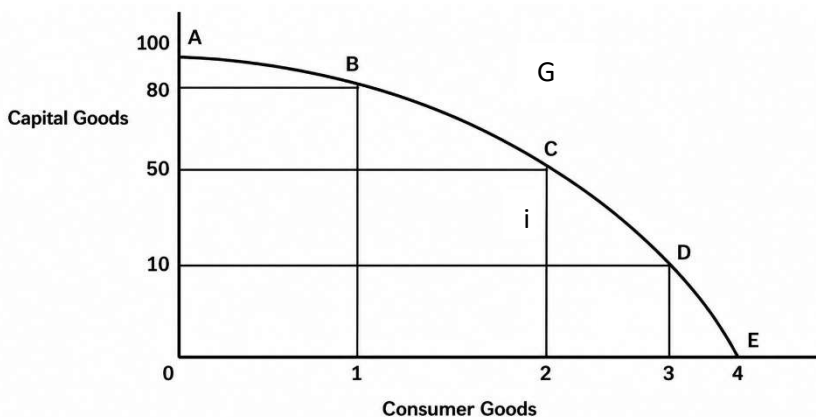
Assumptions of PPF

- ❖ Output is confined to two goods only.
- ❖ The production takes place within a year
- ❖ The economy has limited resources throughout the year, both in terms of quality and quantity.
- ❖ The level of technology remains unchanged throughout the year.

Production Possibilities Frontier Schedule (PPC)

Possible Combinations	Consumer Goods	Capital Goods	Resources Available
A	0	100	100
B	1	80	100
C	2	50	100
D	3	10	100
E	4	0	100

Production Possibilities Curve



The slope of the Production Possibility Curve is graphically represented as Concave to the origin. A straight line and a convex shape to the origin are the extreme scenarios.

The Production Possibility Curve (PPC) is a graph depicting the production schedule. The PPC demonstrates all possible combinations of outputs based on an economy's resource availability and technology. The sloping nature of the PPC indicates that as more of one product is produced, less of another product will be produced. This is due to the scarcity of resources.

X-Axis depicts consumer goods. Y-Axis denotes capital goods. Point A signifies the number of capital goods that can be produced within a year provided that the economy utilizes its resources effectively. Point E denotes the number of consumer goods that can be produced within a year provided that the economy utilizes its resources effectively. Point A is an extreme point because at this point, the economy only produces capital goods.

Similarly, point E is also another extreme point because here the economy produces only consumer goods. Points A and E are the most extreme points where the economy only produces one type of good. Apart from point A and point E, the economy will produce various combinations of capital and consumer goods as illustrated by points B, C, and D.

Points A, B, C, D, and E represent the efficient allocation and proper use of resources. These points show that resources are fully utilized. At point A, the economy produces 100 units of capital goods. No consumer goods are produced at this point. All resources will only be used for capital goods. At point E, the economy produces 4 units of consumer goods. No capital goods are produced at this point.

The economy produces 1 unit of consumer goods as it moves from point A to B. While moving from point A to B, economy loses 20 units of capital goods (100–80) to get additional one unit of consumer goods. In this case, the economy is giving up more capital goods compared to consumer goods. Therefore, the opportunity cost of producing 1 unit of consumer goods is 20. It is calculated as $(100 - 80) / (1 - 0)$.

In moving from point B to C, the economy loses 30 units of capital goods (80-50) for obtaining one extra unit of consumer goods (2-1). Therefore, the opportunity cost of obtaining an extra unit of consumer goods will be 30 units. This is found through $(80 - 50) \div (2 - 1)$. Moving from point C to D, the economy sacrifices 40 units of capital goods (50-10) for obtaining one extra unit of consumer goods (3-2). Therefore, the opportunity cost of getting one extra unit of consumer goods is 40. This is derived from $(50 - 10) \div (3-2)$.

Opportunity cost increases in line with the theory of increasing opportunity cost hence the bowed shape of the PPF toward the origin. Opportunity costs increase from 20 when the country produces goods from A to B. The opportunity costs will be increased to 30 when the country produces goods from point B to C.

Any point inside the PPC can be achieved with the available resources. However, it represents inefficient production because resources are not fully utilized. This can be understood by comparing points C, I, and D.

At Point	Number of Consumer Goods	Number of Capital Goods
C	2	50
I	2	10
D	3	10

Both point C and point I yield the same amount of consumer goods. Point C produces more capital goods than point I. At point C, 50 units of capital goods are produced, while point I produces only 10 units. Point C outperforms point I by generating greater output with the same number of resources. Point D is superior to point I as it produces more consumer goods, while both points maintain the same level of capital goods. Any point inside the PPC is feasible, but it does not represent the optimal use of resources. Accessing any point inside the PPC is possible with current resources and technology.

However, it is considered inefficient. Points beyond the PPC boundary are unattainable with existing resources and technology. The PPC highlights the efficient production levels in an economy and differentiates between attainable and unattainable combinations.

A point outside the frontier, like point G_1 , indicates an unreachable production level with current resources. In contrast, a point inside the frontier, like point I, indicates inefficient use of resources. PPC does not tell us which combination to choose.

Can the PPF appear as a straight line? If resources can be used equally well for different purposes, the PPF becomes a straight line. In this case, the opportunity cost remains constant at every point on the PPF.

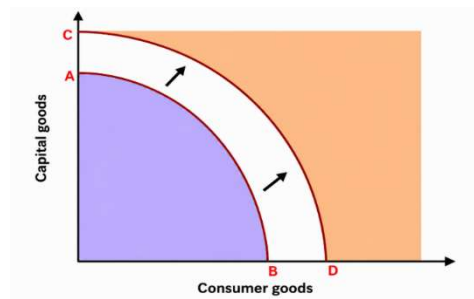
Possible Combinations	Consumer Goods	Capital Goods	Available Resources
A	0	100	100
B	1	80	100
C	2	60	100
D	3	40	100
E	4	20	100

Can PPC Shift in Short run?

Resources are fixed in both quality and quantity in the short term, so the PPC can't shift. But resources aren't fixed in the long run. This means that the PPC will shift outward when resources increase or when training and new technology improve labour productivity. Technological progress increases output and moves the PPC outward and vice versa. In the short term, PPC won't shift, but in the long term, it will. PPC will shift in the long run for the following reasons:

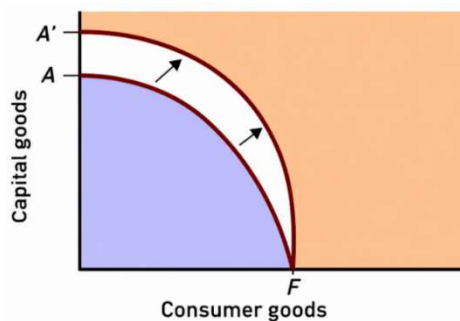
- More resources available
- Fewer resources available
- Changes in resources and technology that only help consumer goods
- Changes in resources and technology that only help producers' goods.
- Changes in resources and technology that help both producer and consumer goods

(a) Increase in available resources



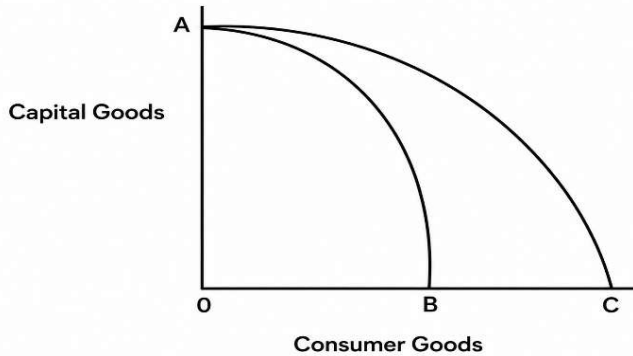
Upon an increase in available resources, PPC shifted from AB to CD. The PPC will shift outward if people work more hours. The PPC shifts outward when the labour force gets bigger or healthier, when the skills of the labour force improve, or when other resources become more available, like when new oil discoveries are found. Here, the economy can produce more consumer and capital goods. The reverse will occur if available resources decrease. When resources go down, the PPC is shifted inward. For instance, Israel invaded Iran in 2026, setting oil fields on fire and destroying a lot of Iran's physical capital. PPC will shift from CD to AB. With fewer resources available, the economy will produce less output. For instance, see the US-Israel-Iran war.

b) Change in resources, technology, that benefits capital goods only.



An improvement in the technology used to make more capital goods, like better software for designing heavy machinery. In this context the economy will be able to produce more capital goods”.

(c) Change in resources, technology, that benefits consumer goods only



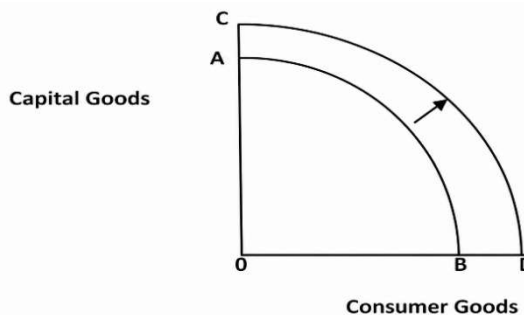
The PPC rotates outward along the consumer good axis when a technological discovery only helps consumer goods, like crops that are more resistant to disease. In this context the production of consumer goods increases from OB to OC without affecting capital goods.

(d) Technological Change

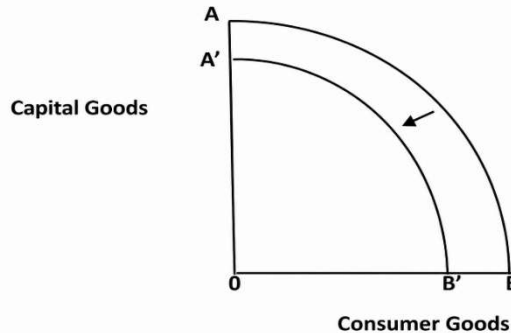
A technological advancement that optimizes resource utilization may expand the economy's PPC. Due to the discovery of technology; it is easier to make both consumer and capital goods. The internet, for example, has made it easier for businesses to make both consumer and capital goods. In this context, PPC will shift from AB to CD (refer a).

(e) Any change in economic growth will shift PPC

If there is expansion or contraction of economic growth, it alters the nation's productive capacity. As a result, PPF will shift. For instance, if there is economic growth of an economy, PPC will shift towards the right from AB to CD and the economy can be able to produce more of capital and consumer goods.



On the contrary, if there is economic slowdown of an economy, PPC will shift towards the left from AB to A'B' and the economy can able to produce less of capital and consumer goods.



Opportunity Cost

When we talk about the cost of something, what do we mean? What must be sacrificed to obtain something? Opportunity cost refers to the value of the next best alternative that is given up when a particular choice is made.

- ❖ Opportunity cost is like the chance you missed.
- ❖ Opportunity cost can occasionally be measured in monetary terms, although as we'll see, money is typically a component of opportunity cost.
- ❖ Everybody has a different opportunity cost. The only person who can determine which alternative is best is the one making the decision.
- ❖ To figure out opportunity cost, you need time and information.
- ❖ Opportunity costs change depending on the situation.
- ❖ Opportunity cost is influenced by the alternatives that are available to an individual. For some students, Tuesday night is a more preferable time for studying compared to Saturday night. This is because there are usually fewer attractive activities on Tuesday. As a result, the opportunity cost of studying is lower on a Tuesday compared to a Saturday.

Economic Systems (Market Economy, Command Economy, Mixed Economy)

All economies in the world, regardless of whether they are developed, developing, or least developed, face resource scarcity and unlimited wants. As a result, there is a trade-off between which goods to produce and which should not. All economic systems, whether capitalist, command, or mixed economies, face three fundamental problems. These fundamental problems are (i) what to produce (ii) how to produce and (iii) for whom to produce.

Production refers to what the economy is supposed to produce in a certain period. It involves the choice between consumer goods, capital goods, or both. The how refers to how these goods and services are produced in the economy. $Q=f(L, K)$. Here L refers to labour

while K refers to capital. Output may be produced using labour alone, capital alone, or both labour and capital. Finally, the question of for whom involves the production of goods either for the poor, rich, or for all classes of the population.

In the production process, if you use more labour than capital, it is called a labour-intensive technique. On the other hand, in the production process, if you use more capital than labour, it is called a capital-intensive technique. Now we will discuss how the three fundamental problems of an economy are addressed by the capitalist, command, and mixed economies.

Capitalist economy: The capitalist economy is also called the market economy. A capitalist economy is one in which the means of production (such as land, labour, capital, and entrepreneurship) are controlled by private individuals or businessmen. As you know, the aim of a private individual businessman is to maximize profit. Capitalist economies include Switzerland, the United States, the United Kingdom, Sweden, Germany, and France. In a capitalist economy, 90% of the means of production are owned by only 10% of the population. It implies that income and wealth inequality is greater in capitalist countries. In a capitalist economy, the aim is to maximize profit, so they prefer to produce goods using technology. Capital is plentiful, and a capitalist country spends a significant amount of money on research and development. As a result, they produce innovative products.

Features of Capitalist Economy

- ❖ **Private property:** Property rights are strong in a capitalist economy. You can buy and sell the property.
- ❖ **Profit motive:** As 90% of wealth is confined to only 10% of people, their objective is to maximize their profit by minimizing the cost
- ❖ **Market Mechanism:** The prices of the product is determined through demand and supply.
- ❖ **Government role:** Government plays a limited role by maintaining police, defense, infrastructure development, law and order, etc.
- ❖ **Decision-making:** Private people or businessmen used to make decisions.
- ❖ **Inequality:** Inequality is more
- ❖ **R&D expenditure:** They spend a lot of money on R&D. As a result, they minimize their cost of production
- ❖ **Nature of competition:** Competition is more

Command Economy

Command economy can be defined as an economy where the factors of production such as land, labour, capital and enterprise are controlled by the state. It is an economic system where the state controls the means of production, consumption, and allocation of

resources. This form of economic system does not have a profit motive. It has a welfare motive. It is also referred to as the planned economy.

Features of a Command Economy

- ❖ **Ownership of Resources:** The state is the owner of the means of production. Private property is discouraged. You are not encouraged to hold private property. You can't buy or sell private property.
- ❖ **Decision-Making:** Government makes all the decisions
- ❖ **Motive of Production:** As the government is the owner of the means of production, the government objective is not profit motive, but rather it is welfare-oriented.
- ❖ **Price Control:** The government sets prices. Price is not determined by the market forces such as demand and supply. The government decides the wage rate.
- ❖ **Availability of Goods:** Goods are not plentifully available. Consumers have fewer options as goods are limited.
- ❖ **Inequality:** Inequality is less. It is due to the government deciding wages.
- ❖ **R&D Expenditure:** They spend less on R&D. As a result, the cost of production is high. As the wage rate is the same, it does not provide any incentive to work more. Productivity falls, inefficiency increases. State-owned enterprises are poorly run. As a result, there is a decline in profit.
- ❖ **Nature of Competition:** It is limited as state-run enterprises are poorly run.

Mixed Economy: A mixed economy is one in which the means of production (such as land, labour, capital, and entrepreneurship) are controlled by both the state and private individuals. The government's aim is welfare-oriented, while private people's aim is profit-oriented. A mixed economy combines the best features of capitalist and command economies. India is the perfect example of a mixed economy. Both the private and public sectors move hand in hand. In this system, both the government and the private sector play important roles in decision-making.

Key Features

- ❖ **Ownership of Resources:** Both the government and private individuals are the owners of the factors of production.
- ❖ **Coexistence of Public and private Sector:** Both government and private companies operate together.
- ❖ **Market Mechanism:** Prices are determined through demand and supply. But the government intervenes and regulates the prices of goods and services whenever it is required.
- ❖ **Social Welfare Measures:** The government prioritizes rural areas, which are often overlooked by the private sector.
- ❖ **Nature of Competition:** Competition is more prevalent than in a command economy.
- ❖ **Inequality:** It is less than in a command economy.

- ❖ **Motive of Production:** The government's objective is the welfare motive, and the private people's objective is the profit Motive.

	Capitalist economy	Command Economy	Mixed Economy
What to Produce	Market forces i.e. supply and demand decides which goods to produce and how much to produce. Firms produce the goods as per the needs of the consumer	Government decides the goods to be produced and what quantity	Both market forces and government decides the goods and quantity to be produced.
How to Produce	Whether the goods to be produced by using labour or machinery or both. In capitalist economy goods are produced by using more machinery rather than labour. Wage rate is high in capitalist economy. Their objectives are to minimize cost and maximize profit.	Goods are produced by using more of labour force as government objective is to provide employment	Goods are produced by both labour and capital. While producing goods government will use more of labour and less capital as government objectives is employment generation. On the other hand, private people will use more of machinery and less labour as their objective is maximization of profit not employment generation
For whom to Produce	Consumption depends on individual disposable income. Production of goods goes to those who are willingness and ability to pay	Government used to produce goods for all the sections of the society. Government is the owner of factor of production	Private people used to focus on high net worth individual as their objective is maximization of profit not maximization of welfare. Government used to focus the low-income group as government objective is maximization of welfare not maximization of profit

In contemporary world, all the economies are some kinds of mixed economies. Both market and command economies do not exist in pure form. Only the degree differs.

Review Questions

1. Analyze the fundamental problems of an economy and compare how capitalist, command, and mixed economies address these issues.
2. Elucidate as how does three types of Economic Systems were able to tackle the central problems of the economy?
3. Analyze the importance of the Production Possibility Curve (PPC) in identifying efficient combinations inside the frontier and unattainable combinations outside the frontier.

Demand and Supply Analysis

“Climbing to the top demands strength, whether it is to the top of Mount Everest or to the top of your career.”

— A. P. J. Abdul Kalam

Demand

Why does Ice cream become costlier during a heat wave compared to the rest of the year?

Why are roses more expensive on February 14 than at other times of the year?

Why do football players earn less than cricket players?

All the above-mentioned questions are answered by demand. It explains the relationship between the price of a commodity and the quantity consumers are willing and able to buy at different prices, given that other factors remain unchanged or constant. Demand will prevail if the following three criteria are fulfilled

(i) Willingness to buy

(ii) Willingness to pay

(iii) Ability to pay

For demand to exist all three specified conditions must be fulfilled; failing to meet any one of them means demand cannot be claimed. Demand in economics is crucial for understanding consumer behavior in the marketplace, which in turn affects prices, production, and resource allocation. It reveals not only what consumers desire and their willingness to pay but also guides producers in making informed decisions regarding the type and quantity of goods to produce, as well as the timing of production, thus preventing overproduction and shortages. Given limited resources, demand analysis helps allocate

them efficiently by prioritizing high-demand items, signaling producers to focus on specific goods and services. Additionally, demand analysis plays a vital role in forecasting future trends, enabling predictions of consumer behavior, market growth, and investment opportunities.

Quantity Demand:

Quantity demanded is the amount of a good or service people are willing to buy at a given price.

What is the Law of Demand?

The Law of Demand states that demand for a product varies inversely with price, assuming other factors remain unchanged.

$$Q_d = f(P)$$

Q_d is quantity demanded, and P is price.

The price here is an independent variable, whereas the Q_d is a dependent variable. Price variation will definitely lead to a change in quantity demanded, but the opposite is not true.

When Price changes, Q_d also changes. Price changes mean increase or decrease in price.

When $P \uparrow Q_d \downarrow$ and When $P \downarrow Q_d \uparrow$

There exists a negative or inverse relationship between price and quantity demanded. An increase in the price of a good reduces demand for it, while a decrease in price increases its demand. Note in demand equation the independent variable changes. The dependent variable remains same. For instance, Q_d is dependent variable and Price (P) is independent variable.

If we say $P = f(Q_d)$. It does not fall under the Law of Demand.

Other factors remaining the same means.....

- ❖ Consumer's income remains same during a period of time
- ❖ Taste and preferences of the consumer remain unchanged.
- ❖ Prices of related goods (substitutes or complementary goods) remain same
- ❖ No expectations regarding future changes in prices and income
- ❖ Number of buyers in the market remains the same.

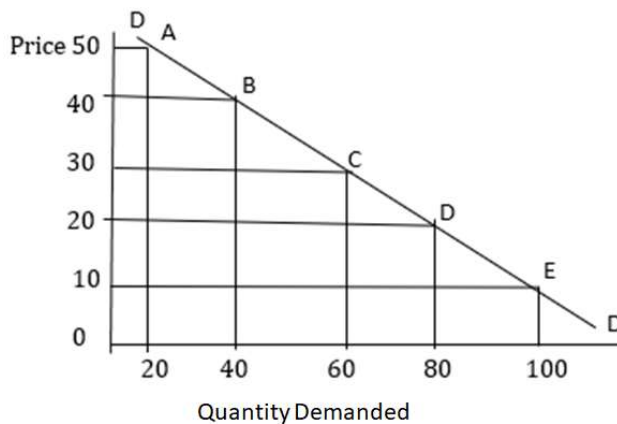
Demand can be expressed through a demand schedule and a demand curve.

Demand schedule: The quantity of goods consumers are willing to purchase at different prices; when presented in tabular form, it is specifically called a demand schedule.

Price (Rs)	Quantity Demanded of Ice cream
10	100
20	80
30	60
40	40
50	20

Demand curve: The graphical representation of a demand schedule is called a demand curve.

In a scientific experiment, the dependent variable is plotted on the Y axis, and the independent variable on the X axis. In the case of economics, the dependent variable is plotted on the X axis and the independent variable along the Y axis. In both cases, the downward sloping demand curve is obtained. In economics, the quantity demanded is presented on the X axis, and the price on the Y axis, for simplicity.



Quantity demanded is presented in X axis and price is presented in Y axis. When price of Ice cream is Rs 10, a consumer buys 100 Ice creams. When price increased to Rs. 20, consumer buys 80 Ice creams. Similarly, when price is Rs. 50, consumer purchases 20 Ice creams. If different quantities and different prices are connected, we will obtain a demand curve. For instance, connecting points from A to E will give us demand curve. In this figure 'DD' represents the demand curve. From the figure it is seen that the quantity demanded and price of ice creams have an inverse relationship. Therefore, if the price rises, quantity demanded falls, and if the price falls, quantity demanded rises.

Suppose the initial price is Rs 50 and consumer demands 20 ice creams. If price falls to Rs 40, consumer demand increased to 40. Likewise, if price falls to Rs 10, consumer demand increased to 100 ice creams.

Why quantity demand increases when the price falls? This is due to following two reasons.

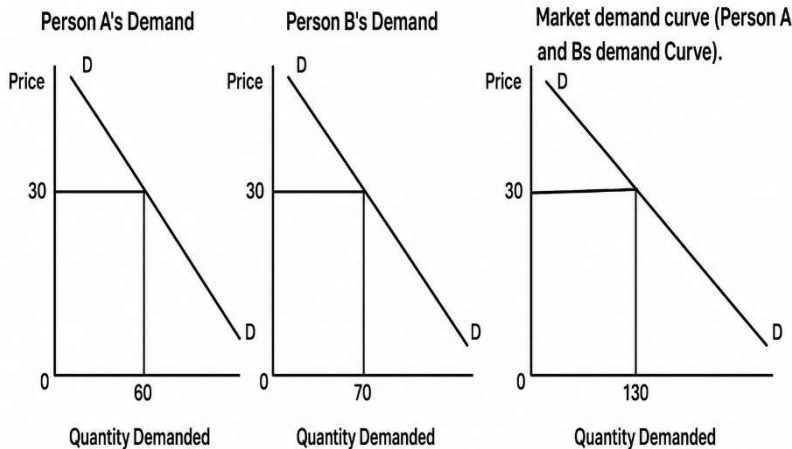
(i) The substitution effect of a price change- If Coca-Cola prices falls and Pepsi prices remain same; Coca-Cola becomes relatively cheaper as compared to Pepsi. Consumers will be interested to purchase more Coca-Colas it becomes cheaper. Hence, consumers will substitute Pepsi for Coca-Cola. This is referred to as the substitution effect resulting from a price change.

(ii) The income effect of a price change- Roshni earns Rs. 300 weekly from her part-time job in India. Rs 300 is Roshni's money income. Money income refers to the total amount of money a person earns during a specific time frame. In the scenario presented, if an ice cream costs Rs 100, Roshni can purchase 3 ice creams with her total income. However, if the price of ice cream decreases to Rs 60, Roshni's purchasing power increases, allowing her to buy 5 ice creams per week. Roshni's nominal income is constant at Rs 300 per week; however, a decrease in ice cream prices has enhanced her real income, which reflects her purchasing power. Consequently, the demand for ice cream is expected to rise due to the positive income effects associated with the price decrease.

Market Demand Curve

The market demand schedule refers to the summation of individual demand schedules of all the individuals in such a way that their demands will be combined horizontally. Contrary to the steepness shown in the individual demand schedule, the market demand schedule shows greater flatness. The reason behind it lies in the fact that the change in prices impacts market demand more as compared to individual demand. In order to construct a graph of market demand schedule, the process will be based on two persons only as the whole market. Let's assume that buyers 1 and 2 purchase 60 and 70 ice creams, then the market demand curve is 130 ice creams.

Note: Market demand curve is always flatter as compared to individual demand curve.



Change in quantity demanded

Changes in quantity demanded occur solely as a result of changes in price only, provided that all other factors affecting demand remain constant. A change in quantity demanded is defined as a shift along a demand curve. In the case of change in quantity demand, all non-price determinants have remained constant. The quantity demand affected due to change in price only. The non price factor remains same means (i) Income of the consumer remain same during a period of time (ii) Tastes and preferences of the consumer remains same (iii) Price of substitutes and complement goods have not changed (iv) Expectations of the consumer remains same (v) Number of buyers in the market remains same.

In simple terms, this kind of change is depicted graphically as a shift from one point to another along the demand curve. The term change in quantity demanded is associated with the rise or fall in the amount of a particular good or service that is demanded, which is caused only because of a change in price.

Summary: So far as change in quantity demand is concerned, all other determinants of demand (Non-Price Factors) remain constant. Changes in quantity demanded arise due to changes in price alone, assuming that all the other determinants of demand have remained constant.

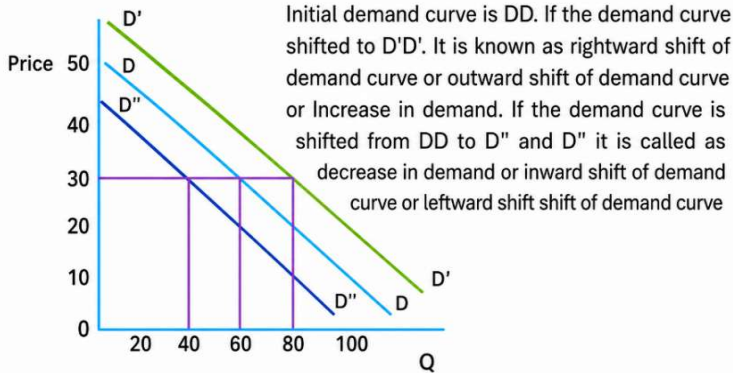
What will be the effect if there is a change in other non-price determinants? This is termed as change in demand. Other non-price determinants means excluding price.

Change in Demand or Shift in demand

A change or shift in demand means an increase or decrease in demand. The demand curve will shift upward or downward due to change in demand. Here, non-price factors are responsible for a change in demand, while the price itself remains unchanged. In other words, change in demand occurs when other factors are not remaining same (i.e. changes).

Note: other factors means other than price. Other factors may not remain same in the real

world during a period of time. Change in demand occurs if one of the determinants of demand changes (here price remains same).



Rightward shift in the demand curve often referred to as an outward shift or increase in demand. In this figure, demand curve shifted from DD to D'D'. Leftward shift in the demand curve, also referred to as an inward shift, signifies a decrease in demand. In this figure, demand curve shifted from DD to D''D''.

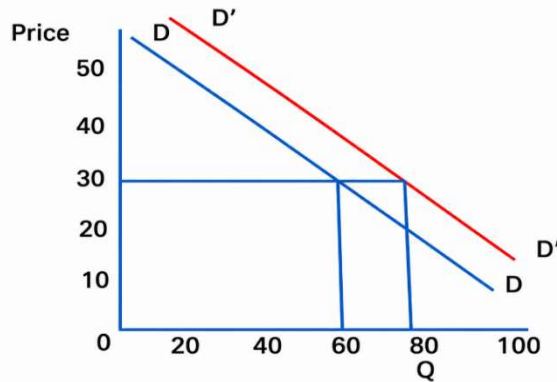
One of the determinants of demand changes means other factor changes. These factors include

- (i) Changes in the income of the consumer
- (ii) Changes in the taste and preferences of the consumer
- (iii) Changes in the prices of related goods (substitute and complementary goods)
- (iv) Changes in the expectations regarding future changes in prices and income
- (v) Changes in the number of buyers

Note: The presence of non-price factor will shift the demand curve upward or downward. Change in demand takes place when non-price factors vary and price remains constant.

Changes in Income of the consumer –Changes in income of the consumer means increase or decrease in income of the consumer. With an increase in consumer income, individuals are able to purchase more ice creams with given price. The demand curve shifts to the right from DD to D'D'. D is the market demand curve for 'Ice creams' with a given level of money income.

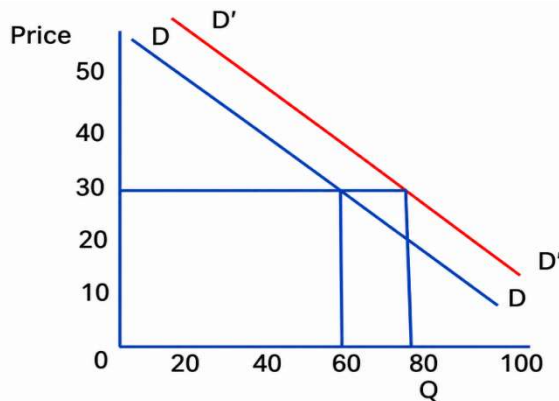
In the real world, an increase in income does not uniformly lead to higher purchases of goods; it depends on whether the goods are normal or inferior



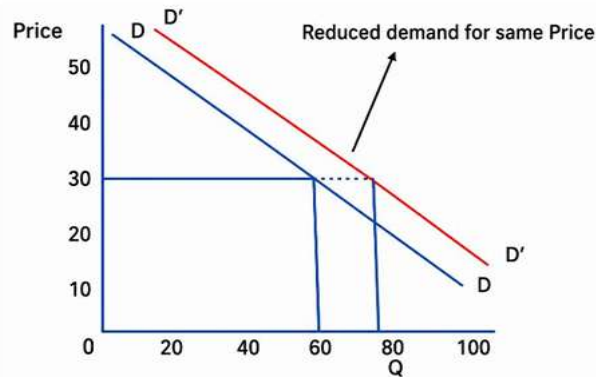
For a Normal Goods

Goods are termed as Normal goods if an increase in income results in the increase of demand for such goods. Most of the goods fall under the category of Normal goods. Some examples of Normal goods are branded dresses, LED television, car, house, pizza, and so on.

If your income is Rs. 10,000 then you will be able to buy 60 ice creams at the price of Rs. 30. But after the income becomes Rs. 20,000, you will be able to purchase more ice creams; thus, leading to the rightward shift of the demand curve from DD to $D'D'$.



For Inferior goods: A good is classified as an Inferior Good when an increase in income leads to a decrease in the demand for that good. Examples: unbranded perfumes, used furniture, used clothing, and low-quality bread. As the income of the consumer increases, consumers moved to more nutritious food and hence the demand for low-quality products falls.

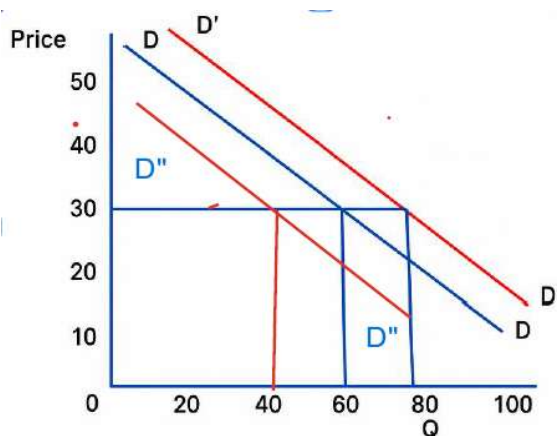


Assume that X is a poor person and his/her income is Rs. 10,000 per month. When low-quality bread is priced at Rs. 30, X demands 80 units. The initial demand curve of poor person is $D'D'$. With increase in income to Rs. 20,000 per month the demand for low-quality bread decreases from 80 units to 60 units. In this figure, X is buying 20 units less of low-quality bread with increase his/her income. With increase in poor persons income, the demand curve shifted to the left from $D'D'$ to DD . Now the question arises why the consumer buys less with an increase in income. The consumer is transitioning from a preference for low quality to high quality bread.

Note: In case of inferior goods, consumer will buy less with increase in income whereas, in case of normal goods consumer will buy more with an increase in income.

Increase or decrease in Number of Buyers in the market

An increase in the number of buyers in the market leads to a rightward shift of the demand curve. Increase in buyers leads to higher demand for goods, shifting the demand curve from DD to $D'D'$. Conversely, a decrease in the number of buyers results in a leftward shift of the demand curve from DD to $D''D''$.

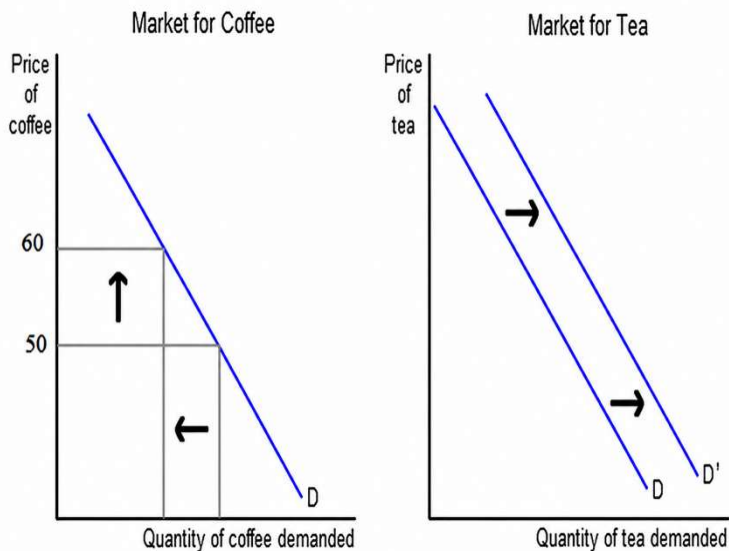


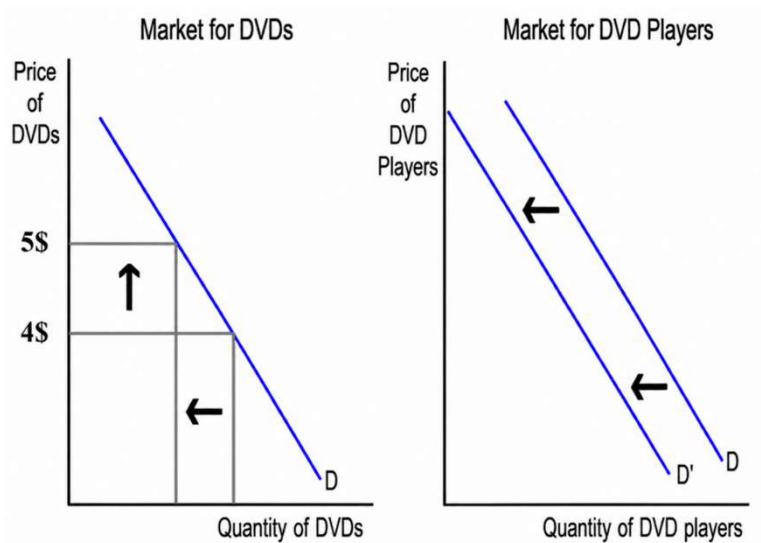
Changes in the Prices of related goods (Substitute and Complementary goods)

Substitute Goods - an increase in the price of one good, result in an increase in the demand for another good. For example, Coffee and Tea are treated as substitute goods. If the tea price will increase, then demand for tea will fall and demand for coffee will increase. Similarly, substitute goods are Pen and pencil, coke and Pepsi, SBI and PNB bank, IIMA and IIMB, Amul paneer vs Heritage Paneer, Taj Krishna vs Taj Banjara hotel etc.

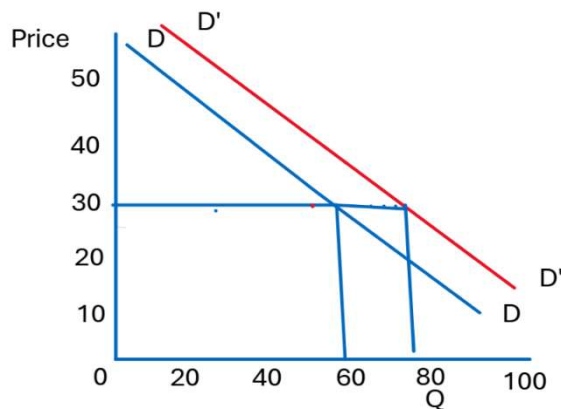
Complementary Goods-complementary goods are those goods which are consumed together. Example includes Tea and sugar, Car and Petrol, coke and pizza, milk and cookies, DVD and DVD Player, computer hardware and computer software, pen and ink, airline tickets and rental cars, rice and curry, and bread and butter. Remember complementary products are consumed together. An increase in the price of one results in a decrease in the demand for the other. For instance, if computer hardware price will increase don't you think computer software demand will fall.

Price of Coffee Rises: Substitute Good of Tea

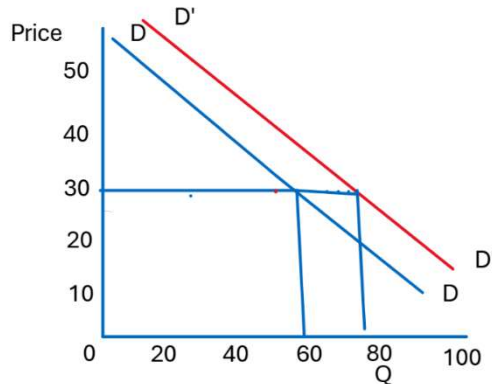


Price of DVDs Rises: Complementary Good of DVD Player**❖ Changes in tastes and preferences of the consumer-**

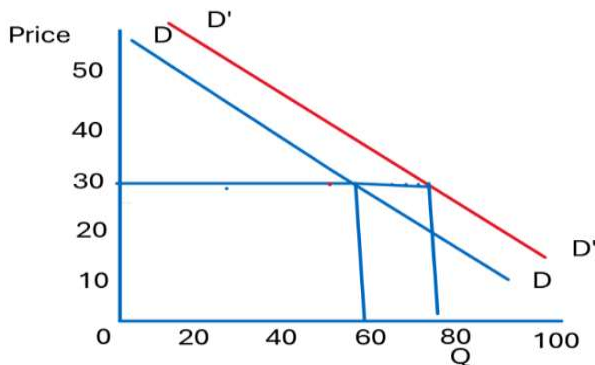
Taste is nothing more than your likes and dislikes of a consumer. A change in taste for a particular good would shift that good's demand curve rightward. Increase in taste and preferences will shift the demand curve towards the right and vice versa.

**❖ Changes in consumer expectations related to future prices and income changes.**

A change in consumer income expectation can shift the demand curve. For example, a consumer who heard about a pay hike, the demand curve will shift to the right from DD to D'D'.



Changes in price expectations by consumer can lead to a shift in the demand curve. If you will expect the price of onion will increase significantly tomorrow you will buy more onion today. In this context, demand curve will shift towards the right from DD to D'D'.



Expectations

- Higher expected future prices will lead to increased current demand.
- Lower expected future prices will lead to a reduction in current demand.
- A higher expected future income will increase current demand for normal goods.
- A lower expected future income will reduce current demand for normal goods.

Changes in the number or composition of consumers-

Changes in population composition can influence demand, even if total population size is constant. For instance, an increase in the teenage demographic may boost pizza demand, while a rise in the student population could enhance fast food demand.

Exceptions of the Law of demand

In this context, quantity demanded remains unchanged despite price increases. It is opposite to the law of demand, which typically states an inverse relationship between price and quantity demanded under ceteris paribus conditions.

❖ Prestige goods/Veblen goods/snob effect:

Veblen goods are premium products, luxury goods. High-price goods serve as status symbols, making them more desirable among high-income consumers. For example, diamonds, high-end cars, premium apartments, etc., may be prestige goods because the higher the prestige value of diamonds, and high-end cars, premium apartments, the higher the desirability. Veblen goods have an upward-sloping demand curve. Rich people may not buy diamonds when the diamond price falls. Veblen goods are like Giffen goods. Veblen goods relate to expensive luxury items where higher prices can make them more desirable. Veblen good refers to prestige goods.



❖ **Giffen goods:** This refers to a specific category of inferior goods known as Giffen goods, identified by economist Robert Giffen. It is a special type of inferior goods on which consumer spends a large proportion of income when its price rises, quantity demand also increases, and vice versa. The concept of Giffen goods pertains to low-income, nonluxury products that have limited close substitutes. Even though a few close substitutes are available but their prices are very high and poor people can't afford them.

Exceptions to the Law of Demand are illustrated by Giffen goods, which experience an increase in demand as their prices rise, defying typical economic principles. Examples of Giffen goods include bread, rice, wheat, and bajra where poor people used to spend a large proportion of their money on these essential items.

Even though the prices of these products increase poor people have no other option to go for substitutes as substitute product prices are very high. This tendency is found for low-income groups of people. In the context of Giffen goods, the law of demand is invalid, resulting in an upward-sloping demand curve. In the case of Giffen goods, an increase in price may lead to an increase in the quantity demanded, resulting in an upward-sloping demand curve.

All Giffen goods are classified as inferior goods, but the reverse does not hold true.

Note: inferior goods have many substitutes but Giffen goods have few substitutes. Even though few substitutes are available for Giffen goods it is not affordable.

- ❖ **Speculations:** Consumers' expectations regarding future prices influence their current purchasing behavior. If they anticipate price increases, they tend to buy more despite high prices. Conversely, if they expect prices to drop, they may buy less even when prices are low.
- ❖ **Ignorance of the customer:** If consumers are unaware of price increases, they may continue purchasing the same or more, leading to unexpected demand levels. This illustrates how a lack of information can influence consumer decisions.
- ❖ **Necessary goods:** Necessary goods deviate from the law of demand, as their sales persist despite rising prices. These essential items, such as food and medicine, are crucial for daily life. Even though price will increase consumers will buy the same amount
- ❖ **Psychological bias of customer:** Different customers have different perceptions about the price of the product. Some customers have the perception that a lower price means bad quality which may not be true always. Therefore, even low-price quantity demand may not increase.
- ❖ **Brand loyalty:** Buyers are biased towards a particular brand and will always choose it regardless of price or competition. In this case customer may prefer it over other competing product.
- ❖ **Emergency Situation:** during emergency situations such as floods, wars, earthquakes, and the COVID-19 lockdown period, the availability of goods becomes scarce and uncertain. Therefore, consumers prefer to store large quantities of goods regardless of the price
- ❖ **Addicted items-** Whatever might be the price, the addicted buyer will buy the product. Example cigarettes, drugs, wines etc.
- ❖ **Demonstration effect (Bandwagon effect):** Bandwagon effect is a psychological phenomenon where individuals engage in behaviors or adopt beliefs primarily because they observe others doing so. This effect is prominently observed in political contexts and within consumer behavior, influencing decisions based on popularity rather than individual assessment.

Supply

Demand and supply are two interconnected concepts that reflect different perspectives in the marketplace. Both the supply and demand curves move in opposite directions. Demand focuses on the consumer's viewpoint, while supply emphasizes the producer's standpoint. Supply, like demand, illustrates the relationship between price and the quantity supplied. It is the quantity of a good sellers wish to sell at each conceivable price. Supply can be represented in two forms: the supply schedule and the supply curve.

A decline in demand signals producers to decrease output, while effective supply management is vital for strategic planning and decision-making in organizations. Managers apply supply data to formulate pricing strategies, adjusting prices based on demand and inventory levels. An understanding of supply dynamics is essential for modifying production in response to market fluctuations, such as increased input costs or material scarcity, which may lead to reduced output or alternative sourcing methods. Furthermore, supply analysis aids in inventory management by balancing excess stock, which incurs storage costs and waste, against insufficient stock that can cause stockouts and lost sales. Supply is defined as the relationship between the price of a good and the quantity that producers are willing and able to sell at different prices, assuming other factors remain the same.

Other things remaining constant (or Ceteris Paribus) means....

- (i) cost of production (Inputs prices such as raw material price, crude oil price, wage rate, interest rate etc.) remaining the same.
- (ii) No change in climatic conditions
- (iii) No changes in external factors
- (iv) No changes in producer's expectations
- (v) Technology remains the same
- (vi) No change in Government policy
- (vii) No change in the number of producers in the market

Quantity Supplied:

It refers to the specific quantity of a product that sellers are ready to sell at a particular price.

Law of Supply:

The Law of Supply states that other things remaining the same, the quantity supplied is directly related to its price. Lower price leads to lesser quantity supplied and vice versa. So, when prices are high, the potential to make money increases, which encourages them to produce and sell more. On the flip side, when prices fall, profits shrink. A direct and positive relationship exists between the price of a good and the quantity supplied in a given time period, assuming all other factors remain constant.

$$Q_s = f(P)$$

$$Q_s = \text{quantity Supplied}$$

$$P = \text{Price}$$

When Price changes, Q_s also changes. Price changes mean price increases or decreases.

When $P \uparrow Q_s \uparrow$ and When $P \downarrow Q_s \downarrow$

There exists appositive or direct relationship between price and quantity supplied other things (i.e factors) remaining constant. An increase in the price of a good increases its quantity supplied and an decrease in the price of a good decreases its quantity supplied

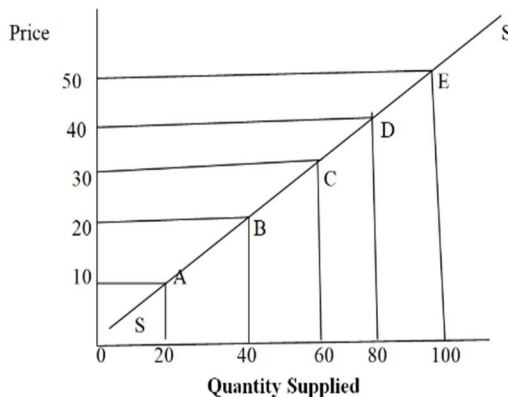
$$P = f(Q_s) \text{ Not under the Law of Supply}$$

Supply Schedule: listing the various quantities that producers/suppliers are willing to produce at different price. Supply schedule refers to a representation of supply information in tabular form. Note: It is the tabular representation of plotting various prices with respect to various quantities.

Price (Rs)	Quantity Supplied of Ice cream
10	20
20	40
30	60
40	80
50	100

Supply curve: The supply curve is the graphical representation of the supply schedule, illustrating a direct or positive relationship between the price of a good and the quantity supplied over a specific time period, assuming other factors remain constant (ceteris paribus). Supply curve is the graphical representation of plotting various quantities with respect to different prices.

X axis shows quantity supplied, and Y axis shows its price. When price increases, quantity supplied increases and vice versa. When the price was Rs. 10, the producer supplied is 20 Ice



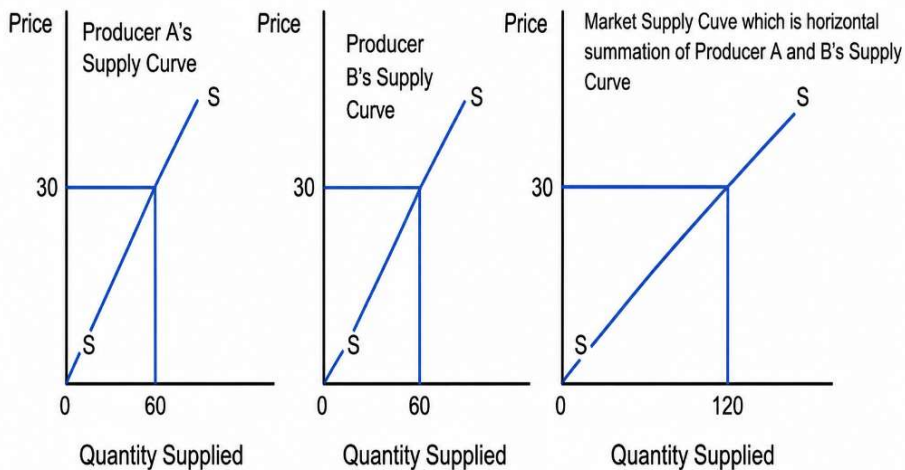
creams, and when the price increased to Rs. 20, the producer supplied 40 Ice creams. Similarly, when price of Ice cream increased to Rs 30 the producer is willing and able to supply 60 Ice creams. The producer is willing and able to supply 100 Ice creams when the price of Ice cream increased to Rs 50. If we will join the points A, B, C, D and E then we will get supply curve of Ice cream. SS is the supply curve which is upward slopping.

It indicates that there exists positive or direct relationship between price and quantity supplied of Ice creams. Conversely, when price was Rs 50, the producers are willing and able to supply 100 Ice creams. When price falls to Rs 40, the producers are willing and able to supply 80 Ice creams. Similarly, when price falls to Rs 10 the producers are willing and able to supply only 20 Ice creams. Note: When price increases, we are moving upward on the supply curve and when price decreases, we are moving downward on the supply curve. The producer is moving A to B to ...E with increase in price and moving E to D to ...A will decrease in price. We are moving along the same supply curve and it is with respect to price changes.

Market supply Curve

The market supply curve is created by the horizontal summation of the supply curves of all producers.

Let's assume that there are two producers in the market and two producers supplies the entire market. We have considered two producers for simplicity. Producer A supplied 60 Ice cream and producer B supplies 60 Ice creams. Market supply curve is a horizontal summation of producer A and B's supply curve that $60+60=120$ Ice creams.



Note: Market supply curve is flatter as it is horizontal summation of individual supply curve.

Change in Quantity Supply: Movement on or Along

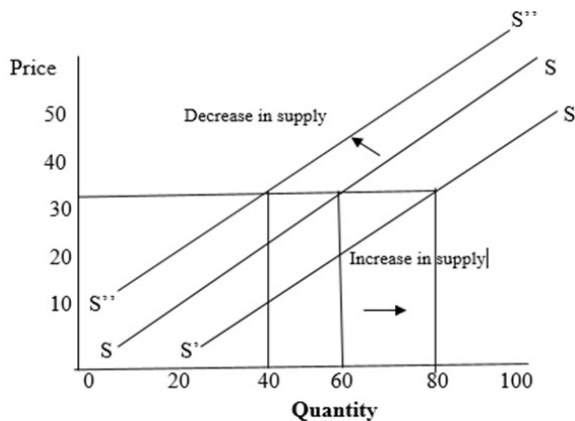
Change in quantity supplied occurs due to a price change of the good while other factors (i.e. non price factors) remain constant. Here, producer is moving along the same supply curve. Producer moves from one point to another point along the supply curve. This

happens due to change in price only. When we are talking about change in price only it implies increase or decrease in price. Other factors i.e other determinants of demand remains same.

Change in Supply: Movement Off (Increase/decrease in supply).

Shift of a supply curve is known as change in supply. Change in supply means increase or decrease in supply. Change of a supply curve occurs leftward or rightward due to changes in determinants of supply other than price. **Note:** Non-price factors shift the supply curve. Here, the price is remaining same.

When producers are optimistic, they increase production, shifting the supply curve to the right. Conversely, pessimism leads to reduced production and a leftward shift of the supply curve.



Let's assume that initial supply curve is SS . Increase in supply means the supply curve shifted towards the right from SS to $S'S'$. The supply curve will shift towards the right when there is increase in producer's optimism. On the other hand, the supply curve will shift towards the left when producers are pessimistic about their return. Note: Rightward shift of the supply curve indicates an outward or increase in supply. When we are talking about change that implies increase or decrease.

Conversely, the leftward shift of supply curve indicates decrease in supply or inward shift of supply curve. Decrease in supply means the supply curve shifted from SS to $S''S''$. Decrease in supply means the producer will produce less as producer's becomes pessimistic. Change or Shift of supply occurs because of changes in one of the determinants of supply other than price. Here, the price remains the same, but other factors change. The change or shift of supply curve occurs because of following reasons

(i) Changes in the cost of production (Inputs prices such as raw material price, crude oil price, wage rate, interest rate)

(ii) Changes in climatic conditions

(iii) Changes in external factors (which is beyond the control of any organizations)

(iv) Changes in producer's expectations

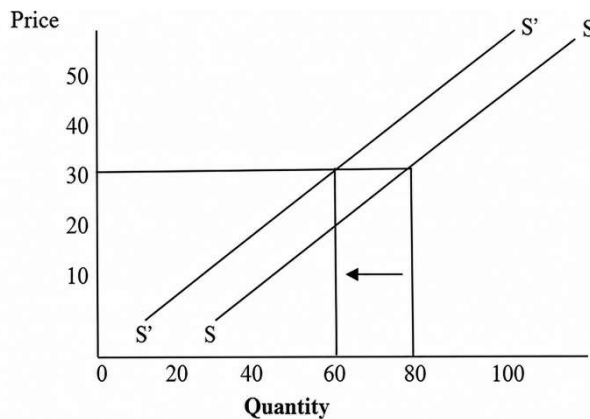
(v) Changes in technology

(vi) Changes in Government policy (Favorable government policy such as reduction of corporate tax and increase in subsidy. Unfavorable government policies such as an increase in corporate tax and reduction of subsidy.

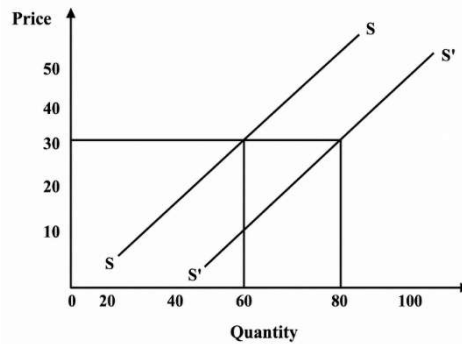
(vii) the number of producers in the market.

Increase in cost of production (Inputs prices)

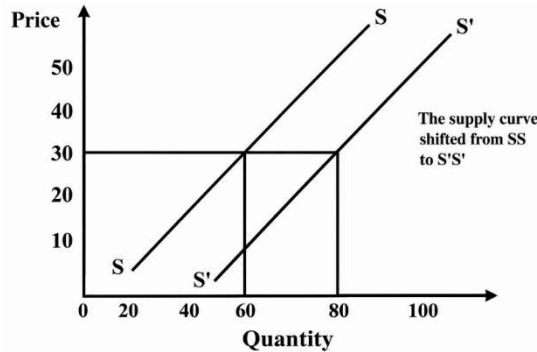
An increase in production costs, such as rising wage rates, raw material prices, and interest rates, leads to reduced profitability for producers. This decline in profit diminishes producers' optimism and results in decreased production. Consequently, the supply curve shifts to the left. For example, if producers initially supplied 80 ice creams at a price of Rs. 30 and then face increased production costs, they may reduce output to 60 units at the same price. The initial supply curve is represented as SS, while the decreased optimism shifts the supply curve to S'S'.



Technological advancements lead to reduced production costs and increased profitability, fostering greater optimism among producers. Consequently, this positive outlook prompts producers to hire additional labor and invest in more machinery to boost production levels. As a result, the supply curve shifts to the right from SS to S'S'.

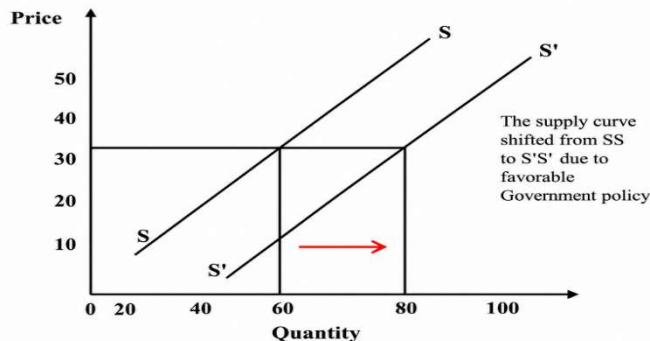


Increase in the number of producers in the market: An increase in the number of producers leads to a rightward shift in the market supply curve, represented as a movement from SS to S'S'. This shift indicates that more producers lead to higher total market output.

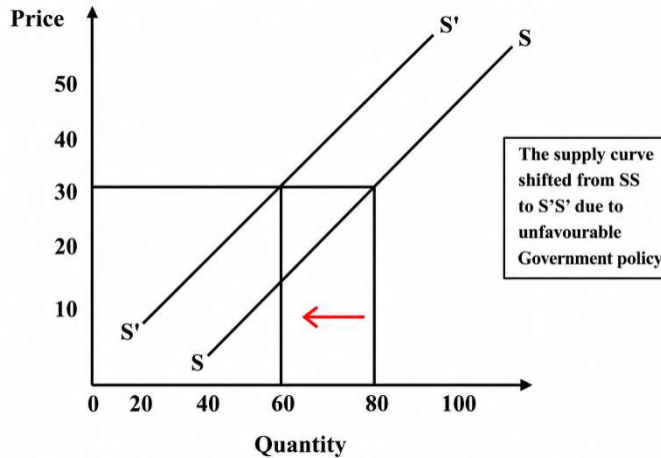


Changes in Government policy

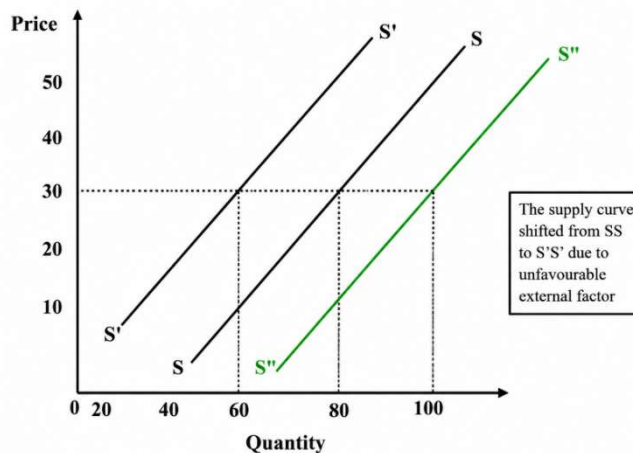
Changes in government policy will shift the supply curve rightward or leftward. Note: any favourable Government policy shift the supply curve towards right and any unfavourable Government policy will shift the supply curve towards the left. Favorable government policies such as reduction of corporate tax and increase in subsidies will shift the supply curve towards the right. Reduction of corporate tax increases firm's profitability. Hence producers become more optimistic.



Unfavorable government policies, such as an increase in corporate tax and reduction of subsidies, shift the supply curve towards the left. An increase in corporate tax reduces firm's profit. Producer optimism declines due to a reduction in output. Earlier, the production was 80 units, but due to unfavorable policy changes, it now produces only 60 units.



Changes in external factors Changes in external factors influence a producer's optimism, resulting in shifts of the supply curve either to the left or right. External factors, which are outside the control of organizations, can adversely impact producer confidence. For instance, events like the COVID-19 pandemic, terrorist attacks, US-Israel-Iran war contribute to a decrease in optimism, leading to a leftward shift of the supply curve from SS to $S'S'$. On the other hand, in case of favourable external factors, such as a reduction in crude oil prices, the supply curve will shift to the right, reducing the cost of production. In this case, the supply curve shifted from SS to $S''S''$.



RECAP

Changes in quantity demanded or changes in quantity supplied are with respect to the price change (price increase /decrease) only. Here, other factors are remaining constant.

Change/shift in demand or supply is with respect to non-price factors. Here, the price is remaining the same or constant. But, due to changes in other factors, the supply or demand curve shifts (I mean rightward or leftward).

The rightward shift of the demand curve signifies an outward shift, indicating an increase or expansion in demand. Conversely, a leftward shift denotes an inward shift, representing a decrease or contraction in demand.

Demand and supply create a market

Buyers and sellers perceive price differently. For consumers, a higher price is detrimental as it leads to a decrease in the quantity demanded. In contrast, producers benefit from higher prices as they can supply more, leading to an increase in the quantity supplied. This creates a conflict between the interests of consumers and producers, raising the question of how this discrepancy is addressed in the market.

Market Equilibrium

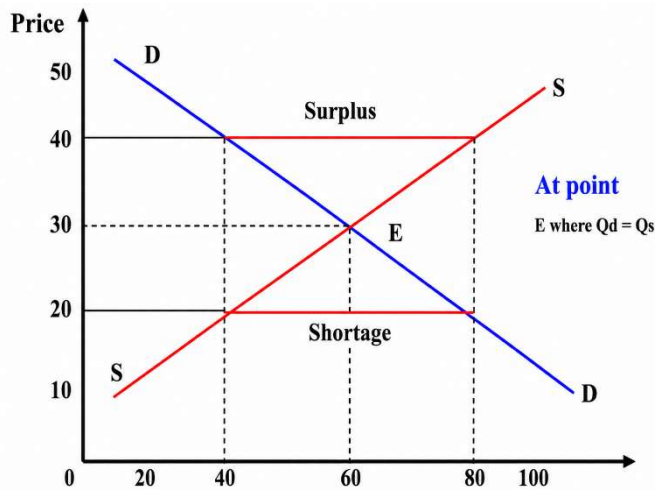
- Equilibrium is important because we never have equilibrium
- Action and reaction takes place between participants in disequilibrium
- The process to have equilibrium keeps the market going on
- Equilibrium: Action and Reaction Comes to a Halt, Tranquility in the Market

Equilibrium is a state of representing a balance of the forces of demand and supply. Equilibrium is a situation where Quantity demanded=quantity Supplied.

There are neither surplus nor shortages and whatever is produced is consumed. At equilibrium both producers and consumers are happy. In equilibrium, the intersection of demand and supply curves determines the equilibrium price and quantity.

When we are talking about demand, we are talking about consumer point of view. On the other hand, when we are talking about supply, we are talking about producers' point of view. Both demand and supply are opposite side of the same coin. With increase in prices producers are happy as their profit margins increases. Whereas, with increase in prices consumer are not happy as it is costlier for them and their purchasing power of money falls. Equilibrium is a situation where both buyers and sellers are happy.

Price (Rs)	Quantity Demanded of Ice creams (In million)	Quantity Supplied of Ice creams (In million)	Surplus/Shortage	Effects on Price
10	100	20	Shortage of 80	Rises
20	80	40	Shortage of 40	Rises
30	60	60	Equilibrium	Remains the same
40	40	80	Surplus of 40	Falls
50	20	100	Surplus of 80	Falls



Quantity demanded and supplied

The X-axis shows quantity demanded and quantity supplied of ice creams, while Y-axis shows its price. The demand curve DD is downward sloping. It indicates negative relationship between price and quantity demanded, assuming other factor remaining same. Conversely, the supply curve SS slopes upward. It reflects a positive relationship between price and quantity supplied assuming other factor remaining same.

At point E, both demand and supply curve intersect with each other. Their intersection determines equilibrium price and quantity. The equilibrium price is Rs 30 and equilibrium quantity is 60 million ice creams. Point E is the equilibrium point where quantity demanded is equal to quantity supplied of 60 million ice creams. At point E, price is stable. Any other points excluding E, price is not stable, price may increase or decrease. At that price

and quantity, the market clears, indicating equilibrium with no shortage or surplus and no price change pressure.

When ice cream prices rise to Rs.40, a surplus of 40 million ice creams occurs as producers supply 80 million while consumers demand only 40 million. This surplus give an indication to the supplier to reduce prices immediately to clear the sales. When price falls producers are pessimistic about production, subsequently decreasing quantity supplied. On the other hand, consumer purchases more ice creams as it becomes cheaper compared to earlier. In one hand, there is reduction of supply of ice creams and on the other hand there is increase in demand for ice creams which results increasing price. The price will continue to increase till the time equilibrium is achieved. When ice-creams prices falls to Rs 20, demand for ice creams rises to 80 million, but supply of ice creams is only 40 million, resulting in a 40-million ice cream shortage. This gap pushes prices higher as producers aim to maximize profits, while consumers seek more products. Prices will continue to rise until equilibrium is achieved where quantity demanded is equal to quantity supplied.

Note: Surplus reduces prices, while a shortage increases them; discrepancies between quantity demanded and quantity supplied trigger price adjustments.

Market Equilibrium (Algebraic Example)

$$\text{Let's assume } QD = 40 - P$$

$$\text{and } QS = 6 + P$$

$$\text{At Equilibrium, } QD = QS$$

$$\text{So ... } (40 - P) = (6 + P)$$

$$34 = 2P \text{ or } P = 34/2 \quad \text{so... } P = 17 \text{ (Equilibrium Price)}$$

To find equilibrium quantity, plug P into either the demand or supply equation.

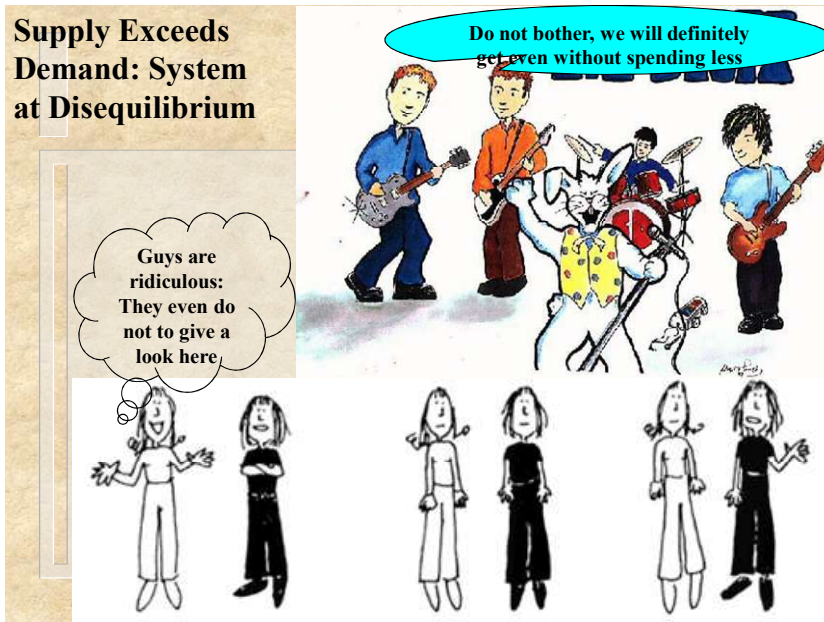
Equilibrium Quantity = $40 - 17 = 23$ (From demand equation)

or Equilibrium Quantity = $6 + 17 = 23$ (From supply equation)

Demand Exceeds Supply: The System is at Disequilibrium



Supply Exceeds Demand: System at Disequilibrium



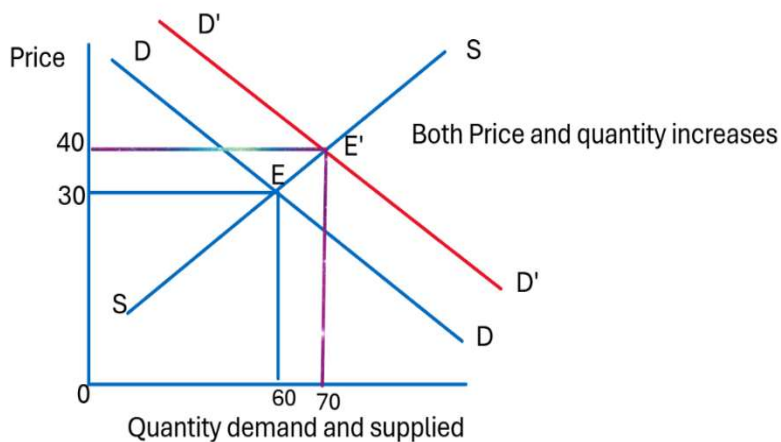
Equilibrium: Tranquility in the System

Quantity Demand equal to Quantity Supply

Lets go together



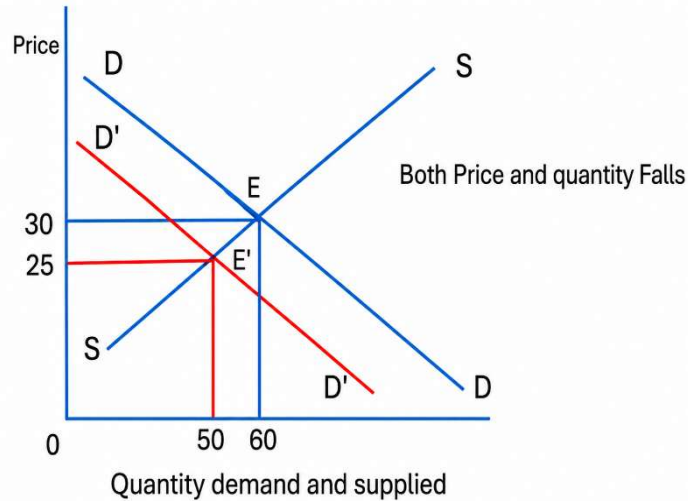
Demand Rises but supply remains the same (Shift of the Demand Curve Upwards):



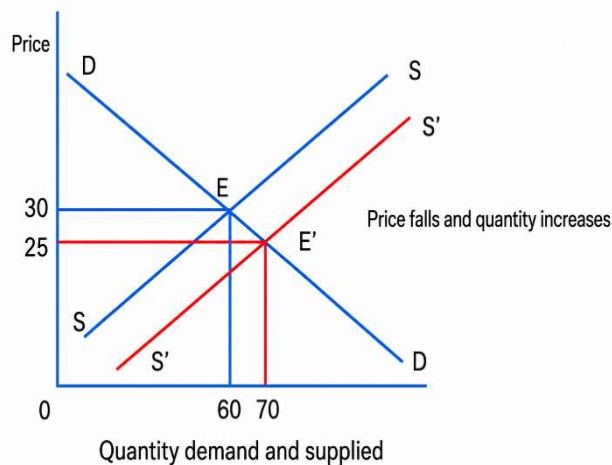
Initial demand curve (DD) and supply curve (SS) intersect at equilibrium point E, with an equilibrium price of 30 and quantity of 60 units. The supply curve remains the same. An increase in demand shifts the demand curve to D'D', altering the equilibrium to E', leading to higher prices and quantities. The extent of these increases depends on the shape and position of the demand curve.

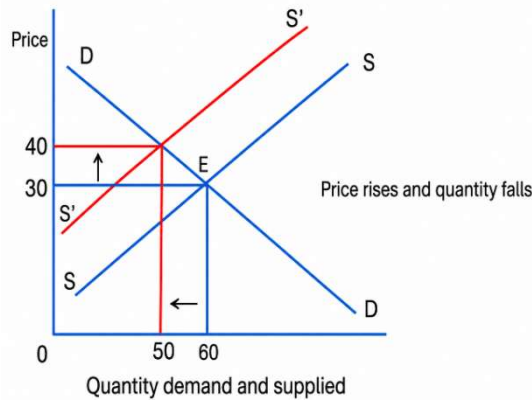
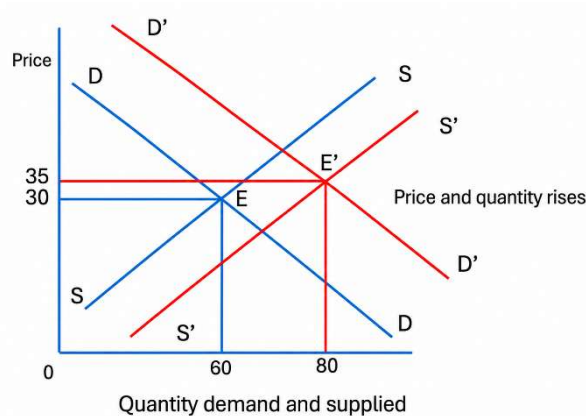
Demand Falls but supply remains the same (Shift of the Demand Curve Downwards):

The initial demand curve (DD) and supply curve (SS) intersect at equilibrium point E, with

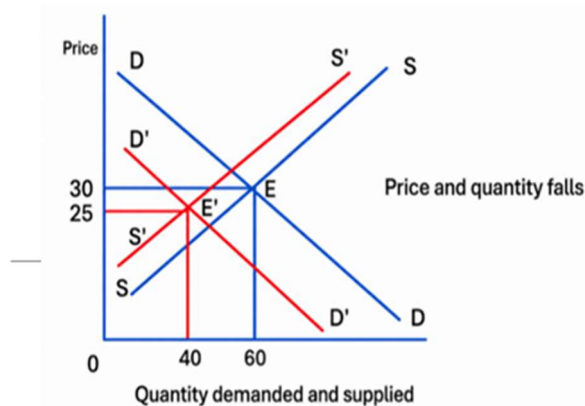


an equilibrium price of 30 and quantity of 60 units. A decrease in demand shifts the demand curve from DD to D'D', leading to a new equilibrium point E'. This decrease results in a reduction in both price and quantity, the extent of which is determined by the shape and position of the demand curve.

Supply rises but demand remains the same (Shift of Supply Curve to the Right):

Supply falls but demand remains the same (Shift of the Supply Curve to Left):**Simultaneous shift of Both Demand and Supply curve (Both Increases)**

Note: It is very difficult to say about the effects on price and quantity. It depends on height and position of both demand and supply curve

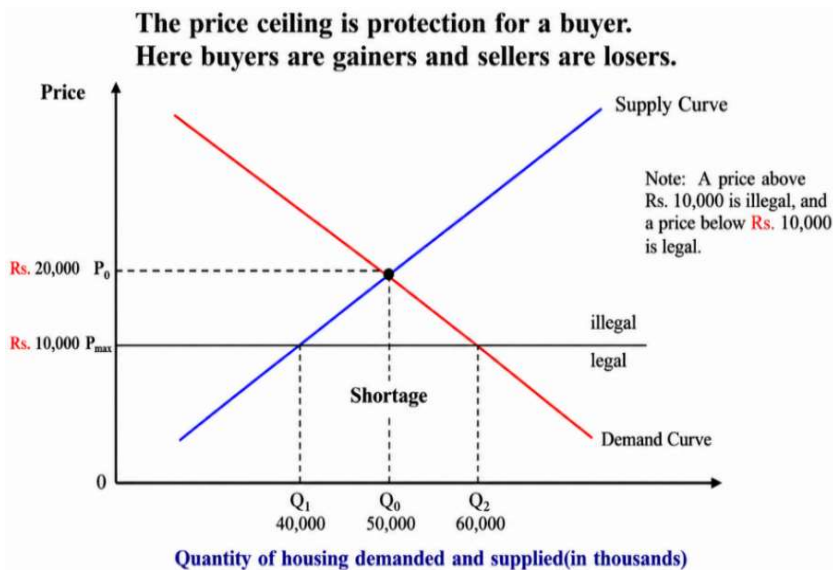
Simultaneous shift of both demand and supply curve (Both decreases)

Effects of Government intervention on Price control

Price ceiling

A price ceiling is a government-imposed limit on the maximum price that can be charged for a product or service. In this case, the government imposes a price ceiling on a good, preventing sellers from charging above that limit. Typically, it is set below the market equilibrium price. The main aim of price ceilings is to keep essential goods and services affordable. They are employed to mitigate the risk of sellers imposing excessively high prices. This is crucial for essential goods in daily life. Price ceilings, such as rent controls and regulated gasoline prices, were implemented during wartime and the 1970s energy crisis. To impact the price ceiling, it should be set below the equilibrium price. The price ceiling is protection for a buyer. Here buyers are gainers and sellers are losers.

P_0 and Q_0 represent the equilibrium price and quantity in the absence of government



Note: Price ceiling is always lower to the market equilibrium price.

regulations. Public officials may impose a price ceiling to maintain prices below the market clearing level. The government has determined that the price P_0 is excessively high and has set a maximum allowable ceiling price, P_{max} , that cannot be exceeded. Producers with higher costs will reduce their output due to the lower price, resulting in a decreased quantity supplied to Q_1 . The consumer desires to purchase quantity Q_2 at a low price, leading to excess demand since supply only meets quantity Q_1 . The excess demand is represented by the difference $Q_2 - Q_1$. Effective price ceilings lead to shortages and must be set below the equilibrium price to have an impact, as seen in examples like LPG subsidies and rent control. Producers face losses due to reduced prices, potentially leading some to exit the industry. While certain consumers benefit from lower prices, those unable to afford the goods at all will be worse off.

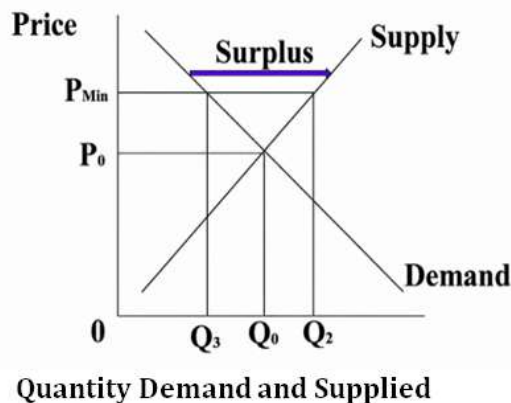
The market clearing price for houses is Rs. 20,000, with a demand and supply of 50,000 units. A rent ceiling of Rs. 10,000 increases demand to 60,000 units but only 40,000 houses are supplied, leading to a shortage of 20,000 units and potentially fostering a black market.

However, price ceilings can lead to various unintended consequences. For instance, they can create shortages if the ceiling is set below the equilibrium price, as suppliers may be unwilling or unable to provide the good or service at that price. Additionally, price ceilings can lead to inefficient allocation of resources, black markets, quality deterioration, and reduced incentives for producers to supply the goods or services in question. Overall, while price ceilings may help certain consumers in the short term by lowering prices, they can have negative consequences for markets in the long term.

Price Floor

A price floor is a government-imposed minimum limit on the price for a product or service, contrasting with a price ceiling. Price floors are generally established above the equilibrium price, where supply meets demand. The main aim of a price floor is to guarantee a minimum price for producers. It assists in guaranteeing sufficient income from goods or services. This is significant in industries where producers may struggle to cover their costs. Common examples of price floors include minimum wage laws, agricultural price supports, and minimum prices for certain commodities.

In some instances, the government sets prices above the market equilibrium level. It may



control agricultural prices to support farmers' incomes. This helps farmers earn more stable and better returns. A minimum price, known as a price floor, is set above the equilibrium price. The price floor is designed to maintain the price of a good above its market equilibrium, serving as a form of protection for producers.

The intersection of demand and supply curves establishes the equilibrium price and quantity. The Government believes that the equilibrium price is too low, offering no incentive for producers to produce. Setting the minimum price the sellers/farmers will get while producing and supplying the product. Now, the surplus, which is the difference

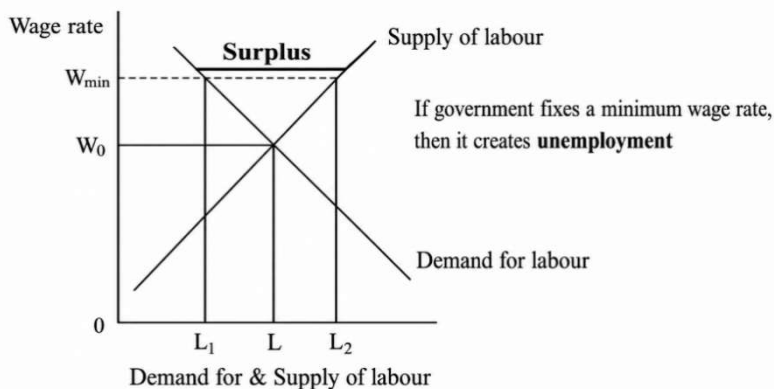
between the quantity supplied (Q_2) and the quantity demanded (Q_3), will be purchased by the government to maintain buffer stock. Here, the government is saying the farmers $P_{\min}$ is the minimum support price for your product and farmers are advised not to sell their product below the $P_{\min}$ price. When price is $P_{\min}$, the supply of food grains has increased compared to earlier. Higher price leads to more production of food grains. On the other hand, the demand for food grains has decreased with increase in price. Hence surplus of foodgrains is created which needs to be purchased by the government to maintain buffer stocks. But if the government won't purchase the food grains at committed price, then farmers will be forced to reduce the price.

Price floors can lead to unforeseen issues, similar to price ceilings. If the minimum price exceeds the equilibrium level, a surplus may occur. Producers may supply more goods than consumers are willing to buy at that price. This surplus can result in inefficient allocation of resources, storage costs, and potentially even government intervention to address the surplus. Overall, while price floors can provide stability and income support for producers, they can also create inefficiencies and distortions in markets if not carefully implemented.

Note: The Price floor is always higher to the market price. Price above $P_{\min}$ is legal and price below $P_{\min}$ is illegal.

Price Floor (Minimum Wage Law)

The supply curve of labor slopes upward as higher wage rates attracts more workers to offer their service. The demand curve for labor slopes downward, indicating that companies require less labor at higher wages. The equilibrium wage rate and employment is determined at the intersection of demand for and supply of labor. At wage rate W_0 , demand for labor is equals to supply of labor, resulting in employment OL . Labor union thinks that wage rate of W_0 is too low and it won't enhance the standard of living of labor. Labor unions advocate for a minimum wage above this equilibrium wage rate of W_0 . Wage rate above $W_{\min}$ is legal and the wage rate below $W_{\min}$ is illegal. With increasing wage rate to $W_{\min}$.



companies are very choosy, as the cost of hiring an employee has increased. On the other hand, higher wage rate attracts more labor to offer their services. In this scenario the

demand for labor is less compared to supply of labor, consequently causing long-term unemployment. The price floor must be above the equilibrium price in order to make it effective.

Elasticity of Demand and Supply and its Applications

Why do we Study Elasticity?

According to the “Law of Demand” price and quantity demanded move in opposite direction assuming other factors remaining the same. When price increases its quantity demanded falls and vice versa. Law of demand does not tell that how much quantity demanded of a product will fall if its price increases. Law of demand doesn't say how much the quantity demanded goes down and how much the quantity supplied goes up when its price goes up. Elasticity concepts can answer these. For example, if the coffee price increases by 10%, how much quantity demanded of coffee will fall and how much quantity supplied of coffee will increase. Elasticity is a critical concept for managers as it illustrates how demand varies with changes in price, income and other factors. By understanding elasticity, managers can determine optimal pricing strategies that enhance sales and profits, deciding whether to adjust prices based on customer responsiveness. Managers can also plan production levels based on demand sensitivity. It is useful in making marketing, discount, and advertising decisions. A manager can easily tell how much demand will drop and how much supply will rise for a product if they know a lot about elasticity.

Elasticity shows how one variable reacts to changes in another variable. Elasticity coefficients are usually expressed in a numerical form.

$$\text{Elasticity} = \frac{\% \text{ Change in response}}{\% \text{ change in stimulus}}$$

Elasticity of demand means expansion or contraction in demand. An expansion in demand indicates a rise in demand, while a contraction in demand means a decrease in demand. Elasticity quantifies the sensitivity of quantity demanded in response to changes in its determinants. Elasticity indicates how a change in one variable affects another variable. For example, a mobile phone discount enhances the sales of mobile phone. Elasticity of demand measures the responsiveness of quantity demanded to changes in various factors. These factors include product's price, the prices of related goods, consumer income and promotional activities.

Types of Elasticity of Demand:

$$Q_d = \int (\text{Price, Income, Price of related goods, Advertisement})$$

When the change in quantity demanded is purely attributed to the change in price, it is known as price elasticity of demand. The change in quantity demanded that results from a change in the income of consumers is known as income elasticity of demand. Cross-price elasticity of demand calculates the sensitivity in the quantity demanded of X when there is a change in the price of Y. Finally, promotional elasticity of demand is the sensitivity of the quantity demanded resulting from advertisement spending.

The Price Elasticity of Demand



Price Elasticity of Demand indicates how responsive the quantity demanded is to changes in the price of the commodity. Price elasticity of demand is represented by P_e . P refers to Price and e refers to elasticity.

$$P_e = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}} = \frac{\Delta Q}{\Delta P} * \frac{P}{Q}$$

ΔQ is the difference between current year and previous year quantity.

It can also be represented through $Q_1 - Q_0$

ΔP is the difference between current year and previous year price.

It can also be represented through $P_1 - P_0$

Small change is noted by Δ

ΔQ and ΔP represents small change in quantity and price

Q_1 and P_1 stand for the current year's quantity and price.

Q_0 and P_0 stand for the previous year's quantity and price.

Price elasticity of demand is usually negative. This is because price and quantity demanded move opposite direction. $\Delta Q/\Delta P$ is negative, but we consider the absolute size of the price elasticity.

Price elasticity of demand Interpretation

Let's assume the price elasticity of demand coefficient is 0.59%. It can be expressed in two ways

- (i) a one percent rise in price leads to a 0.59% fall in quantity demanded
- (ii) a one percent decrease in price results in a 0.59% rise in quantity demanded.

Degree of Price Elasticity of demand

Based on coefficient of demand we have 5 degrees of demand elasticity.

Demand is said to be elastic (Relatively elastic): $\% \Delta Q_d > \% \Delta P$

Demand is said to be inelastic (Relatively inelastic): $\% \Delta Q_d < \% \Delta P$

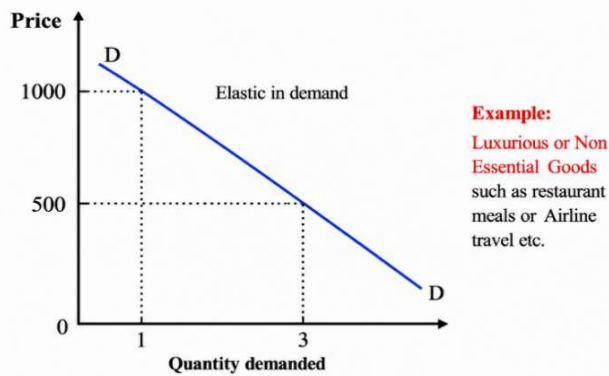
Demand is said to be unitary elastic: $\% \Delta Q_d = \% \Delta P$

Demand is said to be perfectly elastic: $\% \Delta Q_d = \infty$

Demand is said to be perfectly inelastic: $\% \Delta Q_d = 0$

Elastic Demand (Relatively elastic in demand) The demand for a product is said to be elastic when the % change in quantity demanded is > the % change in price. For instance, if price falls by 60% the quantity demanded increases by more than 60%. Here the response of quantity demand is more compared to price. When demand is elastic, the price elasticity coefficient is greater than one. The demand for luxurious or non-essential goods such as car, TV, washing machine, AC, furniture, mobiles, wine, movie tickets, air tickets, restaurant meals, demand for crude oil, electricity, gold etc. is considered to be elastic in demand. When demand for a product is elastic, it suggests that close substitutes are available for that product. If an item has multiple substitutes, its demand curve tends to be more elastic.

$$\Delta Q = Q_1 - Q_0$$



Q_1 is current year quantity. Q_0 is previous year quantity

$$\Delta P = P_1 - P_0$$

P_1 represents the price for the current year, while P_0 denotes the price for the previous year.

Q. Let's assume that initial price of a good is Rs. 1000 which falls to Rs. 500. Initial quantity was 1 units and increased to 3 units. Calculate price elasticity of demand

$$(\text{Ans}) \Delta P = 500 - 1000 = -500$$

$$\Delta Q = 3 - 1 = 2$$

$$(2 / -500) * (1000 / 1) = -4$$

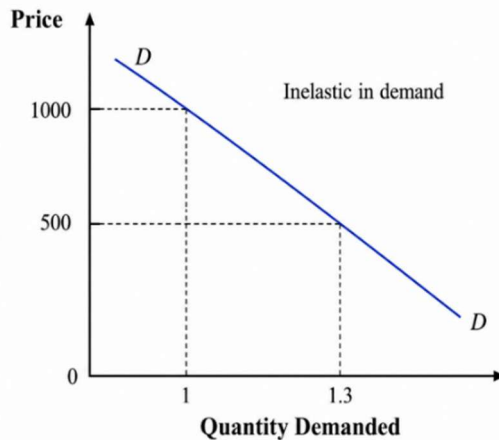
It means if the price rises by 1%, its quantity demand will fall by 4%.

Q. Calculate price elasticity of demand. Initial price was Rs 50 and falls to Rs 25. Initial quantity was 20 and increased to 60. Calculate price elasticity of demand and identify the nature of product.

Price	50	25
Q_d	20	60

Relatively Inelastic in Demand or Inelastic Demand: Elasticity Is less Than 1

The product is considered to have inelastic demand, if the % change in quantity demand is < than the % change in price. In this context, the elasticity coefficient is less than one. For instance, if the price drops by 60%, the quantity demand will increase by less than 60%. The demand for daily necessities like rice, bread, kerosene, and LPG cylinders is inelastic in demand.



Example: Necessary or Essential Goods Such as Gasoline, Matches, LPG cylinder, and Generic medicines etc have inelastic demand for a common man.

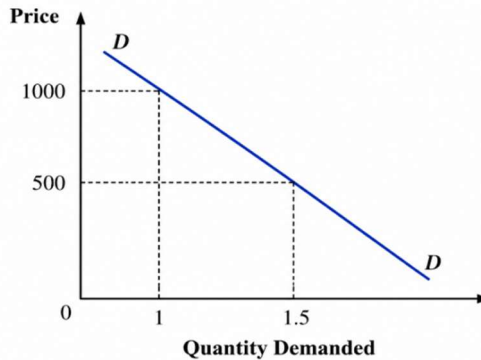
Q. Initial price was Rs 50 and price falls to Rs 20. Initial quantity was 15 and increased to 20. Find out the price elasticity of demand. Identify whether the product is elastic or inelastic or unitary elastic in demand.

Price	50	20
Q_d	15	20

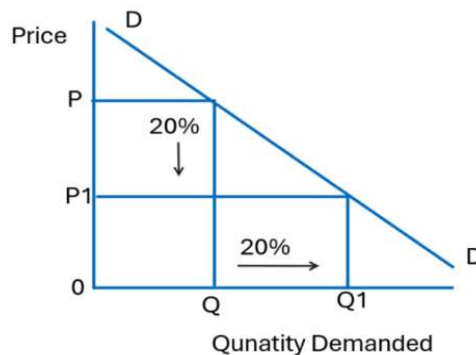
Unitary Elastic Demand: Elasticity is equal to 1

When both changes in quantity demanded and price percentages are the same, then it is clear that the product will have unitary elastic demand. That is, when there is an increase of 1% in the price level, then the quantity demanded will be decreased by 1%. Price and quantity demanded are moving in exactly the same percentage. Therefore, price elasticity of demand equals unity. In today's world, it is impossible for such demand to exist.

Another Example: If price will increase by 20% its quantity demand will fall by 20%.



Examples of unit elastic: Private Education, Radio and Television Receivers (Approximately), etc.



Q. Initial price was Rs 10 and price falls to Rs 5. Initial quantity was 100 and increased to 150. Find out the price elasticity of demand. Find out the price elasticity of demand. Identify whether the product is elastic or inelastic or unitary elastic in demand.

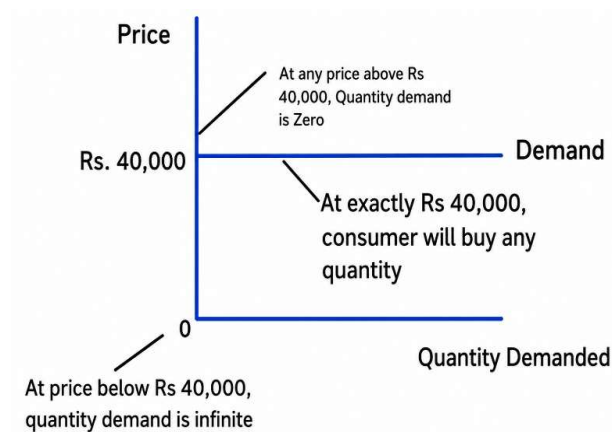
Price	10	5
Quantity	100	150

Critical Thinking Question: Roshni buys Apple every day and spends exactly Rs. 200, regardless of how much it is priced. Find out the price elasticity of demand. Identify whether the product is elastic or inelastic or unitary elastic in demand.

Perfectly Elastic in Demand

In case perfectly elastic in demand elasticity is equal to ∞ . Here, a small change in price causes a very large change in quantity demanded. When the price goes up a little bit the quantity demanded drops to zero. It never happens in real life. So, it doesn't have any real use. The demand curve is horizontal when the product is perfectly elastic in demand.

For example, two stores are selling computers of the same quality, at the same price, and in the same place. Now, one store raised the price while the other store kept it the same. So, the store that raises the price will see a big drop in demand. For instance, the company's total revenue will drop to zero, if the company raises the price of the product by 30%,



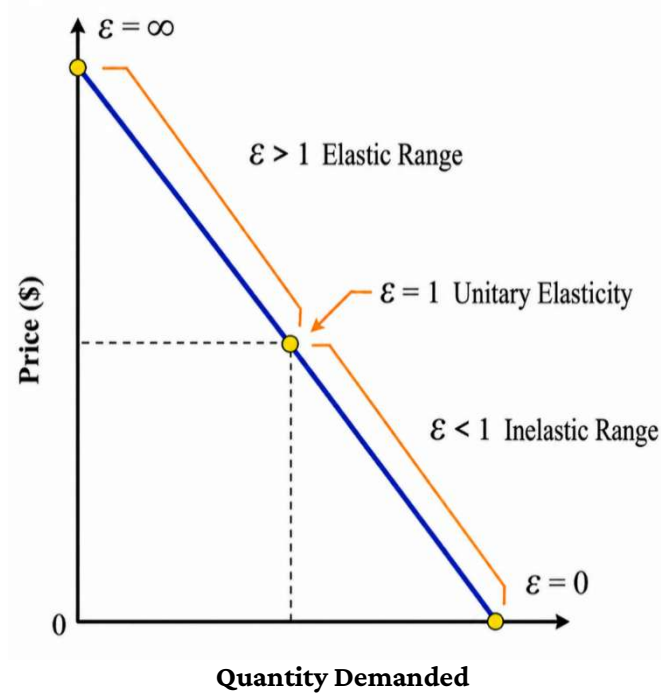
Perfectly Inelastic Demand: Elasticity equals 0

Quantity demanded remains constant irrespective of any change in price. When there is perfect elasticity of demand, the buyer will purchase the same quantity regardless of what the price is. This implies that $DQ/DP = 0$. If the demand curve is perfectly elastic, it will be vertical. Any change in price will have no effect on quantity demanded.



Example: lifesaving Medicine, Salt (basic necessities). The demand for a good like salt and medicine are perfectly inelastic

Price elasticity of demand varies along a straight-line demand curve.



It can be noted from a downwards slope of a straight-line demand curve that as we descend down it, the ratio of P/Q decreases. Demand curve is elastic when the prices are higher and the quantity of demand is lower. In contrast, when quantity of demand is higher and prices are lower, then the demand curve will be considered to be inelastic. Zero elasticity of demand exists in some cases wherein any change in price does not affect the quantity of demand. This phenomenon exists on the upper part of the demand curve while moving down to it makes the demand inelastic.

Methods of Measuring Elasticity of demand

There are three ways to measure elasticity.

- (i) Percentage method
- (ii) Point method
- (iii) ARC method

The percentage method measures how much the dependent variable changes in percentage terms when there is a percentage change in the independent variable.

$$\text{Elasticity} = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}} = \frac{\Delta Q}{\Delta P} * \frac{P}{Q}$$

Point Method (Geometric Method)

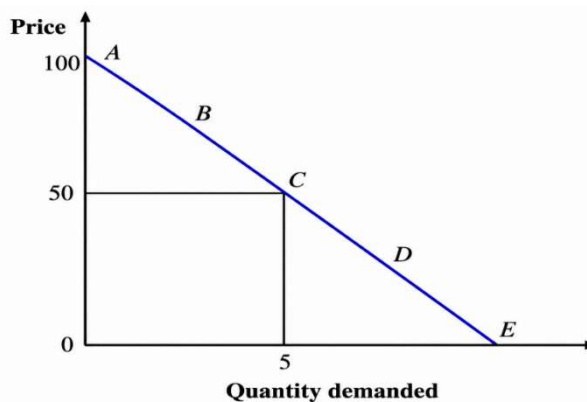
The point method is used when we figure out the price elasticity of demand at a particular point on the demand curve. This method shows that the demand curve's elasticity changes at every point. This method is used when the price and quantity of the good change only a little. This method mathematically defines price elasticity of demand (E_p) as

$$P_{Ed} = \frac{\text{Lower segment of the demand curve}}{\text{Upper segment of the demand curve}}$$

When the demand curve touches the X axis, the elasticity of demand is zero.

When we move upward from the midpoint on a demand curve, elasticity goes above one. This means that demand is more elastic in that area.

Price elasticity of demand becomes infinite when the demand curve meets the Y-axis. It means demand is perfectly elastic at that point.



Calculate elasticity at point C

$$C = \frac{\text{Lower segment of the demand curve}}{\text{Upper segment of the demand curve}} = \frac{EC}{AC}$$

Note: At the midpoint of a straight-line demand curve, elasticity of demand is equal to one.

Calculate elasticity at point D

$$D = \frac{\text{Lower segment of the demand curve}}{\text{Upper segment of the demand curve}} = \frac{ED}{AD}$$

$ED < AD$, hence elasticity is less than 1 (inelastic)

Calculate elasticity at point B

$$B = \frac{\text{Lower segment of the demand curve}}{\text{Upper segment of the demand curve}} = \frac{EB}{AB}$$

$EB > AB$, Hence elasticity is > 1 . It is elastic in nature

ARC Method

The use of the ARC approach is if anyone wishes to calculate the elasticity of any two points. In point elasticity, one can observe the effect on demand as a result of any slight change in price, while arc elasticity focuses on the effect on demand when there is a large change in price.

$$\text{ARC Method} = \frac{\Delta Q}{\Delta P} * \frac{\bar{P}}{\bar{Q}}$$

$$\bar{P} = \frac{P_1 + P_2}{2}$$

$$\bar{Q} = \frac{Q_1 + Q_2}{2}$$

Arc elasticity is always between the point elasticity values measured at the lower and higher price levels, but it doesn't have to be exactly in the middle.

The Determinants of Price Elasticity of Demand

- **Availability of Close Substitutes-** When there are close substitutes, a price increase will cause the consumer to buy less of that good and will buy more substitute goods. Demand will be highly price elastic, when there are close substitutes. When there are no close substitutes, demand will be price inelastic
- **Nature of the commodities (Necessities versus Luxuries):**

For example, when people book vacation flights, they are more sensitive to changes in price than when they travel for business. Also, the demand for luxury goods is elastic, while the demand for basic goods is inelastic (less sensitive to price changes).

- **Proportion of Income Spent on commodity:**

If you spend a lot of money on a good, your demand will be elastic. If you are spending less money on a certain good, on the other hand, the demand is not elastic.

- **Time Horizon**

In the short run, the demand for goods is very inelastic, but in the long run, it is very elastic. People can find out about price changes in the long run, but only a few people can find out about them in the short run. The supply of a product less elastic in short run.

- **Brand image:** If you like a particular brand, then your demand is inelastic in nature, irrespective of the price change.
- **Habit Forming Goods:** Goods such as cigarettes, wine, and drugs tend to be inelastic in demand. With increase in prices the demand won't fall significantly. The demand will fall marginally. Hence, in case of habit-forming goods the demand is inelastic.

Cross price Elasticity of Demand (CP_e)

Cross price elasticity of demand measures the percentage change in quantity demanded of one good due to a 1% change in the price of another good, keeping other factors constant. Let's say there are two goods, X and Y. Then

$$CP_e = \frac{\% \text{ change in quantity demanded of } X}{\% \text{ change in price of } Y}$$

$$CP_e = \frac{\Delta Q_X}{\Delta P_Y} * \frac{P_Y}{Q_X}$$

The cross-price elasticity of demand changes according to the substitute and complementary goods.

If Cross-price Elasticity is Positive (> 0) if and only if the goods are Substitutes

If Cross-price Elasticity is Negative (< 0) if and only if the goods are complements

If Cross-price Elasticity is $= 0$ if and only if the goods are Unrelated. Example, water and computer are not related goods. Likewise, cement and car.

In the context of substitute goods

Let's say that for a lot of people, Pepsi and Coke are very similar. Some customers might switch to Coke if the price of Pepsi goes up. Now, the price of Pepsi and the demand for Coke are both going up. So, CP_e is positive in this context. Tea and coffee are the same way.

In the Context of complementary goods

Complementary products like computer hardware and computer software. If computer hardware price will increase then the demand for computer software will fall. Here increase

in computer hardware price and quantity demand of computer software moves in opposite direction. Hence, the cross-price elasticity of demand is negative. Likewise, tea and sugar.

Economic interpretation of CPe

If the CPe of demand for a table in relation to the price of a chair is -0.45%. This means 1% increase in price of a chair lead to 0.45% decline in the demand for a table.

Income Elasticity of Demand

How much sensitive is the quantity demand due to change in income of the consumer.

Income elasticity of demand (Y_e) = $\frac{\% \text{ change in quantity demanded}}{\% \text{ change in income of the consumer}}$

$$Y_E = \frac{\Delta Q}{\Delta Y} * \frac{Y}{Q}$$

Small change is noted by Δ .

Q_1 and P_1 stand for the current year's quantity and current year's price.

Q_0 and P_0 stand for the previous year's quantity and previous year's price.

ΔQ is the difference between the current year's quantity and the previous year's quantity. It can also be represented as $Q_1 - Q_0$.

ΔY is the difference between the current year's income and the previous year's income. It can also be represented as $Y_1 - Y_0$.

Small change is noted by Δ .

ΔQ and ΔY represent small changes in quantity and income.

The income elasticity of demand varies from commodity to commodity whether it is normal goods, inferior goods and luxury goods

Note: income can be written as Y or I

Normal goods: When people make more money, they want more Normal goods. Branded dresses, LED TVs, cars, and other things are examples of normal goods. For example, if income goes up by 10%, demand goes up by 4%. Demand increases at a slower rate than income. Normal goods are those whose income elasticity of demand is positive but less than one.

Inferior goods: Inferior goods are opposite to the normal goods. In case of inferior goods, rise in consumer income leads to a decline in demand for these goods. The income elasticity of demand is negative in case of inferior goods. For example, low-quality bread, used

furniture, used dress, non branded perfumes etc. As the income of the people increases, they go for organic bread rather than low-quality bread. Hence, demand for low-quality bread falls with an increase in income. Similarly with increase in their income, they go for new furniture not used furniture. The income elasticity of demand is negative for conventional eggs, indicating that they are inferior goods.

Necessity goods: Necessity goods are income inelastic, which means that changes in income don't have much of an effect on how much people want them. Things like food, gas, clothes, utilities, and health care fall into this category. Even when income rises, demand for these goods rises only a little.

Luxury goods: For luxury goods, income elasticity of demand is greater than one. It means as income goes up, quantity demand goes up more than proportionately. For instance, the demand for luxury goods rises by 16% when income rises by 10%. Changes in income significantly influence the quantity demanded. Luxury goods are usually income-elastic. Some examples are sports cars, furs, and expensive food.

Recap

- If the income elasticity of demand is greater than 1, then it is called as luxury goods
- If the income elasticity of demand is less than zero, then the good is an inferior good.
- If income elasticity of demand is positive, the good is classified as a normal good

Interpretation of Income Elasticity

If the income elasticity of demand for BMW cars in India is 2.59. It implies a 1% rise in consumers' income will cause a 2.59% rise in demand for BMW cars. Because the coefficient is higher than one, the good is seen as a luxury item.

For instance, In India, the income elasticity of demand for flour is -0.69. It implies a 1% increase in consumers' income leads to 0.69% fall in demand for flour. Because the coefficient is negative, the good is seen as inferior goods.

ARC Income Elasticity of Demand

Point elasticity of demand can give different values when income changes. The ARC method overcome this by using the average of the initial and final income and initial and final quantity. This gives the same result no matter if income goes up or down.

$$ARC \text{ income elasticity} = \frac{\Delta Q}{\Delta Y} * \frac{\bar{Y}}{\bar{Q}}$$

ΔQ and ΔY are the small change in quantity and income respectively.

$$\bar{Y} = \frac{Y_1 + Y_2}{2}$$

$$\bar{Q} = \frac{Q_1 + Q_2}{2}$$

Promotional/advertisement Elasticity of Demand

Promotional elasticity of demand is how sensitive is the quantity demanded due to changes in advertisement expenditure. It shows how quickly consumer demand changes when advertising expenditure by a company changes.

$$\text{Promotional elasticity of demand} = \frac{\text{Percentage change in quantity demand of } X}{\text{Percentage change in advertisement expenditure}}$$

Elasticity of Supply (Es)

Elasticity of supply tells us how the quantity supplied changes when there is a change in price.

$$ES = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

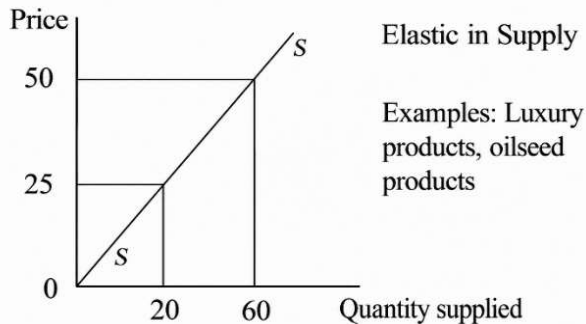
Elasticity of supply is always positive because price and quantity supplied move in the same direction. There are number of factors namely interest rate, wage rate, and input prices which affects the elasticity of supply. Elasticity of supply varies according to the nature of goods. For instance, perishable goods have the least elastic supply.

Elastic supply (Relatively elastic in supply): Elasticity Is Greater Than 1

The product is said to be elastic in supply if the % change in quantity supplied is > % change in price. For instance, if the price goes up by 50%, the quantity supplied increases by more than 50%. If the price elasticity of supply is more than unity, then the product is said to be elastic in supply.

Let's assume that initial price was Rs 25 and increased to Rs 50. Initial Quantity was 20 and increased to 60. Find out whether supply is elastic or inelastic. Here, the price elastic of supply is 2 which means if price will increase by 1% the quantity supply will increase by 2%. The elasticity number is greater than one. Hence it is elastic.

Price	25	50
Q _s	20	60



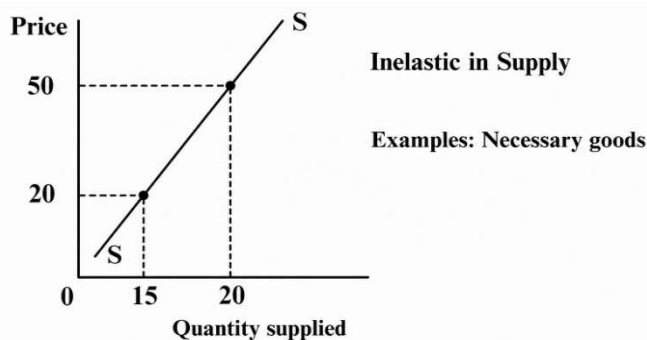
Inelastic Supply (Relatively inelastic in supply): Elasticity is < 1

The product is said to be inelastic in supply if the % change in quantity supplied is $<$ % change in price. If a price goes up by 50%, the quantity supplied will go up by less than 50%. The supply is said to be inelastic, when the price elasticity of supply is less than unity.

Price	20	50
Q_s	15	20

Let's assume that initial price was Rs 20 and increased to Rs 50. Initial Quantity was 15 and increased to 20. Find out whether supply is elastic or inelastic or unitary elastic.

As the price elasticity of demand < 1 , the product demand is inelastic.

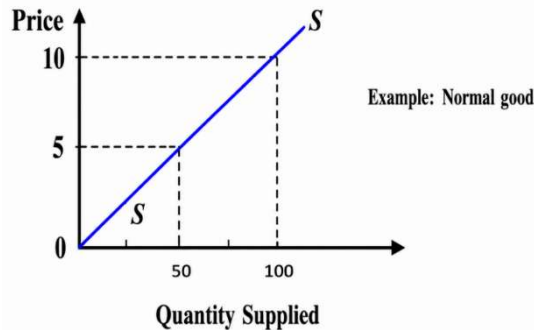


Unitary Elastic in supply: Elasticity is equal to 1

The supply is said to be unitary elastic when the % change in the quantity supplied = the percentage change in price. Hence, the price elasticity of supply is one. This is a unique situation.

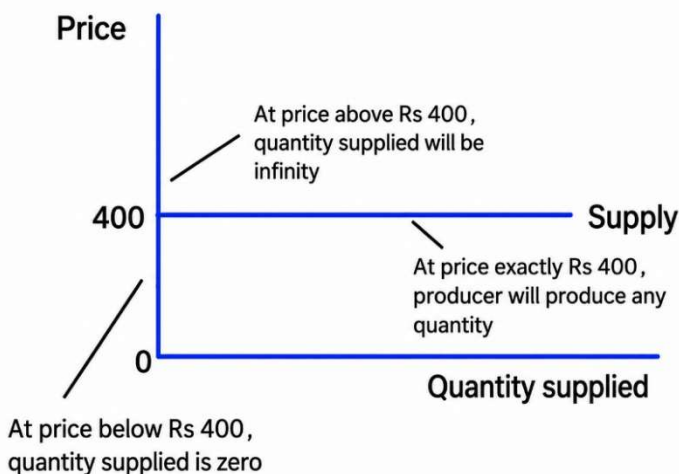
Let's assume that initial price was Rs 5 and increased to Rs 10. Initial Quantity was 50 and increased to 100. Find out whether supply is elastic or inelastic or unitary elastic. Here the supply is unitary elastic as the elasticity coefficient is one.

Price	5	10
Quantity	50	100



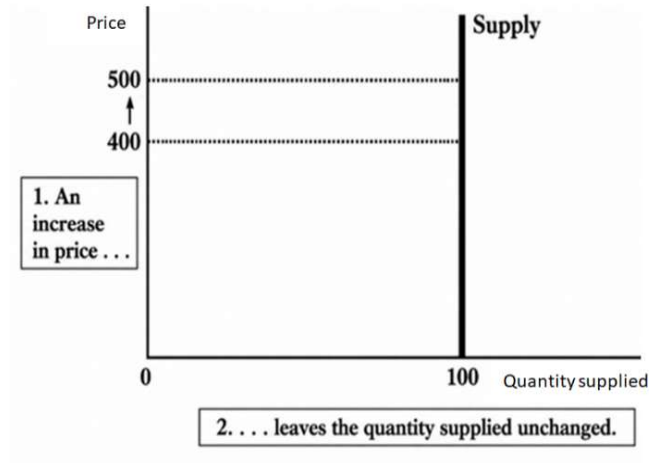
Perfectly Elastic Supply: Elasticity Equals Infinity

A horizontal line that shows a situation in which any decrease in price would cause the amount supplied to decrease to zero.



Perfectly Inelastic supply: Elasticity Equals 0

The supply curve carries a vertical shape in case of perfectly inelastic supply. Any change in price has no effect on the quantity supplied. In this context, the elasticity value is zero. In a perfectly inelastic supply, the producer produces a fixed quantity of goods no matter what the price is.



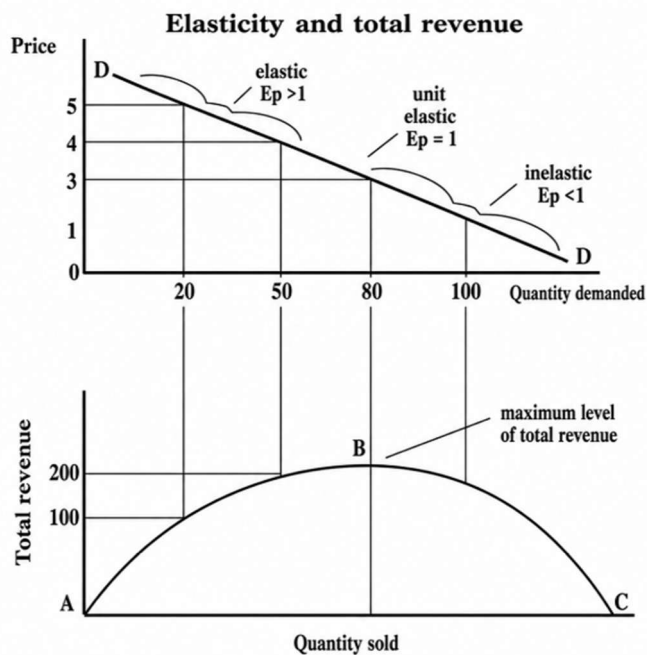
Total Revenue and the Price Elasticity of Demand

Total revenue is the total amount of money a company earns from selling its goods or services over a specific period of time.

$$TR = P \times Q$$

Note: P stands for “price per unit” and Q stands for “quantity”

When the demand curve is inelastic, a rise in price leads to a small fall in quantity demanded. So, total revenue goes up.



The upper segment of the demand curve, represented by AB on the total revenue curve, which is elastic. In contrast, the lower segment of the demand curve, shown by BC portion of the total revenue curve, is inelastic.

If the product is elastic: If the quantity demand is very sensitive to the changes in price, then we can say the product is elastic in demand. If the price goes up, demand will go down, and vice versa. As a manager, how will you make money if the product is elastic? If the manager lowers the price, more people will want to buy it, sales will go up, and profits will go up as well. Let's assume price of a product is Rs. 5 and the quantity demanded is 20 units. $TR = 5 \times 20 = 100$. If the manager lowers the price from 5 to 4, the quantity demanded increases from 20 to 50. Total revenue becomes ($4 \times 50 = 200$).

On the other hand, the inelastic part of the demand curve doesn't change as much when prices change. Cigarettes, for instance. People who smoke or drink won't stop if the price of cigarettes goes up because there is no way to substitute it. So, to make money, a manager should raise the price.

On the inelastic portion of the demand curve, when price is 1, the quantity demanded is 100. Total revenue is $100 \times 1 = 100$. If the price goes up to Rs. 3, the amount of demand is 80, and the total revenue is ($3 \times 80 = 240$). So, if the product is inelastic, the manager can make money by raising the price.

Table

Elasticity For Manager's Product	Price Decision of Manager	Manager's Total Revenue
$E_p > 1$	Price Increase Price Decreases	TR Decreases TR Increases
$E_p < 1$	Price Increase Price Decreases	TR Increases TR Decreases
$E_p = 1$	Price Increases Price Decreases	TR Remains Same

Total Revenue remain unchanged in case of Unitary Elastic Demand

Review Questions

1. How and why do governments intervene in markets? Explain what happens when the government sets the price above or below the market equilibrium price with a suitable diagram. Explain the consequences of government intervention.
2. Graphically illustrate and explain the relationship between revenue and with degrees price elasticity of demand.
3. Analyze the different types of elasticity of demand along with their determinants. Explain how businesses can apply these concepts in their pricing decisions.
4. Using the relevant diagrams, analyze the differences between changes in supply and changes in quantity supplied. Describe the elements that cause the supply curve to shift and move along it.
5. Evaluate how price and non-price factors influence change in quantity supplied and change in supply using suitable graphs.
6. Using the relevant diagrams, analyse the differences between changes in quantity demanded and changes in demand. Describe the causes of changes in the demand curve and movement along it.
7. Evaluate the impact of price and non-price factors on change in quantity demanded and change in demand using suitable graphs.
8. A company produces three products with different price elasticity of demand: one product is **elastic**, another is **inelastic**, and the third is **unitary elastic**. Due to the COVID-19 pandemic, the company's sales volume has declined significantly. As a manager, you have been instructed by the CEO to increase the company's total revenue.
Analyze how price changes would affect the total revenue of each product based on their elasticity of demand. **Evaluate** and recommend an appropriate pricing strategy for all three products to maximize the company's total revenue. Justify your answer with suitable reasons.
9. Briefly explain how you can use various elasticity concepts in managerial decision making. Some have argued that charging higher taxes on cigarettes, in practice, only results in higher government revenue and serves no other purpose. Briefly explain how that could be.

Further Readings

Dominick Salvatore (2008 to 2024), "Managerial economics principle and world-wide applications," Oxford Publication

Pindyck, Rubinfeld, and Mehta (2019), Microeconomics, Pearson Publication

Williom A. McEachern I A. Indira (2012)" Principles of Macroeconomics", Cenage Larning publication

H.L. Ahuja (2007) "Principles of Micro economics," S Chand and Company Limited

Consumer Behavior



What is Consumer Behavior?

Consumer behavior involves the decision-making process by which individuals purchase and utilize goods and services to satisfy their needs and desires. Consumers typically compare various products before making a purchase, evaluating factors like price, quality, usefulness, and their income. Consumer choices are influenced by personal preferences, family, friends, culture, and advertisements. Consumer buying decisions vary due to individual needs and tastes and preferences. Understanding consumer behavior is crucial for businesses to align their products with consumer needs and for economists to analyze market demand patterns. Consumer behavior varies from person to person and time to time. For instance, when Dabur introduced the real orange juice the taste was bitter. The sales volume didn't meet company expectation. Then they did a survey and found that Indians don't like bitter taste. They like sweeter taste. Afterwards company changed the flavor from bitter to sweeter which increases the sales volume. Similarly, 30 years before, LML Vespa dominated the two-wheeler segment in India. But now it is rarely available. Note: the success of any product depends on consumer behavior. If people like a product, then they will buy and the product will be successful and it will remain sometimes in the market. If consumer doesn't like a product, they won't buy that. As a result, price of that product will be low. Consumer behavior affects firms' activities.

Choosing the appropriate type of music can significantly impact consumer behavior in businesses. Slow music fosters a calm shopping atmosphere, leading consumers to spend more time in stores. This leads to additional purchase by the consumers. On the other hand, fast music boosts energy and prompts consumers to shop quickly. Restaurants and shopping malls utilize various types of music to impact customer shopping behavior and influence purchasing decisions. Understanding and predicting consumer behavior is very complex for human being point of view. AI predicts consumer behavior by analyzing extensive data, including past purchases, browsing history, and customer preferences. AI identifies patterns to forecast future buying decisions and product demand. AI assists companies in understanding consumer behavior by analyzing customer data, purchasing patterns, and preferences. It forecasts purchasing likelihood and timing, recommending suitable products, pricing, promotions, and advertisements. This leads to marketing strategies that better align with customer needs and enhance sales.

Consumer behavior rresolving disputes among:

- ❖ **Preference:** Generally, refers to a person's or group's likes, choices, or priorities — essentially, what someone wants or values more when given options. The consumer makes preferences by expressing his or her likes and dislikes. Eg- "I prefer tea over coffee." It leads manager to take the decision what to produce i.e. Product decision
- ❖ **Choice:** is the actual decision someone makes from available options. Means the act of selecting or making a decision when faced with two or more possibilities. "I

chose coffee this morning. Preferences guide choices. But your choice might not always match your preference — for example, due to availability, cost, pressure, etc. Consumer makes the choice from preferred alternatives that suit his or her budget. Helps manager by telling how to produce and taking appropriate pricing decision

- ❖ **Utility**-Utility refers to the level of happiness or satisfaction gained from consuming goods or services. Eating an ice-cream gives you utility because it satisfies your taste. By making a choice he or she maximizes his or her utility or satisfaction. Feedback for managers and helps in taking the product renovation decision

Utility

Utility refers to the total satisfaction received from consuming a good or service. Economic theories based on rational choice usually assume that consumers will strive to maximize their utility. The utility of a good or service is important to understand because it directly influences the demand, and therefore price, of that good or service.

Utility help the consumers to understand how to maximize their utility by allocating their money between multiple types of goods and services as well as help companies understand how to structure pricing.

In practice, a consumer's utility is impossible to measure and quantify. Various schools of thought differ as to how to model utility

The Concept of Utility

Cardinal Utility

Utility is measurable and quantifiable. The imaginary unit of measurement is Utils. Note: The measuring units of utility is known as utils. The Cardinal utility approach assumes that utility levels are measurable in terms of suitable units.

Ordinal Utility

Utility can't be measured. It is Subjective. Only can be ranked or compared.

Cardinal Utility

This approach assumes that utility derived from a particular product is independent of itself. Hence, the utility obtained from the consumption of one product is not dependent on the utility derived from another product.

The Marshallian Cardinal utility analysis is based on the following assumptions

1. **Consumers are rational-** A rational consumer is considered to be a person who makes rational consumption decisions. In other words, the consumer who makes his/her choice after considering all the other alternative goods and services. A



rational consumer should make a choice in such a way it gives maximum satisfaction.

2. **Consumption must be continuous:** Cardinal utility is measurable if consumption is continuous. Not like you will have one bite of Pizza in morning and another bite of Pizza in evening. In this scenario utility can't be measurable.
3. **Units of commodity consumed must be homogeneous** i.e. same quality and quantity. Take one example of Rasgulla. All Rasgulla sizes should be same in quality and quantity
4. **Utility is expressed in monetary term-**Let's assume that you like Panipuri. In your place, one plate Panipuri is Rs 30. If a seller will ask you Rs 300 for one plate Panipuri will you give Rs 300. Obviously, no. One plate Panipuri is worth of Rs 30 only. Hence you will pay Rs 30.
5. **Constant marginal utility of money** means that the utility per unit of money remains unchanged for the customer. This implies that each additional unit of money provides the consumer with the same level of satisfaction.

For example, if a rupee can buy 100g sugar, 200g rice, and 300g wheat with a rupee and the utility received from all these goods is 4 utils, then the marginal utility of money is taken as 4.

Marginal utility of money will remain constant only if the money income of the consumer remains constant.

The marginal utility also increases if the money income of the consumer increases.

6. Diminishing marginal utility

It tells that MU of any good tends to decline as more of the good is consumed over a definite period of time. Therefore, the first unit of consumption for any product is typically the highest, with every unit of consumption to follow holding less and less utility. The consumer will stop consuming the good much before reaching $MU=0$. The law of diminishing marginal utility establishes the relationship between Total utility and Marginal utility. The principle of diminishing marginal utility states that as more of a good is consumed, the consumption of additional amounts will yield smaller additions to the utility. Assume that a person consuming the first cigarette will get a certain level of satisfaction. His total utility goes up when he consumes the second cigarette. Now if he consumes one more, he reaches the satisfaction quantity. Now any additional consumption beyond this point will not add to the person's utility but will make him sick.

Critical thinking question:

1. Do you think Law of diminishing marginal utility will work from 1st pack to 2nd pack of wine?
2. Do you think law of diminishing marginal utility will work in the context of Money

Total Utility and Marginal Utility

Total Utility is the total amount of satisfaction obtained from the consumption of all the units of goods and services during a period of time.

Total Utility of the Consumption of a Good

$$U_x = U_{x_1} + U_{x_2} + U_{x_3} + \dots$$

Total Utility of the Consumption of Different Goods

$$U = U_x + U_y + U_z + \dots$$

Marginal utility is the extra or additional satisfaction that a consumer gets by consuming one more unit of that good

MU_x = Marginal utility of X. Marginal utility of X is the change in total utility resulting from one unit change in consumption of X. $MU_x = \frac{\Delta T}{\Delta X}$. ΔT refers to change in total utility. ΔX represent change in quantity.

The extra satisfaction gained from an additional unit of consumption of the goods is known as the marginal utility. The extra satisfaction gained from an additional unit of consumption of a commodity or service is defined as marginal utility. The utility received by an individual from the consumption of one additional unit of the goods is known as the marginal utility.

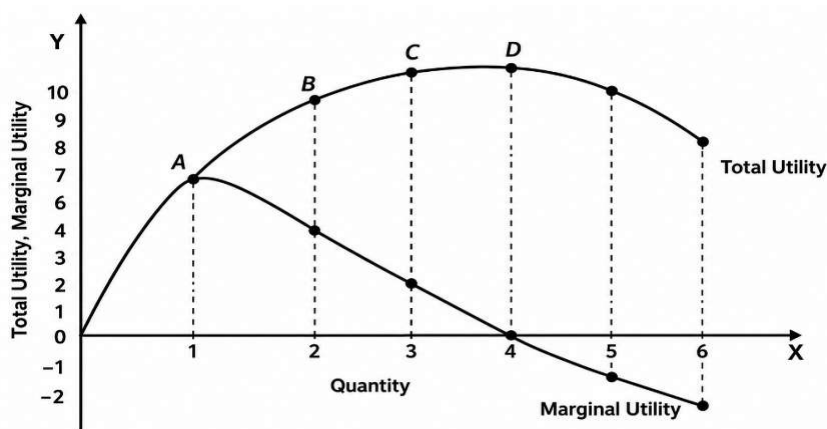
Marginal utility helps the economists find out the quantity of the commodity purchased by individuals.

Note: when a consumer consumes a product that gives no extra measure of satisfaction then marginal utility is zero

- ❖ **Positive Marginal Utility** occurs when the consumption of an additional item increases the total utility. Positive marginal utility occurs when having more of an item brings additional happiness. Suppose you like eating a slice of cake, but a second slice would bring you some extra joy. Then, your marginal utility from consuming cake is positive.
- ❖ **Negative Marginal Utility** occurs when the consumption of one more unit decreases the overall utility.
- ❖ **Zero Marginal Utility:** when consuming more of an item brings no extra measure of satisfaction. For example, you might feel fairly full after two slices of cake and wouldn't really feel any better after having a third slice. In this case, your marginal utility from eating cake is zero.

Total Utility vs. Marginal Utility

Quantity of Cigarette Consumed	Total Utility (Utils)	Marginal Utility (Utils)
0	0	Undefined
1	5	$(5-0)/(1-0) = 5$
2	8	$(8-5)/(2-1) = 3$
3	10	$(10-8)/(3-2) = 2$
4	10	$(10-10)/(4-3) = 0$
5	9	$(9-10)/(5-4) = -1$
6	7	$(7-9)/(6-5) = -2$



Total Utility curve Rises first and eventually falls

Note: When Total Utility & Marginal Utility is equal at 1st unit of consumption

When the first two units of goods are consumed total utility increases. When the third unit is consumed total utility reaches its maximum at point 10. Utility remains constant even after the consumption of the fourth units but when the consumer starts consuming 5th and 6th units of the goods then total utility reduces. In this figure, points A and B have increasing total utility.

Total utility reaches the maximum at point D. A further increase in consumption (5th and 6th units) shows that the total utility diminishing and the total utility curve start declining. At the initial amount of consumption of a good total utility increases and marginal utility decreases. At saturation point, the consumer's marginal utility is zero.

Theoretical relationship between TU and MU

- ❖ At zero consumption, TU is Zero, and marginal utility is undefined
- ❖ At the initial level of consumption of a good total utility increases and marginal utility decreases.
- ❖ Marginal utility decreases if more units of the same commodity is consumed, but total utility increases
- ❖ Total utility is increasing but at a decreasing Rate
- ❖ Marginal Utility becomes 0 at a particular point where the Total Utility will be maximum
- ❖ Once Marginal Utility becomes negative, Total Utility starts falling

Ways to Attain Consumers Equilibrium through cardinal approach

Consumer equilibrium refers to a unique position where the consumer does not want to change his/her consumption

Consumer equilibrium in one commodity context

Consumer will be equilibrium where marginal utility of X is equal to price of X

$$MU_x = \text{Marginal utility of } X$$

$$P_x = \text{Price of } X$$

$$\text{Consumer equilibrium is where } MU_x = P_x$$

Scenarios	Marginal Utility	Price	Equilibrium
1	15	10	$MU_x > P_x$
2	10	10	$MU_x = P_x$
3	8	10	$MU_x < P_x$

If $MU_x > P_x$, then the consumer will purchase more units of X than he/she's required

If $MU_x < P_x$, then the consumer will purchase less units of X than he/she's required

If $MU_x = P_x$, then the consumer will purchase exact units of X he/she's required

Principle of Equi-marginal Utility or Consumer Equilibrium and the Law of Equi-Marginal Utility

Equi-Marginal Utility Law is an advanced version of the law of diminishing marginal utility. Equi-marginal utility concept elaborates the approach of the consumer while allocating his income amongst different goods and services. It defines how the consumer allocates his income from different sources amongst different goods such that he can maximize his total utility level.

The consumer will distribute their money income between goods in such a way that utility derived from the last rupee spent on each good is the same. The Consumer's Equilibrium is the Position where Marginal Utility of Money Expenditure on each good is the same

Assumptions:

- ❖ The wants of a consumer remain unchanged.

- ❖ Consumer has a fixed income
- ❖ The prices of all goods are given and known to a consumer.
- ❖ He acts rationally in the sense that he want maximum satisfaction
- ❖ Diminishing marginal utility for each good
- ❖ Assumption of constant utility of money

Let's assume the consumer's fixed income Rs 13 to be allocated between two goods X and Y. The prices of two Goods X and Y are given as Rs 2 and Rs 5 respectively. The marginal utility of money is constant at Rs 1 = 5 utils.

Marginal Utility of Goods X and Y

Units consumed	MU _x (Utils)	MU _y (Utils)	MU _x /P _x P _x =2	MU _y /P _y P _y =5
1	20	35	10	7
2	18	30	9	6
3	16	25	8	5
4	14	20	7	4
5	12	15	6	3

Let's assume that consumer money income is Rs 13. The consumer will get equal level of satisfaction by consuming 4 units of commodity X and 1 unit of commodity Y. The price is $4 \times 2 + 1 \times 5 = \text{Rs } 13$

When a consumer is buying 4 units of commodity X and 1 unit of commodity Y then the consumer is getting highest level of satisfaction. For example,

$$20 + 18 + 16 + 14 + 35 = 103$$

Any other combination the consumer won't get highest level of satisfaction.

Let's take another scenario. Consumer has Rs 13. The consumer allocates Rs 13 into two goods where the consumer will get same level of satisfaction. If the consumer will spend 2 units of Y that is $(5 \times 2) = 10$ and remaining Rs 3 is used to buy X. The price of X is Rs 2. If the consumer want to buy two units of commodity X the consumer needs Rs 4. But the consumer has only Rs 3. One can say consumer can able to buy 1.5 units of X $(20 + 18/2) = 29$. If we will see how much satisfaction the consumer gets from X and Y that is $35 + 30 + 20 + 9 = 94$. Here the consumer satisfaction is not highest.

(i) $\frac{MU_X}{P_X} > \frac{MU_Y}{P_Y}$. It is worthwhile to spend on X

(ii) $\frac{MU_X}{P_X} < \frac{MU_Y}{P_Y}$. It is worthwhile to spend on Y

Equilibrium condition: $\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} = MUM$

If MU_X / P_X and MU_Y / P_Y are not equal and MU_X / P_X is greater than MU_Y / P_Y , then the consumer will substitute good 'X' for good 'Y'. As a result, the marginal utility of good 'X' will fall.

The consumer will continue substituting good 'X' for good 'Y' till $MU_X/P_X = MU_Y/P_Y$ where the consumer will be in equilibrium.

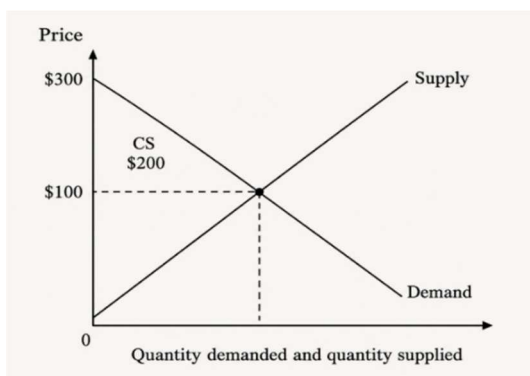
Criticisms of Cardinal Utility Analysis

- ❖ Utility is a psychological concept, can't be measured
- ❖ If consumption is not continuous then cardinal utility can't be applied
- ❖ In many times the marginal utility of money does not remain constant

Consumer Surplus

The Consumer Surplus is the gap between the maximum willingness and ability of a consumer to pay for a commodity and the amount he pays for it. The larger the gap between the price an individual is willing and able to pay for a particular commodity and the market price, the higher the consumer surplus. It refers to how much consumers gain by paying less than their willingness to pay for something. The Consumer Surplus can be computed from the Demand Curve. In simple terms, the Consumer Surplus can be calculated as the gap between the market clearing price and the demand curve. It is simply the gap between Total utility and total payment made on the total consumption of goods. The gap between the total utility received from consuming goods and the total amount of money paid for them is referred to as the Consumer Surplus.

For example, let's say that you bought an airline ticket for a flight to Disney during school vacation week for \$100, but you were expecting and willing to pay \$300 for one ticket. The \$200 represents your consumer surplus.



Ordinal Utility Approach

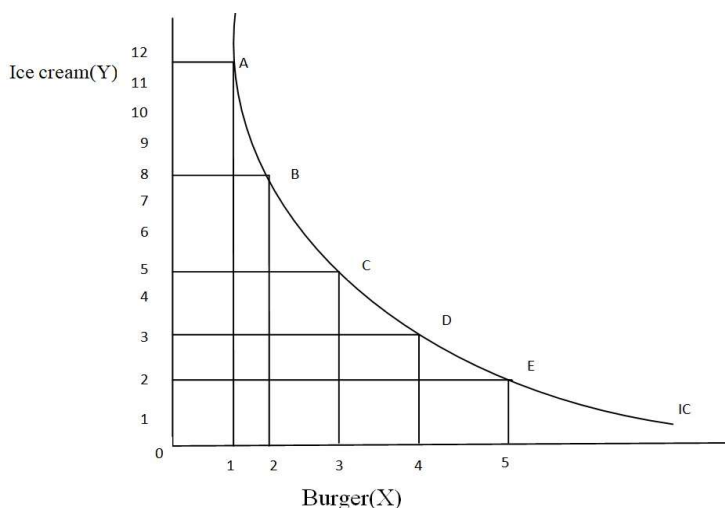
Requires consumers be able to rank their preferences for various combinations of goods. Specifically, the consumer should be able to say whether

- Combination A is preferred to combination B
- Combination B is preferred to combination A. or
- Both combinations are equally preferred

What is an Indifference Curve?

The indifference curve contains all those combinations of goods which yield equal satisfaction to the consumer. Therefore, the consumer treats all combinations along the curve equally good. Since all the combinations of goods give an equal amount of satisfaction, the consumer is indifferent between the consumption of one bundle and the other.

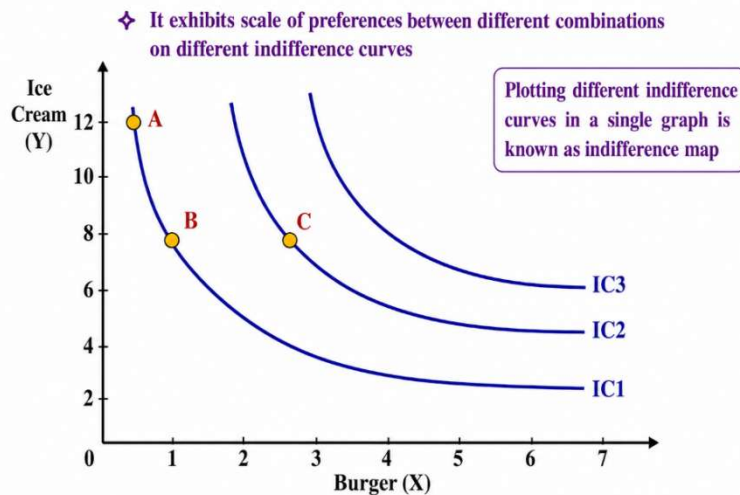
Combination	Burger	Ice Cream	Satisfaction level
A	1	12	50
B	2	8	50
C	3	5	50
D	4	3	50
E	5	2	50



X axis represents Burger and Y axis shows Ice cream. One can take any two commodities of their wish. Though the consumer consumes N number of goods in their day to day life but for simplicity we have taken only two goods namely ice cream and Burger. At point A, the

consumer consumes 12 ice creams and 1 burger, the satisfaction level is 50. At point B, the consumer consumes 8 ice creams and 2 burgers, the satisfaction level is 50. Likewise, at point E, the consumer consumes 2 ice creams and 5 burgers, the satisfaction level is 50. The satisfaction level remains the same as the consumer moves from point A to E or E to A. Hence, the consumer is indifferent among the points. If we will join different combinations where consumer purchases different quantities then we will get indifference curve. The consumer is indifferent as different combinations give same level of satisfaction to a consumer. Hence, it doesn't matter whether the consumer is at point A or C or E points. Thus, along an indifference curve, there is an inverse relationship between the quantity of one good consumed and the quantity of another consumed → indifference curves slope downward. If a consumer buys more of Burger, then the consumer will buy less of ice cream and vice versa due to budget constraint.

Indifference Map

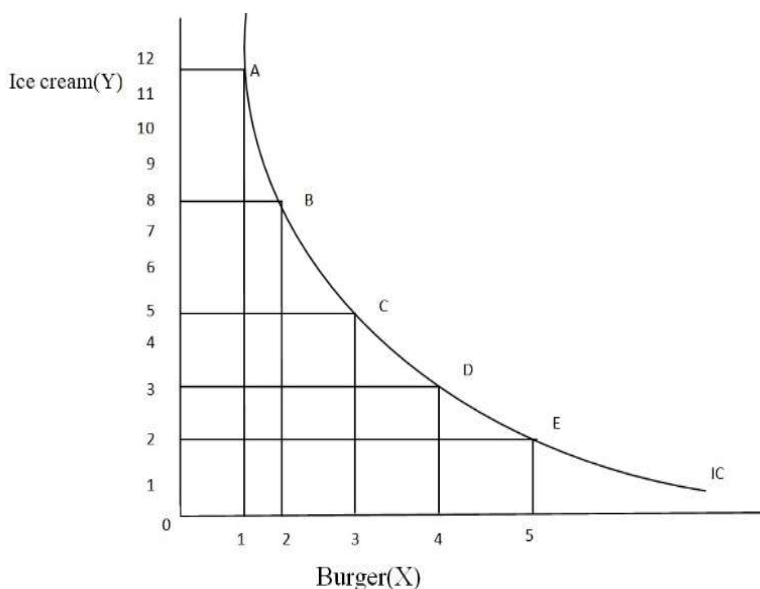


Indifference Schedule

Indifference schedule shows different combinations of two goods that give equal satisfaction or utility to a consumer. When we are plotting two goods in a tabular format it is called an indifference schedule. Here, the combination A to E the satisfaction level remains the same.

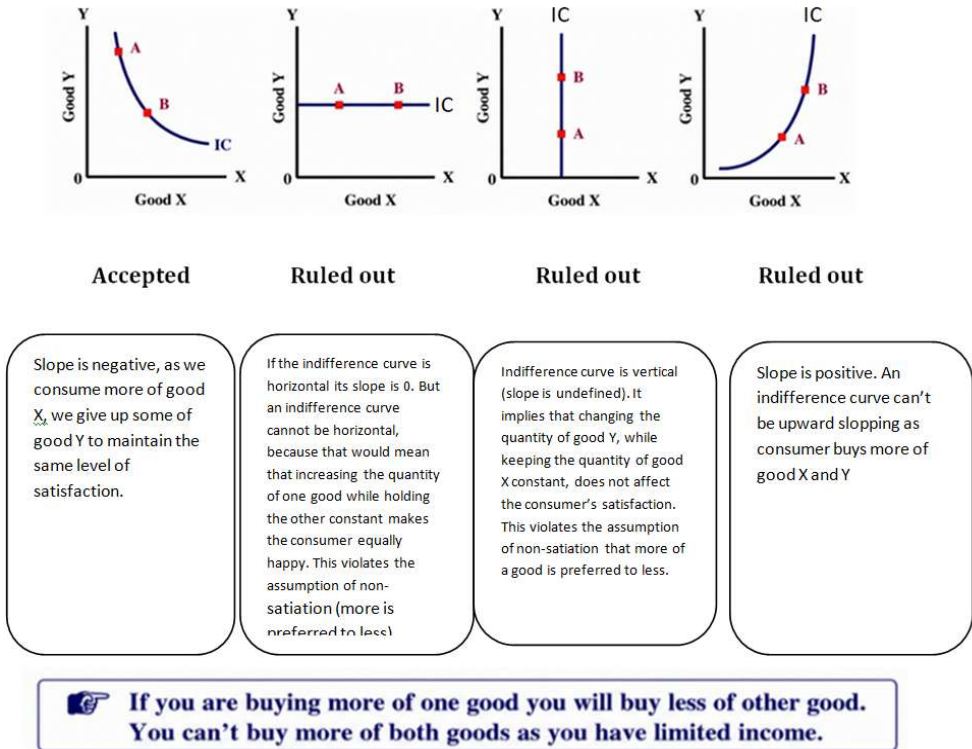
Properties of Indifference Curve

Property 1: Indifference Curves Slopes Downward to the Right



X axis represents Burger and Y axis represents ice cream. If a consumer buys more of Burgers, then the consumer has to buy less of Ice creams and vice versa. This happens because the consumer has a given amount of money. In other words, it can be said that the consumer faces a budget or monetary constraint. With given money, the consumer can't purchase more of both goods. Hence, there is a tradeoff between buying more of one good and less of the other good. As a result, the indifference curve, which is known as the IC curve, slopes downward to the right. For instance, when a consumer buys 12 ice creams, they buy 1 burger. With an increase in the consumer's purchase of 5 burgers, they can purchase only 2 ice creams.

Critical Thinking Question: Why can't an indifference curve be horizontal, vertical or upward slopping curve?



Property 2: The Indifference Curve is Convex to the Origin. It is because of diminishing or decreasing in the marginal rate of substitution (MRS)

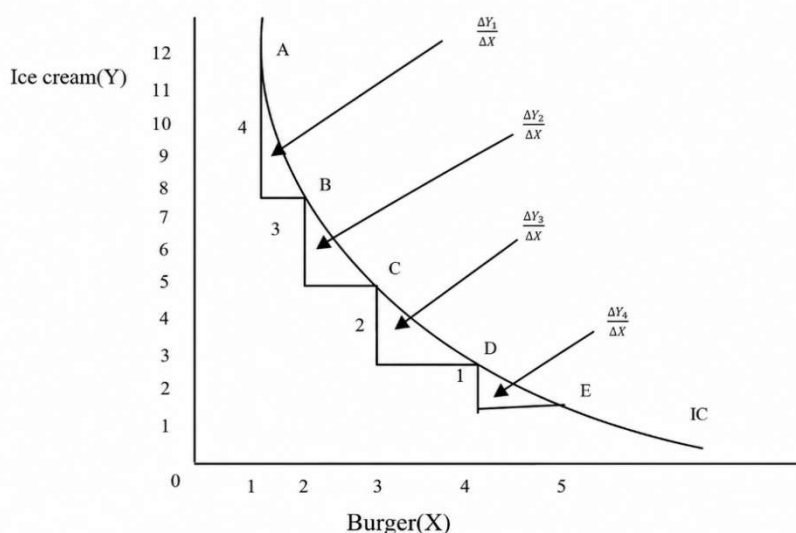
MRS is nothing but exchange of goods. Here for simplicity, we are considering two goods only. ($MRS_{\text{Burger, Ice cream}}$) represents the amount of ice cream the consumer sacrificed to gain one additional unit of Burger so that the level of satisfactions will remain same.

Combinations	Burger	Ice Cream	MRS burger, Ice cream	Satisfaction level
A	1	12		50
B	2	8	4	50
C	3	5	3	50
D	4	3	2	50
E	5	2	1	50

X axis represents Burger and Y axis shows Ice cream. At point A, the consumer consumes 12 ice creams and 1 burger, the satisfaction level is 50. At point B, the consumer consumes 8 ice

creams and 2 burgers, the satisfaction level is 50. When the consumer moves from point A to point B, consumer sacrifices 4 units of ice creams to get an additional one unit of burger. Hence, MRS is 4 when consumer moves from point A to B $(8-12)/(2-1) = -4$. It is usually negative number because it shows the rate at which a consumer is willing to sacrifice a good in exchange for an extra unit of another good to maintain his level of satisfaction. The negative sign reflects the trade off. Mathematically MRS values are negative. However, we are considering its absolute magnitude.

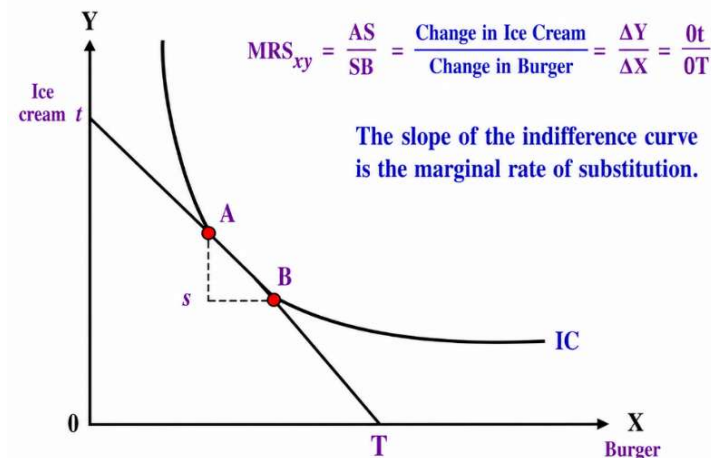
Similarly, when the consumer moves from point B to point C, consumer sacrifices 3 units of ice creams to get an additional one unit of burger. Hence, MRS is 3 when consumer moves from point B to C. Likewise, when the consumer moves from point C to D, consumer sacrifices 2 units of ice creams to get an additional one unit of burger. Hence, MRS is 2 when consumer moves from point C to D. MRS keep on diminishes when consumer moves from point A to E.



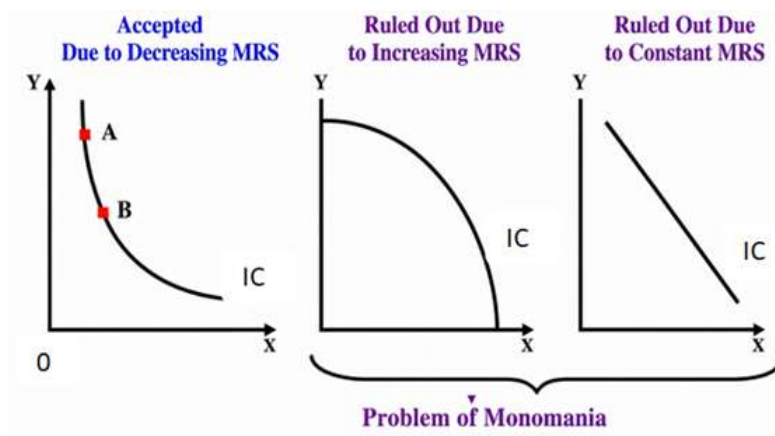
Principle of Diminishing MRS states that MRS of one good (ice cream) will decline as more and more of that good (ice cream) is substituted for another good (burger)

$$\text{Diminishing MRS} = \frac{\Delta Y_4}{\Delta X} < \frac{\Delta Y_3}{\Delta X} < \frac{\Delta Y_2}{\Delta X} < \frac{\Delta Y_1}{\Delta X}$$

Slope of the Indifference Curve

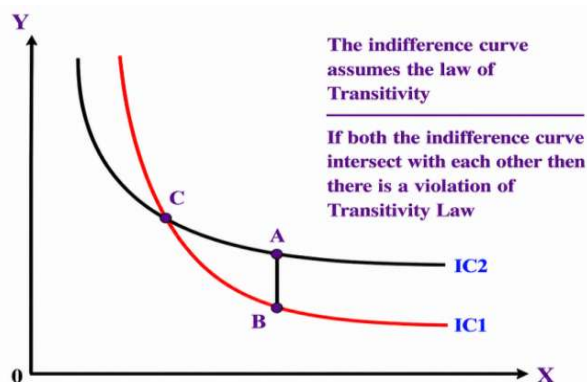


Critical Thinking: Why can't an indifference curve be concave to the origin and downward sloping?



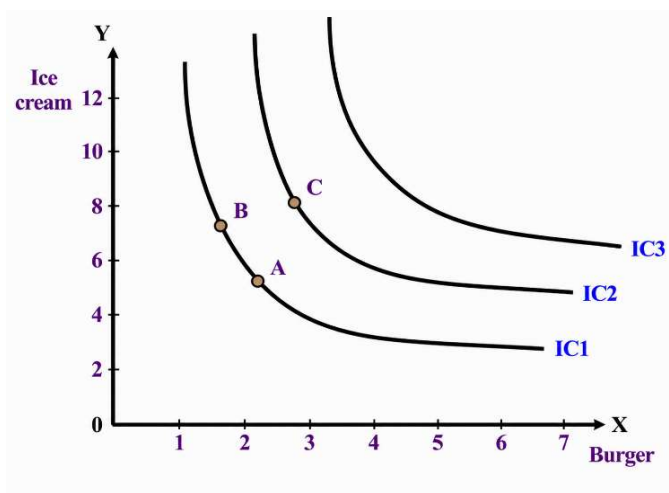
Property 3: Two Indifference curves cannot intersect with each other

Point C and point A lie in same indifference curve (IC₂). Hence, point C and point A gives equal amount of satisfaction. Hence, point C=Point A. Likewise, point C and point B lies in same indifference curve (IC₁). Hence, point C and point B gives an equal amount of satisfaction to a consumer. If C=B and C=A then A=B. But point A lies higher indifference curve compared to B. Higher, the indifference curve higher is the level of satisfaction. The indifference curve assumes the law of Transitivity. In this case, there is a violation of Transitivity Law. Hence two indifference curves can't intersect to each other.



Property 4: Higher Indifference Curve Represents a Higher Level of Satisfaction than the Lower Indifference Curve

In point A and B consumer satisfaction level is remaining same and the consumer is indifferent between the points. Points A and B give same satisfaction to a consumer. Point C lies in IC2 where consumer can purchase more of both the commodities. Hence, higher ISO quant is proffered by a consumer.



Budget Line

Budget line depicts all possible combinations of Goods which the consumer can buy by spending his given income on the goods at their given prices

$$Px \cdot Qx + Py \cdot Qy = M$$

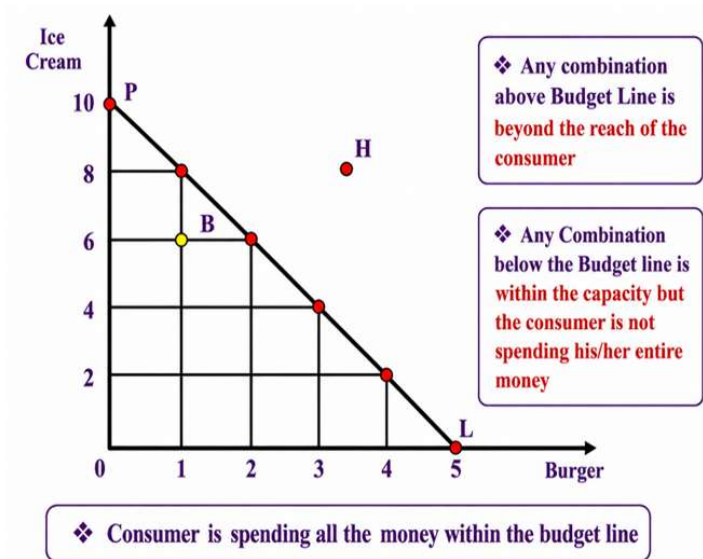
Where,

Px and Py are the prices of commodity X and Y and Qx , and Qy is their respective quantities.

M = consumer's money income

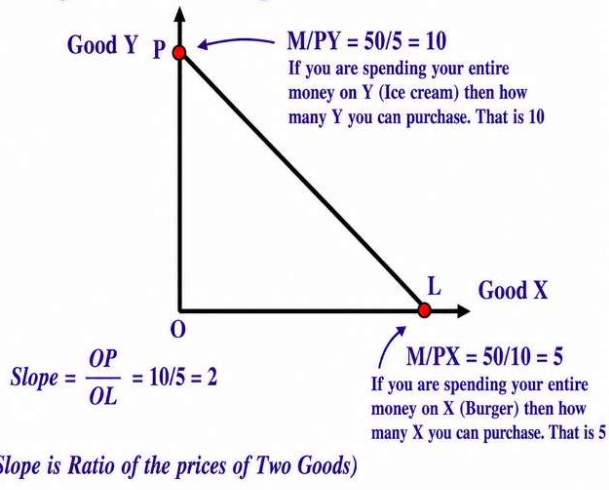
The Budget equation states that the consumer's expenditure on commodity X and Y cannot exceed his money income (M). Budget Line, also called as Budget Constraint

Example: Given consumer Income = Rs. 50, Price of Burger is Rs 10 per unit and Price of Ice Cream is Rs 5 Per unit.



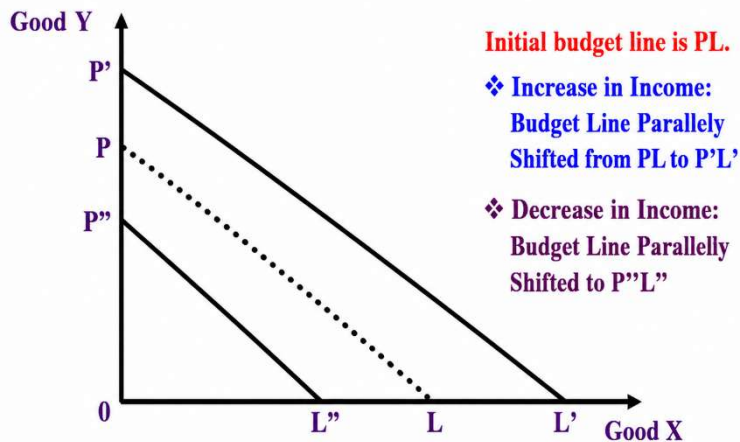
Slope of the Budget Line

Slope of the budget line is equal to relative price of two goods.



Ratio of price of two goods is nothing but the relative price which is P_X/P_Y

Effect of Changes in Income on Budget Line

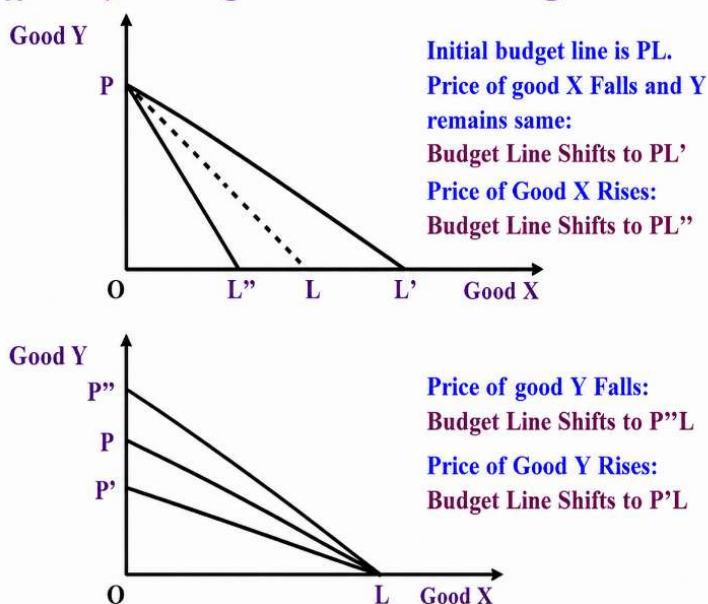


Here the price of both the product remains the same/constant.

Here only income changes

Effect of Changes in Price on Budget Line

Here, the income of the consumer and the price of one product remain the same/constant



Differences between Indifference Curve and Budget Line

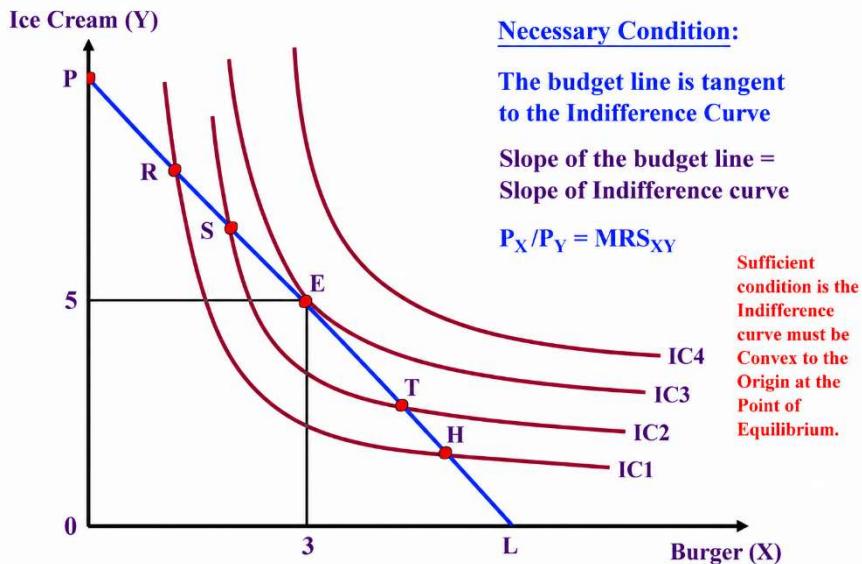
- The indifference curve indicates what the consumer is *willing* to buy
- The budget line shows what the consumer is *able* to buy
- When the indifference curve and the budget line are combined, we find the quantities of each good the consumer is both *willing and able* to buy

Consumer's Equilibrium

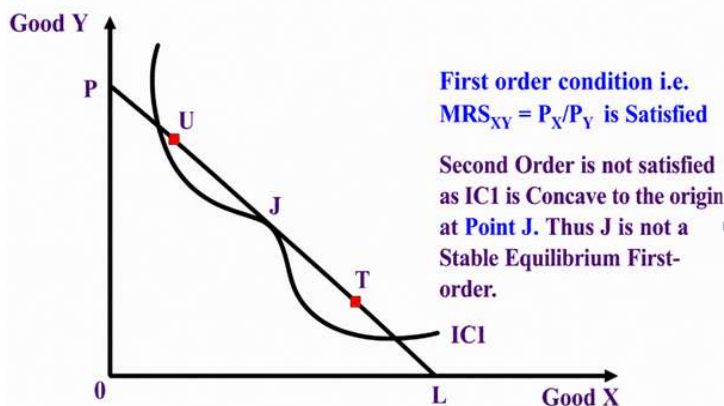
Consumer's Equilibrium refers to the maximum levels of satisfaction. It is a situation where a consumer spends his/her given income on purchasing one or more commodities so that the satisfaction will be the maximum. Once the consumer get equilibrium, the consumer has no urge to change his/her level of consumption. A consumer is at equilibrium where he/she derives maximum satisfaction from the goods and is in no position to rearrange his/her purchase.

Assumptions:

- The consumer has affixed amount of money to spend on two goods. He/she has to spend the entire money on two goods
- Prices of the goods are given and constant
- Goods are homogeneous



Consumer equilibrium refers to a unique position where the consumer does not want to change his/her consumption. PL is the budget line which shows all the combinations of X and Y. The consumer can purchase OL units of goods X if the entire income is spent on X. The consumer can purchase OP units of (goods) Y if entire income is spent on Y. Any other combinations consumer buys different combinations of X and Y. The consumer is in equilibrium at point E where he/she consumes 3 units of Burger and 5 ice creams. At point E both necessary and sufficient conditions are fulfilled. Any other points both conditions are not fulfilled. Sufficient condition is that the indifference curve must be convex to the origin at the point of equilibrium. At the Point of equilibrium E, the indifference curve must be convex to the origin or MRS_{XY} must be falling. Excluding point E, such as point R, S, T and H both the necessary and sufficient conditions are not fulfilled. A rational consumer wants to maximize the utility subject to the budget constraint.

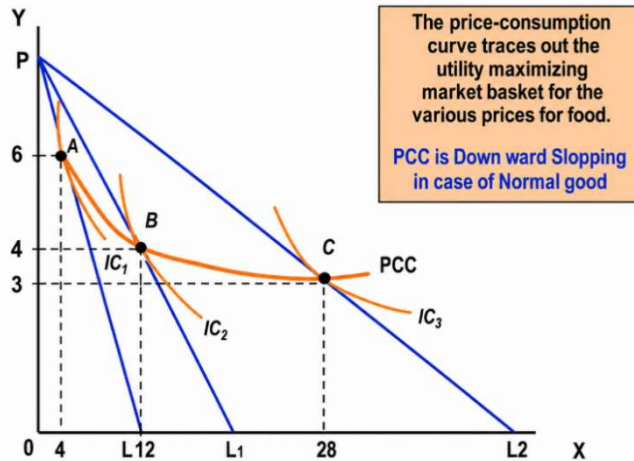


Price Consumption Curve

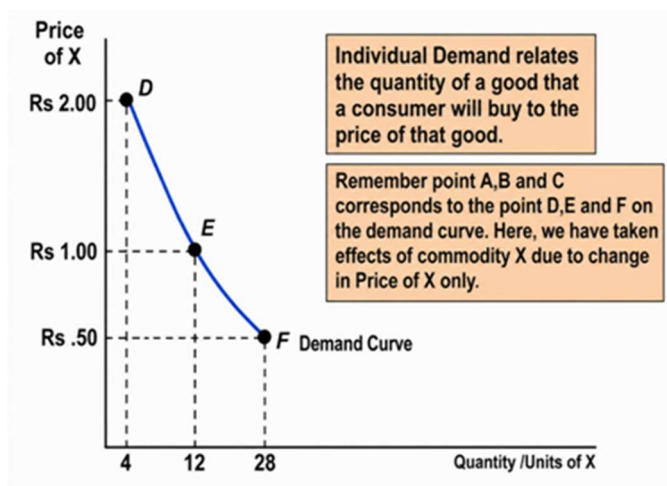
The Price Consumption Curve (PCC) shows all equilibrium points when the price of one good changes, while the other good price remains constant. In this analysis, we focus on two goods, X and Y, to simplify the consumer's behavior, even though consumers typically engage with multiple products in their daily lives. When the price of X changes and the price of Y stays the same. Price of X changes means price of X increases or decreases. But for simplicity, we are assuming price of X falls. When price of X decreases while price of Y remains same, there are two possibilities. Someone may say (i) Consumer will buy more unit of commodity X as it is relatively cheaper and will buy same amount of commodity Y (ii) consumer will buy more of commodity X and will buy less of commodity Y. When the price of commodity X falls, consumers typically purchase more of X and less of commodity Y. In this diagram, PL is the budget line which is a downward slopping straight line.

Assuming consumer income is Rs 20. Initially, the price of X was Rs 2, then decreased to Rs 1, and then further to 50 Paisa. The price of Y remained constant at Rs.2.

In a scenario where a consumer has an income of Rs. 20 and the prices of two goods are both Rs. 2. The consumer achieves equilibrium at point A, where the budget line is tangent to the indifference curve IC1. At tangency, the indifference curve is convex to the origin. At point A, both necessary and sufficient conditions are fulfilled. At point A, the consumer purchases 6 units of Y and 4 units of X. When the price of X falls from Rs. 2 to Rs. 1 and the price of Y stays at Rs. 2, the budget line becomes flatter along the X-axis. The new budget line shifted from PL to PL1, shifting the consumer equilibrium point from A to B. Consumers at point B increased their purchases of X from 4 to 12 units, while decreasing their purchases of Y from 6 to 4 units. The fall in the price of X from Rs. 1 to 0.50 Paisa results in a flatter budget line, now represented as PL2. The consumer is in equilibrium at point C. Here, the new budget line is tangent to the IC3. At point C both the necessary and sufficient conditions are fulfilled and consumer is equilibrium at point C. Consumer purchases 28 units of X and 3 units of Y at point. By joining all the equilibrium points namely point A, B and C, we will get the price consumption curve (PCC) which is downward slopping.



We can derive demand curve from price consumption curve. According to “Law of demand” price and quantity demanded are inversely related. In PCC, we have considered two commodities namely X and Y. To derive demand curve from PCC, whether we need to choose commodity X or commodity Y. Note that commodity Y price remains same and commodity X price changes. Hence, to derive demand curve we are considering commodity X price but not the commodity Y price. Commodity X is represented along the X-axis. Price of commodity X is represented in Y-axis. Initially, consumer income was Rs. 20, with prices of goods X and Y both set at Rs. 2. Consumer purchases 4 units of commodity X with given income and price of commodity Y. Assuming consumer income remains constant and the price of good Y is unchanged, a decrease in the price of good X from Rs 2 to Rs 1 is considered. As the price of commodity X decreases, demand for commodity X rises, while demand for commodity Y falls. Consumer purchases 12 units of commodity X at a price of Rs 1 per unit. Commodity X price drops from Rs. 1 to 0.50 paisa. Commodity X is becoming increasingly cheaper. The consumer purchases 28 units of commodity X. Different quantities and prices were obtained. If points D, E, and H are joined, a demand curve can be derived. The demand curve is downward sloping as price and quantity demanded are inversely related.

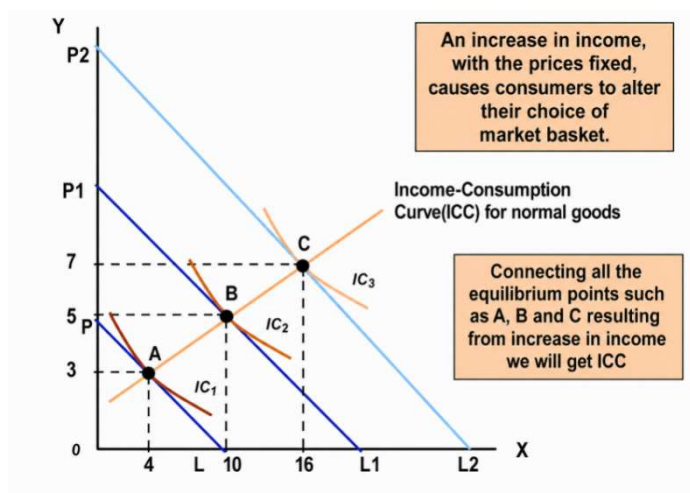


Income Consumption Curve (ICC) for Normal Goods

The Income-Consumption Curve (ICC) shows all the equilibrium points that result from changes in consumer income, holding commodity prices constant.. Let's assume that consumer is dealing two goods namely X and Y. Assuming $P_x = \text{Rs. } 1$ and $P_y = \text{Rs. } 2$, the income of the consumer can be Rs. 10, 20, or 30. P_x and P_y are the price of commodity X and Y respectively. Y is the income of the consumer. Initial consumer income increased from Rs 10 to Rs 20, and then to Rs 30. **Note:** when we are deriving ICC, the income of the consumer changes but price of both commodity such as X and Y remains same. When discussing changes in consumer income, we are assuming an increase in consumer income for simplicity.

Let's take two goods, X and Y for simplicity. Initially, consumer income is Rs.10, with the price of good X set at Rs.1 and the price of good Y at Rs.2. The initial budget line, represented as PL, slopes downward. The consumer is in equilibrium at point, A where the budget line is tangent to the indifference curve (IC_1) and at tangency, the IC_1 is convex to the origin. Point A represents the equilibrium point at which a consumer purchases 4 units of commodity X and 3 units of commodity Y. With an income of Rs. 10, a consumer can purchase 4 units of product X at Rs. 1 each and 6 units of product Y at Rs. 2 each. In this scenario, the consumer allocates all their income between two goods. What happens if the consumer income increases from Rs 10 to Rs.20. Consensus believes that with increase in consumer income, consumer will be able to buy more of both the goods. The budget line will shift to the right, remaining parallel to the original budget line. The budget line moved from PL to $P_1 L_1$, with equilibrium points changing from A to B due to an increase in income.

The consumer is equilibrium at point B, where the budget line P_1L_1 is tangent to the IC_2 and at equilibrium IC_2 is convex to the origin. Point B satisfies both necessary and sufficient condition. Consumer income is Rs 20, with prices of X and Y unchanged. At point B, the consumer purchases 10 units of commodity X and 5 units of commodity Y. If the consumer's income increases to Rs.30, the budget line will shift parallelly to the right. The equilibrium point shifted from point B to C. At point C both necessary and sufficient conditions are fulfilled. Point C is the equilibrium point for a consumer with an income of Rs 30, given that the prices of goods X and Y remain constant. If we connect all the equilibrium points A, B, and C, we obtain the income consumption curve for normal goods.



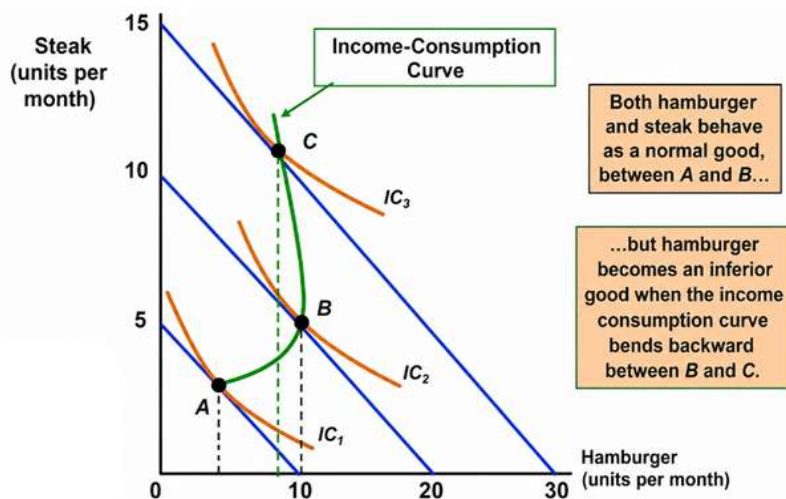
Income Consumption Curve (ICC) for Inferior Goods

X and Y axis represents the quantity of commodity X and Y respectively. Assume: $P_x = \text{Rs. } 1$ and $P_y = \text{Rs. } 2$ and income of the consumer $Y = \text{Rs. } 10$ further increased to 20, 30. When consumer income was Rs 10 consumer was buying 4 units of commodity X and the remaining Rs 6 is used to buy commodity Y. The consumer can able to buy 3 units of commodity Y.

Now the consumer income increased from Rs. 10, to Rs 20 and price of Commodity X and Y remains same. Let's assume that with increase in consumer income consumer is buying 10 units of commodity X ($10 \times 1 = \text{Rs } 10$). Consumer income is Rs 20 and Rs 10 has spent on commodity X. The remaining Rs 10 consumer will spend on commodity Y. The consumer can able to buy 5 units of commodity Y. Note: Both normal goods and inferior goods react differently with changes in consumer income. For instance, for normal goods, with increase in consumer income, consumer will buy more of commodity X and Y. With increase in consumer income, if the consumer buys less, then the good is treated as inferior goods. The movement from point A to point B the consumer is buying more of commodity X and commodity Y. Hence, the good is treated as normal goods. But what will happen even

though consumer income increased from Rs. 20 to Rs. 30, the consumer purchases less of commodity X and more of commodity Y. The movement from B to C, the commodity X has become inferior goods whereas; commodity Y is treated as normal goods.

Note: We can't present both the commodities as inferior goods in a single diagram. We can present one commodity as inferior goods and another commodity as normal goods. However, we can present both the commodity as normal goods



Normal Good vs. Inferior Good

- ❖ For normal goods, the income-consumption curve slopes upward. In other words, ICC has a positive slope for normal goods. More income leads to more demand for goods. In contrast, for inferior goods, the income-consumption curve slopes downward.
- ❖ Therefore, income elasticity of demand is positive for normal goods, while it is negative for inferior goods.

Engel Curve

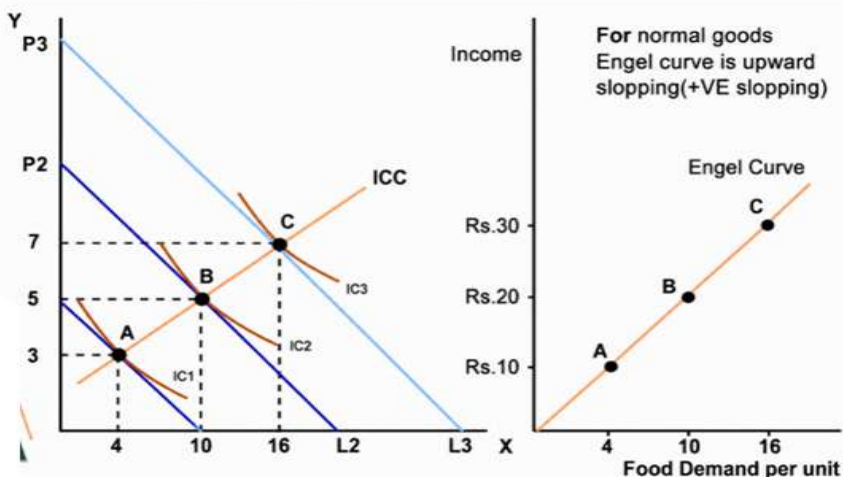
The Engel curve explains how quantity demand for good changes with changes in income. Note: Here the prices of two goods are remaining same. X axis presents the demand for the goods whereas; Y axis represents income of an individual. The Engel curve of an individual can be derived from ICC. On every point of the Engel curve, the consumer is in equilibrium. As a person's income goes up from Rs.10 to 20 to 30, their food spending goes up from 4 to 10 to 16 units. The good is normal if the curve goes up, and it is inferior if the curve goes down.

Engel Curve for Normal Goods

Let's assume initial income of the consumer is Rs.10, and then increased to Rs.20 and then Rs.30. Price of X is Rs 1 and price of Y is Rs 2. If consumer will spend his/her entire income on both the commodities then the consumer can able to buy 4 units of commodity X and 3 units of commodity Y. The consumer is at point A where indifference curve is tangent to the budget line. Likewise, with increase in consumer's income from Rs 10 to Rs 20, with given price of commodity X and Y, the consumer purchases 10 units of commodity X and 5 units of commodity Y. The equilibrium point shifted from point A to point B.

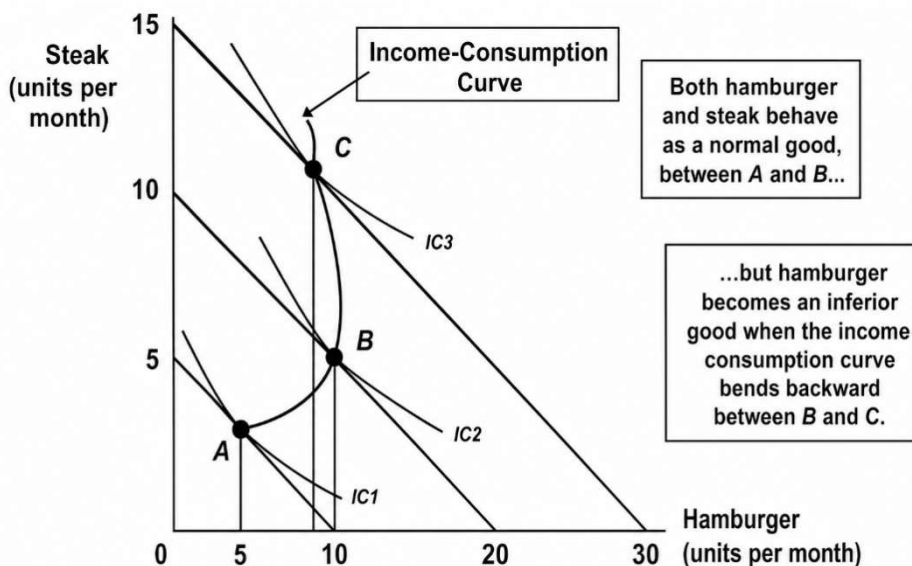
Similarly, if the consumer income increases from Rs. 20 to Rs. 30, with given price of commodity X and Y, the consumer purchases 16 units of commodity X and 7 units of commodity Y. The equilibrium points shifted from point B to C. If we will join the points of A, B and C then we will get income consumption curve (ICC) which is upward slopping.

From ICC we can derive Engel curve. X axis represents commodity X and Y axis represents income of the consumer. Note: we are considering change in commodity X for simplicity. If you will make change in commodity Y then the diagram will be very complex. When consumer income was Rs 10, consumer buys 4 units of commodity X. When consumer income increased to Rs 20, consumer buys 10 units of commodity X. Similarly, when consumer income increased to Rs. 30, consumer buys 16 units of commodity X. If we will join points A to C then we will get Engel curve for normal goods.



Engel Curve for Inferior Goods

Inferior goods are those goods with increase in consumer income, the demand for these goods decreases as consumer moves to better quality goods. Let's assume that there are two goods Steak and Hamburger. Initially both goods are treated as normal goods. With increase in consumer income, consumer purchases more of both the goods. Let's assume that initially consumer income is Rs 10 then increased to Rs 20 then Rs 30. When consumer income is Rs 10, the consumer is equilibrium at point A where budget line and indifference curve tangent to each other and at equilibrium the indifference curve is convex to the origin. With increase in consumer income from Rs 10 to Rs 20, the budget line shifts parallelly. The equilibrium point shifted from A to B where both necessary and sufficient conditions are satisfied. Similar to point A, the consumer purchases more of Steak and Hamburger with increase in consumer income from Rs 10 to Rs 20. At Point B, consumer purchases more of both goods with increase in income. Hence, similar to point A, point B consumer purchases more of Steak and Hamburger. Both goods are treated as normal goods at point A and B. With increase in consumer income to Rs 30, the budget line shifted parallelly. The consumer is at equilibrium at point C. Here consumer purchases more of Steak and less of Hamburger. We can say that though Hamburger was treated as normal goods but with further increase in consumer income this has become inferior goods. Note: we can't show both the goods as inferior goods at single point of time.



Price Effect is the combination of Income and substitution Effect

A decrease in the price of a good leads to two effects. (i) Income and (ii) Substitution effect

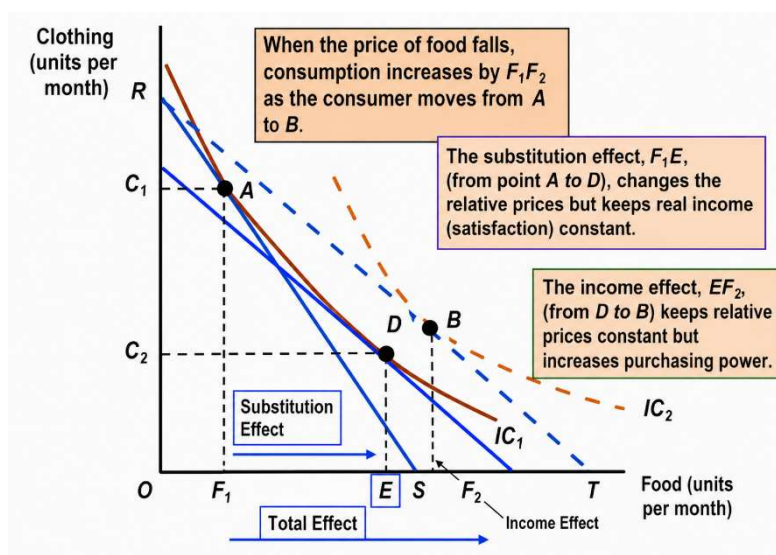
Substitution Effect

- The substitution effect refers to the change in consumption when a good becomes cheaper or costlier compared to other goods, while keeping satisfaction the same.
- Consumers tend to purchase more of a good when its price falls and less when its price rises. The substitution effect always makes people want more of an item when its price goes down.

Income effect

Because the price of a good went down, that good is now cheaper. So, people can buy more goods. The consumer is now better off because they can buy the same amount of goods for less money, which means they have money left over for other goods. The income effect is the change in demand that happens when real purchasing power changes.

Price, Income and Substitution Effects: Normal Good



Let's assume that the initial budget line is RS and there are two goods (food and clothing). Food is measured in X axis and clothing is measured in Y axis. The consumer is equilibrium at point A where IC_1 is tangent to the budget line (RS) and at equilibrium IC is convex to the origin. At point A, the consumer buys OC_1 units of clothing and OF_1 units of food. Let's see what happens if price of food(X) falls and price of clothing(Y) remaining same. Due to fall in price of food, the budget line will become flatter and the new budget line is RT. The

consumer is equilibrium at point B where both necessary and sufficient conditions are equilibrium. IC_2 is preferred to IC_1 due to more satisfaction. Why the consumer is moving from A to B i.e. from IC_1 to IC_2 . The movement from point A to C is because of price effects.

Price effect is the combination of substitution and income effect. First, we will see substitution effect. If price of food falls and price of clothing remaining same, then food will be relatively cheaper compared to clothing. Hence, consumer will substitute clothing with food. In order to see the substitution effect, we have to draw one imaginary budget line which will tangent to IC_1 . Though the consumer is substituting one good for another, the consumer satisfaction level remains same as consumer is in same indifference curve. Now new budget line tangent to IC_1 at point D. The movement from A to D is called substitution effect.

Since the price of food is relatively cheaper, the consumer will substitute clothing for food and consumer real income increases. Consumer purchasing power increases due to fall in price of food. With increase in consumer income, the budget line will shift parallelly. The budget line shifted to RT. The equilibrium point changes from D to B. The consumer is equilibrium at point B on IC_2 . With increase in income, consumer can able to buy more of both goods in the context of normal goods. The movement from D to B is known as income effect.

Note: You can break this diagram into 2 different parts such as substitution and income effect. Then you can combine the two diagrams to get price effect.

Review Questions

1. Evaluate the significance of the ordinal utility approach in explaining consumer equilibrium. Critically examine whether indifference curve analysis offers a more practical and realistic explanation of consumer behavior. Support your answer with appropriate arguments and a suitable diagram.
2. Explain consumer equilibrium with the help of Cardinal and Ordinal measurements of utility.
3. Vikram is working as a CEO in a top 10 investment bank of the world and his income is Rs. 2 crore per annum. Vikram wants to buy laptop and mobile phone. The Price of a Laptop is Rs. 2 lakh, and the price of a mobile phone is Rs. 1 lakh.
 - a. In this context, plot the budget line assuming that there is no changes in the price of laptop and mobile phones. What changes do you see in the quantity of laptop and mobile phones due to increase in Vikram income from Rs. 2 crore to Rs. 3 crore per annum
 - b. Plot the new Budget Line with Vikram's income is Rs 3 crore with no change in prices of mobile phones but the price of laptop increases from 2 lakh to 2.5 lakh.
 - c. What are the necessary and sufficient conditions that are required by a producer to reach the equilibrium position? Explain it with the help of a suitable diagram.
4. Let's assume that Virat's income is Rs. 120 per day. The price of a Burger is Rs. 20, and the price of an Ice-cream is Rs. 12.
 - a. In this context, plot the Budget Line assuming that there are no changes in the price of Burger and Ice-cream. Because of the economic slowdown, Virat's income falls from Rs. 120 to Rs. 60 per day. What changes do you see in the quantity of Burger and Ice cream due to fall in Virat's income
 - b. Plot the new Budget Line with Virat's income is Rs. 60 with no change in prices of Burger (Rs. 20) and price of Ice cream falls from Rs. 12 to Rs. 6.
 - c. What are the necessary and sufficient conditions that are required by a consumer to reach the equilibrium position? Explain it with the help of a suitable diagram.
5. Analyze the concept of consumer behavior in terms of utility
6. maximization by using the Cardinal utility approach
 - a. How does a consumer decide the optimum quantity of a good that he/she selects to consume?
 - b. utilities? How does he/she allocate his/her income for different goods to equalize the marginal

6. Analyze the relationship between price effect, income effect, and substitution effect. Explain how a change in the price of a commodity influences consumer demand using a suitable diagram.
7. Using an appropriate illustration, demonstrate how the demand curve is obtained from the Price Consumption Curve (PCC). Describe the connection between quantity demanded and price variations.

Unit-4

Theory of Production

“Confidence and Hard-work is the best medicine to kill the disease called failure. It will make you a successful person”.

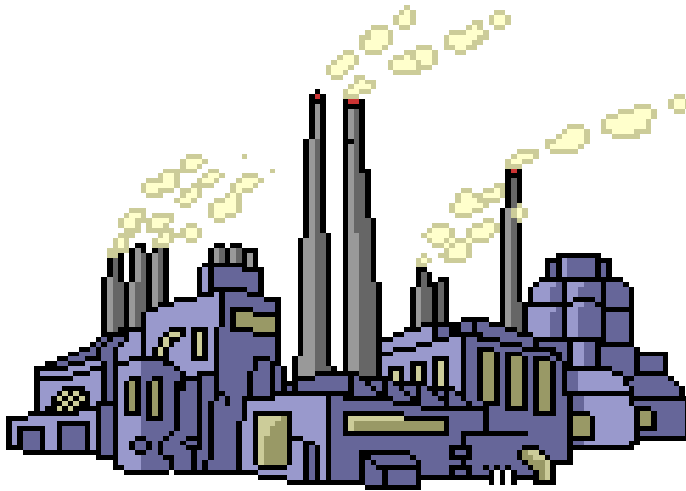
— A. P. J. Abdul Kalam

Production is a process of converting inputs into output through a technical process. Inputs are nothing but the factor of production. There are six factors of production that are used in the production process namely, land, labour, capital, entrepreneur, intermediate inputs, and technology.

Land is immobile in nature and inelastic in supply. Labour can be skilled or unskilled labour. Skilled labour can be found in fields such as health care, engineering, and science. Unskilled labourers are agricultural labourers, sweepers, and office boys. Capital represents machinery, buildings, equipment, and tools that are used in the production process to produce other goods.

Entrepreneurs are decision-makers who take risks and whose main aim is to increase the firm's value and make a profit. Intermediate inputs, such as raw materials, are used in the production process. Intermediate inputs are those inputs that are used to produce other final goods and services. Let's assume the car is the final product. Intermediate products are steel, plastic, rubber, etc. Technology plays a crucial role in the production process. For example, the Italian tire company Pirelli came up with a new technology called MIRS (Modular Integrated Robotized System) that can make a tire in 72 minutes instead of the 6 days it used to take. In developed countries, most of the work is done by technology.

Whether a company uses technology in its production process is related to entrepreneurial decision-making. The implementation of technology in the production process doesn't always need to be successful. For example, General Motors failed to use new technology, like robots to automate its factories. For simplicity, we are using only two inputs in the production process: labour and capital.

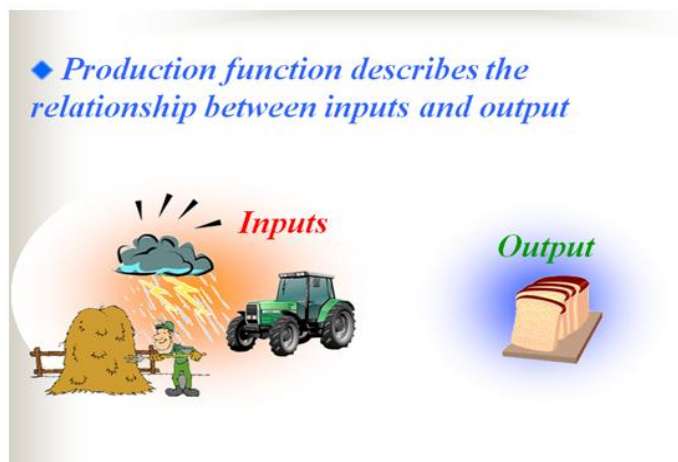


Production Function can be represented by

$$Q = f(L, K)$$

Q is the output produced in the production process. K is the number of machinery (capital) used and L is number of labours used in the production process. When all the inputs are efficiently used, then the maximum amount of output that can be produced. If more labour is used than capital to produce output, it is known as a labour-intensive technique.

On the other hand, if more capital is used than labour to produce output, it is known as a capital-intensive technique. Production functions are classified into short-run (SR) and long-run (LR) categories. In the short-run production function, capital is fixed, and labour is variable, whereas in the long-run production function, both capital and labour are variable.



When looking at production, it's important to distinguish between the short run and the long run production function. The short run refers to the time period during which at least one input is fixed. In the short run, capital is fixed, and labour is variable. The long run, on the other hand, is the time when all inputs can change. More clearly, in the long run, both labour and capital are variable.

The length of the long run depends on the industry's nature. For example, to establish a dry-cleaning industry, the long run might be 1 month, and the short run might be less than 1 week, whereas to establish a power plant, the long run might be 10 years, and the short run might be 5 years.

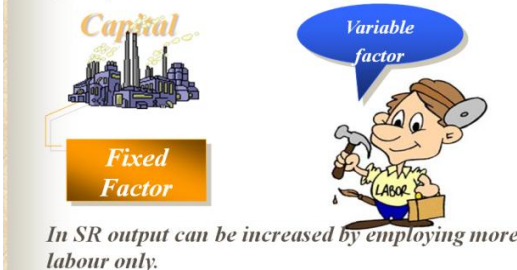
In the short run, a firm can increase output by using a greater number of labours as capital is fixed. In the long run, a firm can increase output by using a greater number of labour and more unit of capital. In the long run, labour and capital are variable. Hence, one firm increases output by using different combination of labour and capital.

Long-run technological changes always lead to better efficiency such that more output is produced with a constant amount of input used. There is no specified number of years – whether one year, five years, ten years, or twenty years between the short run and the long run. In long run, both the inputs are variable.

Production function in the short run(SR)

Assumptions:

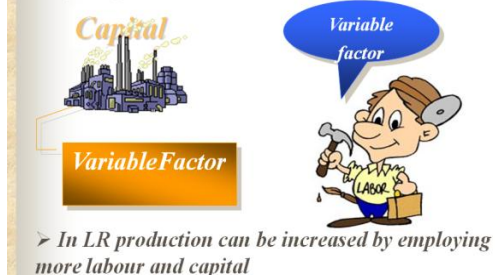
- only two factors are involved → *capital & labour*



Production function in the long run(LR)

Assumptions:

- only two factors are involved → *capital & labour*



Total Product (TP), Marginal Product (MP), and Average Product (AP).

Total Product (TP): Total product is the total amount of output that comes from using all the inputs in production. In the short run, total product is the maximum amount of output that can be produced at different levels of one input (L) while keeping the other input (K) fixed at a particular level. For instance, if we keep capital at one unit ($K=1$) and raise the unit of labour (from 1 to any unit), we will get the total product. In short run, total product is determined by changes in labor inputs because capital is fixed. Hence, in short run $TP = AP \cdot L$. TP is total product or total output of a firm. AP stands for average product. L represents units of labor employed. It is noteworthy to mention here that total product is obtained by adding the marginal products of successive units of labour.

Marginal Product (MP): the word marginal means extra or additional output. Now the question arises how the producer produces extra output. The producer can produce extra output by using extra labour or by using extra capital. If a producer produces extra output

which is related to employing extra unit of labour then we will get marginal product of labour. Similarly, if a producer produces extra output which is related to employing extra unit of capital, then we will get marginal product of capital. In the short run, labour is the variable input and capital is the fixed input. So, in the short run, you will get the marginal product of labour but not the marginal product of capital.

The marginal product is the extra output that comes from using one more unit of a variable input while keeping all other inputs the same.

$$MP = \frac{\Delta TP}{\Delta inputs} \quad \Delta TP \text{ is change in output}$$

$$MP_L = \frac{\Delta TP}{\Delta L} \quad (MP_L \text{ stands for marginal product of labor})$$

$MP_K = \frac{\Delta TP}{\Delta K}$ (MP_K stands for marginal product of capital. ΔK is the change in capital. It is possible only in long run not in short run). The firm use marginal product to decide if it is worth hiring another worker, and to decide how many workers to hire.

Note: in the short run labour is variable and capital is fixed. But in long run both the inputs such as labour and capital are variable.

Average Product (AP)

Average product is the total amount of output per unit of input. Average product=Total product/variable inputs.

$$\text{For Labor} = AP_L = \frac{TP}{L}$$

$$\text{For capital} = AP_K = \frac{TP}{K}$$

In the production process inputs are labour and capital. Labour is denoted as L and capital is denoted as K.

Let's assume there are 5 professors and they published 100 articles. Average publication per faculty is 20. Per unit efficiency is measured through average product.

Average product can't be a negative number because output of a firm can't be negative. The output of a firm can be zero if the firm shut down it's production. The total output of a firm is positive. Likewise, number of labourer employed in a firm is positive.

The Law of Variable Proportion (Short Run Production Function)

The Law of Variable Proportion is known as Law of Diminishing Return or Short run production function. The Law of Variable Proportions tell that

“If a producer increases the units of a variable factor (i.e. Labour) while keeping other factors fixed (Capital), then initially the total output increases at an increasing rate, then output increases at a diminishing rate, and finally the output starts declining.”

According to Law of Variable Proportion, there will be behavior of the output with respect to increase in the input factor, other factors remaining unchanged. It is where we have increasing one factor such as labour while other factors such as capital remain constant. With the increase in the variable factor, there will be a rising rate of the total product.

But beyond a certain point, adding more units of the variable input results in smaller and smaller increases in output (diminishing marginal returns). The law suggests that there is an optimal level of input use that maximizes the output more efficiently. After this point, inefficiencies or diminishing returns begin to occur. For instance, a farmer has a piece of land that doesn't change. The crop yield goes up at first when the farmer adds more fertilizer and labour. At some point, adding more labour or fertilizer results in a smaller increase in output.

Further, adding more units of labour and fertilizers leads to a decline in total output. The law covers many economic activities. For instance, in agriculture, it can be seen when more fertilizer is used; in manufacturing, it can be seen when more workers are hired; and in services, it can be seen when more staff are added to a restaurant. Understanding this concept helps businesses and government officials allocate resources and set production levels more efficiently to achieve the best possible output.

Derivation of the Law:

- ◆ *A single worker can only use part of those fixed resources at a time, even if the fixed factors stay the same.*
- ◆ *When more labourers are hired, they can specialize in different tasks, which makes them more productive. This leads to increase in MP and AP.*
- ◆ *Once the fixed factors are fully used, new workers can only assist the existing workers. As a result, the marginal product begins to decline, lowering the average product and possibly the total product.*

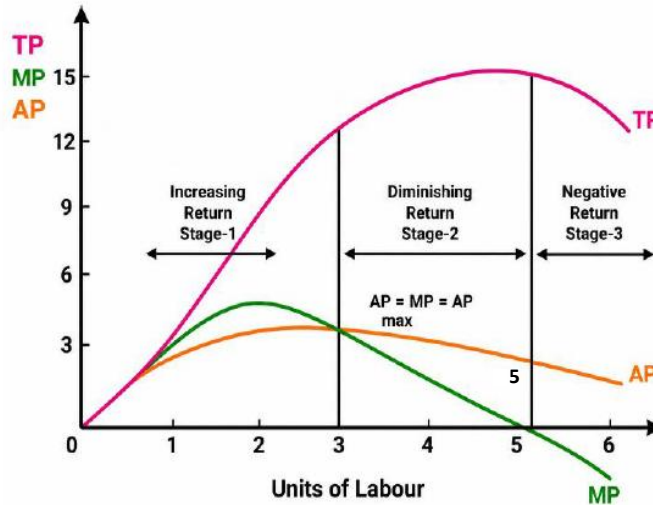
Assumptions

- ❖ The level of technology is assumed to remain fixed. There is not enough time to bring in new machines. Technological improvement requires planning, investment and time, which are possible only in the long run not in the short run.
- ❖ One input is kept constant and do not change in quantity (Here it is capital). There isn't much time to change all of the inputs. In the short run, one input stays the same because it can't be changed quickly.
- ❖ Units of the variable factor are homogeneous (same quality and efficiency). For instance, in scenarios where labourers are the variable factor, it is assumed that all labourers have same speed, and productivity. It means any changes in output are

attributable solely to the number of labourers rather than their individual quality differences.

- ❖ Possibility varying the proportions: In production, you can vary one input, such as labour, while holding others, like machinery, constant, enabling various input combinations.
- ❖ The law applies in the short run. In the short run, one factor remains fixed (such as machinery), while only variable factors (like labour) can be adjusted. The law operates in the short run as it necessitates one fixed input and one variable input, which is only possible in the short run.
- ❖ No training is provided to the labourers used in the production process. If training is provided, there is a greater likelihood that negative returns to the factor will not occur. Excluding agricultural labour, manufacturing and service sector labourers used to get some training. Hence, the law of variable proportion occurs in the field of agriculture.

Units of Labour Used	Total Product (TP or Output)	(MP _L)	(AP _L)
0	0	–	—
1	3	3	3
2	8	5	4
3	12	4	4
4	14	2	3.5
5	14	0	2.8
6	12	-2	2



From the above diagram, three stages of Law of Variable Proportion are observed.

- **Stage 1:** This is the phase of increasing returns to a factor, where output rises at an increasing rate as more units of the variable factor (labour) are used.
- **Stage 2:** This is the stage of diminishing returns to factor, where output continues to increase, but at a slower and slower rate as more units of the variable factor (labour) are added.
- **Stage 3:** This is the stage of negative returns to factor, where adding more units of the variable factor (labour) leads to a decline in total output.

Stage 1: Increasing Return to Factor

- TP increases at an increasing rate. MP & AP Increases.
- Stage 1 ends when AP reaches maximum and where $AP = MP$

Why MPL increases at stage 1

That is because of fixed inputs are more as compared to variable inputs

- **Division of labour:** In the first stage of the law of variable proportions, workers specialize in different tasks, which makes them more efficient and productive. Division of labour means the work is distributed among different workers as per their expertise. These results increase in output.
- **Specialization:** specialization lets workers focus on certain tasks, which makes them more efficient and increases output.
- **Efficiency increases:** efficiency goes up at first because better use of resources and better coordination lead to higher productivity.

- All the above-mentioned factors lead to increase in productivity.

Reason: Fixed factor is abundant in comparison to variable one. So better utilization of fixed factor takes place with the increase in amount of variable factor.

Stage 2: Diminishing Return to Factor

MPL diminishes, TP continue to increase at a decreasing rate until it reaches its maximum points

MP and AP falls but remains positive

Second stage ends where TP reaches maximum and MP is zero

Why MP_L diminishes?

Variable input is comparatively more compared to fixed inputs

The fixed input is utilized up to the maximum capacity

TP reaches its max point when $MP=0$

This is the end of stage 2. Here fixed inputs are utilized to its optimal capacity

This is the most rationale stage of production. Here TP is maximum

Reason: Increasing utilization of fixed factor and variable factor is getting less and less aid from the fixed factor.

Stage 3: Negative Return to Factor

- Since MP is negative, TP diminishes from the Max. Here AP also declines but AP can't be Zero.

Since MP is negative, no producer will prefer this stage.

Reason: The variable factor is too high compared to the fixed factor. Impairs the efficiency of the fixed factor. Too many cooks spoil the broth

- The law of diminishing returns says that when you add more units of a variable input, the extra output you get from each unit gets smaller. Rational producers, whose goal is to make as much money as possible and work as efficiently as possible, may not want to work in this phase for a number of reasons:
- In phase 3, each additional unit of the variable input (such as labour) contributes less to the total output compared to previous units. This means that the cost per unit of output starts to increase, reducing profitability.
- Cost Inefficiency: As productivity per unit of input declines, the cost-effectiveness of production decreases. This can lead to higher average costs of production, reducing profit margins.

- Producers aim to maximize profits. In phase 3, where marginal returns diminish, continuing to produce at the same level may not yield sufficient additional revenue to cover the increased costs. Rational producers may choose to reduce production or find alternative ways to improve efficiency
- Operating in phase 3 may indicate falling behind competitors who are achieving higher productivity and lower costs.
- In summary, rational producers typically avoid operating in phase 3 of diminishing returns because it signifies reduced efficiency, higher costs, and potentially lower profitability.

Long run Production Function

There are only two factors of production that are used in the production process. These two are labour and capital. There is the long run production function where both the labour and capital may vary. In the long run, the factors of production have nothing that is fixed since both vary hence the firm may increase its output by using different quantities of both the labour and capital.



ISO Quant

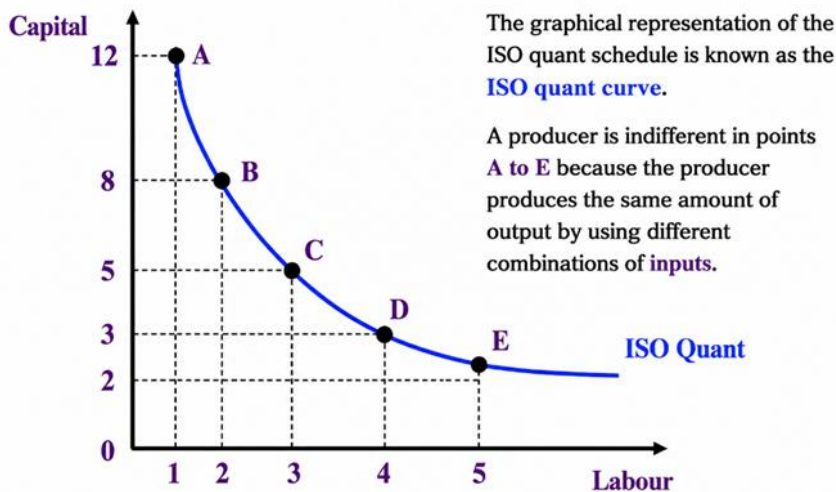
An ISO quant shows all the possible combination of two inputs that gives same level of output to a producer. In this case, labour and capital are the two inputs. When we are talking about ISO quant then we are talking about producer point of view. On the contrary, when we are talking about indifference curve then we are talking about consumer point of view. As different combination of labour and capital provides same level of output to a producer, the producer become indifferent in different points. Hence, ISO quant is also called as production-indifference curve.

ISO Quant Schedule

It is a tabular representation of various inputs or factors of production (labour and capital). Each of the factor combinations A, B, C, D, and E producer produces the same level of output. For example, output is 20 units at different combinations from point A to E.

Factor Combinations	Labour	Capital	Output
A	1	12	20
B	2	8	20
C	3	5	20
D	4	3	20
E	5	2	20

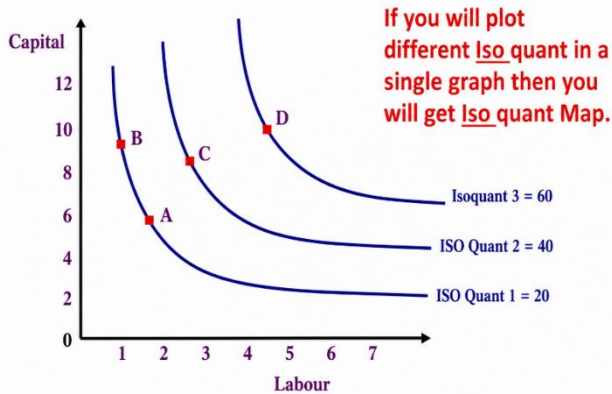
ISO Quant Curve



Labour is presented in X axis and capital in Y axis. For combination of factors in A, the producer produces 20 units of output by using 1 unit of labour and 12 units of capital. For combination of factors in B, the producer produces 20 units of output by using 2 units of labour and 8 units of capital. For combination of factors in E, the producer produces 20 units of output by using 5 units of labour and 2 units of capital. Here we can say that the producer does not mind any point A, B, C, D, and E.

There is an inverse relationship between the quantity of one factor employed and the quantity of another along an ISO quant curve. Hence, an ISO quant curve is downward slopping.

ISO Quant Map



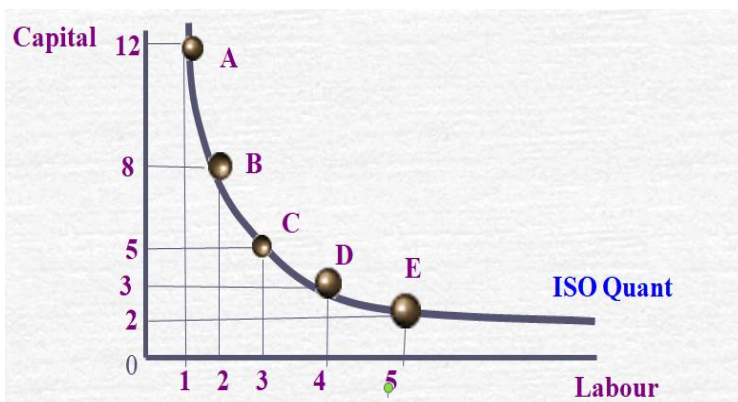
ISO Quant vs. Indifference curve

Though ISO Quant is similar to indifference curves, there are some differences between them.

- ❖ Indifference curve shows various combinations of two goods which gives same satisfaction to a consumer whereas, ISO Quant shows various combinations of two inputs which give same output to a producer.
- ❖ In indifference curve, we studied consumer behaviour whereas, in ISO quant, we studied producer behaviour
- ❖ In indifference curve, we are dealing two goods (X and Y) whereas, in ISO quant, we are dealing with two inputs such as Labour and Capital
- ❖ The indifference curve does not define the level of utility or satisfaction in quantitative terms whereas, ISO quant does

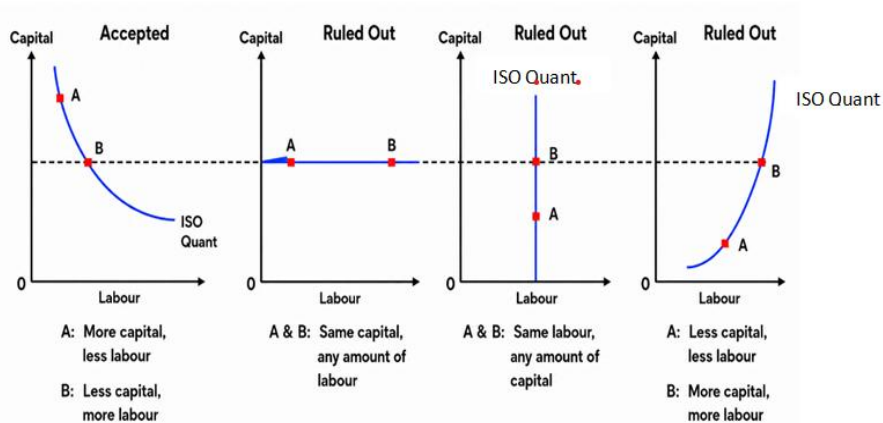
Properties of ISO Quant

(i) ISO Quant Slope Downward to the Right



An ISO quant slopes downward to the right, indicating a trade-off between factors of production. At combination A, 12 units of capital and 1 labour produce 20 units of output. In combination B, 8 units of capital and 2 labours yield the same output. It implies that if one factor of production is used more, the other is used less. This happens due to scarcity of resources.

Why can't an ISO quant be horizontal, vertical, or upward sloping?



(ii) ISO Quant is convex to the origin

The reason why ISO quant is convex to the origin is due to diminishing Marginal Rate of Technical Substitution (MRTS). This concept is quite analogous to Concept MRS. MRTS is basically about substituting inputs. It means the extent to which the substitution of factors of production occurs without changing the output produced. MRTS of labour to capital implies the number of capital units that could be reduced when labour increases by one additional unit.

Marginal Rate of Technical Substitution (MRTS)

Factor Combinations	Units of labour	Units of capital	$MRTS_{LK}$	Output
A	1	12	-	20
B	2	8	4	20
C	3	5	3	20
D	4	3	2	20
E	5	2	1	20

$$MRTS_{LK} = \frac{\text{Change in Capital inputs}}{\text{Change in labour inputs}} = \frac{\Delta K}{\Delta L}$$

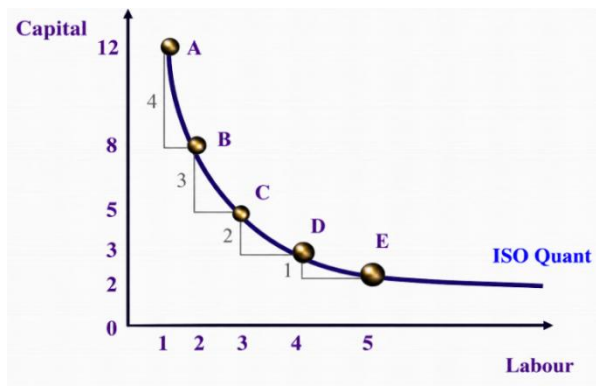
ΔK is the small Change in capital and ΔL is the small change in labour
MRTS at a point is the slope of the ISO quant at that point.

The Slope of the ISO Quant shows the rate at which one input may be substituted for the other keeping the same amount of output constant.

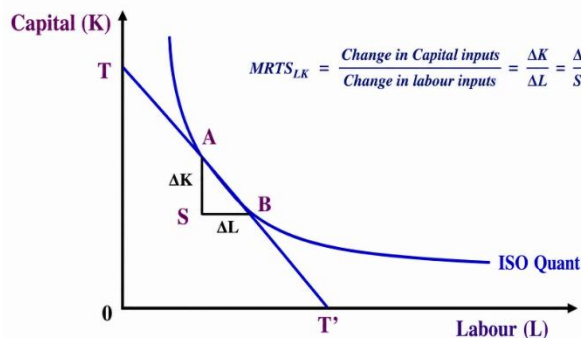
At point A, producer uses 1 unit of labour and 12 units of capital for producing 20 units of output. Producer uses 2 units of labour and 8 units of capital for producing 20 units of output at point B. So when the producer shifts from combination A to B, (12-8 = 4) units of capital get substituted by (2-1 = 1) unit of labour. So MRTS = 4 at combination B.

Similarly, when the producer moves from combination B to C (8-5=3) units of capital are replaced for (3-2=1) unit of labour. Therefore, MRTS is 3 at combination C. Likewise, when the producer moves from combination C to D, MRTS is 2. MRTS is keep on diminishing as a result ISO quant is downwards slopping. When producer moves from combination A to B, B to C and C to D and D to E, MRTS falls from 4 to 3, 3 to 2 and 2 to 1.

Diminishing Marginal Rate of Technical Substitution (MRTS)

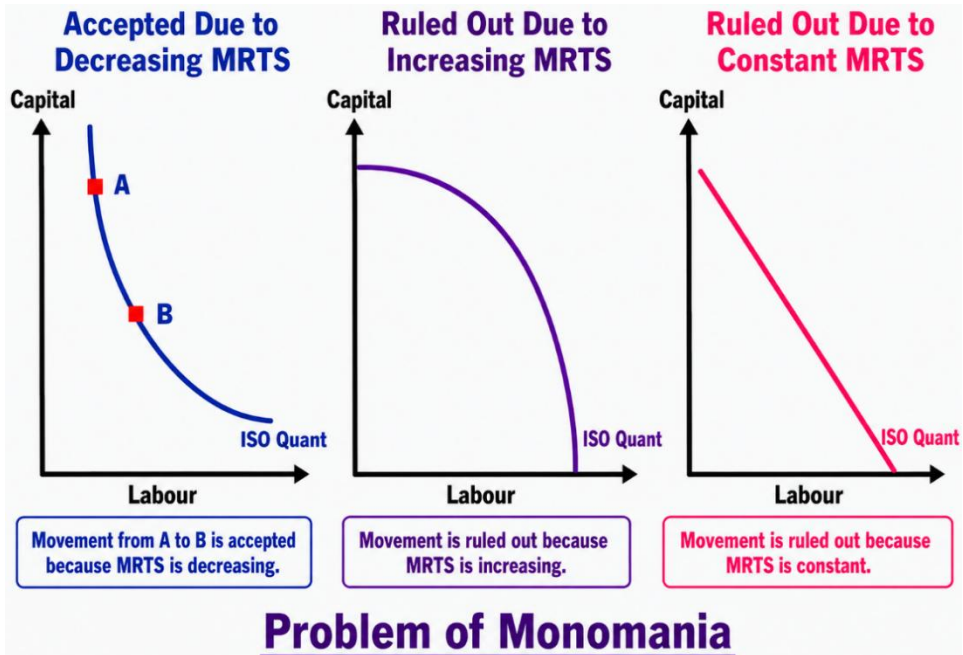


The ISO quant is convex to the origin due to diminishing or falling Marginal Rate of Technical Substitution. As the producer proceeds from A to B, B to C, C to D, and finally D



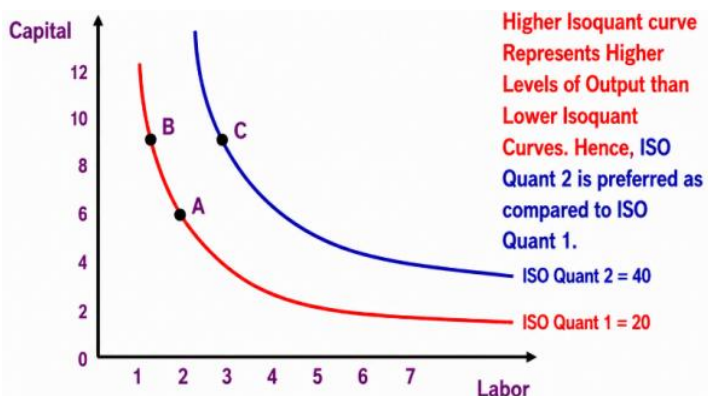
to E, the MRTS drops from 4 to 3, 3 to 2, then 2 to 1. The marginal product of labour decreases if the amount of labour increases at the expense of capital. Similarly, if the amount of capital increases at the cost of labour, the marginal product of capital declines.

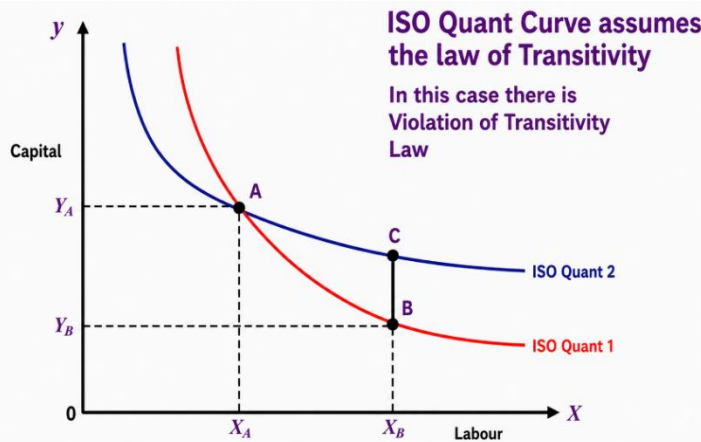
Why can't an ISO Quant be concave to the origin and downward sloping?



(iii) Higher is the ISO quant higher is the output

If we will compare point A, B and C, at point C the producer uses more labour and capital to produce output. More labour and capital leads to more output. Hence, point C the producer produces more output compared to point A and B. Hence, point C is preferred by a producer compared to other points. Point A and B, the producer is indifferent as producer lies in same ISO quant. Different points on a same ISO Quant provides same level of output and the producer is indifferent.



(iv) Two ISO Quant cannot Intersect with each Other

Let's assume ISO quant 1 and ISO quant 2 intersect with each other at point A. Points A and B lie in ISO Quant 1. Point A and Point B provide the same output to a producer. Hence, the producer is indifferent at point A and point B.

Likewise, point A and C lie in ISO Quant 2. As both points lie in the same ISO quant, they provide the same output to a producer. Hence, the producer is indifferent between points A and C in ISO quant 2. In ISO Quant 1, $A=B$ and ISO Quant 2, $A=C$. It is clearly visible that A is common.

Hence $B=C$. But point B lie in ISO quant 1 and point C lie in ISO quant 2. Hence, B and C are not equal. If you will put one line point C produces more output by using more capital. You also know that a higher ISO Quant is preferable to a lower ISO Quant. Hence, we can say that two ISO quant can't interact with each other.

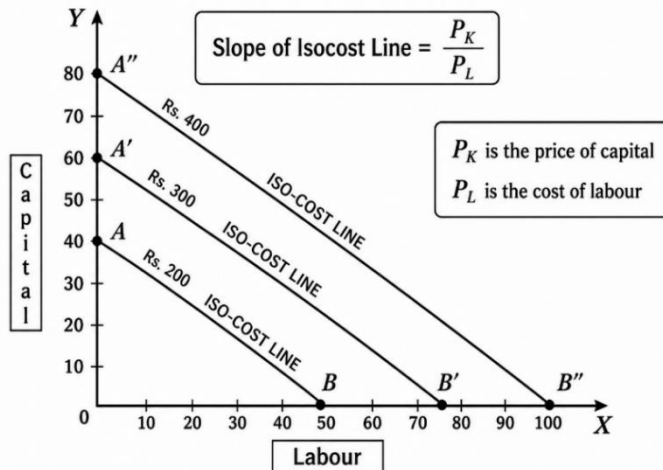
Different ISO Quants

- ☐ Right Angle: Inputs are used in Fixed Proportions and therefore are not substitutable. Example: Yeast and Flour for a specific type of bread. Tires and a Battery for an automobile
- ☐ Straight Line: Inputs are perfectly substitutes. MRTS is constant. Example: Honey and Brown sugar are nearly perfect substitutes in the area of baking. Natural Gas and Fuel Oil in Energy production.
- ☐ Downward sloping and Convex to Origin: Common situation in which inputs are imperfect substitutes. Labour and Capital are imperfect substitutes in general.

ISO-Cost Line

Various combinations of two inputs, such as labour and capital, yield the same cost to a producer. Here, the producer is indifferent at any point of the same ISO cost line. Example: XYZ company has Rs 200 which need to be spend on two inputs namely labour and capital.

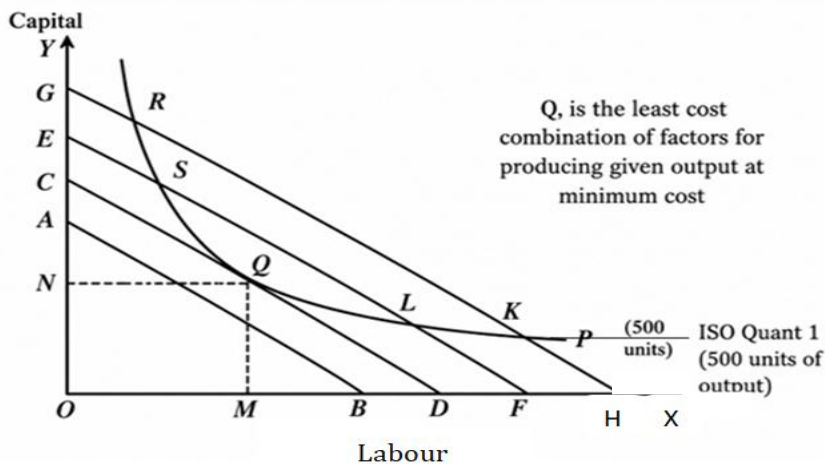
The price of capital is Rs 5 per unit, while the price of labour is Rs 4 per unit. At point A, company spent entire money on capital and purchases 40 units of capital. At point B, company spent entire money on labour and purchases 50 units of labour. If we will join the two extreme point A and B, then we will get ISO cost line of 200. AB is the ISO cost line where the company spent Rs 200. Any combination within AB line gives same cost to a producer. Further, the company wants to spend Rs 300 and Rs 400, respectively.



Producers Equilibrium or Least Cost Combination of Factors

The producer's goal is to make as much money as possible. To make the most money, the manager:

- (i) Wants to keep costs as low as possible for a certain level of output
- (ii) Desire to get maximum output for a given cost



The Firm wants to maximize the level of output with minimum cost. The least-cost combination happens when the ISO quant curve and the ISO cost line touch each other.

We can say the producer will be in equilibrium and least cost combination is achieved when both necessary and sufficient conditions are met.

Necessary condition

- The ISO quant curve touches the ISO cost line at one point.
- The ISO quant and ISO cost line have the same slope.

Sufficient condition

- At equilibrium, the ISO quant is convex to the origin

The Y axis shows capital and the X axis shows labour. Let's assume that the producer wants to produce 500 units of output.

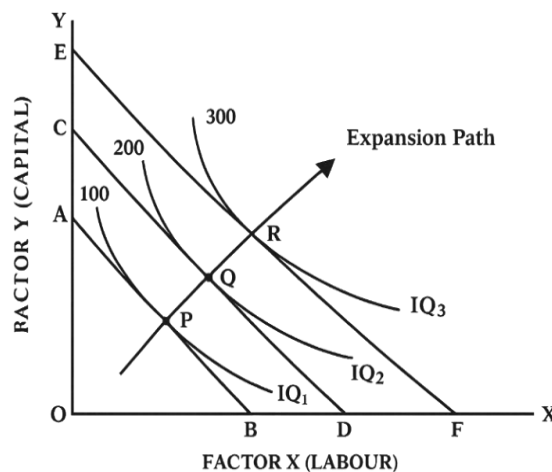
This level of output can be produced by various combinations (R, S, Q, L, K) of labour and capital on the ISO quant curve.

At point Q, where the ISO quant curve touches the ISO cost line, it is clear that the producer is in equilibrium and costs are at their lowest. At point Q, the isoquant and ISO cost have equal slopes, and at equilibrium ISO quant is convex to the origin. At point Q both necessary and sufficient conditions are fulfilled.

Any other points such as R, S, L, K lie on the higher ISO cost line for the same level of output. Hence, the producer prefers only point Q where output is maximum and the cost is minimum. So, point Q is the least cost combination.

Expansion Path

Expansion path is known as scale line. In expansion path, prices of two factor inputs remain constant. AB represents the ISO cost line. IQ₁ is the ISO quant curve of 100 units of output. The expansion path connects all the cost minimizing input combinations. The expansion path connects all the equilibrium points where cost is minimum. The cost will be lowest when the ISO quant line touches the ISO cost line at a certain output level. In order to



produce 100 units of output, the least cost combination is P. If the firm wants to expand its scale (Output to 200 units) the firm requires more labour and capital. Here, the least cost combination is Q. If you will join all the least cost combination points such as P, Q and R, then you will get an expansion path.

Expansion path shows how firms adjust their input mix as they increase production to maintain efficiency and minimize costs. It's a practical application of economic theory that helps firms make informed decisions about their production processes.

Returns to Scale

Returns to scale shows how output changes when all inputs increase in the same proportion. It is the rate at which output increases as inputs are increased proportionately. Change in output due to change in scale Change in Scale means Increase in the Scale i.e. all inputs or factors are increased together in the same proportion. It is the quantitative change in the output of a firm or industry resulting from a proportionate increase in all inputs. It focuses on the overall relationship between input and output when all factors are increased simultaneously.

Different Types of Returns to Scale

Returns to scale are classified as

- ☛ Increasing Returns to Scale (IRS)
- ☛ Constant Returns to Scale (CRS)
- ☛ Decreasing Returns to Scale (DRS)

Machine Labour Output

$$1 + 1 = 100 \text{ units}$$

$$2 + 2 = \text{What will be the output?}$$

If the output is equal to 200, then it is called as CRS

If the output is more than 200, then it is called as IRS

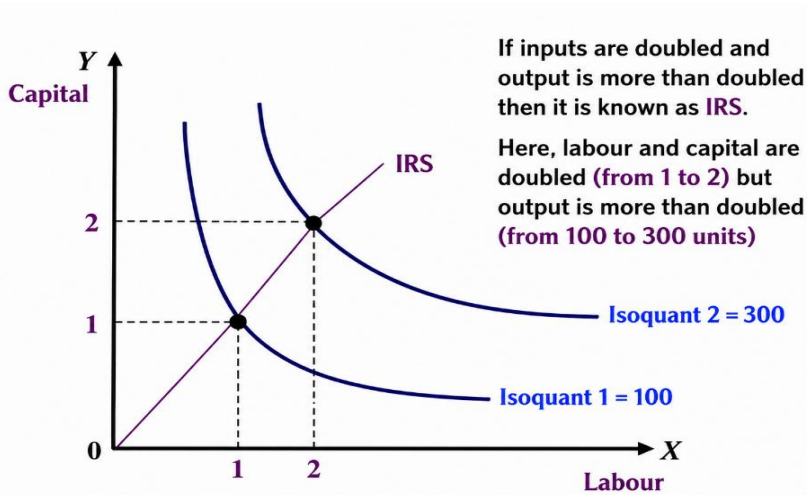
If the output is less than 200, then it is called as DRS

Increasing Returns to Scale (IRS)

The output increases greater proportions as compared to inputs.

$$\begin{aligned} Q &= f(L, K) \\ &= f(\alpha L, \alpha K) \\ &= \alpha^\beta f(K, L) \end{aligned}$$

When $\beta > 1$, then we have IRS



In the long run, some benefits can be expected from the business, concerning the savings in costs and increases in production. This is the advantage obtained when producing only one commodity, and the concept is called economies of scale. The economies of scale involve a situation whereby an increase in the production level leads to a reduction in the cost of producing each unit. For instance, a plant doubling the quantity produced would not require double the number of administrators in order to produce, which would reduce the cost per unit of production. Economies of scale have a strong relationship with increasing returns to scale.

Benefits/advantages of producing in large quantities

- ❖ Increased Specialization of labour
- ❖ More efficient use of large pieces of machinery
- ❖ Technology improvement will enhance the output and reduce the per-unit cost of production
- ❖ Training is given to the employee to enhance their skill so that they can produce more output.
- ❖ Transportation benefits
- ❖ Bargaining power benefits
- ❖ Volume of raw material discounts
- ❖ Managerial skills benefits

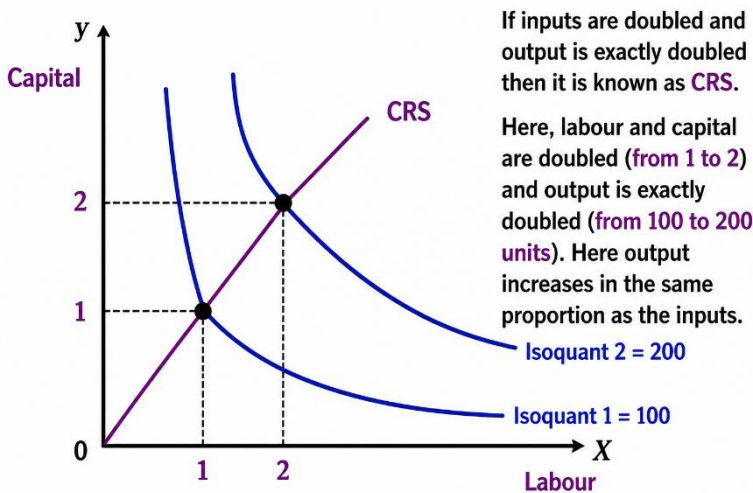
Constant Returns to Scale

Constant returns to scale happen when a firm's output increases in direct proportion to the rise in inputs. Can be Stated as:

Proportionate Increase in Output = Proportionate Increase in all inputs

- If inputs will increase by 10% the output will exactly increase by 10%
- Likewise, if input will decrease by 10% the output will exactly decrease by 10%
- It means change in input and change in out is at same proportion

$$\begin{aligned} Q &= f(L, K) \\ &= f(\alpha L, \alpha K) \\ &= \alpha^\beta f(K, L) \end{aligned}$$



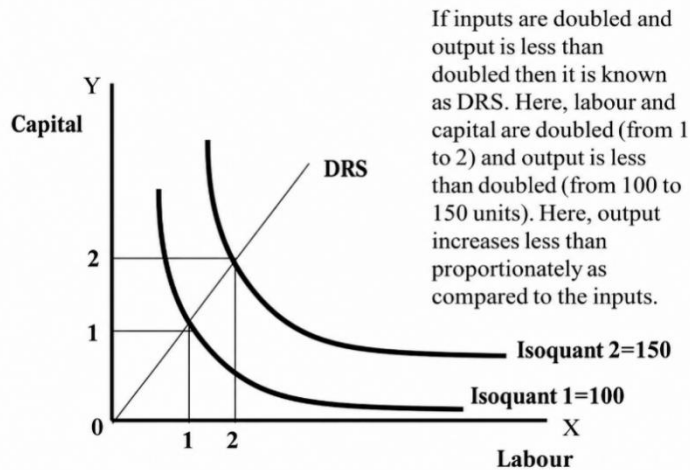
When $\beta = 1$, then We have CRS

Decreasing Returns to Scale:

Output rises but at a slower rate compared to the overall increase in inputs.

$$\begin{aligned} Q &= f(L, K) \\ &= f(\alpha L, \alpha K) \\ &= \alpha^\beta f(K, L) \end{aligned}$$

when $\beta < 1$, then we have DRS



Decreasing return to scale indicates that the disadvantages that the firm faces while generating any output during the long run. During the long-run, firms can become too large and this will cause them to create too many layers of management and departments. Such problems can lead to poor communication, inefficiency, delays in decision-making, and poor output generation as well. The reason for the latter is that the level of output will increase at a smaller rate than inputs.

Decreasing returns to scale are directly linked to diseconomies of scale. Here the average cost per unit increases. It means the disadvantages occurring to a firm while expanding its output in the long run. When input prices remain the same, decreasing returns to scale leads to higher long-run average costs (diseconomies of scale).

Coordination problems that are natural to any large organization can lead to diseconomies of scale. With increase in scale of operation, management becomes difficult which leads to increase in cost of production. Communication between workers, management, and various functional areas is becoming increasingly challenging. This can lead to extended decision implementation times and increased input costs.

Reasons:

- ☞ Increasing Difficulties of Management/mis-management
- ☞ Increasing Difficulties of Co-ordination
- ☞ Increasing Difficulties of Control
- ☞ Less incentives
- ☞ No proper recognition of talent and skills
- ☞ No motivation

Application of Returns to Scale

- ☛ Returns to Scale Affects Company's cost

With Given Input Prices:

- ❖ Company having IRS, will have declining cost per unit of output as output increases.
- ❖ Company having DRS, will have increasing cost per unit of output as output increases.
- ❖ Company having CRS, will have constant cost per unit of output as output increases.
- ☛ It affects the company's ability to compete in global markets. Industries experiencing increasing returns may be able to produce goods more efficiently at larger scales, enhancing their export competitiveness.
- ☛ IRS is a important phenomenon in the automobile industry. (The Global Six, Business Week, Jan 25, 1999)
- ☛ Companies most aware of the Returns to scale: General Motors, Ford, Daimler-Chrysler
- ☛ Returns to scale help businesses determine the most efficient scale of production. By analyzing whether they are experiencing increasing, constant, or decreasing returns, firms can adjust their production levels to minimize costs and maximize output.
- ☛ Returns to scale provide a framework for businesses,
- ☛ Policymakers, and economists to analyze production
- ☛ Efficiency, plan investments, and formulate strategic decisions that optimize resources and maximize returns in various economic contexts.

Cost Theory

“Arise! Awake! Stop not Until the Goal is Reached”.

— Swami Vivekananda

Cost refers to the expense incurred in the production of goods and services. For a firm point of view cost includes rent, labour, raw material, machinery and day to day operating expenses. Cost plays an important role in managerial decision-making process. A company can make profit by maximizing the output and minimizing the cost. In order to survive in the competitive world, each and every company must try to minimize the cost.

If cost of the product will be low then price will be low and then the firm can sell the product at lower price. Controlling and minimizing cost helps the manager to enhance their profitability. In order to minimize the cost managers are following cost benefit analysis. Manager must try to minimize unnecessary expenses.

Due to globalization competition has increased among the companies. Hence cost management is required for the survival in business. During COVID-19 scenarios, the entertainment industry, hotel industry and aviation sector reduces their cost for survival. They fired their staff. The Airline Industry reassessed costs following the September 11, 2001 terrorist attacks.

For instance, American Airlines eliminated all food service on domestic flights in 2002. Big Companies such as General Motors and General electronics pressured their suppliers to cut costs. During COVID-19 scenario the situation were cut costs or else ready for shut down of your business. Understanding costs helps managers in optimizing limited resources for maximum output. Successful managers need to assess the cost-to-revenue ratio to evaluate the company's profitability. For business or managerial economics perspective, the analysis of cost is very important. It helps manager to make two important decisions:

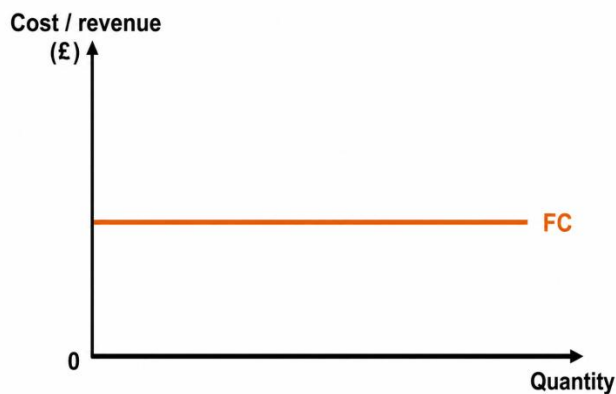
- (i) Whether to produce or not and
- (ii) How much to produce when a decision is made to make it





Fixed Cost:

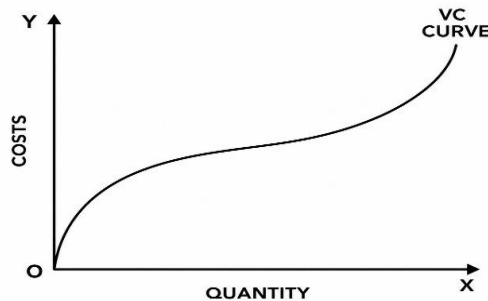
The cost of employing fixed factors refers to fixed cost. It doesn't vary in the short run. No matter how much output the firm produces fixed cost remain same. Even if the firm is not producing any output but the fixed cost must be paid. Fixed cost include contractual rent, insurance, Interest on loans, property taxes, plant construction, telephone tower construction, dam construction, purchase of AC, Laptop, Smart board, and salary of the top management etc. Whether company is producing any output or not company has to pay rent, insurance, salary to the top management etc. The company can avoid fixed costs if it is going out of business or closing the factory. Fixed costs are graphically appeared as follows:



Variable cost

The cost incurred when employing variable factors is known as variable cost. It varies directly with the level of output. Variable costs are costs that change depending on how much is produced in the short and long run. Variable cost includes wages, salary, purchase of raw material, fuel, power, labour cost, excise taxes, packaging costs, heating and lighting, advertisement expenditure cost etc. Variable cost changes with the level of

output. Example: In a bakery, increased bread production leads to higher flour costs. Producer will get variable cost when some amount of output is produced. Variable costs can be illustrated graphically.



Semi variable cost: Electricity cost. If you are not using electricity, you are paying the minimum charge. If you are using more you have to pay more.

Opportunity Cost

Opportunity cost is the cost of next best alternative foregone. It is also called as opportunity lost. Let's assume that you got two jobs. One job is in HSBC Bank and another job is UBS. You are not Shakti man. Hence, you can't join two jobs simultaneously. The company you are not joining is opportunity cost. Likewise, the opportunity cost of going to university is the money you would have earned if you worked instead. The opportunity cost of losing four years of salary while doing your graduation degree in United States. Opportunity cost is subjective in nature. It varies from person to person. In order to calculate opportunity, cost we need more time and proper information. The calculation of opportunity cost will be wrong without proper information.

Social Costs: It is the cost to the society. These costs are created by an individual or companies. The Cost that the Society bears because of individual and Companies activities. Air pollution, Noise, Garbage etc. are the costs for Societies. Let's assume that industrialist is throwing waste into the river. Poor people are using river water for bathing, cooking etc. and frequently going to the hospital and becoming poorer and poorer. As a result, inequality of an economy increases. It is cost to the society. Cutting down the number of trees in the forest creates environmental pollution as a result social cost increase.

Sunk Costs: Expenditure that has been incurred and is non-recoverable. It means the money a business can't get back from what it spent. You have booked a flat in a high-rise apartment building near the catchment area, but there is a legal action going on. You can't get that money back.

Your building has no alternative uses. This is sunk cost. After 10th exam, you decided to join in science stream. But after 1 year you changed your mind to go with commerce stream. Hence, the money you have spent you can't get it back. It is known as sunk cost. Research and development expenditure is sunk cost. If you fail to develop that, your entire money will go. Note: the product is called as sunk cost if it has no alternative uses.

Explicit Costs: Explicit costs refer to expenses that involve actual payments to others. It is the total of all expenses that have been paid. So, an explicit cost is the money a company pays to use an input that someone else owns. Explicit costs are money spent on purchase of raw materials, renting a building, the amount spent on advertising etc. It also called as accounting costs.

Implicit Costs: The cost of inputs used by self-employed individuals that are not recorded in their accounting books. For example, using your own land, labour, machinery in your production process. Though you are not paying any money to your own land, labour, machinery etc. but they have some market value. Implicit costs are not shown in accounting statements.

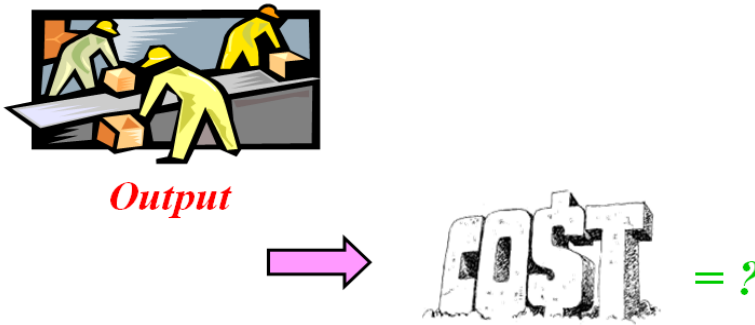
Accounting Costs: Accounting costs refer to the actual monetary expenses incurred in the operation of a business. Accounting costs refer to the monetary expenses required to obtain a specific benefit. Accounting costs are reflected in a company's financial statements.

Accounting costs, also known as explicit costs, refer to the direct, out-of-pocket expenses associated with a business's operations. From a firm point of view accounting costs are production costs, lease payments, marketing budgets and payroll, rent, raw material and electricity. These are the actual costs involved in the manufacturing, marketing, and delivery of products. Accounting costs refer to the cash payments made by a firm. From a student point of view accounting cost of attending a college includes tuition fee, hotel room charges, books, food and other expenditure.

Economic Costs: Economic cost is the combination of explicit cost and implicit cost. The economic cost of a college includes both the accounting cost and the opportunity cost. Economic cost includes both accounting costs and opportunity costs, which, although not explicitly documented, are important for decision-making.

Operating cost: Operating cost are costs associated with the day-to-day operations of a business. It includes salary, wages, rent, cost of raw material, electricity, transport and fuel. These costs can be either fixed or variable depending on the unique situation. A manager can't avoid operating cost.

Cost function describes the relationship between cost and output.



Measure of costs

- ◆ Output changes → Cost changes

Change in cost can be expressed in three ways:

- ◆ Total cost (TC)
- ◆ Average cost (AC)
- ◆ Marginal cost (MC)

Total cost (Total Fixed Cost)

- ◆ It is the whole number of payments *to all factors* used in producing a given amount of output (Q). Total cost is the combination of total fixed cost and total variable cost.
- ◆ Total fixed cost (TFC): is the whole amount of payments *to fixed factors*. Total variable cost (TVC): is the whole amount of payments *to variable factors*.

Assume two factors only: Capital and labour. You know in the short run, Labour is variable input and capital is fixed input. Hence in the short run, we have some variable cost and some fixed costs.

$$\text{Total Cost}(TC) = TFC + TVC$$

TFC is total fixed cost and TVC is total variable cost

Total fixed cost: is constant and independent of output

Total variable cost: Is dependent of output

Average cost (average total cost (ATC))

It is the cost per unit of output. It is nothing but the total cost divided by the level of output

$$AC = TC/Q$$

TC is total cost and Q is quantity of output

Average cost (AC) is the combination of average fixed cost and average variable cost

$$ATC = AFC + AVC$$

ATC is average total cost, AFC is average fixed cost, AVC is average variable cost

Average fixed cost (AFC): it is nothing but fixed cost per unit of output. Average fixed cost is equal to total fixed cost divided by the output. $AFC = TFC/Q$

Average variable cost (AVC): the variable cost per unit of output. Average variable cost is nothing but total variable cost divided by output. $AVC = TVC/Q$

◆ *Average Total Cost:*

$$ATC = \frac{TC}{Q} = \frac{TFC + TVC}{Q} = AFC + AVC$$

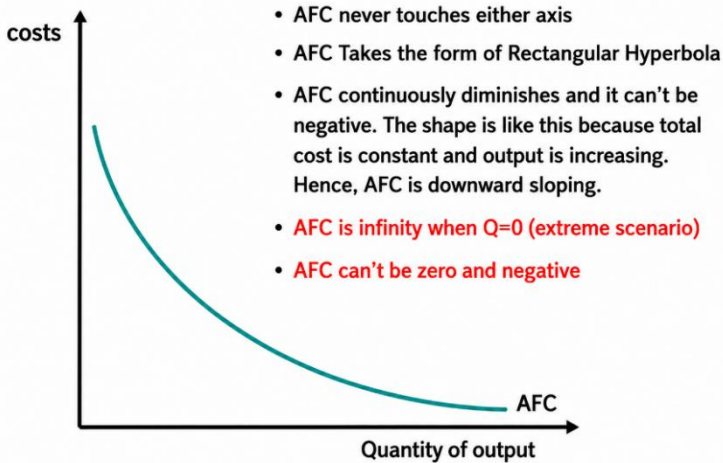
Average Fixed Cost

Average fixed cost refers to fixed cost per unit of output. The average fixed cost per unit goes down when the number of units produced goes up. The average cost per unit goes up when the firm makes fewer units. Let's assume a firm pays Rs. 100000 as rent and produces 1000 laptop. Then the average fixed cost Rs. 100.

Total fixed cost (TFC)	Output(Q)	Average Fixed Cost (AFC)
10	0	-
10	1	10
10	2	5
10	3	3.3
10	4	2.5
10	5	2

$$AFC = \frac{TFC}{Q}$$

Where q represents output



Average Variable Cost

$$AVC = \frac{TVC}{q}$$

Where q represents output

AVC stands for Average Variable Cost. TVC stands for Total Variable Cost.

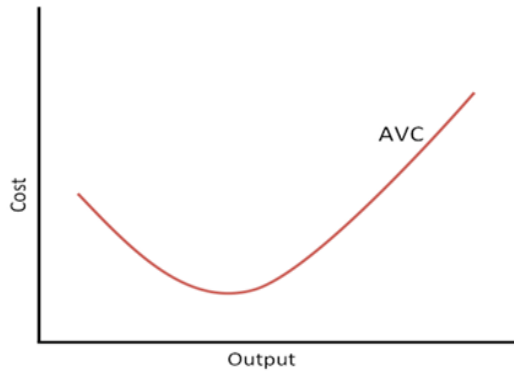
AVC is the cost of each unit of output that changes. In other words, it is the cost per unit of output. Initially AVC decreases as output increases due to the occurrence of increasing returns to factor. Production increases at an increasing rate. Hence cost falls. It reaches minimum. AVC Rises steeply because of the operation of diminishing returns to factor.

Output increases at a diminishing rate (Hence, cost increases).

The firm uses average variable cost to figure out when to stop producing in the short run. A company continues production if it gets a good price that is higher than the average variable cost and covers the fixed costs. If the company gets a price that is lower than the average variable cost, it will stop producing because it can't pay for both fixed and variable costs.

The average variable cost curve looks like a U-shaped. When output is low, average variable cost is pretty high.

Note: with increase in production, average variable cost goes down, reaches a minimum point, and then rises.

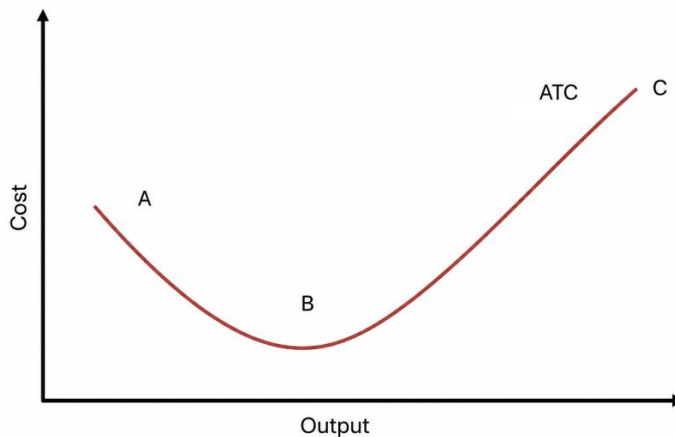


Average Total Cost (ATC or AC)

Total cost is the sum of total fixed cost and variable cost. Likewise, average total cost is the combination of average fixed cost and average variable cost. The law of variable proportions can also explain why the average cost curve has a U shape.

In the beginning, when output goes up, the average cost goes down because of the law of increasing returns to factor. Average cost goes down and reaches the minimum points. After that if a firm will produce more and more goods, the average cost goes up. It is due to law of diminishing returns to factor.

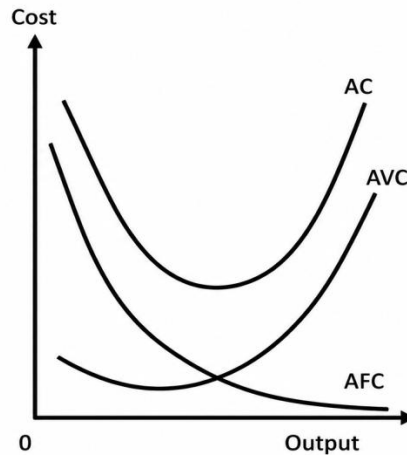
Points A to B represent Average fixed cost (AFC) and points B to C represent Average variable Cost (AVC). Note that ATC and AVC is U-shaped where, whereas AFC is not.



$$ATC = \frac{TC}{q} = \frac{TFC + TVC}{q} = AFC + AVC$$

Some Points:

- **In the beginning, AVC and AFC Fall, so AC Falls**
- **AVC rises but AFC steeply falling outweighs the rise in AVC which leads to fall in AC**
- **AVC rises steeply offsetting the fall in AFC which leads to rise in AC**

**Marginal Cost (MC)**

Marginal cost can be defined as the increase or extra cost that comes as a result of the production of one more unit of goods. Marginal cost can be defined as the difference in total costs as a result of production of an additional unit of output. $MC = \frac{\Delta TC}{\Delta Q}$. TC stands for total cost. Q stands for output. ΔTC is change in total cost. ΔQ refers to change in quantity.

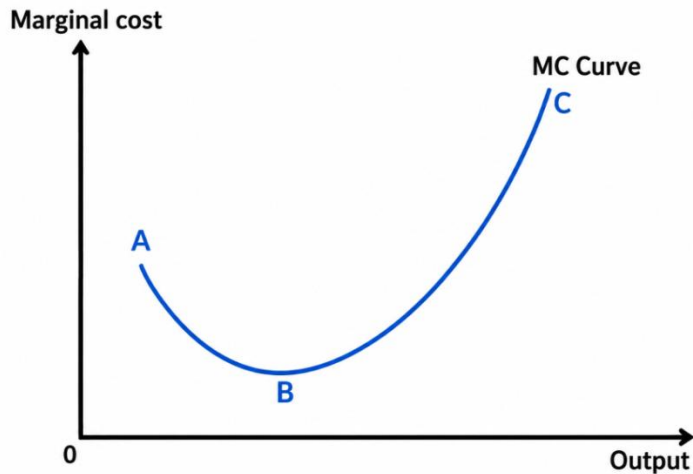
This measure does not depend on fixed cost. The slope of the total cost function gives us the marginal cost. Since there is zero output, the marginal cost would be undefined. With an increase in output to 1 unit, the marginal cost would be 250 $(1250-1000)/(1-0)$. Similarly, if the number of units increased to 2, then the marginal cost would be 200 $(1450-1250)/(2-1)$.

Computation of Marginal Cost

Output	Total Cost	Marginal Cost
0	1000	–
1	1250	250
2	1450	200
3	1600	150
4	1800	200
5	2060	260
6	2360	300

Another example: Let's assume that 10 labour produced 20 bags and you paid Rs. 10,000 to them. You hired one more unit of labour. The additional labour produced 1 bag and you paid Rs 800 to that person. Now your total cost has increased from Rs 10,000 to Rs.

10,800. Like out your total number of bags(output) has increased from 20 to 21 unit. Now we will calculate the marginal cost. The cost of hiring one additional unit of labour is (Rs 10,800-Rs 10,000)/ (21-20) =Rs 800

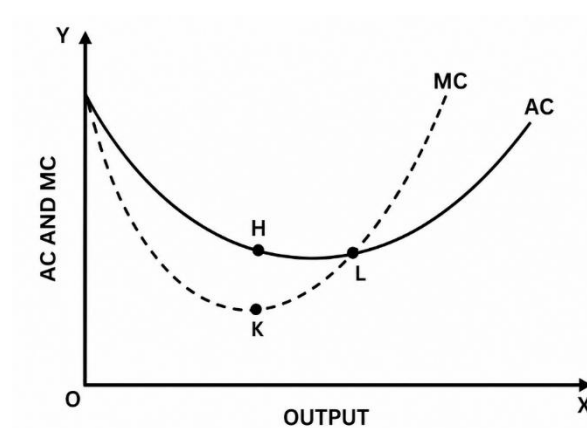


The short-run marginal cost curve 'U'-shaped. Why?. That is due to the operation of the "law of variable proportions".

The short-run marginal cost curve (SMC) curve is a U-shaped. It is because of the law of variable proportions. To understand the U-shape of the SMC curve, it can be divided into three distinct segments according to the law of variable proportions.

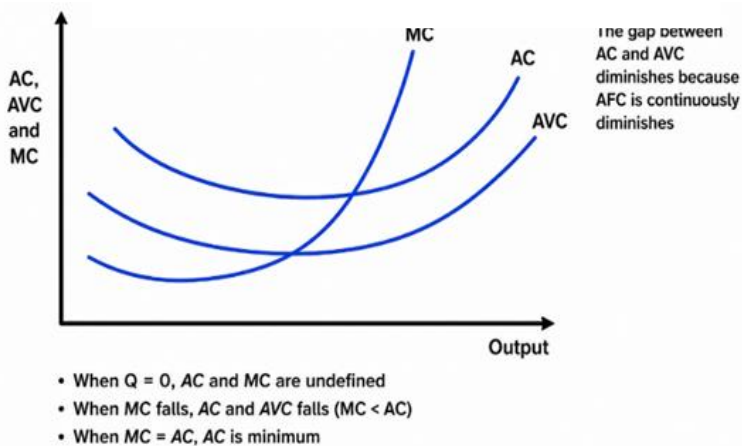
1. The AB portion where MC is falling represents the increasing returns to the factor.
2. MC stops falling and reaches its minimum point 'B'. Point B corresponds to constant returns to factor.
3. After minimum point B, the BC portion where MC curve is rising corresponds to the decreasing returns to the factor.

Relationship between Average Cost and Marginal Cost



- When output is zero, average cost (AC) and marginal cost (MC) are undefined
- When MC declines, AC also declines
- When $MC=AC$, AC is minimum
- When MC rises, AC also rises ($MC > AC$)

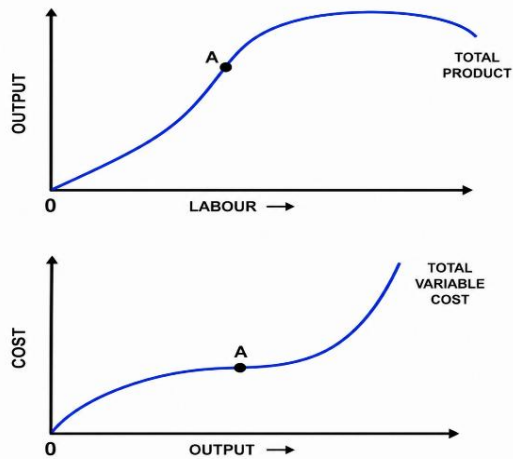
Relationship between AC, MC and AVC



Short Run Cost

In the short run, firm can produce output by using labour and capital. The input labour is variable in nature and capital is fixed in nature. Hence, in short run we have both variable cost and fixed cost. Variable cost is associated to labour and fixed cost is associated to capital. Variable cost: wages paid to labour, Raw Materials cost, Chemicals etc. Fixed cost: Capital equipment and building, and top management salary etc.

Relationship between short run production function and short run cost

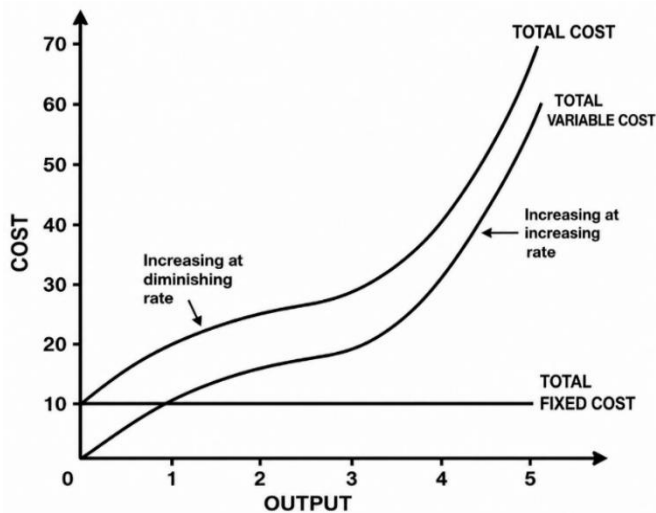


- ❖ When output (Q) is zero, the total variable cost is also zero.
- ❖ When output increases at an increasing rate, the variable cost increases at a diminishing rate (that is because of increasing return to factor)
- ❖ After point A, output increases at a diminishing rate, the variable cost increases at an increasing rate (that is because of decreasing return to factor)

The relationship between output, TFC, TVC and TC

Output (Q)	Total fixed cost (TFC)	Total Variable Cost (TVC)	TC
0	10	0	10
1	10	10	20
2	10	15	25
3	10	18	28
4	10	30	40
5	10	60	70

Short run Total Cost, Total Fixed Cost and Total Variable cost



Important Points

- Short run total cost (STC) is function of Output
- STC is obtained by adding up total fixed cost curve (TFC) and total variable cost (TVC)
- The distance between TVC and TC remains constant because TFC is constant.
- The distance between TFC and TC is Increasing as the Output Increases because of Variable TVC

Relationship between Average Cost and Marginal Cost

- When Marginal Cost < Average Cost, Average cost Falls
- When Marginal Cost > Average Cost, Average Cost Rises

Marginal-Average Relationship is Mathematical Truism.



Sachin Tendulkar: The Greatest Batsman the World has Ever Produced.

Batting Average (Test): 57

Next Innings (Marginal Score): 50 Less Than the Average Score

Average Score will Fall

Generalization :

If $\text{Marginal} < \text{Average}$ then Average will Fall



Batting Average (Test): 57

Next Innings (Marginal Score): 100 Greater Than the Average Score

Average Score will Rise

Generalization :

If $\text{Marginal} > \text{Average}$ then Average will Rise



**Batting Average
(Test): 57**

**Next Innings
(Marginal Score): 57
equal to Average
Score**

**Average Score will
be Same**

Generalization :

**If $\text{Marginal} = \text{Average}$
then
Average is Same**

Exercise: Using the calculated values draw the cost curves of FC, VC, MC, AFC, AVC and ATC

Output (Q)	Fixed Cost (FC)	Variable Cost (VC)	Total Cost (TC)	Marginal Cost (MC)	Average Fixed Cost (AFC)	Average Variable Cost (AVC)	Average Total Cost (ATC)
0	50	0	50	—	—	—	—
1	50	50	100	50	50	50	100
2	50	78	128	28	25	39	64
3	50	98	148	20	16.7	32.7	49.3
4	50	112	162	14	12.5	28	40.5
5	50	130	180	18	10	26	36
6	50	150	200	20	8.3	25	33.3
7	50	175	225	25	7.1	25	32.1
8	50	204	254	29	6.3	25.5	31.8

Long run cost curves

Long term is when all inputs are variable. As a consequence, long term is the period where all costs are variable. One can argue that in the long term, there is no fixed cost at all. Talking about long term cost is nothing but talking about long term variable cost. The long run total cost (LTC) is the least cost of production for producing different levels of output. LTC begins from the origin because of no fixed cost in the long run. LTC assists us in calculating the long-term average and marginal cost curves of a firm.

$$LAC = \frac{LTC}{Q}$$

$$LMC = \frac{\Delta LTC}{\Delta Q}$$

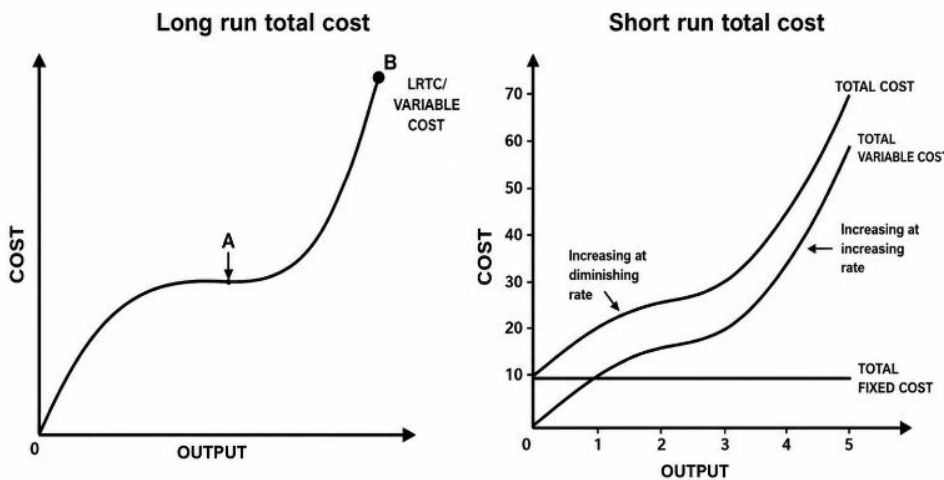
LMC looks at how much LTC changes for every unit of output that changes. Both short run and long run average cost curves are U shape. Only the sources are different. LAC is U shape because of increasing, constant and decreasing return to scale. SAC is U shaped because of increasing return to factor and decreasing return to factor.

The long run average cost curve shows how the average cost of production varies with level of output.

Short run average cost curve represents the relationship between average production costs and output levels, assuming capital is held constant.

The long-run marginal cost curve shows how the long-run total cost changes when one unit of output is added.

Long run Total Cost (LTC) vs Short run Total Cost (STC)



Note: In the long run there is no fixed cost. Hence all the cost is variable in the long run. Hence we may say long run total cost is the long run variable cost

Reason behind the shape of the LTC

The LTC curve rises at a diminishing pace until point A due to increasing returns to scale. Increasing returns to scale imply that when there is an increase in the levels of the inputs, then the level of the output will increase by a greater proportionate amount. This means that if we were to double the level of the inputs, the output will be more than doubled. Given that the price levels of the inputs remain unchanged, the cost per unit will reduce.

After point A, LTC increases at an increasing rate. That is because of decreasing return to scale. Decreasing return to scale means output increases less than proportionately in comparison to inputs. Keeping the input prices remaining same/constant, as output grows less than proportionately compared to inputs, the per unit cost increases.

Long Run Average Cost (LAC)

- LAC represents the long run total cost divided by the level of output.
- In long run, a company has different plant size having corresponding short run average cost curves
- LAC shows the lowest average cost for making all levels of output.
- LAC is the point where the short-run average cost curves touch each other.

Why LAC is U-shaped in General?

- ❖ Each part of the long-run average cost curve shows a different return to scale.
- ❖ Falling part of LAC implies increasing returns to scale (Economies of scale). When companies become more efficient at production, they can lower their costs through economies of scale. Here, the cost per unit goes down.
- ❖ Rising part of LAC implies decreasing returns to scale (Diseconomies of scale). Diseconomies of scale refers to the disadvantages the company is experiencing while expanding its scale of operation in the long run. Due to diseconomies of scale the cost of production increases
- ❖ Lower part of LAC implies constant returns to scale (Exhausted Economies of Scale)
- ❖ The LAC is U shaped in economic theory because of the assumption that technology is unchanged.

Breaking LAC into Three Parts such as Figure A, B and C

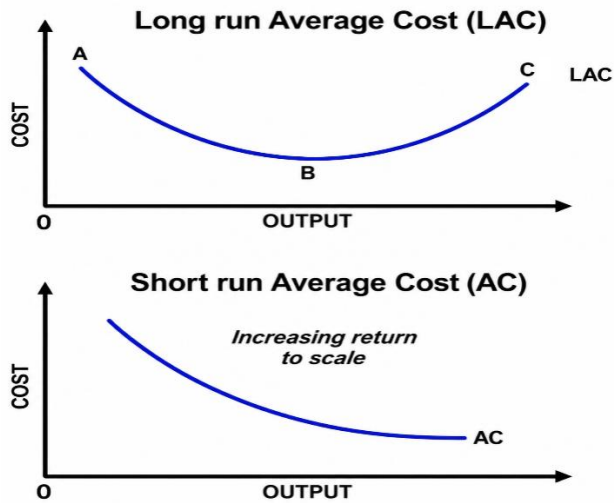


Figure A

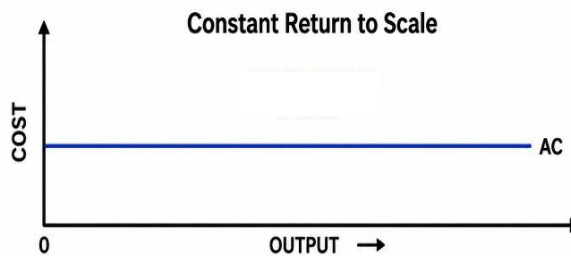


Figure B

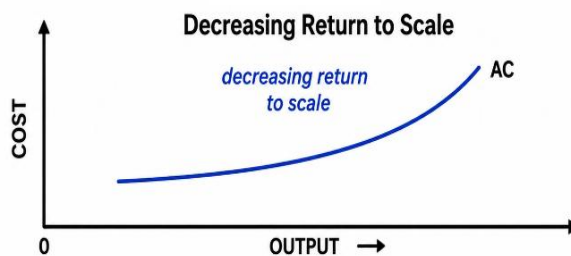
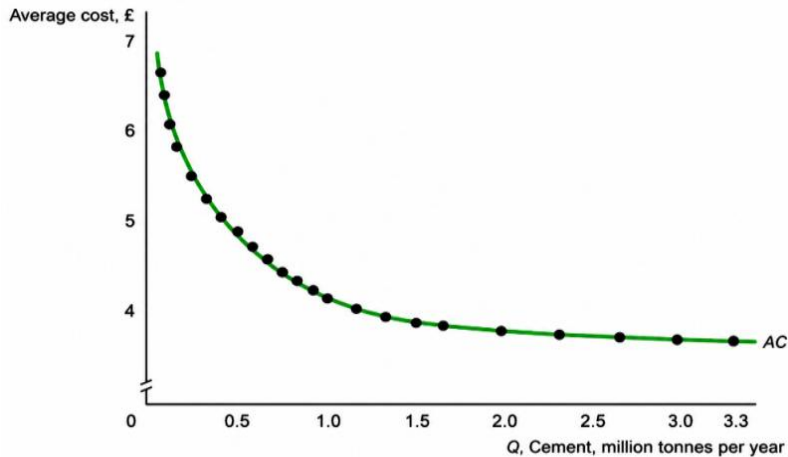


Figure C

Point A to B corresponds to increasing return to scale. Increasing return to scale means lower cost. Constant return to scale means cost per units are the same for any level of output. Decreasing return to scale means increasing cost. By joining the Figure A, B and C, we will get LAC. There are three possible shapes of LAC. It can be U shape, L shape and constantly declining.

L-Shaped Average Cost: Frequently Confronted

- In some industries fixed start-up costs generate falling average cost or increasing returns to scale.



- This seems to be the case in the cement industry

Average Cost of Cement Companies

In the beginning, Average cost falls rapidly. Average cost remains flat throughout after a point or slope gently downwards at its right hand. No rising portion. Here AC decreases initially because of IRS and AC is constant because of constant returns to scale.

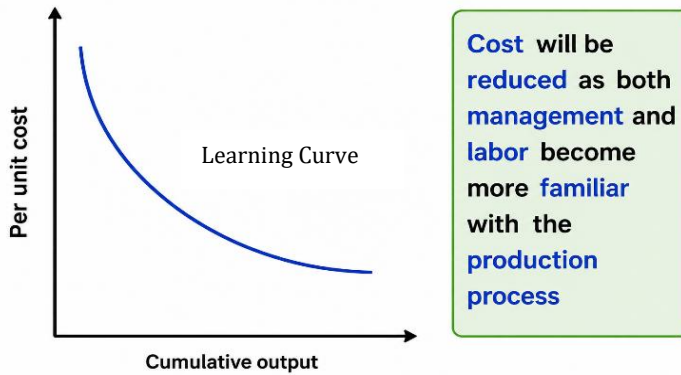
Reasons of L-Shaped LAC: Resolving the Contradiction between Theory and Empirical Evidence

(1) Technological Progress

- The LAC is U-shaped in economic theory because of the assumption that Technology is unchanged
- In real world Technological Progress takes place over period of time and thus LAC will shift downward overtime
- Technological progress is rapid enough to reduce unit costs outweighing the rise in unit costs due to the management problem

Learning by Doing (Learning Curve)

A company learns to make a good more efficiently as it makes more of it, which lowers the cost per unit.

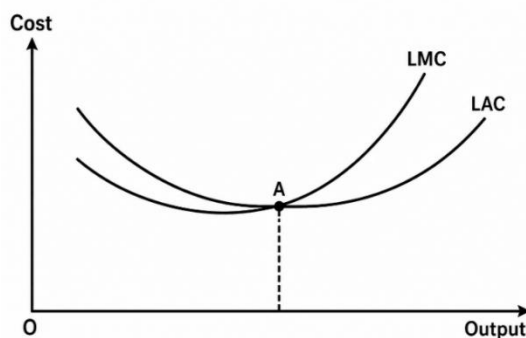


The learning curve is an indication of the relationship that exists between the output produced by a firm and the quantity of input required per unit of output produced. The more management and workers get familiarized with the process of production, the lower the marginal and average cost incurred to produce a certain level of output. This occurs because, at first, workers will take longer times to perform different activities, but as they acquire experience, their speed increases. Management learns how to plan the process of production more effectively through material flow and manufacturing organization.

The learning curve is essential for firms aiming to estimate the production costs of new products. Managers utilize learning curve information to assess the profitability of production operations and strategize the necessary scale and cumulative output to achieve positive cash flow.

Long run Average and Marginal Cost

Average cost curve is U shaped irrespective of time horizon. In other words, both SAC and LAC curves are U shaped. SAC is U shaped due to increasing and diminishing return to a



factor. On the contrary, LAC is U shaped because of increasing, constant, and decreasing return to scale.

- When output is zero AC and MC are undefined
- When MC falls, AC also falls ($MC < AC$)
- When $MC = AC$, AC is minimum
- When MC rises, AC also rises ($MC > AC$)

At point A two curves intersect with each other. Here the long run average cost curve reaches its minimum point.

When the LAC is falling, the LMC is below it. On the other hand, when the LAC is rising, the LMC is above it.

Optimum Plant, Optimum Output, Optimum Company

- **What is optimum plant?**

The optimum plant where the lowest point of the SAC matches exactly with the lowest point of LAC. It is called as least-cost Plant.

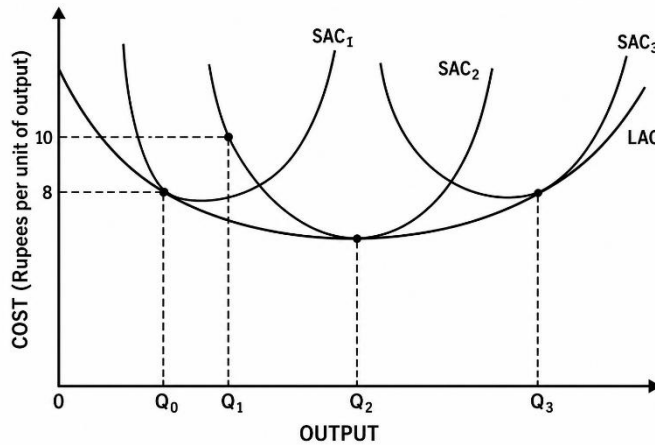
- **What is optimum output?**

The plant being efficiently utilized produces the output called as optimum output

LAC is Envelope of Short Run Average Cost Curve

- LAC is not tangent to the minimum points of the short-run average cost curves
- When the LAC is falling, LAC will be tangent to the falling portions of the SACs
- When the LAC is rising, LAC will be tangent to the rising portions of the SACs
- The LAC curve remains below all short run average cost curves.
- Larger output can be produced at the lowest cost in a larger plant. Smaller output can be produced at the lowest cost in a smaller plant.
- LAC is envelope of short run average cost curve.

- The minimum point of the short run average cost curves does not align with the long run average cost curves due to the influence of economies and diseconomies of scale.



Assume that the firm is uncertain about the future demand of its product. Due to increase in uncertainty firm is not sure about future demand. Hence, firm is considering three alternative plant size; small, medium and large size plants. SAC_1 , SAC_2 and SAC_3 represents short run average cost curve of small, medium and large size plants. If the firm wants to produce less output, the firm will choose smaller size plant (SAC_1). If the firm wants to produce large amount output, the firm will choose larger size plant (SAC_3).

To produce Q_0 units of output, the firm will choose small size plants where the cost of production is Rs 8. To produce Q_0 units of output, if the firm will choose medium size plants, then there is underutilization of resources. It is not a rational decision for a manager. Any how the firm is operating in smaller size plant to produce Q_0 unit of output. To produce little extra output the firm should operate in same plant size. If the firm manager will think to go for medium size plant, then it will be costly as labour will move from small size plant to medium size plant which will increase the cost of production.

To produce Q_2 units of output, the firm prefers medium size plants. If the firm will choose large size plant for Q_2 units of output then the cost will be more. For same output lesser cost is preferred. If firm will choose large size plant to produce Q_2 units of output then it is also called as underutilization of resources. To produce Q_3 units of output the firm will choose larger plant size. Small plant side can't produce more output and it is not cost effective also.

Now the question arises which plant size the manager will choose to produce Q_1 units of output. Whether the manager will choose small or medium size plant. If firm is choosing SAC_1 then the per unit cost is less than Rs 8. On the other hand, if firm is choosing SAC_3 then the per unit cost is Rs 10.

Less cost is always preferable for same amount of output. Any how the firm is operating in SAC_1 and to produce little more output there is no need to change the plant size.

The size of the plant of the firm can vary over time. It is always interested in the plant that reduces the average cost of production. LAC is an envelopment of short run average cost curve SAC1, SAC 2, and SAC3.

Envelope means surrounded by short run average cost curves. The long-run average cost curves show economies of scale at first, but then they show dis-economies at higher levels of output.

Note that LAC never lies above any of the SAC curves. With economies and dis economies of scale, the minimum point of the short run average cost curve do not lie on the long run average coost curves.

Return to factor Vs. Return to scale

Return to factor

- It is in short run production
- Here the proportion between fixed and variable input changes
- There are 3 types of return to factor (IRF, DRF, NRF)
- In case of return to factor, there are some fixed factors and some variable factor

Return to Scale

- It occurs in the long run production
- Here the proportion remains constant
- There are three types of return to scale (IRS, CRS and DRS)
- In this case all inputs are variable

Return to Scale vs. Economies of Scale: Let's Start with an Example

You studied for 1 hour and scored 10 Marks in Math

You studied for 2 hours and scored 20 Marks in Math

Here your return from the input is constant. It is a constant return to scale only. Return to scale is with respect to input only. Return to scale means output in terms of physical quantity.

Now you studied 2 hours but you scored 40. It is like a miracle. How it is possible? This might be because your parents are encouraging you to study, they might have told you the reality of life, they might have given you one new laptop with your favourite brand, new headphone, and one single room with new AC, all the food you like is given to you. Some of your best friends are encouraging you on regular basis. Because of all the factor, the miracle occurs. This is nothing but economies of scale. Economies of scale means many other factors (Internal and external) factors other than inputs. Internal factors are new AC, New Laptop, headphones etc and external factors are here your friends

Return to scale and economies of scale

- In a dairy firm, milk production depends on cows, feed, land, and labour rather than equipment. For instance, a firm with 50 cows relies on manual milking. If all inputs are doubled, milk production can also double, indicating constant returns to scale with increased production capacity through more cows.
- Large dairy firms can opt for milking machines to enhance efficiency. Transitioning from 50 to 100 cows allows a firm to reduce its average production cost of milk from Rs. 10 per liter to Rs. 7 per liter, demonstrating economies of scale.

Economies of Scale vs. Economies of Scope

What is Economies of Scale?

Over time, there can be certain benefits for the industry/Firm regarding cost reductions and output growth. These benefits when it comes to producing a single good are referred to as economies of scale. Cost per unit falls with an increase in the amount of production. Economies of Scale is where an increase in production by two times results in less than double costs. Similarly, increasing the output by three times leads to less than three times costs. In other words, whenever there is a doubling of output, the cost increase by less than twice the amount. Similarly, in the case of a threefold increase in output, there is less than thrice cost increase.

These benefits can be internal and external. Economies of scale occur in the long run and hence all inputs are variable. Economics of scale refers to the benefits a firm gets due to large-scale operation

E.g. Economies of Scale is a railroad track, which carries 100 trains per day. The total cost will be higher when it carries one train.

Economies of scope: if the government will allow goods and passenger train in the same track and increase the number of trains then per unit cost will decrease

Internal and External Economies of Scale

(a) Internal Economies of Scale

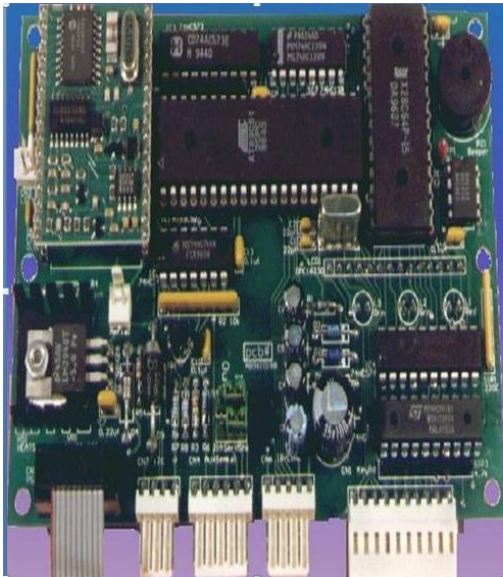
Internal factors are controlled by a firm/ industry. The benefits/advantages the firm is getting while producing a single product in the long run is called internal economies of scale. The benefits on which the industry has control are called internal economies. Advantages accrue to the company due to the changes within the company. Cost per unit declines.

Advantage of Internal Economies of Scale

- ❖ Production economics of scale- Operating on a large scale allows workers to specialize in more productive activities.
- ❖ Research and Development expenditure: more efficient use of research & development expenditure

- ❖ Improvement in technology: better and specialized machines (Gujrat Ambuja case study example replacement of V belt to flat belt)
- ❖ Tanning: to enhance knowledge and skills of the employee
- ❖ Labour economies: Specialized Labours
- ❖ Replacement of coal for crushed sugar cane ang ground not husk
- ❖ Transportation economics of scale-Efficient transportation. More than 200 KM distance they preferred by train, water ways
- ❖ Inventory economics of scale-Ware housing (inventory management)
- ❖ Quality control-every half an hour. 48 times per day
- ❖ Managerial economics of scale: Managers can enhance the production process by adjusting the combination of inputs used to generate outputs.
- ❖ Financial economies: Large firms receives loans easily compared to smaller firms and receives bank loan at lower interest rates than the smaller firms.
- ❖ Marketing economies: spread high cost of advertising
- ❖ More bargaining power: The firm can acquire production inputs at lower costs by purchasing in large quantities, allowing for better price negotiations.
- ❖ Advertisement cost- Due to frequent advertisement, cost per unit decreases

PC Production



- ◆ **R&D costs are much lower per unit when mass produced.**
- ◆ **Bulk purchasing applies.**
- ◆ **Other reasons?**
- ◆ **4-function calculator would cost about Rs 2500 if not for economies of scale.**

(b) External Economies of Scale

External factors are not controlled by the firm/industry. The benefits appropriated to the industry due to the factors on which the industry has no control on it. External economies of scale occur outside of a firm.

Advantages of External Economies of Scale

The benefits appropriated to the industry due to the factors on which the industry has no control on it

- ❖ Local skilled labour force available
- ❖ Area with good transportation network
- ❖ Area with good communications network
- ❖ Government policy (reduction of tax rate) and increase in subsidy
- ❖ Decrease in crude oil and raw material price

Note: In case of economies of scale LAC continuously declines till it reaches minimum.

Diseconomies of Scale

The diseconomies of scale are the problems that a business encounters when it tries to produce more. This happens when a doubling of production leads to an increase in the cost by more than double. Thus, the cost per unit will rise.

Internal Diseconomies of Scale

- Managing a larger firm may become relatively more complex as compared to a smaller firm.
- The advantages of bulk purchasing diminish when specific quantities are reached.
- Available supplies of key inputs may become limited, resulting in increased costs.
- Communication begins to break down
- Management is out of touch with production process
- Decisions are bogged down by too much red tape
- Poor labour relationships may develop
- Mismanagement
- Inventory mis management
- Employee employer relation
- Less incentive
- No proper recognition to talent and skills

External Diseconomies of Scale

- ❖ Local labour force becomes scarce. Higher wages are needed to attract workers.
- ❖ Rents rise as area grows.
- ❖ Transportation congestion occurs as area grows.
- ❖ The U-shaped average cost curve illustrates a firm's experience of economies of scale at low output levels and diseconomies of scale at higher output levels.
- ❖ When output increases and costs rise proportionately, neither economies nor diseconomies of scale are experienced. Economies of scale occur when the marginal cost is lower than the average cost (both are declining). Due to the presence of diseconomies of scale marginal cost is greater than average cost.

Economies of Scope

Economies of scope refer to a condition where a single firm's combined output exceeds the total output of two separate firms producing individual products. Economies of scope denote the cost reduction a firm experience when producing multiple products simultaneously rather than producing them separately.

The advantages occurring to the firm or industry while producing more than one output (multiple output) in the long run is known as economies of scope.

For example, Gujrat Ambuja case example (a) dry cement (b) Semi dry cement (c) Weight cement.

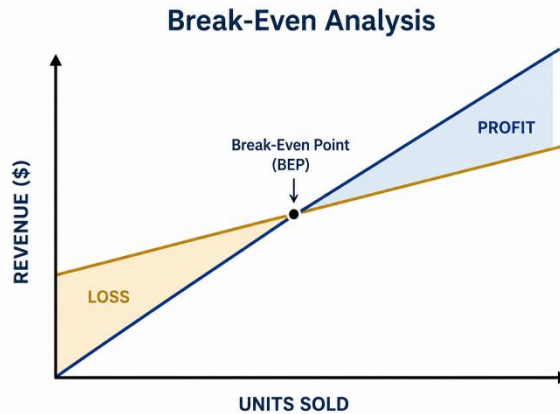
Another example: is Tata Group which produces cars, SUV, buses, and trucks. The techniques employed in producing both commodities are sufficiently similar.

Why Farmers does not produce a single variety of rice production?

Whether Economies of scale or economies of scope will occur first?

Break-Even point

Total revenue is upward-sloping because of the increase in quantity sold. Total cost is not starting from Zero as it is the combination of fixed and variable cost. Before reaching breakeven points, I mean the initial years in the business the company incurs losses, then achieves breakeven point then makes a profit. In this diagram, E is the breakeven point.



Marginal and Total Revenue

Marginal revenue is the extra or additional revenue results from the sale of additional units of output.

$MR = \text{Change in total revenue} / \text{Change in quantity}$

Total revenue is the total earnings of a firm by selling all the goods and services.

Q. Hypothetical Financial Statement of Tata Sky (FY 2007, Figures are in Million). Find out the shutdown point of Tata Sky and the number of subscribers when Tata Sky will reach the break-even point.

Particulars	Amount (Rs Million)
Total Revenues	6130
Income towards Subscriber Acquisition	700
Income from Operations	4520
Total Expenditure	7310
Expenditure towards Subscriber Acquisition	2330
Expenditure towards Operations	5980
Reported EBITDA	-3180
Reported EBITDA Margin	-55%
Number of Subscribers	8 Million

Shut down point if the price is less than the minimum point of Average variable cost

$$AVC = TVC/Q$$

No of subscribers is 8 million. $AVC = 2330/8 = 291$ million

$$AR=P$$

$$AR=TR/Q$$

$$6130/8=766$$

Here $p >$ minimum point of AVC. Hence Tata Sky will not shut down.

The breakeven point is a point where total revenue is equal to the total cost. Here there is no profit and no loss. Each and every firm used to come across the break-even point after some years of production. Before the breakeven point the company used to make a loss and after the breakeven point, there are many chances that the company will make a profit.

$$\text{Break - even point } TR = TC$$

$$TR = P * Q$$

$$P * Q = TC$$

$$P * Q = TFC + TVC$$

$$P * Q - TVC = TFC$$

$$P * Q - AVC * Q = TFC$$

$$Q(P - AVC) = TFC$$

$$Q = TFC / P - AVC$$

$$\text{Breakeven quantity} = TFC / P - AVC = 5980 / 766 - 291 = 12.54 \text{ million}$$

If the number of Tata Sky subscribers will reach 12.54 million then Tata Sky will reach the break-even point

Review Questions

1. Describe how a company's long-term cost structure is affected by economies and diseconomies of scale. Use suitable examples or diagrams to support your explanation. Explain how these factors affect production efficiency and the overall cost of production.
2. Illustrate the concept of a production function and explain the differences between short-run and long-run production with the help of appropriate diagrams.
3. Evaluate the importance of economies of scale and economies of scope in improving business performance. Critically assess whether a firm can benefit from one without experiencing the other, giving suitable justification and examples.
4. From the following table, calculate the different costs.

Q	TFC	TVC	TC	MC	AC	AFC	AVC
0	20	0					
1	20	10					
2	20	15					
3	20	18					
4	20	30					
5	20	60					

Note: Q represents quantity, TFC stands for a total fixed cost, TVC is total variable costs, TC is the total costs, MC is marginal cost, AC is the average cost, AFC and AVC are average fixed cost and average variable costs respectively.

5. A manufacturing company plans to expand its production in the long run by increasing all factors of production, such as labour, machinery, and factory space. As a production manager, explain how the **long-run production function** helps the company understand changes in output when all inputs are varied. **Illustrate your answer with a suitable diagram and explain the stages of returns to scale.**
6. Differentiate between returns to a factor and returns to scale by using suitable diagrams. Explain how changes in input combinations influence output in the short run and long run.
7. Discuss the relationship between long run cost curve and returns to scale with suitable graphs?
8. Illustrate graphically and discuss the various stages of the “Law of Variable Proportions?”. In which stage a rationale producer wants to operate.
9. Differentiate between economies of scale and economies of scope with suitable examples. Explain whether both can occur simultaneously in a firm and how they contribute to business growth.
10. Examine how the average cost curves for the short and long run relate to each other. Explain why the LAC curve is called the ‘envelope curve’ with a suitable diagram.
11. In short run production the marginal product ultimately goes on diminishing from zero to negative. Substantiate it.
12. In short run production the marginal product ultimately goes on diminishing from zero to negative. Substantiate it. In which stage a rational producer wants to continue its production.

13. Distinguish between return to factor and return to scale. Explain AP and MP relation in the short run production

Unit-5

MARKET STRUCTURES

A market is a place where buyers and sellers interact to exchange goods and services. Market structure refers to the nature and intensity of competition that firms face when making pricing and output decisions. Four distinct types of market structure are typically recognized, each defined by specific characteristics that influence the competitive dynamics and economic outcomes within a market.

These types include

- (i) Perfect competition
- (ii) Monopoly
- (iii) Monopolistic competition
- (iv) Oligopoly

Characteristics of perfectly competitive markets include a large number of buyers and sellers in the market providing identical goods. There is also free entry and exit in such a market structure. Monopoly represents a market structure whereby there is only one seller supplying the total output to all the consumers. Here, mono means one and poly stands for seller; therefore, a monopoly is one-seller market structure. In the monopolistic structure, a firm provides a product with no close substitutes making it very difficult for new entrants to join the market.

As they are producing differentiated products they are treated as price makers. This market structure enables easy long-term entry and exits from the industry. Oligopolies are characterized by few numbers of firms that dominate the market. These firms' produces homogeneous or differentiated products. Though entry into the market is possible in the long run but it is very difficult as fixed cost is very high. As there are few large firms the competitions among the large firms are too high.

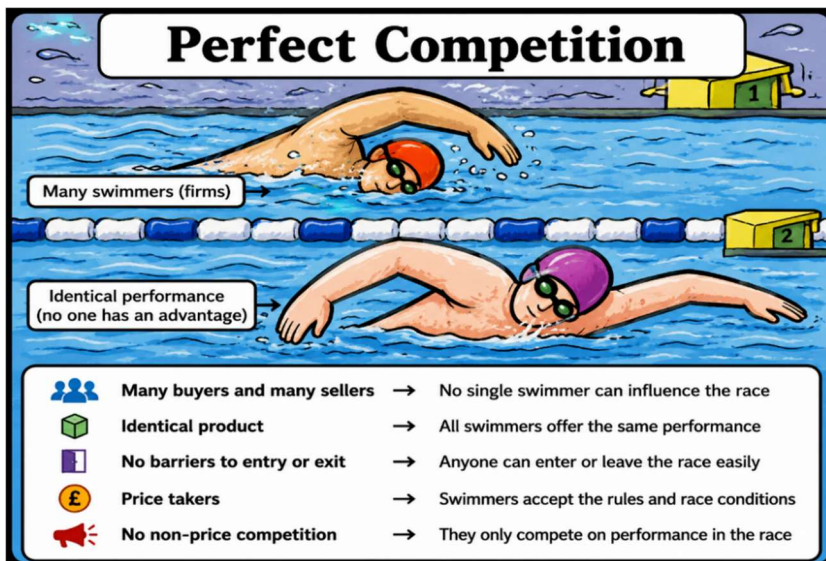
Imperfect competition includes monopolistic competition and oligopoly, as opposed to perfect competition. Perfect competition lies on one end of the market structures continuum, while the other end is taken up by monopoly. Monopolistic competition and

oligopoly lie somewhere in between. Understanding these different market structures is extremely important when examining market behaviors and formulating policies.

The market structure may be affected by different considerations, such as the number of firms and their sizes, kinds of products, barriers to entry and exit, and information about the market. It may be categorized based on certain aspects, namely pricing power, number of firms, kinds of products, entry barriers, and purposes of non-price competition.

Market Structure

Perfect Competition Eg. Stock price/Parts of agriculture	Monopoly Eg. Public Utilities	Monopolistic Eg. Retail trade	Oligopoly Eg. Computers, oil, steel
No. of Buyers: Many No. of Sellers: Many	No. of Buyers: Many No. of Sellers: One	No. of Buyers: Many No. of Sellers: Many	No. of Buyers: Many No. of Sellers: Few
Type of Product: Homogeneous/identical	Type of Product: unique product (No substitutes are available)	Type of Product: differentiated	Type of Product: Homogeneous/differentiated
Barriers to entry and exit: No	Barriers to entry and exit: Yes/very high	Barriers to entry and exit: low entry barriers.	Barriers to entry and exit: with high entry barriers.
Power of the firm over price – None	Power of the firm over price – Considerable	Power of the firm over price – Some	Power of the firm over price – Some
Firms act as - a price taker	Firms acts as - a price maker	Firms acts as a price maker	Firms acts as a price maker
Non-Price Competition: None	Non-Price competition: None	Non-Price competition: Yes	Non-Price competition: Yes



Perfect Competition Assumptions:

- ❖ A perfectly competitive market characterized by many buyers and sellers.
- ❖ In a perfectly competitive market, individual buyers and sellers cannot affect product prices due to their small size relative to the market. Consequently, any single firm's change in output does not impact the overall market price of the product.
- ❖ Each firm in this market produces identical or homogeneous products, such as Basmati rice, produced by numerous farmers. Consequently, buyers cannot distinguish between products from one firm and those from others, leading to buyer indifference.
- ❖ Under perfect competition, resources, including workers, can move freely across locations.
- ❖ In the long term, the firm can easily enter or exit the industry.
- ❖ In perfect competition, buyers and sellers have complete information regarding prices and costs. Because of this, consumers do not pay more than the market price. Producers supply goods that accurately meet demand.
- ❖ Price differences in the market are quickly corrected, leading to a uniform price for the product.

The perfectly competitive market does not exist in the contemporary world. Part of the agricultural product and stock price captures some features of the perfectly competitive markets.

- Examples: Parts of agriculture
- Number of producers: Many
- Power of firm over price: None
- Type of product: Standardized
- Barriers to entry: No barriers
- Non-Price competition: None

Market Structures: Perfect Competition

When we say “*Identical*” that is exactly what we mean.

1. The consumer does not perceive any difference, real or imagined, between the product that is produced by the large number of different suppliers.

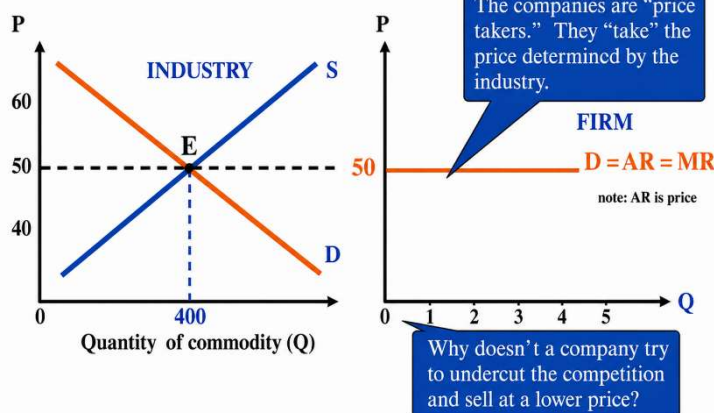
2. Products are perfect substitutes

How large is “*large*?”

There is no set cutoff. There need to be enough companies so that each one is so small that it has no “*power*” to determine its own price.

The company in the industry is the price taker! The industry is the price maker and the firm/company is the price taker!

The equilibrium Price and Output Determined by a Perfectly Competitive firm



A firm refers to a specific business entity that produces goods or services, whereas an industry is a broader category comprising multiple firms that generate similar or related products. For instance, e.g., the auto industry, telecom industry, the steel industry, etc.. Tata motors is a company which falls under automobile industry. Similarly, Airtel is a firm or company which falls under telecom industry. Now the question arises whether Reliance industries is an industry or a firm. It is a firm. Before your name you can write PM Roshni.

Under perfect competition, the price in the market is determined by the point at which demand and supply intersect. Market demand curves are drawn by summing up the demands of consumers horizontally in the market. Assume that we have three individual consumers namely; A, B and C. If Consumer A uses 20 Ice creams while B and C use 30 and 10 ice creams respectively, then the market demand curve will be $20 + 30 + 10 = 60$ Ice cream.

We have taken three consumers for simplicity and these three consumers assumed to represent the entire market demand curve. The market supply curve represents the total

supply of goods in the market, derived by horizontally summing the individual producers' supply curves.

Let's assume that there are three individual producers such as A, B and C. Producer A produces 20 Ice creams and producer B and C produces 30 and 20 ice creams. The market supply curve for ice creams is $(20+30+20) = 70$ ice cream. We have taken three producers for simplicity and these three producers assumed to represent the entire market supply curve.

The letter "D" shows the demand curve in the market while the letter "S" indicates the supply curve in the market. They help in finding the equilibrium price and quantity for the product. The market demand curve and market supply curve intersect each other at point E. The equilibrium price for the product is 50 per unit and quantity produced by the industry is 400 units. Point E is equilibrium points as prices are relatively stable there.

If the price is higher than equilibrium price, that is, Rs. 60, the supply exceeds the demand. Therefore, the price of the commodity will keep on falling until the equilibrium is attained at point E. When the price falls, demand rises while the supply falls to achieve equilibrium at price Rs. 50, when $Q_d = Q_s$. However, if the price is lower than the equilibrium price, for example, at Rs. 40 per unit, the $Q_d > Q_s$, hence the price rises.

Whatever the price is decided by the industry the firm will sell the product at that price. Hence, we can say that perfectly competitive market industry is the price makers and firms are the price takers. Whatever the price the industry will decide the firm will follow that price.

Why is it possible? Well, under perfect competition, a large number of sellers operate and each of them is relatively smaller than the other sellers operating in that particular market environment. Hence, the actions of a single firm cannot result in price increase. Being price-takers, the demand curve for firms is horizontal in relation to X axis. D represents demand curve and AR stands for average revenue curve which is nothing but the price. On the other hand, MR is marginal revenue curve where marginal revenue means the additional revenue earned from the production of extra unit of output.

$$MR = \frac{\Delta TR}{\Delta Q}$$

Let's assume that when output is zero average revenue is Rs. 50 and marginal revenue is undefined. When output increased to 1 unit, the AR is Rs.50. Now we need to calculate marginal revenue. We can derive marginal revenue from the total revenue (TR). When output is 0, TR is price multiplied by quantity that is $(50 \times 0) = 0$. When output increased to 1 unit, TR is $(50 \times 1) = 50$. The change in total revenue is $50 - 0 = \text{Rs } 50$ and change in quantity is $(1 - 0) = 1$. We can say that when output increased to 1 unit the MR is Rs. 50. Let's assume that output increased from 1 to 2 units, in this context how much is the marginal revenue. When output is 1 unit the total revenue is $\text{Rs } 50 \times 1 = \text{Rs } 50$. When output is 2 unit the total revenue is

Rs. $50 \times 2 = \text{Rs } 100$. Marginal Revenue (MR) is calculated as the change in total revenue divided by the change in quantity.

That is $\left(\frac{10}{2-1}\right) = 50$. You will get the marginal revenue of Rs. 50 if output increased from 2 to 3 units and so on. Hence $AR = MR$ and the demand curve is horizontal in a perfectly competitive market.

Critical Thinking Questions and suggested Answers

Q1. Can one Perfectly competitive firm charge higher price above the market price?

(Ans) Firms cannot price their products above the market price due to product homogeneity, as doing so would result in the loss of all customers.

Q2. Can one Perfectly competitive firm charge below the market price?

In a perfectly competitive market, firms have no incentive to sell below the market price, as they can sell any quantity at the prevailing price. Consequently, they face a horizontal, infinitely elastic demand curve. For instance, basmati rice producer can sell as much rice as desired at the market price of Rs.50. Thus, the firm acts as a price taker, and the marginal revenue (MR) from selling an additional unit remains constant at Rs.50.

Profit Maximization with price taker firms

- ❖ Individual firms cannot affect the market price, i.e., they are “price takers”
- ❖ Only decision is what quantity to produce
- ❖ Result is a perfectly elastic demand curve
- ❖ Market price is set by the interaction of supply and demand curves.
- ❖ The firm faces a perfectly elastic demand curve, $\eta = \infty$
- ❖ As a consequence, marginal revenue (MR) = Price (P)

Equilibrium (or Profit Maximizing) Conditions:

A perfectly competitive firm reaches equilibrium when necessary and sufficient conditions are satisfied. If one condition satisfied and other is not, then the perfectly competitive firms can't reach the equilibrium.

1. **The first order conditions.** It is also known as necessary condition where

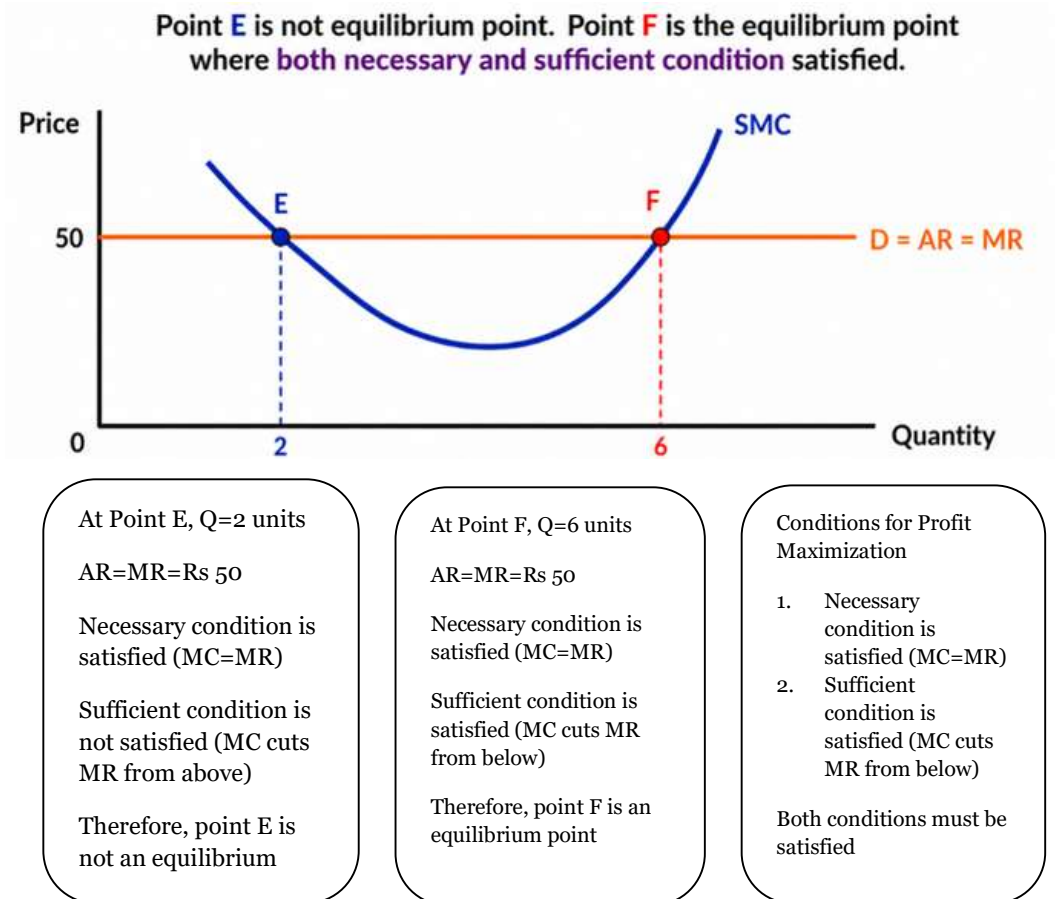
Marginal revenue (MR) = Marginal cost (MC)

2. **The second order condition.** It is known as sufficient conditions

(a) MC cuts MR from below.

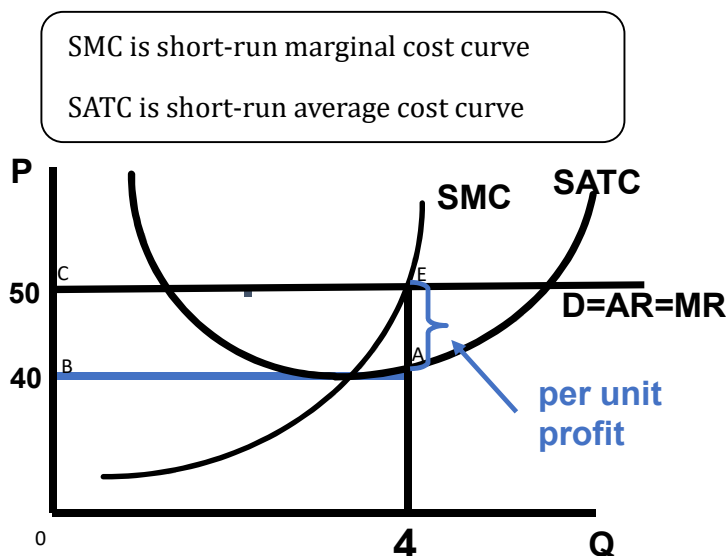
(b) At the point of equilibrium, the marginal cost (MC) curve must be rising. In this scenario, the slope of Marginal Cost (MC) is greater than the slope of Marginal Revenue (MR).

Since $P = MR$, the profit max condition for a price taker is $P = MC$



Note: Average revenue (AR) is price. SMC refers to the short-run marginal cost curve. Point F is the equilibrium points where both necessary and sufficient conditions are fulfilled. Here, the firm can sell 6 units of output when the price of a product is Rs. 50 per unit.

Short-run price and output determination of a perfectly competitive firm (profit case).



At the short run Equilibrium

- Firm produces 4 units of output
- The best level of output is at point E where $MR=MC$ and MC cuts MR from below
- Price=Rs 50 per unit
- Average total cost (ATC)=Rs 40 per unit

- Per unit profit=per unit price-average total cost
- Per unit profit=Rs 50-Rs 40=Rs 10
- Total profit=per unit profit*output produced
- Total profit= 10×4 =Rs 40
- The firm is in short run equilibrium where both necessary and sufficient conditions are met and here the firm makes economic profit

In this diagram X axis shows output and Y axis shows price. The profit maximising quantity is where $MR=MC$

$$\text{Profit} = \text{TR} - \text{TC}$$

TR=Total revenue which is equal to price multiplied by quantity

TC= Total cost equals fixed costs plus variable costs. To compute total cost, multiply average total cost by quantity.

At what price firm can sell 4 units. A firm can sell 4 units of output at each price is Rs 50.

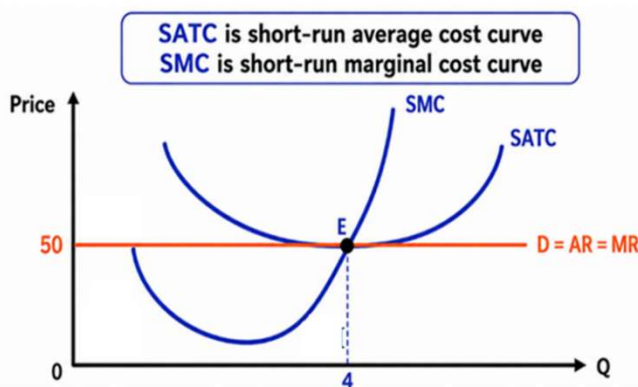
The optimal output for an enterprise in the short run is where marginal revenue is equal to marginal cost in the short run. One may denote the short run marginal cost by MC or SMC. The optimal output for the enterprise occurs at point E. This is where the marginal cost curve (MC) intersects the marginal revenue curve (MR). In point E, the firm produces four units at a price of Rs. 50 each. In Point E, Price= Rs. 50 & ATC =Rs. 40.

Short run average total cost curve is popularly called as ATC. The profit made by the firm is Rs. 10 per unit, and the total profit comes out to be Rs. 40 denoted by area EABC. This profit is the maximum profit possible for the firm as the firm would increase production up to 4 units because in that case $MR > MC$.

The increase causes total revenues to rise more than total costs, resulting in higher total profits or lower total losses. If $MC > MR$, and price equals Rs. 50, then the organization needs to produce less to reduce total costs more than total revenue; this will maximize profits or minimize losses. The appropriate quantity of production will be at $MR = MC$.

In a perfect competition model, since the demand faced by the firm is horizontal or infinitely elastic, then $AR=MR$, and thus the best point to maximize output will be when $AR=MR=MC$. The firm will make maximum profits at point Q=4, where $P=MR=MC$.

Price and output determination of a perfectly competitive firm in the short run (normal profit or break-even case).

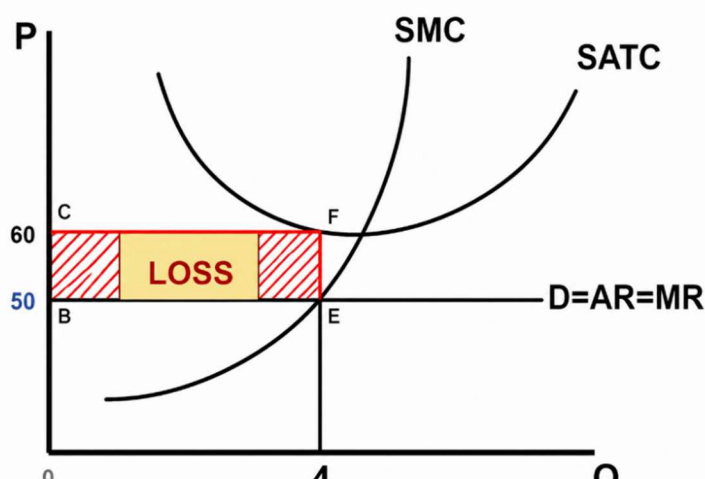


- Best level of output is point E where both necessary and sufficient conditions are satisfied
- The best level of output is 4 units. Price=AR=Rs 50 and output is 4
- The breakeven point is at point E, where the company makes neither profit nor loss
- Here total revenue=Total cost and Average revenue (AR) =Average cost (AC)
- Total Revenue=Price*quantity ($50 \times 4 = 200$) Total cost=Average cost *quantity ($50 \times 4 = 200$)

In the above figure, the X-axis shows Quantity, while the Y axis shows Price. The ideal output is at point E, where both necessary and sufficient conditions are satisfied. At point E, producers produce an output of 4 units, and the Price of each unit is Rs 50. This point is called the break-even point. The Break-even point is where the producer neither earns profit nor incurs a loss. Producers are neither gaining anything nor losing anything at point E. At the break-even point, Total Revenue = Total Cost ($TR = TC$) and Average revenue = average cost. Total Revenue = Price * quantity ($50 * 4 = 200$) and Total cost = Average cost * quantity ($50 * 4 = 200$). When the demand curve is tangential to ATC curve, then Normal profit takes place.

To optimize profitability, a firm should increase output when marginal revenue (MR) exceeds marginal cost (MC), as this adds more total revenue than costs incurred. Conversely, if MC is greater than MR, the firm should reduce output to lower total costs.

Short-run price and output determination of a perfectly competitive firm (Loss case).



$TR = P * Q$ ($50 * 4 = 200$). Total Cost = $AC * Q$ ($60 * 4 = 240$). Here the total cost is > total revenue ($240 > 200$). Hence, the firm incurs total losses of $240 - 200 = \text{Rs } 40$.

In the graph shown above, the X-axis denotes the quantity of output and the Y-axis denotes the price. A perfectly competitive firm faces a perfectly elastic demand curve. The average revenue curve is called the price. The upward-sloping MC curve or the marginal cost curve denotes the supply curve. Two criteria have to be fulfilled for finding out the optimal output of perfectly competitive firms. One of the criteria is $MR = MC$. Second Order Condition states that (a) MC must cut MR from below, and (b) MC should have an upward-slope at the point of equilibrium. MC slope > MR slope. Here, the MR and MC curves meet at point E. This point represents optimal production level, at which the company produces just four units. To determine loss, the cost curve should remain above the price line. In the figure above, the

per-unit cost equals to Rs 60 and price to Rs 50. So, the business experiences a loss of Rs 10 per unit. In order to calculate total losses suffered by the business, we should multiply losses per unit with total units produced. Consequently, total losses experienced by the firm equal to Rs 40. The total loss incurred by the firm can be shown in the area of BCFE.

Note whether a firm makes profit, loss or breakeven, everything we are comparing with respect to best level of output.

Important point to remember

A perfectly competitive firm can experience profits, break even, or incur losses in the short run. It all depends on the height of the ATC curve at the best level of output.

If $P = ATC$ at the best level of output, a perfectly competitive firm will break even.

If $P > ATC$, at the best level of output, the perfectly competitive firm will incur a profit

If $P < ATC$, at the best level of output, the perfectly competitive firm will incur a loss

If $P < \text{minimum point of AVC}$, the perfectly competitive market (PCM) firm will be forced to shut down their operations in the short run as it cannot cover its fixed and variable costs.

Critical Thinking Questions

Q1. Should a firm operate at a loss in the short run? Will it continue its operation or shut down when the demand curve falls between the AVC and the ATC

(Ans) The firm will continue production as long as the price (P) is greater than the average variable cost (AVC). If P is less than the average total cost (ATC) but greater than AVC, it covers all variable costs and part of fixed costs.

- Variable costs are covered by revenue
- Some, but not all, of your fixed costs are covered by revenue
- The remainder of the fixed costs need to be paid “out of pocket.”

Q2. When the demand curve is just tangent to the AVC

- ❖ Variable costs are covered by revenue
- ❖ None of the fixed costs are covered by revenue
- ❖ All of the fixed costs need to be paid “out of pocket.”

Q3. What should you do?

Go out of business –No.

Q4. When the demand curve/Price falls below the AVC. What should you do?. Go out of business –Yes/NO.

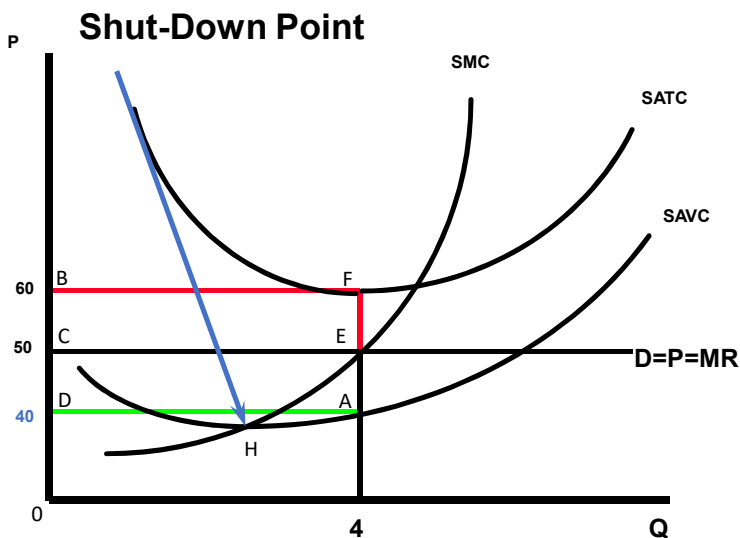
(Ans) Let's have a look with one example. Suppose the average fixed cost is Rs. 50. Your daily average variable cost is Rs. 10

Let's assume that you are selling the product at Rs. 11. By selling it at Rs. 11 you are covering your variable cost but not the fixed cost. Hence, you are making a loss as that loss is due to fixed costs. This situation is tolerable for a firm.

On the other hand, let's assume that you are selling the product at Rs. 9 and the average variable cost of producing that product is Rs. 10. Tell me whether you are making a profit or a loss. You are making a loss in both fixed as well as variable costs. It is not at all tolerable for a company. All expenses are paid from the pocket.

- Not all variable costs are covered by revenue.
- None of the fixed costs are covered by revenue
- The remainder of the variable costs and all of the fixed costs need to be paid "out of pocket."

The firms need to shut down the business as the firm is unable to cover its fixed and variable cost.

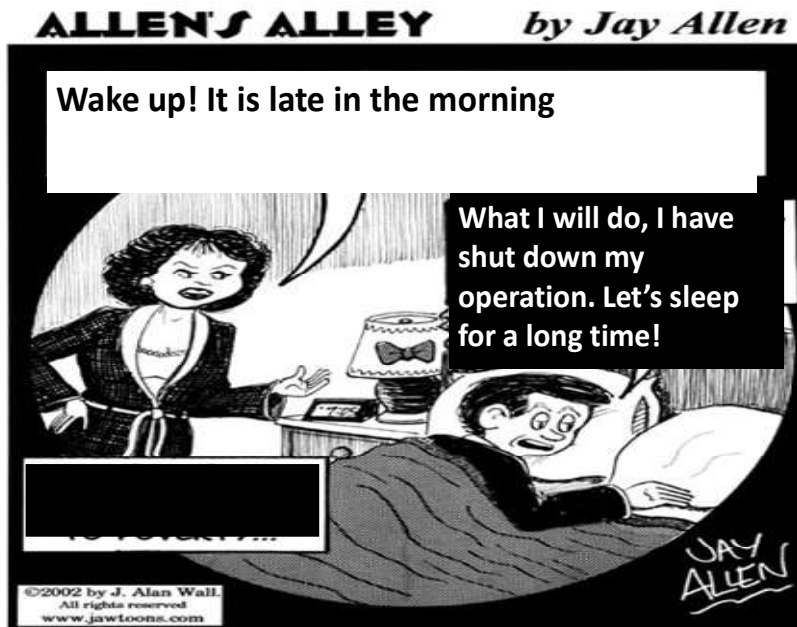


In a hypothetical scenario where the market price of a product is set at Rs 50, the best level of output is determined at point E. Here, the necessary and sufficient conditions for perfect competition are met, with the firm producing 4 units at Rs 50. At this output level, the price=MR=MC. However, the average total cost (ATC) is Rs 60, leading to a loss of Rs 10 per unit. Consequently, the firm incurs a total loss of $(10 \times 4) = \text{Rs. } 40$, represented in the area BFEC.

If a firm stops production, it incurs a fixed cost loss (FA) of Rs 20 per unit, resulting in a total loss of $(20 \times 4) = \text{Rs } 80$ when production is stopped at 4 units. It is shown in the area of FADB. At this output level, the price is Rs 50, while the average variable cost (AVC) is Rs 40. The firm must determine whether it can offset part of its fixed costs. By continuing production at the best output level, the firm minimizes its losses. If the market price of a product drops below Rs 40, the firm becomes indifferent to production, as at point H, where price equals average variable cost ($P=AVC$), total losses equal the total fixed costs, regardless of production.

Therefore, point H is identified as the shutdown point, below which the firm cannot cover its variable costs. The firm can limit its losses by going out of business. Note: if the price is equal to or below the minimum AVC, the firm should shut down the business. If the firm cannot cover fixed and variable costs, it is better to shut down the business.





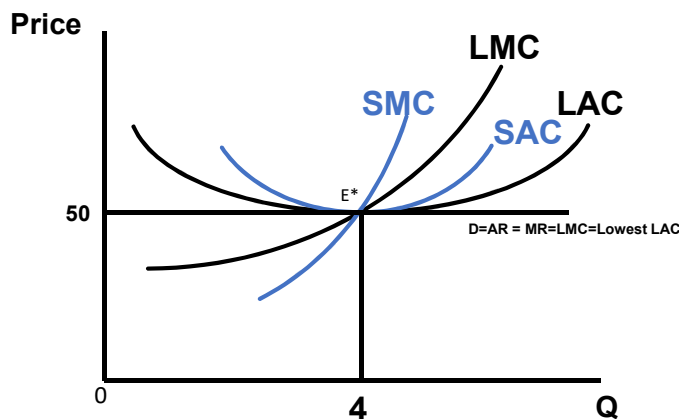
Long-run price and output determination in a perfectly competitive market

All input and production costs are variable in the long run. This will give the firm an opportunity to choose its plant size when it produces at its efficient production level. The most efficient production level is reached when the price equals the long-run marginal cost of the firm. The most efficient plant is the one that just makes contact between the short-run and the long-run average cost curves.

When there is profit being made by the firms, new firms would come into the market in the long run. The market supply of the product would increase as well as its price would decrease until no more profit remains for any firm to make. Those firms that make losses will be forced to leave the market in the long run due to lack of profit. The reduction of supply will continue until all firms earn normal profit.

These firms operate at the lowest point on the long-run average cost graph. Such a point is usually represented by E^* . At this stage, the firm earns only normal profit which is the result of the equilibrium situation where total revenues and total costs are equal. The most optimal output level for the firm in perfect competition when the price is $P=50$ will be 4 units, and this is the level marked by point E^* . At such a point, the price (P) is equal to marginal revenue (MR) and lowest long-run marginal costs (LMC), which is also equal to the lowest average cost (LAC). Due to no entry barriers in the market, the firm operates at the point where there are no profits or losses.

In perfect competition, firms attain the state of long-run equilibrium. As such, firms earn zero economic profits in perfect competition or break-even. In such a case, it means that the owner of the business gets a normal return on capital invested. If there is zero economic profit, it means that the total revenue equals total cost.



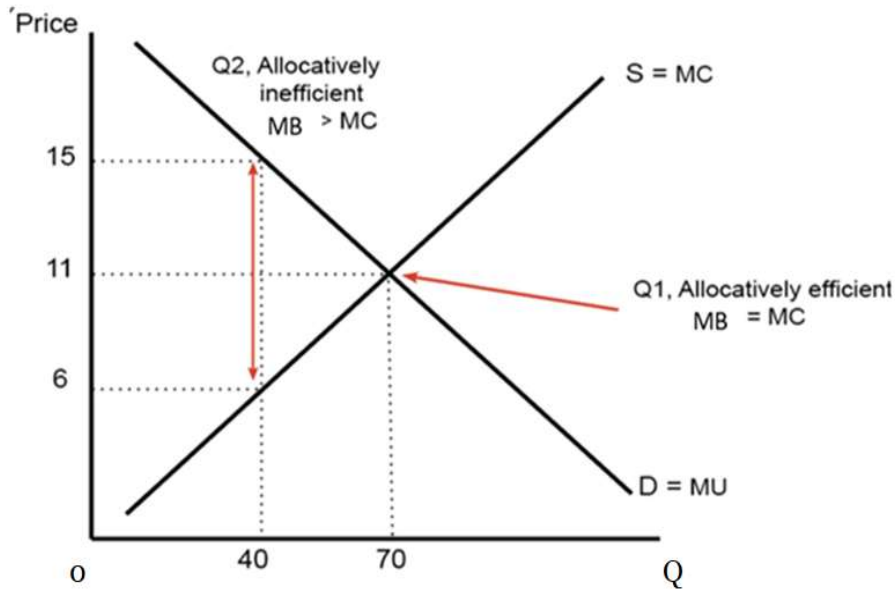
Allocative Efficiency

The two forms of efficiency are used to determine how the market functions. First is termed productive efficiency while the other is referred to as allocative efficiency. Productive efficiency involves the ability to produce goods at minimal costs. Productive efficiency can be attained by a firm in case its production involves the lowest average cost in the long run and price equals average cost. In the long run, perfectly competitive firms produce at a minimal average cost. Allocative efficiency results where marginal cost equals marginal benefit. This is a situation where price equals marginal cost of production of the product. Marginal cost is the cost involved in producing additional units of the product. Marginal benefit is the additional benefit that a consumer obtains from using another unit of the product.

In the presence of allocative efficiency demand=Supply or marginal benefit is equal to marginal cost. It will occur when there is no government intervention. Allocative efficiency occurs in highly efficient markets. It means there is the absence of asymmetric information.

The same information is available to the buyers and sellers. In the presence of allocative efficiency, the resources are utilized properly and there is no wastage. The market will have allocative efficiency where the total surplus is maximum. Whenever we are taking about total surplus is maximum that implies producer surplus is more and consumer surplus is more.

Consumer surplus is more and producers' profits is more. Allocative efficiency is achieved when consumers and producers can access information freely, eliminating asymmetric information, which enables optimal decision-making in purchasing and production. It requires consumers to have choices that enhance their satisfaction. This efficiency arises when market resources are allocated in accordance with consumers' desires and needs.



Note: in the presence of allocative efficiency total welfare to the society is maximum. But in reality, allocative efficiency rarely occurs because buyers and sellers don't have the same available information. Government intervenes in the markets in the form of taxes. As a result, price increases. With the increase in price Consumer and producers surplus decrease and allocative inefficiency occurs.

Consumer surplus is an idea that comes out of the demand curve. This can be seen through the region which lies below the demand curve and above the market equilibrium point. The consumer surplus is an indication of the difference between the price paid and the maximum amount a customer would pay for a product. If a consumer wants to purchase an air ticket and pays Rs 20,000 against the willingness to pay Rs 50,000, the consumer surplus will be Rs 30,000.

- Producer surplus is referenced in economic theory as the area situated below the market-clearing price and above the supply curve. It quantifies the difference between the lowest price at which a producer is willing to supply a product and the price actually received by the producer. This concept illustrates the benefit to producers in terms of additional revenue generated beyond their minimum acceptable price.

Monopoly

Mono stands for one while poly stands for sellers. Therefore, monopoly can be defined as one seller. Monopoly is an individual seller operating within the market with whatever products produced by them having no close substitutes. For this reason, monopolies operate as price makers in the market. In monopoly cases, a firm takes control over the market, and other competitors are prevented from joining in the market. Perfect competition on the other hand is characterized by firms operating as price takers. A company is considered as a monopoly if...



***“Isn’t there some other company
that makes software more cheaply?”***

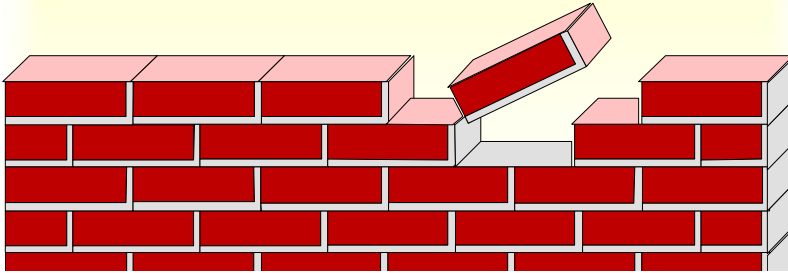
.....a firm must: be a sole provider of goods/services in the market.

.....Have no substitute for their product/having no close substitutes.

.....and entry into the market is restricted.

Why Monopolies Arise?

Entry barriers are the primary cause of monopolies.



Barriers to entry have following sources:

- ❖ **Ownership of a key resource** e.g. a toll highway, Railway and Postal Service (Postal and telecom services before 1990). Example. DeBeers diamond company
- ❖ **Government licensing and foreign trade barriers:** The government grants exclusive production rights to a single company through a patent for a specific good, such as a new drug.
- ❖ **Formation of a cartel;** e.g. OPEC. Group of firms joining their hands and deciding the price.
- ❖ **Patent:** Patent ownership grants a firm exclusive right to produce a product, functioning as a monopoly while the patent is active, as illustrated by the example of Xerox.
- ❖ **Technological superiority:** A firm can maintain monopoly power if its technological expertise surpasses that of its competitors over time, as illustrated by IBM's historical dominance in the computer business due to its technological superiority.

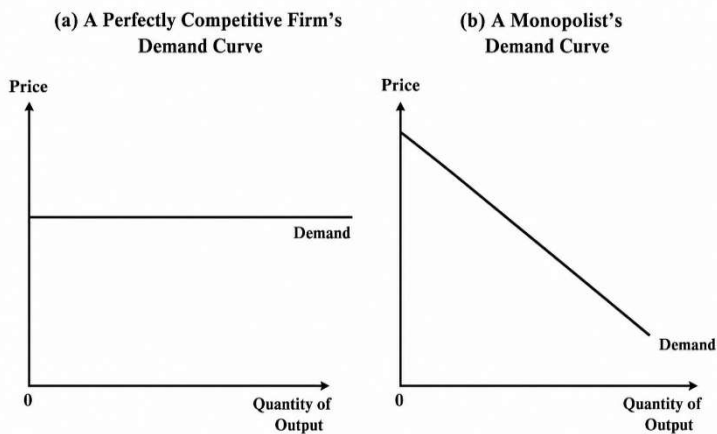
Natural Monopolies

An industry is termed a natural monopoly when a single firm can efficiently serve the entire market at a lower cost than multiple firms. This happens because large-scale production reduces costs. Notable examples of natural monopolies include electricity and gas supply services.

Perfectly Competitive firm versus Monopoly

Perfectly Competitive Firm	Monopoly Firm
There are many sellers in the market.	There is only one seller (sole producer).
The firm faces a horizontal demand curve.	The firm faces a downward-sloping demand curve.
Firms are price takers.	The monopoly firm is a price maker.
The firm can sell any quantity at the same market price.	The monopoly firm must reduce price to increase sales.

Demand curve of a perfectly competitive firm and a Monopoly



The Monopoly's firms total revenue (TR) $= P \times Q$

TR represents total revenue. P shows price and Q represent quantity

❖ A Monopoly's firms Average Revenue (AR) $= TR/Q = P$

❖ A Monopoly's firms Marginal Revenue (MR) $= \frac{\Delta TR}{\Delta Q}$

Δ Represent a small or marginal change

ΔQ represent change in quantity

ΔTR represent change in total revenue

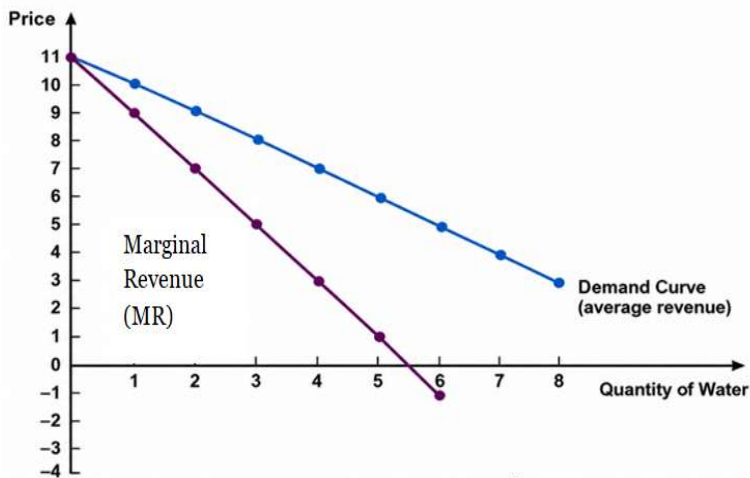
- ❖ *Marginal revenue is the extra or additional revenue which the seller obtains by selling one additional unit of good.*
- ❖ Average revenue is the firm's earnings per unit sold, equivalent to the product's price.

A Monopoly's Total, Average, and Marginal Revenue

Quantity (Q)	Price (P)	Total Revenue (TR = P × Q)	Average Revenue (AR = TR / Q)	Marginal Revenue (MR = $\Delta TR / \Delta Q$)
0	11	0	–	–
1	10	10	10	10
2	9	18	9	8
3	8	24	8	6
4	7	28	7	4
5	6	30	6	2
6	5	30	5	0
7	4	28	4	–2
8	3	24	3	–4

Demand Curve and Marginal Revenue Curves for a Monopoly...

A monopolist's marginal revenue is always lower than the price of its product



MR represents marginal revenue and AR represents average revenue. The monopoly is the sole supplier of its product and there are no other substitutes for it. Being the only supplier in the market, the firm faces the market demand curve. The market demand curve shows how the monopoly can relate its price to the amount that it can sell.

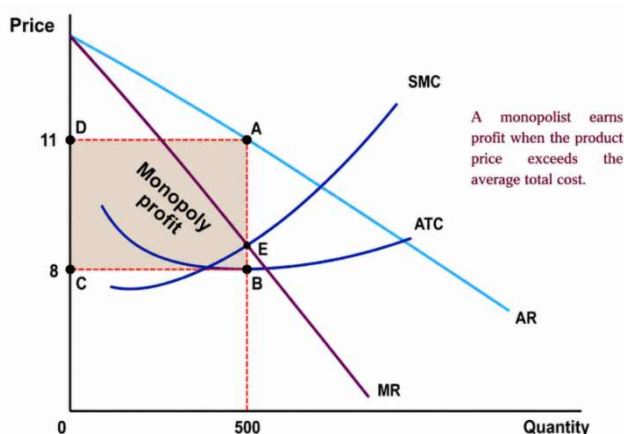
The monopoly can either produce less quantity for a higher price or more quantities for a relatively low price. Generally, the monopolist produces less quantity at a higher price than in perfect competition. Monopoly refers to a condition where the firm is the sole provider of

a particular good or service without any close substitutes in the market. Consequently, the firm will have to supply for all the requirements in the market. The demand curve of the monopoly slopes downwards, which means price will decline when quantity increases. This makes the price of the product greater than the marginal revenue. Therefore, the marginal revenue curve lies below the demand curve. Unlike perfect competition, a monopoly is profitable in the long term since it cannot be entered by any other firm in the market.

Profit Maximization or Equilibrium of a Monopoly Company/Firm in short run

- ❖ A monopolist maximizes profit at the output level where marginal revenue equals marginal cost. This is the first-order (necessary) condition for profit maximization.
- ❖ MC will cut MR from below. Slope of MC must be greater than Slope of MR- 2nd Order condition (Sufficient condition)

Short Run price and output Determination under Monopoly (Profit)



Demand Curve - AR Curve

MR – Marginal Revenue Curve

Point to be noted: The MR curve is always half way below the AR curve.

A monopolist will maximize his profits when $MR = MC$.

The slope of the MC is greater than that of MR. The optimal output level in the short run is 500 units, represented by the point E, where the marginal cost is equal to marginal revenue ($MC=MR$). Where there are outputs that are less than $Q<500$, the monopolist optimally produces more output since the marginal revenue (MR) is higher than the marginal cost (MC). In the case of an output higher than $Q>500$ where the marginal cost (MC) is higher than marginal revenue (MR), the firm will be able to maximize profit by decreasing output. At a level of 500 units, the price is 11, while the average total cost is Rs. 8, thus making point A on the demand curve. The profit that is obtained per unit by the monopolist is Rs 3. This results into a total profit of Rs 1500.

Although the monopolist may make economic profits in the short-run, he/she may also make normal profits or losses just like any other perfectly competitive firm, depending on how high the ATC curve is positioned at the most profitable output level. Economic profits will only be made if the price exceeds average total costs.

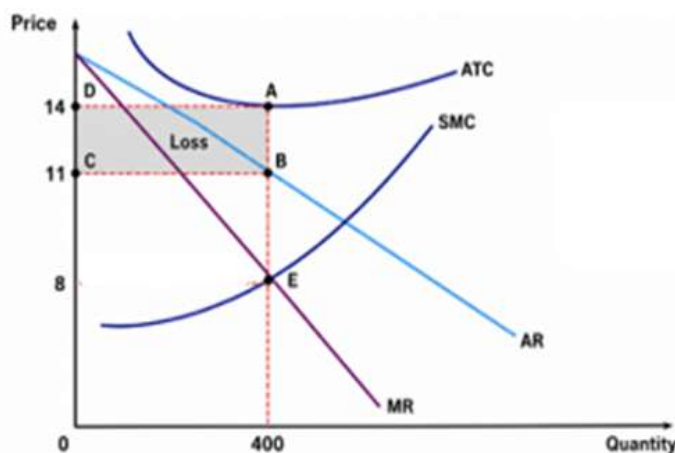
If $P > ATC$ at the best level of output, the monopoly incurs a Profit

If $P < ATC$ at the best level of output, the monopoly incurs a Loss

If $P = ATC$ at the best level of output, the monopoly breaks even.

In the short run, a monopolist aims to maximize profit or minimize losses, similar to a perfect competitor.

Short Run price and output Determination under Monopoly (Loss)



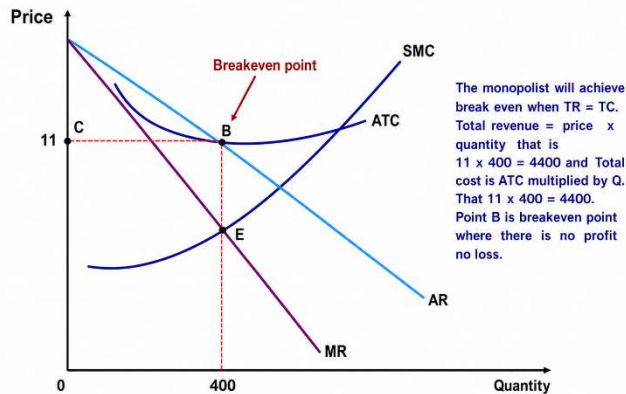
The monopolist will incur loss if $ATC > Price$. Price is Rs 11, ATC is Rs 14. Per unit loss is Rs 3 and total loss is $(44 \times 3) = \text{Rs } 1200$

In the X axis represents quantity and Y axis is the Price. In economics X axis we are presenting quantity and Y axis for price for simplicity. But in pure science it is reverse. Note that marginal revenue (MR) curve is half of average revenue (AR) curve. For instance, if AR is Rs 100 then MR will be Rs 500. First draw the AR curve, AR curve is downward sloping.

Next plot the MR curve, which is half of AR curve. Make sure that you draw a larger diagram to avoid any confusions and mistakes. The MC curve, which is also called as the supply curve is drawn in such a way that it always slope upwards. The MC curve cuts the MR curve at point E. The most suitable point for producing output lies at point E. At point E, monopoly produces 400 units of output at what price? To find the price, the most suitable point of producing output line must meet the AR curve, as AR is price. So, monopoly is selling 400 units of output at Rs 11 per unit of output. To see the losses, costs must be greater than the price.

Now draw the ATC curve which is like U shape. Now drag the best level of output which will touch the ATC curve. In this diagram cost is Rs 14 and price is Rs 11. Monopoly incur the loss of Rs 3 per unit. Total loss is loss per unit multiplied by total quantity. In this diagram total loss is $(14-11) \times 400 = 1200$. Another way we can find the same result. Profit = TR - TC. TR is total revenue. TC is total cost. Total revenue is nothing but $AR \times Q$. Here AR is Rs 11 per unit. TR is $11 \times 400 = 4400$. $TC = TFC + TVC$. It can be also $AC \times Q$. Hence total cost is $14 \times 400 = \text{Rs } 5600$. In this diagram monopoly loss is Rs 1200. Note whether a company is making profit or loss everything we are comparing with respect to the best level of output.

Short Run price and output Determination under Monopoly (Break even point)



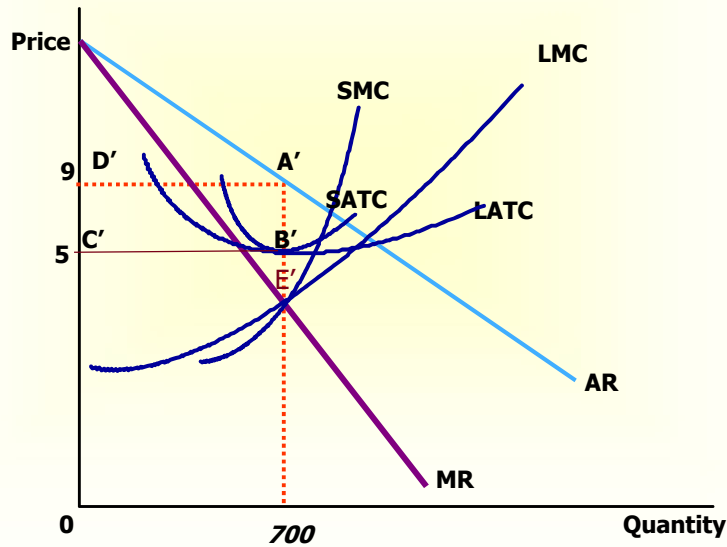
Price and output determination under Monopoly in the Long Run

In the long run, all the inputs and cost of productions are variable. In a monopoly, market entry is restricted, allowing the monopolist to sustain economic profits over the long term, unlike in a perfectly competitive market.

It should be noted that profits are considered as economic profits. It is unlikely that monopolies will operate at the bottom of their LAC since there is no chance of entering by new firms. The optimal output for the monopoly in the long run will be 700 units, which will be attained at point E', where $MR = LRMC$. The price is determined at 9.

In the long run, the monopolist chooses the most efficient plant size, where the short-run average cost touches the long-run average cost at an output of 700 units. At the optimum scale of operation at B', the monopolist achieves a best output level of $Q=700$, with SATC equaling LAC at a cost of 5.

In the long run, the monopolist earns a profit of 4 per unit; totaling 2800. It is shown in the area of A'B'C'D'. In long-run equilibrium, a monopolist is simultaneously in short-run equilibrium. At this point, marginal revenue is equal to short-run marginal cost ($MR = SMC$). Being in short-run equilibrium does not necessarily imply that the firm is also in long-run equilibrium.



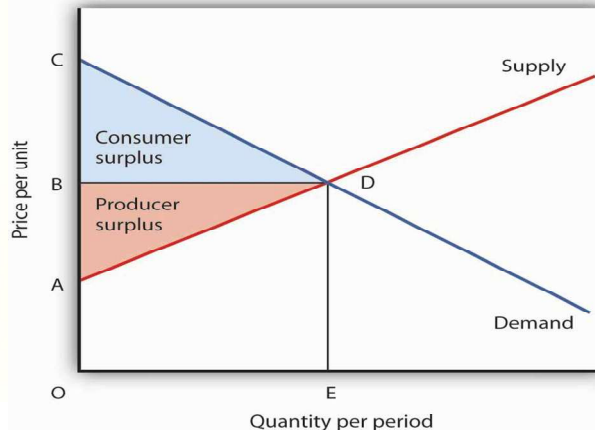
Dead Weight Loss

Dead weight loss is nothing but loss to the society. Loss to the society is nothing but loss of output. Whether this loss arises in a perfectly competitive market (PCM) or a monopoly. Note PCM works at full capacity level. They produce more output with less cost. On the contrary, monopoly produces less output as a result monopoly prices are always higher as compared to PCM. When we are talking loss to the society from where the loss is coming. Loss is coming from the producer and consumer side. Before discussing dead weight loss, we need to recall consumer and producer surplus first for easy understanding.

Consumer surplus is described as the gap between how much a consumer can afford to pay and what they end up paying. In the case that an individual is prepared to pay more than the equilibrium price, consumer surplus will be higher. Consumer surplus measures how much extra happiness that consumers enjoy when they pay less than what they can afford. Consumer surplus arises from the demand curve. That part of the graph above the market equilibrium and below the demand curve is referred to as the consumer surplus. As an illustration, let's suppose that you have bought an air ticket worth \$100 for your visit to Disney during the period of your school break, yet you had been prepared to pay \$300 for one air ticket. Your consumer surplus amounts to \$200.

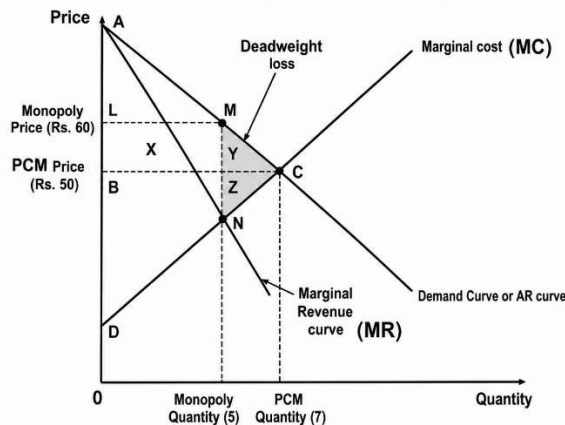
The producer surplus is the difference between the minimum price and the actual price received by a producer. Producer surplus comes from the supply curve. It is shown visually as the region below the equilibrium price and above the supply curve. The producer surplus rises as prices rise because there is a greater incentive to manufacture more items.

Producer and consumer surplus



Price and marginal cost are equal in a market with perfect competition. However, price and marginal cost are not equal in a monopoly. Prices set by monopolies are always higher than the cost of production. The monopoly raises prices and produces fewer commodities as a result of its monopoly power. Because they are paying more, the customers are in a worse situation. A monopoly has a negative impact on society as a whole. A monopoly creates a barrier between the consumer's willingness to pay and the producer's cost because it sets its price above marginal cost. This gap reduces the quantity sold below the socially optimal level.

The Inefficiency of Monopoly...



Now let's discuss why there is dead weight loss. Does it happen due to monopoly or perfectly competitive market? Assume the competitive market and monopoly have similar cost curves. AR and MR are the Average Revenue curve and the Marginal revenue curve for the monopoly respectively. The AR curve is called the demand curve (Industry demand curve in PCM). The demand curve in PCM is downward sloping. The MC curve is the monopoly marginal cost curve which is upward sloping.

It is considered the supply curve of the industry for PCM (the supply curve of the industry is upward sloping). In order to maximize its profits, the monopolist produces when $MR=MC$ and the price charged by him is Rs 60 with an output of 5 units. Now let us see what quantity a perfectly competitive firm produces.

A perfectly competitive firm (Industry) attains equilibrium position when the demand and supply are equal. In this case, AR curve is considered as demand curve while MC is considered as the supply curve of the industry under PCM condition. Both AR and MC curves intersect each other at point C and perfect competitive market price and output are Rs 50 and 7 respectively.

The area under the demand curve and above the market price level is consumer surplus. If we talk about a perfectly competitive market, then consumer surplus area would be ABC. The area above the supply curve and below the market price level is termed producer surplus. In PCM, the producer surplus area would be BCD.

You know a monopoly charges a higher price than the cost of production. Due to higher prices charged by the monopoly, the consumer will pay more and consumer surplus will come down. Now, the area of consumer surplus is ALM in Monopoly. Consumer surplus has come down because of higher prices charged by the monopoly. Consumer surplus $X+Y$ is being lost.

The producer surplus in a perfectly competitive market was BCD. Due to the price increase in the monopolistic market, producer surplus increases and becomes LMND. The loss in producer surplus is represented by $X-Z$. Loss in consumer surplus is represented by $X+Y$. Therefore, the deadweight loss will be $Y+Z$. Deadweight loss is the social loss. This occurs due to losses incurred by producers and consumers. Total revenue in a perfectly competitive market is $\$60 \times 5 = \300 in the case of monopoly. In PCM, total revenue is $\$50 \times 7 = \350 .

Price Discrimination

The practice of charging various prices for one item based on different buyers is termed price discrimination. Price discrimination occurs in an imperfect market scenario. If a company faces a downward-sloping demand curve, it will enjoy partial control over pricing and price discrimination. In a perfectly competitive market, it would be impossible for a firm to indulge in price discrimination as its demand curve would be perfectly horizontal.

Note: It is not considered price discrimination, if price differences are due to variations in the cost of providing a good or service. If pricing differences are not based on cost differences, price discrimination is considered to be present. In a market with perfect competition, price discrimination is impossible, but in an imperfect market (monopoly), it is possible. Price discrimination is possible in a monopoly since a firm is a price maker.

Conditions for Price discrimination

- ❖ Price elasticity must vary across different markets.

- ❖ Market imperfection
- ❖ The seller must have some monopoly power. The firm must have some control over its price
- ❖ Resale should not be possible. Buyers in low-price markets must not be able to sell to the buyers in high-price markets. It shouldn't be feasible to resell.
- ❖ Buyers must have a different price elasticity of demand
- ❖ The seller should have some control over how much of the product is supplied.
- ❖ The seller must be a price maker, not a price taker

Degrees of Price Discrimination

Three levels of price discrimination exist. Price discrimination in the first, second, and third degrees.

First-degree price discrimination

In this type, the seller prices the product at its maximum possible price for each unit sold, and thus companies are able to maximize their profits. This is because the seller knows the entire shape of the demand curve and the consumer's willingness to pay. For example, surgeons or lawyers etc. The patient requires a unique bone marrow transplant. Patients are going to pay whatever price is charged by the doctor, since it is a unique procedure, and the doctors can perform it only at a very high price. Likewise, in case of land disputes, lawyers charge what is the consumer's willingness to pay.

Not a commonly seen price discrimination strategy.

First-degree price discrimination is often criticized because it requires firms to have accurate knowledge of each consumer's demand curve to charge the maximum price for every unit sold.

Second Degree of price discrimination

Second-degree price discrimination involves a firm charging varying prices based on the quantities or blocks purchased, rather than setting individual prices for each buyer as in first-degree price discrimination.

Different prices are based on the quantity purchased (500g, 1kg, 5kg, 10kg). Consumers will benefit from bulk purchases. It is the concept of buying more of the same product at a lower price. E.g. Opera tickets, Internet usage, FMCG products. It involves setting prices based on the quantity purchased to capture some of the consumer surplus. By utilizing second-degree price discrimination, the firm aims to capture some consumer surplus and reduce excess inventories during periods of low demand.

Third degree of price discrimination

Most common form of price discrimination. Divides markets into different sub-markets and charges a different price in each sub-market. Different prices according to respective price elasticity of demand. Examples of third-degree price discriminations are

- ❖ Electricity companies charge lower price to residential user than the commercial user
- ❖ Companies charge a lower price in abroad than at home
- ❖ Entertainment companies charge lower price in the afternoon than in the evening
- ❖ Service industries charge lower prices for children and older people
- ❖ Airlines charge different prices for Economy and Business class travel
- ❖ Milk man charges more money to residential users compared to the sweet shops
- ❖ Hospitals charge different amounts of money for the same disease to different persons

Conditions for third degree

- ❖ Effect segregation prohibits resale from a low-price market to a high price market.
- ❖ Different sub markets must have different elasticities

Equilibrium condition for Third Degree:**1st order condition**

$MR_A = MC$ (determine the output which will be sold at sub market A)

$MR_B = MC$ (determine the output which will be sold at sub market B)

2nd order condition

Slope of $(MR_A) < \text{slope of } (MC)$

Slope of $(MR_B) < \text{slope of } (MC)$

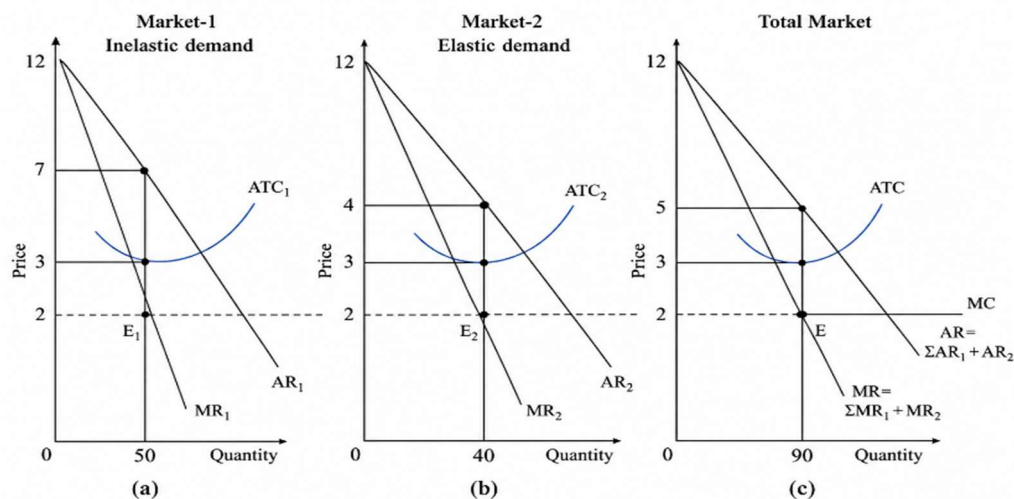
Equilibrium of Discriminating monopolist

- ❖ The First Problem. How much to produce?

Solution: Aggregate MC = Aggregate MR

- ❖ The Second Problem. How to share the produced output between two sub markets

Solution: $MR_A = MR_B$



Note : MC is ₹ 2 in market-1, market-2 and total market

ATC is ₹ 3 in market-1, market-2 and total market

The monopolist splits the whole market into two markets named as market 1 and market 2. The question which rises here is on what criterion monopoly divides the market? The monopolist divides the market according to the elasticity of the demand for the product. We can take the example of market 1 product that has inelastic demand while in market 2 there is elastic demand for the product. Panel (a) shows AR_1 and MR_1 as average and marginal revenue of market 1 respectively while panel (b) shows AR_2 and MR_2 as average and marginal revenue of market 2 respectively.

The total market demand curve is D. It is obtained from the horizontal summation of market 1 and 2 demand curve. Note that AR curve is the demand curve. AR_1 is the demand curve of market 1. AR_2 is the demand curve of market 2. The total market demand curve is $(D_1 + D_2)$ or $AR_1 + AR_2$.

Similarly, the total marginal revenue curve is the horizontal summation of MR_1 and MR_2 . Let's assume MC is constant for simplicity. The best level of output of a monopoly is 90 units is given in point E in panel C where $MR = MC = ₹ 2$. Put it simply when marginal cost is ₹ 2 the monopoly produces 90 units of output. If you will see in panel C the AR curve is not straight line. The reason is in panel C AR consists sum of AR_1 and AR_2 . In market 1 product is inelastic in demand whereas, in market 2 the product is elastic in demand. Hence the AR curve has broken into two different parts not a straight line. As AR curve has broken the MR curve has broken into two different parts. But for simplicity we have considered a straight line.

Now the monopolist would like to sell the quantity of 90 units into the sub markets where the market 1 is inelastic in nature while market 2 is elastic in nature. Now the monopolist sells more in market 1 since the demand for its product is inelastic in nature and lesser in market 2 since the demand for the product is elastic in nature. Suppose that the monopolist

sells 50 units of the commodity into market 1 and the remaining 40 units into market 2 such that $MR_1=MR_2=MR=MC=Rs.2$

Let's assume that average total cost (ATC) in total market is Rs. 3 at the best level of output. Then profit is $(5-3)=Rs. 2$ per unit and total profit is $2*90=Rs. 180$.

While exercising price discrimination let's see how much profit monopoly will earn. In market 1, the price is Rs. 7 and ATC is Rs. 3. Then profit is $7-3=Rs. 4$ per unit and total profit is $4*50=200$.

Likewise in market 2 price is Rs. 4 and ATC is Rs. 3. Then Rs.1 is the profit per unit and total profit is $1*40=40$.

What is the total profits in market 1 and market 2 together. That is $200+40=Rs. 240$. We can say, while exercising price discrimination the total profit is Rs. 240.

Let's assume firm is not exercising price discrimination then what is the total profit. Price is Rs. 5 and ATC is Rs. 3. Then profit is Rs. 2 per unit and total profit is $2*90=180$.

Rs. 240 total profit is better than Rs.180 total profit. Hence, we can say that monopoly should exercise price discrimination. By exercising price discrimination monopoly total revenue and profit will be more.

Summary: The total revenue in sub market 1 is $P*Q$ that is Rs 350($7*50$). The total cost in sub market 1 is $AC*Q$ that is Rs 150 ($3*50$). Profit is the difference between TR and total cost. The sub market 1 makes the profit of Rs 200.

Similarly, the total revenue in sub market 2 is Rs 160($4*40$). The total cost in sub market 2 is $AC*Q$ that is Rs 120 ($3*40$). The sub market 2 makes the profit of Rs 40 (Rs 160-Rs 120). If we will add the profits of two sub markets such as market 1 and 2, then total profit is Rs 200+Rs 40=Rs 240.

In total market the total revenue is Rs 450 and total cost is Rs 270. The profit in total market is Rs 450-Rs 270=Rs 180. Monopoly is making profit of Rs 180 without price discrimination. When monopoly is going for price discrimination total profit in two sub market is Rs 240. Hence, monopoly should go for price discrimination. While doing price discrimination monopoly is doing better.

Monopolistic Competition

Monopolistic competition can be defined as the type of market structure in which there is a presence of many firms selling differentiated products. Differentiation helps differentiate the product of one firm from the products sold by other firms in the market. Even though they are similar products; however, they are regarded as close substitutes for each other. A differentiated product is a product that is unique in its own way as compared to other similar products.

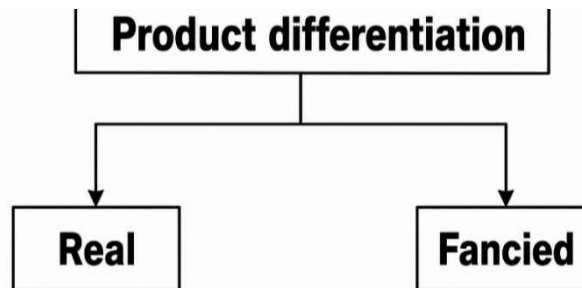
Due to differentiated product, a monopolistic firm is treated as price maker. But their pricing power is limited due to the presence of competition. The monopolistic competitive firms are toothpaste, cigarette, Shampoo, saving cream, market for bicycle, cotton and textile industries, fast-food industries, restaurant industry, shoe store, petrol stations, beauty salons etc. This market structure enables easy long-term entry and exits from the industry.

Firms in various industries possess monopoly power over competitors due to product uniqueness, superior location, enhanced service, and a wider range of product varieties. However, the market power is constrained by the presence of numerous close substitutes. In monopolistic competition, sellers diversify their competition strategies beyond price, similar to firms in an oligopoly. These factors include (i) aggressive advertising (ii) product development (iii) better distribution, and (iv) post-sale services. Product differentiation is emphasized, with limited price variation due to the presence of market substitutes. Pricing strategies are independent of the actions and reactions of other firms.

The three distinguishing characteristics of Monopolistic Competition

- ❖ Many sellers and many buyers
- ❖ Differentiated products
- ❖ Easy entry of new firms is expected in the long run.

Differentiated Products



What is Real Product Differentiation?

It appears when the inherent characteristics of the products are different

Includes

- ❖ Product Quality
- ❖ Location
- ❖ Services

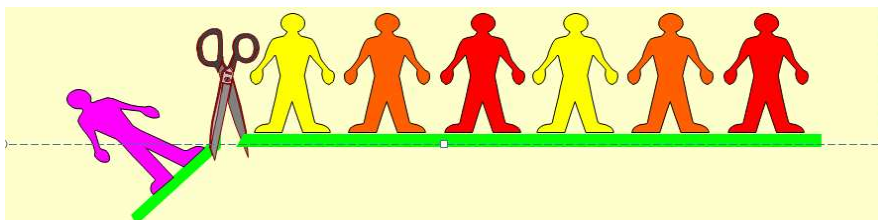
What is Fancied Product Differentiation?

It happens when products are actually similar, but advertising makes consumers believe they are different.

- ❖ Advertisement
- ❖ Difference in Packaging
- ❖ Difference in Design
- ❖ Brand Name

Free Entry and Exit

- ❖ Monopolistic competitive firms can freely enter into and exit from the market without any restrictions.
- ❖ The market adjusts the number of firms until economic profits reach zero.



Monopolistic Competition vs. Perfectly competitive market

Similarity: A monopolistic competitive market shares similarities with a perfectly competitive market in two main aspects. (a) large number of firms (b) no restrictions of entry.

Differences: Monopolistic competitive firm differs from perfectly competitive firm in terms of product differentiation. Each firm sells a product that differs in terms of quality, appearance, and each firm is the sole producer of its own brand. The firm's monopoly power is significantly constrained by the presence of numerous close substitutes.

For instance, Hindustan Lever is the only company that manufactures Pepsodent, giving it monopolistic power. However, the monopolistic power enjoyed by Hindustan Lever is limited since consumers can easily switch to other brands in case of an increase in the price of Pepsodent. This is because the differences among the various toothpaste brands do not matter to consumers who consider toothpaste to be toothpaste.

Under monopolistic competition, firms offer products that are differentiated from each other. For instance, different brands of toothpastes, salons etc. Due to this reason, it is impossible to draw a single market demand and supply curve. In a perfectly competitive

market drawing a single market demand curve was possible as the products belongs to different firms are homogeneous.

In a monopolistic competitive market, there is no single price in the market. Instead, there is a range of prices for different products. In monopolistic competition, a firm offers slightly differentiated products, resulting in a downward-sloping demand curve. If we will compare AR and MR curve of monopoly with monopolistically competitive firms, the later AR and MR curves are flatter compared to the former. This is due to availability of many close substitutes in a monopolistically competitive firm. As a result, the demand curve of monopolistically competitive firm is sensitive to the price change.

Competitive firms have prices equal to marginal cost, but for monopolistic competitive firms, prices are higher than marginal cost. Monopolistic competitive firms gain from selling an additional unit of output at prices that exceed the marginal cost, and hence profits increase.

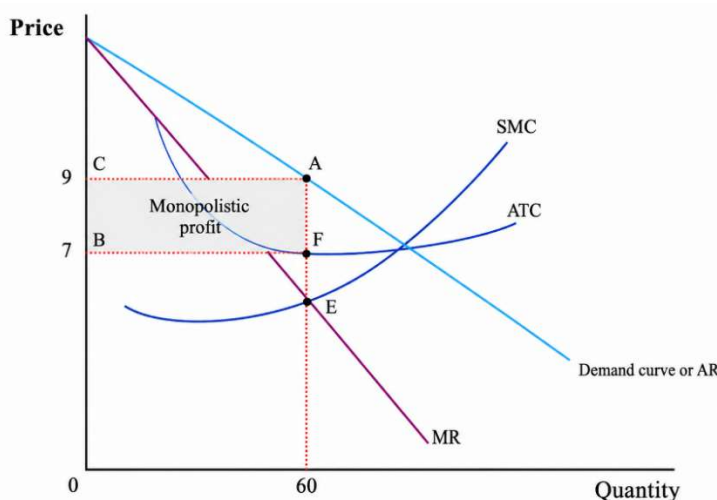
Monopolistic Competition (Short-Run Equilibrium)

- ❖ Monopolistic firms operate under a downward-sloping demand curve.
- ❖ The MR curve lies always below the AR curve. MR curve is always half of the AR curve.
- ❖ Maximizing Profits: Firms determine production levels by equating marginal cost with marginal revenue, setting prices at the maximum level that the market will bear for those quantities.

Profit Maximization or Equilibrium of a Monopolistic Firm

- ❖ A monopolist maximizes profit where $MR=MC$. This is the basic necessary condition required for profit maximization.
- ❖ The 2nd-order condition is called the sufficient condition. It requires the MC curve to cut the MR curve from the below. It also indicates that the slope of the MC is greater than the slope of MR.

Price and output Determination under Monopolistic Competition in the Short Run



X axis shows quantity and Y axis show price. Both AR and MR curves slope downwards. Both AR and MR are relatively flatter when compared to AR and MR curves of Monopoly. The most appropriate output for the monopolistically competitive firm is the output that is determined at point E where both the conditions are satisfied. When we say that the conditions are satisfied at point E, then it means that the monopolistically competitive firm will produce an output of 60 units and the price at point E is Rs.9 per unit. It is because AR is equal to price in economics. Point E satisfies the necessary and sufficient conditions. When the output is below 60 units, then marginal revenue is more than marginal cost and there will be a rise in total profits through increased output.

For an output level of 60 units, the price is Rs 9 per unit and ATC is Rs 7 per unit. The monopolistically competitive firm earns a profit of $(9-7) = \text{Rs } 2$ per unit. The total profit will be the per unit profit times quantity of output. Total profit will be Rs 120 (2×60). The total profit is represented by area AFBC.

A monopolistic company can make profit in the short-run period, although it can also break-even or make loss, like any perfectly competitive firm and monopoly. This all depends on the position of the ATC curve, whether it is high or low, when reaching the optimum level of production.

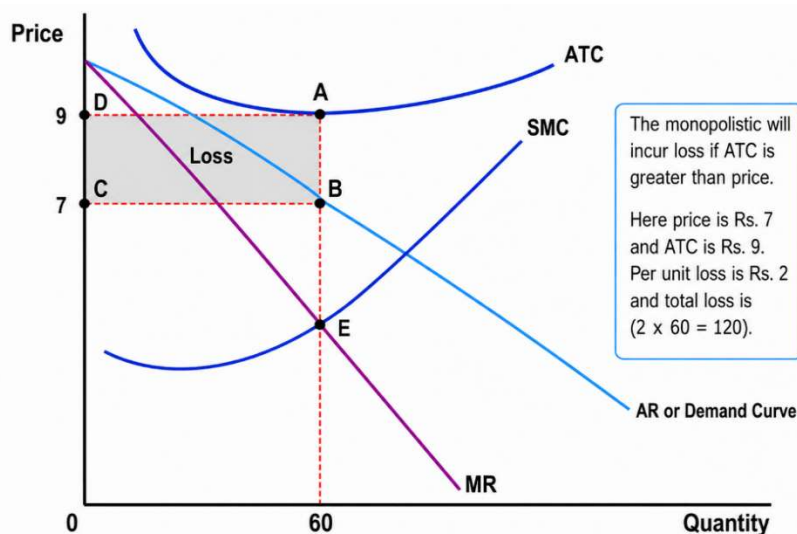
If the average total cost exceeds the price at the best output level, a monopolistic firm will incur a loss.

If the average total cost is less than the price at the best output level, a monopolistic firm will incur a profit

If the average total cost and prices are equal at the best output level, a monopolistic firm will incur a breakeven point

Note: Profit is the difference between TR-TC. TR is total revenue and TC is total cost. We can compute total revenue by multiplying average revenue with quantity. We can get total cost by multiplying average cost with quantity. If total revenue of a firm is greater than its total cost, the firm will make a profit.

Price and output determination under Monopolistic firm in the short run (Loss)

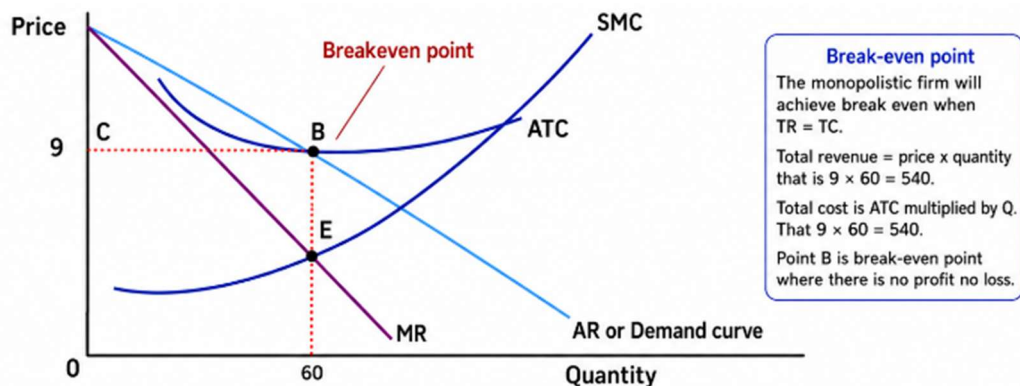


Quantity is presented in X axis and price is in Y axis. AR is average revenue curve. It is also known as price. It is similar to demand curve. MR is marginal revenue curve. MR is always half of AR. SMC is known as short run marginal cost curve. You can write MC curve also.

When we are writing LMC that represent long run marginal cost curve. MC curve remains same irrespective of SMC or LMC. Both MR and MC curve intersect with each other at point E where the output is 60 units and price is Rs 7 per unit. The cost curve is above price. Pls. note whether the company is making profit or loss or breakeven everything is compared with best level of output.

When output is 60 units the price is Rs 7 but the cost is Rs 9 per unit. Cost is more than price. The firm is incurring loss of Rs 2 per unit and total loss is $(2 \times 60) = \text{Rs } 120$ which is shown in the area of ABCD.

Price and output determination under Monopolistic firm in the short run (Break even points)



Price and output determination under Monopolistic firm in the long run

New firms are encouraged to enter markets due to profits, leading to a leftward shift in the demand curve for existing firms. Long-run equilibrium is reached when no further incentives exist for new entries, as existing firms achieve break-even.

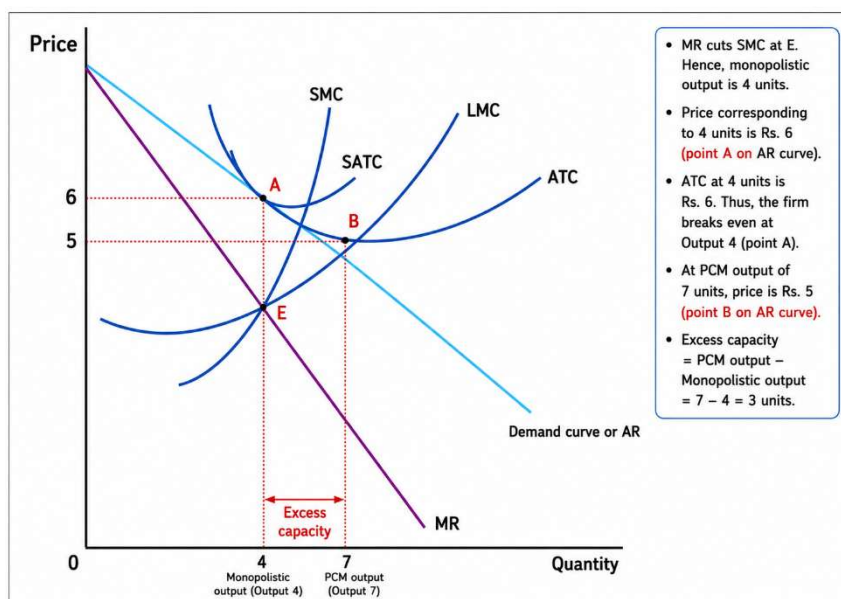


Diagram: X-axis represents quantity whereas Y-axis represents price. Monopolistic competitors might enjoy profit in the short run. This lures many new competitors into the market gradually. Due to increased number of firms in the market, demand falls causing profits to decrease for each company. Hence, in the long run, monopoly competition earns only normal profit.

Despite being considered as price-makers and producing differentiated products, they are not able to fully benefit from their market situations because of the low entry and exit barriers. They operate at the decreasing portion of the long-run average cost (LAC) curve, but do not produce at its lowest level, unlike the firms operating under perfect competition.

The demand curve tangents the LATC and SATC curves at point A, where marginal revenue equals long-run marginal cost and short-run marginal cost at point E. A monopolist sells 4 units at Rs. 6 each and breaks even in the long run, covering its costs.

The monopolistically firm produces to the left portion of the LATC curve. These firms are unable to produce at lowest point of the long run average total cost curve.

This is because they spent huge amounts on advertisement expenditure of the products and also designing of the product. In perfectly competitive markets, production occurs at the lowest point of the cost curve due to homogeneous products and lack of advertisement expenditure. In case of monopolistic competition, the cost average is higher than in perfect competition. The monopolistic firm will produce four units of output at a price of Rs 6 per unit while perfectly competitive firm will produce seven units of output at a price of Rs 5 per unit.

In the long run, a monopolistically competitive firm produces 4 units of output. At point E, where $MR = LMC = SMC$. Price and average cost are identical at this point ($P = LATC = SATC$). At point A, the price is Rs. 6. As a result, the firm only makes a normal profit after covering costs.

The best output level in perfect competition is 7 units. This is shown at point B, where $MR = LMC$. At this point, Price equals average cost ($P = LAC$). In the long run, the perfectly competitive firm price is Rs 5

Monopolistic has the capacity to produce 7 units but produces only 4 units: this creates excess capacity. The monopolistic is charging more price because of advertisement expenditure, marketing expenditure, and research and development expenditure.

What is Excess Capacity?

A company has excess capacity if it can reduce its average cost by raising its output.

Oligopoly

An oligopoly is a market structure in which only a small number of firms operate. These firms may produce either identical or differentiated products. Another term for oligopoly is rivalry among the few big firms. We can refer to it as pure oligopoly if the products are homogeneous (e.g., steel and aluminum). The products that differ from each other, like cars and cement, are referred to as differentiated oligopolies. Different products are called as heterogeneous products.

Duopoly, which occurs when there are just two producers or sellers of a product. The simplest form of oligopoly is a duopoly. Most people consider duopoly as a special case of oligopoly. While it is possible to enter the oligopoly industry, it is extremely challenging. The market is already dominated by a few large firms, and entering into the oligopoly market requires significant investment.

These companies have control over resources and a high level of brand loyalty. In addition, the fixed cost is very high in the oligopoly market. A few examples of oligopoly industries are Two-wheelers, steel, aluminum, telecoms, airlines, cars, and the cement industry. There are very few firms in an oligopoly market. As a result, one firm's activities have an impact on the others. These firms produce and sell either similar or different products.

For instance, Hyundai promptly follows Maruti's introduction of a price rebate to boost sales. Oligopolists always favors nonprice competition, such as advertising spending, product difference, and services they offer, because price competition results in a massive price war. For instance, Coca-Cola launched a significant advertising effort in response to Pepsi's massive campaign in late 1990. Only a few firms compete with one another under an oligopoly, and each firm must carefully analyze how its actions will impact its rivals and how those rivals will respond.

In such a market, price war is prevalent and has disastrous effects for the industry as a whole because each firm's activity depends on the actions and reactions of other firms. As a result, in an oligopoly market, the price remains same for a while. An oligopolistic firm is aware that other firms in the market are impacted by its choices. Thus, every firm considers the possible responses of its rivals. Before setting prices, altering items, allocating funds for advertising, and selecting the degree of service to offer, it takes this into account. Compared to other types of market structures, management decision-making in an oligopoly is far more complicated due to this interdependence.

Non pricing Strategy

- Advertisement
- Extending the service/warranty
- Longer time opening hours
- Keeping the price the same for next year
- Post sale service

Features or Characteristics of Oligopoly Market Structure

- ❖ **Few firms:** Only a small number of firms operate in the oligopoly market. There are approximately 10 firms that dominate the markets

- ❖ **Interdependence:** Each firm's actions influence the decisions of the other firms in an oligopoly. Put differently, strategic interactions are the main emphasis of pricing strategies. Because price competition can result in a disastrous price war, every firm attempts to implement several forms of non-price competition. Every firm takes rivals' activities and responses into account.
- ❖ **High entry barriers:** New firms find it challenging to enter the market. Though entry is possible in the long run, but it is extremely challenging due to the huge fixed cost.
- ❖ **Product type:** Products can be different (differentiated) or similar (homogeneous).
- ❖ **Price rigidity:** Prices typically hold steady for extended periods of time.
- ❖ Prevalence of non-price competition

Sources or Causes of Oligopoly

- ❖ **Economies of scale:** One significant factor contributing to a market's small number of firms is economies of scale. In the presence of economies of scale, the average costs falls rapidly with the expansion of output. Because of economies of scale, large firms have an advantage over small firms.
- ❖ **Barriers to entry:** Due to entry barriers, there may only be a few firms in the industry. Barriers come in two varieties: (A) technological and (B) legal. For small firm, there are technological barriers to entry. They are therefore unable to compete with the big firms. When the government restricts the entry of new firms to prevent damaging competition among the firms, a legal barrier is created. For instance, the governments of many nations only allow a few companies to offer airline or long-distance phone services. Another form of government-imposed legal barriers to firm entrance is patents. A small number of firms hold patents that grant them the sole right to manufacture a good or to employ a specific manufacturing technique.
- ❖ **Strong Brand Loyalty as a Barrier to Entry:** Because of their unique quality and brand, established firms may have loyal consumer base that is extremely difficult for new firms to match.
- ❖ **Few companies may own the scarce resources-** The limited resources may be owned by a small number of firms. The whole supply of raw materials needed to produce the product may be owned or controlled by a small number of firms.
- ❖ **A large amount of fixed cost** – Entering an oligopoly business (such as the automotive, steel, aluminum, and similar industries) typically requires large capital investments and specialized inputs, which serve as significant natural barriers to entry.

- ❖ These are not only the cause of oligopoly, but they also serve as long-term obstacles to new firms entering the market.

Collusive vs non-collusive oligopoly

Collusive oligopoly

Firms may occasionally attempt to eliminate the uncertainty associated with acting independently by entering into price agreements with one another. Collusive agreements are one way to avoid the uncertainty that results from oligopolistic dependency. Price leadership and cartels are the two main types of collusion. Both forms suggest secret agreements, as open collusive action is typically illegal in most countries today. Collusive oligopoly is a market structure where few firms agree to minimize competition by forming a cartel, determining output quotas, and fixing market prices.

Non-collusive oligopoly: When firms in an oligopoly market compete with each other, it is known as a non-collusive oligopoly. Non collusive oligopoly is similar to non-cooperative games. Each firm independently sets its price and output policies, allowing them to compete for market share. Notable models illustrating this competition include Augustin Cournot's Model, Bertrand's Model, and Sweezy's Kinked Demand Curve Model.

Pricing Strategy followed in an oligopoly market

In such market structure, the action of each firm affects the other firm's decision. In other words, pricing strategies focus on strategic interactions. Therefore, price competition can lead to ruinous price war and as such each firm tries to adopt various non-price competition. In general, firms follow

- ❖ **Limit pricing-** If new firms are not restricted, the industry will not stay oligopolistic in the long term. Limit pricing is a strategy where an existing firm sets a low price to discourage new entrants, sacrificing short-term profits for potential long-term gains.
- ❖ **Predatory pricing-** When a large firm enters a market, it may set very low prices to attract customers and discourage new competitors, as seen with Reliance Jio. Predatory pricing is a strategy that involves setting prices very low to eliminate competitors or hinder new entrants in the market. This pricing strategy, known as predatory pricing, involves a dominant firm setting prices very low to eliminate competition. It is illegal under antitrust laws. Predatory pricing is an anti-competitive strategy employed by a firm to monopolize the market by eliminating competition. In the case of predatory pricing firms are selling goods at a price lower than the cost of production. Reliance Jio's predictor price benefits consumers in the short run by requiring lower payments. In the long run, it is detrimental to the consumer.
- ❖ **Gillette pricing-** Gillette pricing involves selling the main product at a low price or a loss while generating profit from associated items. Companies may sell main products at low prices while charging more for complementary goods, encouraging continued

purchases. Once the consumer buys the product then recurring expenditure is there. For instance, printers, water purifiers, AC, DTH, saving razors etc.

- ❖ Dominant firm pricing
- ❖ Pricing determined by cartel etc

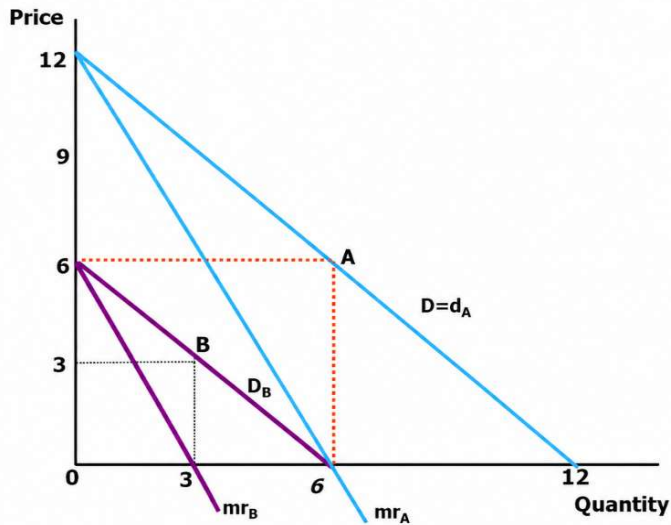
Cournot's Duopoly Model

Cournot's duopoly model outlines the decision-making process of two firms regarding production quantities, with each firm considering the other's output as constant. Each firm selects its output level to optimize profit, guided by their beliefs and strategies. The process continues until both firms achieve a stable production level where neither desires to alter its output. Cournot's duopoly model is the oldest theory in oligopoly market. Cournot's duopoly model explains two firms competing with each other by producing the same product, like water from a spring. The Cournot model illustrates the interdependence among oligopolistic firms.

Assumptions of Cournot Model

- ❖ There are only two firms in the market (duopoly).
- ❖ Both firms produce identical products.
- ❖ The model is relevant for firms producing identical or standardized goods, assuming no cooperation or cartel formation among them.
- ❖ Identical Costs: In Cournot's model, both firms are assumed to have identical cost conditions, facing equal production costs for each unit. This means each firm faces equal production costs for every unit produced. Neither firm has a cost advantage, resulting in equal competition in production decisions.
- ❖ The duopolistic will decide about his output by assuming that the rival will keep his output constant
- ❖ Firms are competing on the quantity of output produced not on price
- ❖ The market price depends on the total output of both firms.

❖ In this model, quantity is the primary decision variable



It is postulated in the Cournot's Model that there are two firms producing identical spring water. Customers used to bring their own bottles; hence, the two firms did not have production costs, i.e., the marginal cost was zero. According to Cournot's Model, both firms aim at making maximum profits by taking into account that the other firms would maintain their production constant at the current levels. Eventually, a series of reactions and actions between both firms end up with each of them producing one third of the total industry production.

Graph D is the demand curve for spring water in the market. At first, the only producer of spring water in the market was firm A. It will, therefore, be facing the whole demand curve in the market. In this case, the demand in the market (D) is equivalent to that of firm A (d_A). The marginal revenue of firm A can be denoted as MRA . As mentioned above, marginal cost is zero meaning that the MC curve is drawn on the horizontal axis. In this case, for simplicity, marginal cost is zero. Firm A earns maximum profits where marginal revenue is equal to marginal cost, which is zero. Firm A earns Rs. 36 as total revenues from selling six units of spring water at Rs. 6 each. Point A is the point at which total revenue is maximized with the price elasticity being one.

At some point in time, firm B joins the industry. The assumption here is that firm A continues to produce 6 units of output. The demand curve for firm B can be calculated using the total demand curve of the market by deducting the output produced by firm A, which equals 6 units ($12 - 6$). The demand curve for firm B is thus 6 units. The marginal revenue curve for firm B is denoted as MRB .

Firm B realizes its maximum total profits when marginal revenue of firm B (MRB) becomes equal to marginal cost (MC), and this leads to a maximum profit condition of $MRB = MC = 0$. Firm B produces three units selling at Rs. 3, which represents the middle point on its

demand curve (dB). In case firm B continues to produce and sell three units, firm A will respond to this situation. The demand for firm A is equal to market demand less three units produced by firm B.

Marginal revenue is half of average revenue. Firm A maximizes its profit by producing 4.5 units. Firm B then responds again and maximizes its profit based on the new demand curve. Until each duopolistic supply has one third, four units each (and two thirds, or eight units collectively), this kind of reaction will persist.

Reaction Curve

Firm A's Reaction Curve

When

$$Q_B = 0 \Rightarrow Q_A = 6$$

$$Q_B = 3 \Rightarrow Q_A = 4.5$$

$$Q_B = 6 \Rightarrow Q_A = 3$$

$$Q_B = 12 \Rightarrow Q_A = 0$$

Firm B's Reaction Curve

When

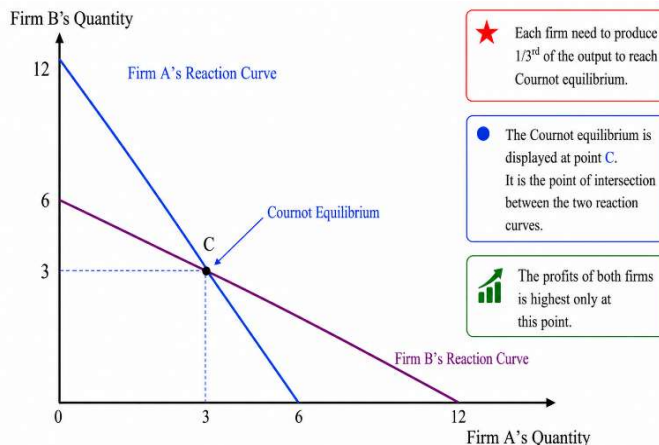
$$Q_A = 0 \Rightarrow Q_B = 6$$

$$Q_A = 3 \Rightarrow Q_B = 4.5$$

$$Q_A = 6 \Rightarrow Q_B = 3$$

$$Q_A = 12 \Rightarrow Q_B = 0$$

Here, Q_A = Output of Firm A and Q_B = Output of Firm B.
The values in red show intercepts where the other firm's output is zero.



Limitations of Cournot Model

- ❖ In real life, some of the model's assumptions are not very realistic. First, it makes the assumption that each of the two firms determines its output on its own. This is unlikely to happen because firms typically take into account one another's behavior.
- ❖ Opponents of the approach wonder how frequently oligopolies compete on quantity rather than price.

- ❖ The Cournot model focuses on how firms decide output, not price. Price is determined indirectly from total output and market demand.
- ❖ The Cournot model makes the assumption that all products are identical. Cournot used competition between two firms selling spring water as the basis for his approach. Even basic goods, like bottled water, are not exactly the same, though. Complete homogeneity is uncommon in practice because different firms offer differences.

Price Leadership Model

Price leadership is an important form of collusive oligopoly. One firm sets the price underprice leadership, and other others follow suit. The firm that sets the price is called the price leader, and the others who follow it are called followers. Price leadership can arise through either secret agreements or open agreements. In most cases, formal (open) agreements to establish price leadership are prohibited. Thus, informal and secret agreements among firms can lead to price leadership. The leader's price is accepted by follower firms. At that pricing, they might not earn maximum profit. Still, they follow it because they prefer not to compete with the leader or with each other. As a result, there is less price rivalry and the market is more stable.

Types of Price Leadership

Price leadership is of various types

- (1) Price leadership by a low-cost firm
- (2) Price leadership of the dominant firm
- (3) Barometric price leader ship

Price leadership by a low-cost firm- The low-cost firm established a lower price than the high-cost to maximize profits. High-cost firms are unable to charge more for their goods. As a result, they are compelled to accept the lower price that the low-cost firm has set. As the leader, the low-cost firm sets the pricing. It must set a price that allows the high-cost firms that follow it to make some profits as well.

Firm produce homogenous products. Firm have different cost structures. The low-cost firm sets the lower price for the product by equalizing its MR and MC. The high-cost firm follows this price. At this profit, the high-cost firm (the follower) does not maximize profits. However, prefers to follow the price leader in order to avoid the price war

Price leadership of the dominant firm: It's possible that the dominant firm produces a sizable share of the industry's total output. The market's overall demand is known by the dominant firm which is leading the market. The small firm marginal costs are known to the dominating firm. To maximize its own profits, the dominant firm decides the price. Follower firms accept the price set by the dominant firm and adjust their output accordingly.

Barometric price leadership- Every firm choose to follow a single leading company. This firm is often well-established, experienced, and reputable. It has superior market insight and predictive capabilities regarding future changes. It sets a price that other firms follow it.

Price Leadership of a dominant firm

D_T (ABCFG) indicates the total market demand curve. In an oligopoly market, the focus is on a homogeneous product. ΣMC_F represents the total marginal cost associated with all the combined follower firms. It is obtained by adding their marginal cost curves horizontally. Follower firms behave similarly to perfect competitors. They produce at a level where the price established by the leader matches the sum of marginal costs (ΣMC_F). $D_T - \Sigma MC_F$ results in D_L . D_L represents the demand curve encountered by the dominant firm in a market.

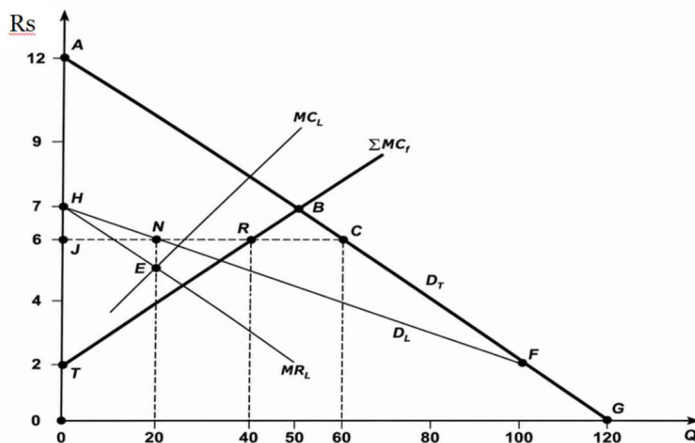
The leader sets the price at Rs7, resulting in follower firms supplying 50 units (HB). This leaves no output for the leader to supply. If the leader sets the price at Rs6, followers will supply 40 units ($\Sigma MCF = JR$).

This leaves 20 units (RC = JN) for the leader to supply. This point is N on the D_L curve. At a price of Rs2, there is no supply from the followers, indicated by a total marginal cost (ΣMCF) of 0 at point T.

At point F, leaders face the entire range of market demand. The leader's demand curve is represented by D_L (HNFG). Its marginal revenue curve is called MR_L . The marginal cost curve of the leader is MC_L .

It selects the output where $MC_L = MR_L$ in order to maximize profit. The price is fixed at Rs6 at this output level, which is equivalent to point N on the demand curve D_L . The followers produce forty units (JR) when the price is Rs6.

In order to satisfy the entire market demand, the leader then provides the final 20 units (RC or JN).



Note: The market share of the Leader firm is higher. The follower firms are not motivated to produce when the price is lower because it is more expensive.

Cartel

An agreement between a number of firms in the same industry to cooperate rather than compete is known as a cartel. A cartel is a group of firms that have joined together to control their production, sales, and prices in an effort to gain monopolistic benefits. They may fix prices, limit output, or divide markets to increase their joint profits. Cartels enable firms to control prices and reduce market uncertainty. If the firms in the oligopoly markets produce only homogeneous product, then cartel formation is possible.

Note: cartel formation is not possible for differentiated products. For instance, the firms which are producing cement and steel can't form a cartel. Cartel formation arises because of following two reasons.

- (i) To reduce the uncertainty arising from the interdependence of oligopoly firms and
- (ii) To maximize joint profits. Cartel aims at joint profit maximization. Since agreements that increase prices and limit customer choices frequently hurt consumers, many nations prohibit or outlaw them.

Cartels come in two varieties. market-sharing cartel and centralized cartel.

The most well-known kind of cartel is a centralized one. For example, Organization for Petroleum Exporting Countries (OPEC). A formal agreement between firms in an oligopoly is called a centralized cartel.

Together, they determine a monopoly price for the product. The total output is distributed among member firms, with established rules for profit sharing. Centralized cartels focus on maximizing joint profits.

The Centralized Cartel

- They establish the overall output and set the selling price for the product. The maximum combined profit is shared among all member firms.
- Allocate the production among the members of the cartel
- They have the authority to have the access to the cost figures of individual companies
- They estimate and calculate the market demand

Market sharing cartel: A market-sharing cartel is an arrangement where members divide the market amongst themselves. Each firm is granted exclusive rights to sell within a designated geographic region. DuPont has been linked with such arrangements in specific contexts. The agreement outlines the selling limits for each member at a predetermined

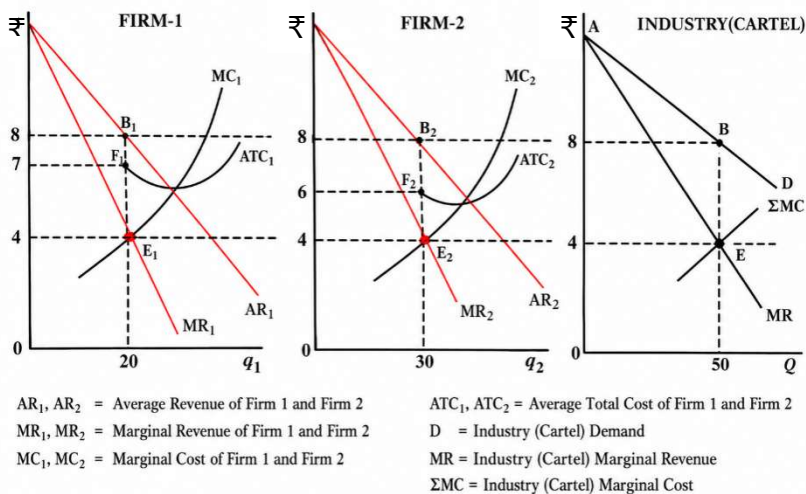
fixed price. Allocation of quota-share is based on the basis of cost of the company and its bargaining skill. During the bargaining process two things are taken into consideration

- (i) Past levels of sales (ii) Productive capacity

Some Cartels: Examples

- OPEC: The root cause of the Oil Crisis in 1973
- The Cartel of GM, DC and Ford
- The Japanese Automobile Cartel

The Centralized Cartel



The overall demand in the market is shown by the curve D . MR represents the marginal revenue curve for a product. The product is identical and is manufactured by both firms in the centralized cartel. ΣMC represents the marginal cost curve of the entire cartel. It is calculated by adding the two firms' marginal cost curves horizontally. Let's say the price is set at Rs. 8 by the cartel authorities. 50 units of its output are sold by the cartel authorities.

Firm B maximizes its total profit when the marginal revenue of firm B (MR_B) equals marginal cost (MC), which results in a profit-maximizing condition where $MR_B = MC = 0$. Firm B's production is 3 units at a price of Rs. 3, which corresponds to the midpoint on its demand curve (dB). If firm B continues producing 3 units, then firm A will react to this eventuality. The demand function for firm A is the market demand minus 3 units produced by firm B.

If MC_1 exceeds MC_2 at any given level of production, then the cost can be minimized by the cartel. This is because the cartel can transfer production from firm 1 to firm 2. As long as MC_1 and MC_2 do not equal each other, this action will continue. Given that the cartel price is Rs. 8, the profit made by firm 1 per unit is Rs. 1. The total profit for firm 1 stands at Rs. 20.

Cartel success story

- ❖ Prices have been successfully raised by international cartels. For instance, the international Bauxite Association tripled the price of bauxite in the middle of the 1970s.
- ❖ OPEC: OPEC mostly controls the market's supply of oil, which drives up prices. The overall supply decreases when member nations decide to cut back on their oil output. This shortfall drives higher prices if demand stays the same.

The oil crisis of 1973 is a prime illustration. OPEC nations reduced their oil exports during this time. As a result, the world's oil supply drastically decreased and prices rose considerably.

- ❖ A cartel known as Mercurio Europeo maintained the price of mercury near monopoly levels from 1928 until the beginning of 1970.

Cartel failure story

However, most cartels have failed to raise prices.

1. An international copper cartel operates till today, but it has never had a significant impact on copper prices. Cartel's attempt to drive up the prices of tin, coffee, and tea has failed.

Problems with the Cartel

- ❖ Inherently unstable given the different cost structures of different companies
- ❖ Low-cost firms have a high tendency to either publicly split from cartels or cut their prices.
- ❖ Low-cost firms may violate the contract by providing secret discounts.
- ❖ The cheated companies together may decide to wage a war against the company that is cheating

Conditions for Cartel Success

Why do some cartels thrive while others don't? Cartels must meet certain requirements in order to succeed. For a cartel to succeed, it needs

- ❖ **Get agreement on prices and production levels:** Firms try to reach an agreement on prices and production levels. However, this is not easy. Each firm may have different costs and expectations about demand. They may also have different goals. As a result, they often prefer different price levels.
- ❖ **Prevent cheating by cartel members:** Additionally, each cartel member will be motivated to "cheat" by slightly lowering their price in order to gain a larger market share than they were given. Offering a price slightly below the fixed price allows cartel members to boost sales and profits, despite close monitoring of each firm's pricing. A single firm can achieve increased sales through additional services, secret rebates, and

various concessions. As profitability rises due to increased prices, sellers may be tempted to provide a secret discount.

❖ **Restrict the entry of new competitors**

If a cartel cannot prevent new firms from entering the market, these firms will ultimately lower prices, reduce economic profits, and disrupt the cartel's operations. The cartel's profit leads to increased market entry and supply, resulting in lower prices. Therefore, obstacles that prevent new firms from entering the market are essential to a cartel's success.

- ❖ **Number of firms in the cartel:** The more firm in an industry, the more challenging it is to negotiate a fair distribution of output among them.

Kinked Demand curve

Price rigidity in an oligopoly market is explained by the kinked demand model. It makes the assumption that if a firm raises its pricing, its rivals won't follow suit and it will lose clients. However, it gains little advantage if it cuts its price since others would follow suit. This maintains prices steady and causes a "kink" in the demand curve.

This concept is still applicable in today's world in sectors like telecom, aviation, and retail, where firms avoid frequent price changes to prevent price wars and maintain stable profits. Skipping the kinked demand curve leads to a gap in understanding why prices in oligopoly markets tend to remain stable. Without it, it is more difficult to comprehend why firms avoid changing prices even when costs change. Additionally, you would no longer be able to understand how firms respond to rivals' actions, particularly the fear of price wars. As a result, your understanding of real-world markets like telecom, airlines, and retail would be incomplete.

The kinked demand curve is studied by MBA students because it describes how firm act in highly competitive, interrelated markets. It explains why firms frequently maintain fixed prices rather than making frequent adjustments, even when cost changes.

This guidance assists future managers in grasping competitor reactions, like responding to price cuts while disregarding price increases, thereby helping to prevent detrimental price wars. The concept is notably applicable in industries like telecom, airlines, and retail. Pricing strategy is crucial in these sectors, requiring firms to anticipate competitors' actions.

These factors assist in maintaining profits and safeguarding market share. Kinked means the prices are sticky or inflexible. Oligopoly industries often exhibit sticky or inflexible prices. Oligopolists tend not to adjust prices in response to changes in economic conditions.

Each oligopolist anticipates a particular response from their competitors. If a firm lowers its price below the prevailing price, competitors are expected to follow suit with similar reductions. Competitors typically do not follow a firm's price increase above the prevailing level.

Each oligopolist firm anticipates that while competitors will not follow a price increase, they will respond to any price decrease. Competitors' varying responses to price increases and cuts result in different elasticities of demand. When prices exceed the prevailing level, demand tends to be more elastic. When prices fall below the prevailing level, demand becomes relatively inelastic.

An oligopolist gains little market share by reducing prices below current levels. Raising the price above the prevailing level leads to a significant fall in sales. Due to this, the firm is hesitant to modify the current pricing structure.

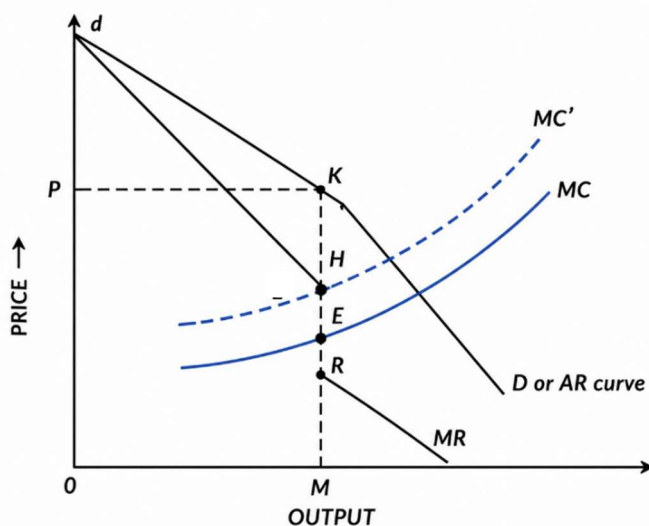
Price decrease: If an oligopolist reduces its price below the prevailing level (OP) to boost sales, competitors may become concerned about customer loss. Buyers may switch to the firm that lowers its price. To retain customers, they must quickly match the price cut. Due to rapid competitor responses to an oligopolist's price reduction, sales gains for the oligopolist will be minimal.

The firm's sales increase not by taking customers from competitors. Rather, the increase occurs as a result of a surge in overall market demand when the product's price decreases. A slight increase in sales following a price reduction by an oligopolist indicates that demand is relatively inelastic to price changes. This indicates that demand is relatively inelastic. The segment of the demand curve (KD) that is situated below the current price OP exhibits relative inelasticity. Oligopolists can only achieve a minimal increase in sales despite lowering prices.

Price increase: If the oligopoly raises its price above the current price, it will lead to a considerable decline in the number of sales of the company. The customer can shift to other companies willing to take them. These companies do not need to increase their prices since there is no reason for them to follow their competitor's actions. When an oligopoly raises its price, it can keep only some customers. Such clients usually show loyalty to the company or have high brand loyalty. Any considerable decrease in sales by increasing prices above the current price level suggests elasticity of demand. Demand elasticity refers to that segment of the demand curve (dK) which is above the current price level OP.



The kinked demand curve hypothesis posits that the demand curve for an oligopolist exhibits a "kink" at the prevailing price level. The demand curve is elastic above the prevailing price level and inelastic below it.



A kinked demand curve, denoted as dD , features a distinct kink at point K , indicating a change in elasticity of demand at that point. The prevailing price is OP , and the firm is producing and selling OM units of output. In the demand curve, the upper segment (dD) is characterized by relative elasticity, while the lower segment (KD) exhibits relative inelasticity.

In an oligopoly, the price OP is likely to remain stable or rigid because no member gains from lowering or raising the price.

The kinked demand oligopoly will make profits maximized at the current prices level. In order to find the profit maximizing level of production combination, the graph for the marginal revenue (MR) derived from the kinked demand curve (dKD) is presented. Clearly, the MR curve is discontinuous, especially when examining its vertical portion. The portion above the discontinuous section of the marginal revenue graph is parallel to the upper part of the demand graph dK . The other part is parallel to the lower portion of the demand graph KD . This discontinuity is measured on the basis of elasticities of dK and KD at point K .

The more significant the difference between the elasticities, the more extended the discontinuous segment. The curve of the marginal revenue is related to the curve of the kinked demand having a discontinuous portion or gap HR . When the curve of the marginal cost of an oligopoly crosses at point E on the discontinuous portion of the MR curve, then the price and quantity will not change. An oligopoly maximizes his profit at the price OP .

In case there are changes in production costs, the price will be unchanged as long as the marginal cost (MC) curve cuts through the gap HR on the marginal revenue (MR) curve. In case the marginal cost (MC) curve moves upwards to MC' as a result of higher costs, both

the price and the level of output will remain unchanged because the new MC' curve cuts through point E' on the MR curve.

Exceptions

- ❖ **Rise in cost:** If the oligopolist industry experiences a cost increase, prices are unlikely to remain rigid. In an oligopoly, when one firm raises its prices due to increased industry costs, it can anticipate that other firms will simultaneously follow suit. Consequently, the segment of the demand curve above the prevailing price will become less elastic.
- ❖ **Increase in demand:** When demand increases, prices are likely to rise. In an oligopoly, an increase in demand leads an oligopolist to anticipate that raising prices will prompt competitors to follow suit. Thus, the upper segment of dK on the demand curve will become less elastic

Criticism of Kinked Demand curve

1. There is not much evidence that oligopolist readily matched price cuts not price increase
2. It didn't explain at what price the kink will occur

Pricing in Practice

- ❖ **Peak load pricing-** A pricing strategy known as "peak load pricing" involves charging higher prices during times of peak demand. The same goods or services are priced lower during off-peak times. For example, the price of electricity is higher in the summer than in the winter. You might have realized that for same unit of electricity consumption your electricity bill is more in summer compared to winter. Similarly, summertime hotel rates are higher than wintertime rates. Similarly, when a new film is released, its price is high at first, but as time goes on, they drop.
- ❖ **Two-part tariff-** A two-part tariff is a pricing strategy where customers pay a set upfront fee to obtain a good or service. They also pay an additional fee for every unit they use. This pricing strategy is frequently used by monopolistic and oligopolistic firm to increase profits. Club memberships and the British Library are a few examples of two part tariff. Amusement parks, which charge an entry ticket plus fees for rides. Club memberships, in which there is a joining fee followed by extra fees for usage or services. Software and gyms are examples of subscription services with a set monthly cost and additional fees for more features or usage.
- ❖ **Prestige pricing:** purposefully charging high prices to draw in clients who value prestige. Assume for the moment that you have bought an apartment in Bandra, where Salman Khan resides. You will gain a lot of respect from others if you tell them that you own a three-bedroom flat in Bandra from the Hirandani group. Rich people typically prefer to spend more for goods even while comparable goods are offered for less money. They evaluate the price in relation to the quality of the product or

service. Similarly, if you own a Rolls-Royce Phantom Century Edition automobile that costs 29 crore ex dealerships, just think of the status that comes with it.

- ❖ **Price lining:** this is the process by which a company sets a target price and then creates a product that will enable the company to maximize its overall profit at that price. For instance, the Tata Nano.
- ❖ **Skimming:** this is the practice of initially setting a high price for a product and then progressively reducing it. For durable products like computers, televisions, and washing machines, this pricing technique is frequently employed. A company can target consumers who are prepared to spend more for the goods by establishing a higher starting price. In order to boost sales and deter new businesses from entering the market, the company will subsequently cut prices.
- ❖ **Predatory pricing:** A company that uses predatory pricing sets a very low price for its goods or services. The aim is to force competitors out of the market or to prevent new firms from entering. A dominant firm uses this pricing strategy to maintain extremely low prices in order to eliminate competitors. Antitrust laws prohibit predatory pricing. A firm that uses predatory pricing is engaging in anti-competitive behavior. Gaining monopoly power in the market is its primary goal. Additionally, it compels additional rivals to leave the market. It is accomplished by setting prices that are lower than the costs incurred by the firm during the production process. Reliance Jio, for instance. In the short term, the consumer benefits from predictor pricing since they have to spend less. However, it is detrimental to the consumer in the long run.
- ❖ **Price matching-** As a seller, I am telling you this is the best price I am offering to you and you will not get this product with this price anywhere. If you are getting the product lesser price anywhere else then I will match that price.
- ❖ **Auction pricing-** Instead of setting a fixed price, sellers invite buyers to bid and use the level of demand to decide the final selling price. In auction pricing, the final price is decided through competitive bidding. Let's say the starting bid for Virat Kohli is ₹3 crore. The bidding increases to ₹20 crore as multiple IPL teams compete for him. The player is acquired by the team with the highest bid.
- ❖ **Bundling** – Companies group two or more products together into a bundle and sell them at a single price. Usually bundling prices are lower than buying them separately. In this pricing method, customers buy a group of products for one combined price. For instance, a fast-food restaurant offers a combo meal that includes a burger, potato fries, and cold drinks for Rs 225.

Review Questions

1. Enumerate the features of a market structure that practices price discrimination and illustrate different degrees of price discrimination with the help of suitable graphs.
2. Do a comparative analysis of Monopoly versus Perfect competition with the help of a suitable diagram
3. Discuss the economic logic behind the Cartels that aim for joint profit maximization with the help of suitable graph
4. “A perfectly competitive firm can earn super normal profit in the long run”. Is the statement true or false?. Explain
5. Define price discrimination and analyze the conditions required for its successful implementation. Examine whether Monopoly firms benefit from price discrimination and explain the reasons with suitable examples and graph.
6. Explain the implication of price rigidity in Oligopoly market with the help of a suitable diagram. Do you think this theory works in real world.
7. Delhi is known for its wide-ranging variety of restaurants located across the city. Connaught place is one of the most popular places to eat out with good number of quality eating –houses. Entry into the restaurant business is relatively easy with hundreds of restaurants opening each year. Since each restaurant offers a somewhat differentiated product, many advertise their existence, location, and menu together with usual claim of superiority over all other restaurants in the same class.

Given the market situation, can you explain how each restaurant determines the level of price and output in the short run?

8. Explain the pricing strategies that the firms generally follow in such market structure with reference to the Indian mobile telecommunications industry. Why do you think that prices followed by firms in such an industry are somewhat rigid? Explain with the help of a diagram.
9. Price discrimination is possible in a Monopoly, not in a Perfectly Competitive Market”. Build your arguments with the help of demand curves in the two market conditions
10. Discuss the Kinked demand curve model and how this model accounts for price stability in an Oligopoly market. Analyze how the kinked demand curve influences firm behavior and pricing policies in an oligopolistic market. Support your answer with an appropriate diagram.
11. Recently, Jio increased tariffs by 25%, and Airtel and other telecom operators increased their tariffs by 20%. Discuss the Price Leadership model under Oligopoly in this scenario.
12. In a perfectly competitive industry, change in industry equilibrium affects the firms’ equilibrium, but a change in the firm equilibrium does not affect the industry equilibrium. Is this statement true or false? Justify your answer in favour of your argument.

13. Analyse whether Perfect Competitive Market is better than Monopoly or Monopoly is better than Perfectly Competitive Market. How prevalent are these conditions in the real World?. Justify your answer in favour of your argument.
14. Does a Monopoly earn any loss during short run?. Justify your answer.
15. What is the implication of free entry and exit for Perfectly Competitive Market? Justify your answer.
16. In a perfectly competitive market, if the demand curve lies between the Average Variable Cost (AVC) and the Average Total Cost (ATC), a firm must decide whether to continue operations, shut down, or achieve a breakeven point. Justify your answer in favour of your argument.
17. Build your argument whether price discrimination is possible in Perfectly Competitive market or Monopoly. Explain the different degrees of price discrimination with the help of suitable examples.
18. Evaluate the effectiveness of cartels by discussing their success and failure stories. Critically examine whether cartel formation is feasible for differentiated products and assess the efficiency of centralized cartel output distribution in two different markets with suitable justification.
19. Analyse the concept of deadweight loss and compare its existence in perfectly competitive and monopoly markets. Explain why monopoly creates a higher deadweight loss than perfect competition with the help of a suitable diagram.
20. Evaluate the impact of deadweight loss on economic welfare under perfectly competitive and monopoly markets. Which market structure leads to greater inefficiency? Justify your answer with suitable arguments and diagram.

Factor Markets and Welfare Economics

“Believe in yourself, and the world will be at your feet”.

— Swamy Vivekananda

A factor market is known as inputs markets. Inputs are nothing but factor of production. Land, labour, capital, Entrepreneur and technology are used as inputs in the production process. In factor market the factor of production such as land, labour, capital, entrepreneur and technology are bought and sold. Household is the owner of factor of production. Household means an individual or group of individuals staying together and sharing a common roof and kitchen. For, instance, your family is treated as household. Household supplies the factor of production to the firm. By using the factor of production, firm produces goods and services and supplies to household. As firm borrows or hires factor of production from the household, firm makes factor payments to the household in the form of rent, wage, interest and profit. Note: Household is the consumer unit and firm is the production unit.

Factor Pricing

Rent

Land is a free gift by the nature to the society. The society has not incurred any cost to acquire the land. Whatever might be the price of land; it is not producible by man. The supply of land is inelastic in nature. One can't take land from one place to another place. We can say that the supply of land is immobile too. The productivity of land can be enhanced through various ways such as

- (i) Proper irrigation facilities
- (ii) Use of quality seeds and fertilizers
- (iii) Improved farming methods
- (iv) Land conservation
- (v) Use of modern technology

(vi) Training and education.

Population growth is leading to a daily increase in the demand for land. Because there is a limited amount of land available for use compared to demand, there is a cost associated with using it. The price of using land, also known as land rent. Rent is a payment made by the tenant to the land owner for the use of land. The supply of land is described as perfectly inelastic, meaning it does not change regardless of price variation. Consequently, the income generated from land primarily relies on the demand for it. The reward for land is referred to as rent.

Determination of Land Rent

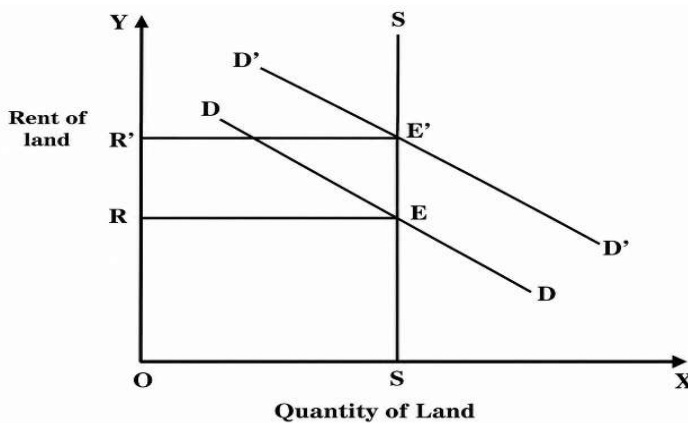
Rent is characterized as the price of factors such as land, which have a perfectly inelastic supply. The amount of land available to the society remains constant regardless of fluctuations in its price or rent. This reflects the inelastic nature of land supply, where price changes do not impact the total quantity supplied. The supply curve shows the supply of land which is perfectly inelastic in nature. It means the supply of land remains fixed at OS, irrespective of rental rates.

Demand for land is determined by its income effect. An increase in the income from one additional unit of land plays an important role in this process. Such an increase in income is known as the marginal revenue product of land. DD stands for the demand curve for land. The demand curve for land slopes downwards from left to right. The law of diminishing marginal return to land can be used in connection with the above diagram. Law of diminishing marginal return to land states that the more land is used, the less additional output will be produced from each additional unit of land.

In a competitive market, the rent of land is determined based on supply and demand dynamics. It is assumed that all land is homogeneous, meaning it is equally fertile. It is assumed that land serves the sole purpose of producing food grains. It has no alternative usages. At the beginning, the demand curve for land is represented by DD, which is based on the demand for food grains. At point E, the DD demand curve intersects the SS supply curve. As a result, the rent of land is determined at OR. The OR is the rent of land per acre and the total earnings of land are equal to ORES

Due to an increase in demand for food grains caused by the rise in the population, the demand for land would eventually become high as well. Consequently, the demand curve for land shifts from DD to D'D'.

New demand curve cuts the fixed supply curve SS at point E'. The equilibrium moves from point E to point E'. Rent of land rises from OR to OR'. The demand for land causes the rise in rent of land.



The reward for labour is referred to as wages. Labour can be skilled and unskilled labour. Training is provided to the skilled labour. No training is given to the unskilled labour. Skilled labour is seen in the manufacturing and service sector whereas, unskilled labour is found in the field of agriculture.

Labour has certain unique features that should be understood. Unlike land and capital, workers often come together to form trade unions. Unions seek to negotiate with employers for higher wages and better working conditions. Labour is inherently linked to the labourer, unlike land and capital, which are separate from their owners. Workers have some freedom to choose their working days. Employees have the option to take leave or remain absent from work within specified limits. Fourth, workers have the option to decide on increasing their labor supply. Fourth, unlike other factors of production, worker might choose to work longer hours during the week.

There are two types of wage rates: nominal and real. The wage in monetary terms is known as the nominal wage. The amount of goods and services a person can buy with their money is termed as real wage. When prices rise in the economy, the purchasing power of the same money wage decreases, meaning it can buy fewer goods and services. The increase in prices while money wages remain constant will result in a decrease in real wages. The real wages represent income levels adjusted for the effects of inflation. The real wage is symbolized as $W_r = \frac{W}{P}$. W_r represent for real wage. W and P represent the money wage rate and the general price level.

Definition of Real Wage

Real wage is defined as the complete wage paid to workers which includes both the money wage and non-money components of wage linked to some jobs. Some organization workers earn some non-money wages like flexible work hours, free dresses, free accommodation, free hospital service, free honeymoon trips. Consequently, real wage is the sum of money wage and non-money wages. Wage rates just like other production inputs are subject to the laws of demand and supply. Labour does not satisfy direct people's needs; unlike products, workers cannot satisfy people's needs directly. Labour is demanded since it can produce commodities that meet people's wants. The level of demand for products determines the demand for labour force. The demand for a particular type of labour is positively related to demand for the products it produces. Demand for products is based on their utilities for consumers. Similarly, the demand for labour depends on its marginal revenue product.

Interest

The reward for capital is referred to as interest. Interest has been defined and interpreted in various ways. Interest refers to the rate of return on capital. Interest is the price paid by borrowers to lenders for using their savings funds. An interest rate is the rate at which interest is paid by a borrower to the lender for the use of borrowed money. It is expressed as a percentage per annum. Physical capital is bought using money. For example, a factory owner may borrow money to buy machines. The profit earned from the machines is associated with the interest on borrowed money. Therefore, the interest rate can be understood as the return on money invested in physical capital. Economist always distinguishes between the nominal and real interest rate.

Nominal interest rate

When people are talking about interest rate they usually talk about nominal interest rate which is not inflation adjusted. It is the rate of interest, that investor paid for the borrowed money. Let's assume an individual deposited Rs. 10, 000 in SBI for one 1 year. At the end of the year, the depositor received Rs. 11,000. The principal amount is Rs. 10,000 and interest rate is Rs. 1,000. In this situation, the nominal rate of interest stands at 10% annually.

Real interest rate

The real interest rate is inflation adjusted. Real interest rates are obtained by subtracting inflation from nominal interest rates. Let's assume that depositor received 10% from the fixed deposit and inflation rate during that period is 6%. Then the real interest rate is 4% ($10\% - 6\%$).

Classical and Keynesian theories of interest rates provide frameworks for understanding how interest rates are determined. Classical theory determined the rate of interest through saving and investment. Keynesian theory determines the rate of interest through demand for and supply of money.

Saving

Saving is that portion of the disposable income that is not spent but saved. As income rises, savings rise and vice versa. Saving and income have a positive relationship. Savings may either be positive or negative. If savings turn out to be negative, then we refer to this situation as dis-saving. Changes in the interest rate affect saving. An increase in the interest rate promotes saving, while decrease discourages saving.

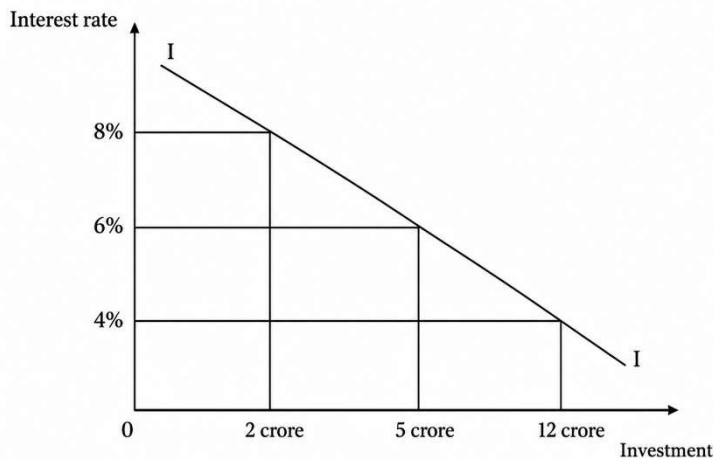
Investment

In finance, investment means expenditure incurred by an individual to purchase some financial assets. These assets can be stocks, commodities, currencies, bonds etc. In finance, investment means investment in financial assets. But in economics point of view, purchasing stock, bonds, currencies are not treated as investment. This type of investment won't enhance income and employment of the country. Economists believe that when one individual purchase the shares, other one would sell them. In this context, there is change in ownership of assets.

It does not create any capital assets. In economics, investment means investment in physical capital rather than investment in financial assets. In economics, investment means investing in plants, machines, equipment, tools, trucks, new factories and so on that creates income and employment of an economy. In economics, investment means investing on physical capital rather than financial assets. Physical capital such as:

- Buying new Machinery
- New plants(factories)
- New Homes
- Inventories (Goods produced by firms)

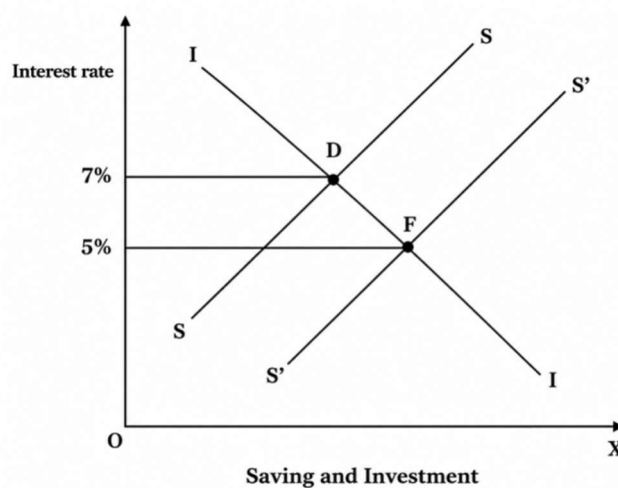
Investment Demand Curve



Investment is presented in X axis and interest rate is shown in Y axis. The investment demand curve is II. When interest rate is 8%, then 2 crore investment is undertaken by an investor. When interest rate falls from 8% to 6%, then 5 crore investment is undertaken by an investor.

Similarly, investment increased to 12 crore when interest rate falls to 4 %. It is noteworthy to mention here that with decrease in interest rate the rate of borrowing will be cheaper for an investor. As a result, investor will borrow more and invest more.

Classical theory of interest rate determination



X axis represent saving and investment and Y axis represent interest rate. The savings curve slopes upward because it is positively related to interest rate. An increase in interest rates tempted households to increase their savings. An investment curve is downward slopping because investment and interest rate are inversely related. Investment increases when cost of borrowing decreases. When the country's income is Rs. 400 crore, the saving curve is SS and investment curve is II. The saving and investment curve intersect with each other at point D and the market clearing interest rate is 7%. With increase in country's income to Rs. 500 crore, the saving curve shifted from SS to S'S'. The new saving curve shifted towards the right. Now new saving curve and investment curve intersect with each other at point F. Equilibrium point shifted from D to F. When income is Rs. 500 crore, the market clearing interest rate is 5%. With increase in savings rate of interest falls.

Keynesian Liquidity Preference Theory of Interest

Anything that is widely accepted as payment for goods and services is referred to by economists as money.

Keynesian theory of the demand for money

The amount of money that people wish to keep in cash or checking accounts is known as demand for money. Checking accounts means savings bank accounts. The demand for highly liquid assets like cash and checking account deposits by individuals, companies, and the government is referred to as money demand.

Demand for money is like demand for any other assets and is influenced by expected return, risk, and liquidity. Risk means uncertainty. Liquidity means the assets that can be easily converted into cash without decreasing its face value. Money is the most liquid asset compared to other assets such as stocks, bonds etc. For example, you have 1 crore worth of cash, and your friend has 1 crore worth of house. Whose asset is more liquid?. Here cash is 100% liquidity compared to the house. To sell this house your friend may take 1-6 months or more than that. It depends on the locality and the price you are quoting etc. The liquidity of an asset indicates the ease with which it can be converted into the medium of exchange without a significant loss of value.

Note: whenever we are talking about liquidity, means we are talking in a relative sense. Keynes identifies three primary reasons for the demand for money namely transactions, precautionary and speculative demand for money.

Income has an impact on transactions and the precautionary demand for money. Interest rates have an impact on the speculative demand for money.

$$M_d = M_t + M_p + M_{sp}$$

M_d represents total demand for money in an economy

M_t represents transaction demand for money

M_p represents precautionary demand for money

M_{sp} represents speculative demand for money

Hence $M_d = f(Y, r)$

Y is income and r is rate of interest. The total demand for money is influenced more by interest rate

Transaction Demand for Money

Why People Need Money: One reason people hold money is that it can act as a means of payment in making transactions. You keep some money every day for making payments for purchasing coupons for food, travelling costs, and for watching movies. Point to note: When your income increases, there will be an increase in the demand for money in transactions, and vice versa.

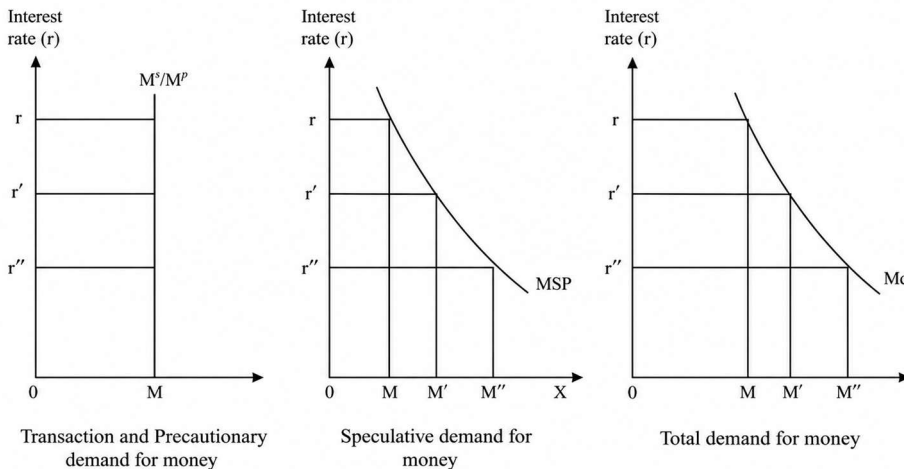
Let's assume that interest rate is 8% and it went to 10%. Can you tell your landlord that I will pay the house rent after two months as the interest rate has increased, I have deposited all of my money in the bank. You can tell your house owner but you know the consequence. Hence, the transaction demand for money cannot be postponed. It doesn't depend on the interest rate.

Precautionary demand for Money

Precautionary demand for money is a function of income, just like transaction demand. Higher the money you have, the more money you can save for the future. People keep extra cash on hand to cover unforeseen expenses. For example, a person may need money for surgery if the health condition is critical. The future is unknown and uncertain. Hence, people used to save money for the future. Future, there might be an increase in children's education expenses, health care expenses etc., Hence people used to save money. People tend to demand money for the so-called "Precautionary motive" because they wish to make some provision against unforeseen contingencies.

Speculative demand for Money

The motive behind keeping cash or money instead of investing them into other financial assets is called the speculative motive for money. This is because an individual expects that the future changes in interest rates might lead them into thinking that it would be beneficial to hold onto their money rather than investing it. As the interest rate increases, the opportunity cost of holding money increases and, thus, the amount of money demanded falls. On the contrary, if there is a fall in the interest rate, then the quantity of money demanded increases, while the opportunity cost of holding money falls. At low interest rates, the individual can take more loans from the commercial bank because it is relatively cheaper and seek out investments.



M_t and M_p is represented by transaction and precautionary demand for money respectively. MSP stands for speculative demand for money. M_d represents the total demand for money. On the X-axis we have the demand for money while on the Y-axis we have interest rate. Graphing the transaction and precautionary demand for money along with interest rate, assuming that they are interest-inelastic in nature. The transaction and precautionary demand for money are plotted vertically.

The transaction and precautionary demand for money stays the same regardless of the interest rate; that is, OM. It should be noted that the transaction and precautionary demand for money depends on the level of income and not the interest rate; hence, any changes in interest rates do not impact them.

The speculative demand for money is shown against the interest rate. The speculative demand for money is indirectly proportional to interest rate. The total demand for money is the summation of all transaction, precautionary, and speculative demands. M_d is the total demand for money in relation to interest rate. The total demand for money decreases with increasing interest rates.

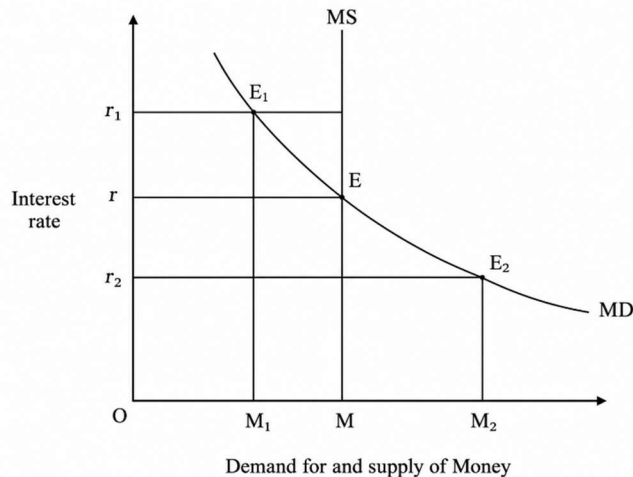
Supply of Money

The amount of coin and currency in circulation as well as the total value of all bank checking accounts make up a nation's money supply. Checking accounts in banks includes saving bank accounts and current account. The total amount of money available in an economy over a given time period, such as a year, is referred to as the supply of money. Money supply includes all the money in circulation and money held in banks and other financial institutions. During one year, supply of money is constant and it is not affected by the interest rate. Hence money supply curve is vertical. Central Bank (RBI in India) is the main source of the Money supply. But it will narrow view if we will tell RBI is the only source of money supply. Apart from RBI, depository institutions (Commercial banks) which accepts deposits and give loans also source of money supply. Public also supplies money. For instance, if you will store all the money at home then money supply will be affected. What we are doing-we are keeping extra money in commercial banks and banks gives loans to other person and credit creation occurs. In that way people also the source of money supply. NBFIS such as IDBI and NABARD are the source of money supply. A rise in the money supply necessitates that central banks expand the quantity of money in circulation.

The Keynesian theory of Determination of interest rate

X-axis is that for money supply and demand whereas Y-axis is used for the interest rate. The money demand MD curve has an inverse relationship with the interest rate. It can be noted that money demand curve is the total sum of transactions, precaution and speculation demands on money influenced largely by the interest rate. Supply of money for the short-run in the Keynesian economics approach has been assumed to be constant. In one year, the amount of money supplied remains constant. Money supply in the case of India is controlled

by RBI, i.e., the central bank of India. The central bank keeps the money supply fixed despite fluctuations in the interest rate in any period of time, i.e., it is vertically oriented with respect to the interest rate. The money supply is the responsibility of the central bank in any nation. Demand and money supply meet at a particular point E to determine the interest rate r .



When money supply and demand are equal at point E ($MD = MMS$). Interest will be stabilized when the money supply and demand are equal. For any other points where the money supply and demand are unequal ($MD \neq MMS$), interest will not be stabilized. This means that when money supply is more than demand, the interest will either rise or fall. Money supply and demand determine interest rates. Excess in money supply reduces interest rates while excess in demand increases interest rates. When there is high interest rate r_1 , money supply is MMS and demand is OM_1 . At point r_1 , the money supply is greater than the demand.

This leads to a drop in the interest rate from r_1 to the equilibrium level of r . On the other hand, when the interest rate drops to r_2 , then the demand for money becomes OM_2 and the supply of money becomes MMS . This implies that $OM_2 > MMS$. This results in an increase in the interest rate level up to the equilibrium level of E where the interest rate level is r . Note that a fall in the interest rates from r to r_2 leads to an increase in the demand for speculative money.

One may observe that an increase in the money supply caused by the monetary authorities without an increase in the demand for money MD will cause interest rates to decrease. On the other hand, if the money demand MD increases without any change in the supply, the interest rates will increase.

Why does the demand curve for money exhibit a downward-sloping nature? This is due to the fact that there exists interest sensitivity in the money demand function. Thus, an increase in the interest rate causes people to economize on their money balances used in

transactions by transferring their money balances to investments offering higher interest rates.

Liquidity Trap

A liquidity trap arises when low interest rates do not stimulate spending or investment, resulting in economic stagnation. Deflation is a key factor that contributes to this situation. A liquidity trap arises when consumers and investors choose to hoard cash over spending or investing, despite low interest rates. This happens during a recession or when people expect economic conditions to worsen, causing them to prefer holding money. This phenomenon can hinder monetary policy effectiveness, resulting in economic stagnation.

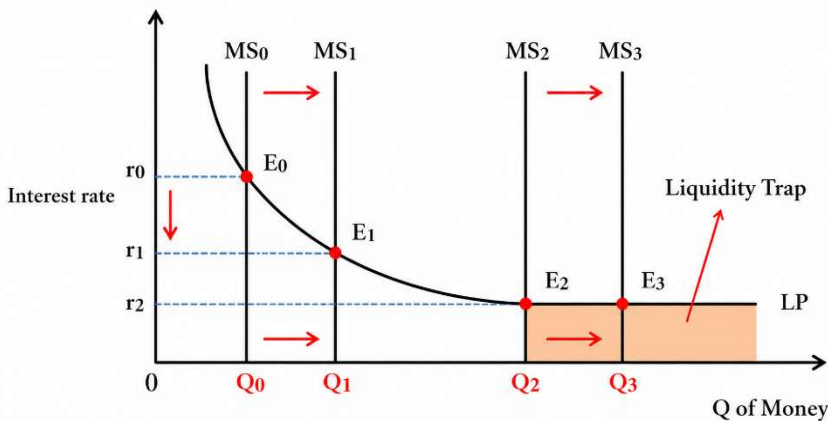
A liquidity trap occurs when interest rates decline to near zero, resulting monetary policy loses its effectiveness. People prefer to hold cash instead of spending or investing it. Increasing the money supply does not significantly boost economic activity. In this scenario, the opportunity cost of holding money is indicated as zero. When the nominal interest rate approaches zero, the cost of holding money becomes negligible. Because of this, banks, companies, and people frequently choose to keep cash rather than spend or invest it.

Even in cases where individuals have more money than they need to conduct normal business, this is still the case. There have been two notable instances of a liquidity trap throughout history, namely the Great Depression in America during the 1930s and the prolonged period of economic downturn in Japan during the last decade of the 20th century. The economic stagnation in Japan since the 1990s can be considered an example of a liquidity trap.

Characteristics of a liquidity trap

- Very low interest rates or near 0%.
- An economic slowdown or recession in the economy.
- Increased tendency of people to save rather than spend.
- Very low inflation or a fall in the general price level (deflation).
- Limited effectiveness of expansionary monetary policy in stimulating economic activity.
- **Preference for liquidity:** Individuals and businesses prefer to hold cash rather than invest in bonds or other securities that offer very low returns.

The X-axis denotes the amount of money, and the Y-axis denotes the interest rate. The overall demand for money has a negative slope. The initial amount of money supplied is denoted by MS_0 . The intersection between the demand for money and supply of money occurs at point E_0 . At this intersection point, the interest rate becomes r_0 . Now let us assume that the demand for money is unchanged but the Reserve Bank of India increases the supply of money from MS_0 to MS_1 . The result would be that there is a fall in interest rates from r_0 to r_1 . Further increase in the money supply from MS_1 to MS_2 would mean that there would be a fall in interest rates from r_1 to r_2 when the demand for money is unchanged. An increase in the supply of money to MS_3 results in constant interest rates at r_2 .



Profit

The reward for entrepreneur is referred to as profit. Profits are closely connected with entrepreneurs and their activities. Economists have expressed varying opinions regarding profits throughout history. They have described the meaning, source, and significance of profits in various contexts. Till today, economists have not reached a consensus on the true nature and origin of profits.

Entrepreneurs are viewed as organizers and coordinators of the factors of production. Entrepreneurs earn profit by performing entrepreneurial functions, as noted by some economists. Others believe profit is a reward for taking risks and facing uncertainty in business. The entrepreneur oversees the company, making key decisions about pricing and production. An entrepreneur makes money by taking risk and dealing with uncertainty in business.

Business decisions might not always be the right ones, and future market conditions are uncertain. Price and output decisions made by the entrepreneur can occasionally result in profits or losses. Schumpeter identifies the entrepreneur as an innovator, with profits serving as a reward for the introduction of innovation. Keynes argues that profits arise from favorable changes in the general price level.

Schumpeter's Innovations Theory of Profits

According to Joseph Schumpeter, the main function of an entrepreneur is to bring new ideas and innovations into the economy. He believed that profit is the reward for performing this function. Innovation can be divided into two categories namely first and second type of innovation. First types of innovations focus on reducing production costs. The first type of innovation involves employing new machinery and enhanced production techniques. It involves finding cheaper production techniques, using new sources of raw materials, and adopting better ways to manage the firm. The second type of innovations increases product demand. In this category, the focus is on the introduction of new products, product variations or designs, improved advertising methods, and the discovery of new markets. Successful innovation leads to profit by either reducing production costs or increasing product demand. Profits arise from successful innovations, either through reduced costs below the market price or increased sales at better prices.

When an entrepreneur introduces a new innovation, they initially hold a monopoly position, as the innovation is exclusive to them. It allows them to achieve substantial profits. Other firms may adopt the same innovation to capture market share. As competition rises, profits generally decline considerably. If a law allows, and the entrepreneur successfully patents an innovation, they will continue to earn profits. Innovations drive profits, which are crucial for entrepreneurs to foster economic development in a country.

The marginal productivity theory of distribution

The key point of marginal productivity theory of distribution is that a factor of production is priced by the amount of marginal productivity. To be more precise, the compensation for factors depends upon their marginal productivity. Thus, factors of production have to be paid in accordance with their contribution to the output. Initially, the marginal productivity theory was created in order to explain the process of payment of wages for the labour force. According to the theory, workers had to be paid depending upon their contribution to production. However, further on the theory was extended to cover the payment for new factors of production, such as land and capital.

The idea of marginal productivity appeared thanks to the economists Ricardo and West. According to the theory, the payment for land could be explained with the help of the marginal productivity theory. The rent theory of Ricardo was based upon the marginal productivity theory.

In the beginning, the concept of marginal productivity was not commonly acknowledged. During the late 19th century, the marginal productivity theory of distribution gained attention. This idea was reintroduced by economists including Leon Walras, Philip Wicksteed, William Stanley Jevons, and John Bates Clark. Alfred Marshall and John Hicks contributed to the popularization of the theory.

Clark's Version of Marginal productivity theory

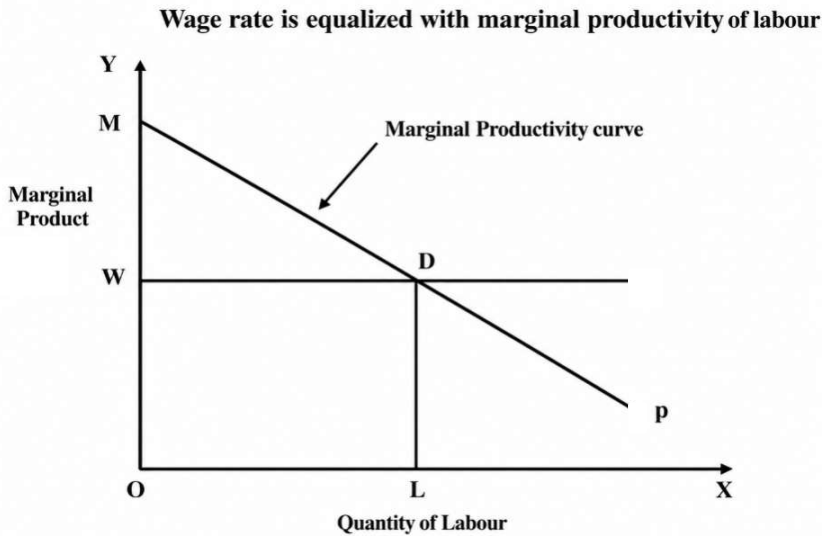
Marginal productivity theory based on following assumptions.

- (i) The author assumes that the population remains unchanged over time.
- (ii) The amount of capital remains unchanged.
- (iii) Production techniques do not change over time.
- (iv) The presence of perfect competition in the factor market.
- (v) Both labour and capital were assumed to move freely from one place or use to another.
- (vi) Labour is treated as a homogeneous factor. It indicates that all workers have similar skills and productivity.

Rational entrepreneurs aim to maximize profits by effectively utilizing their fixed capital. Entrepreneurs hire workers as long as their contributions enhance profits relative to the available capital. When the number of workers increases while capital remains unchanged, the marginal productivity of labor for an individual firm or industry will decline.

Rational entrepreneurs will continue to hire more labor units as long as the marginal product added by an additional unit exceeds the wage rate paid. The employer is in equilibrium when a worker's wage matches the worker's marginal productivity.

X axis presents the units of labour and the Y axis shows the marginal product of labour. The MP curve shows that the marginal product of labor decreases as more workers are employed. Let's assume that OW is the current wage rate. In this wage rate, OL units of labour will be hired by the entrepreneurs or employer. This occurs because, at the OL level of employment, the additional output produced by labour matches the wage rate OW. The employer will not hire more than OL units of labour. This happens because, after OL, the extra output generated by labour is less than the wage rate OW. Hiring more labour beyond OL, would result in financial losses for the employer. Employers earn maximum profit when the additional output from labor is equal to the wage rate (OW).

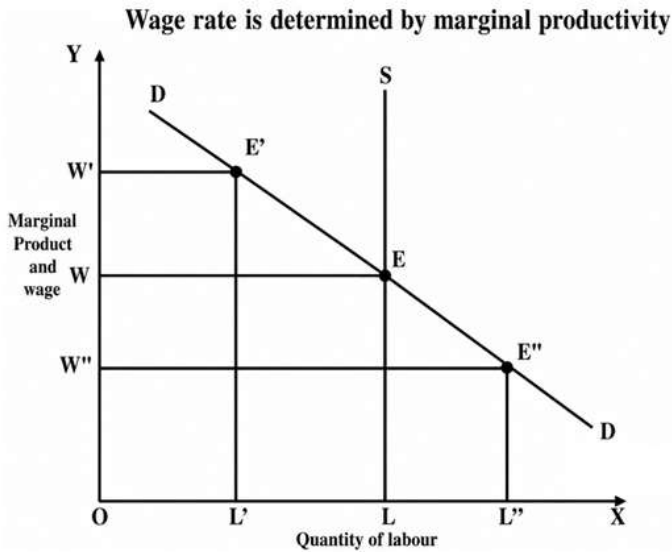


In the case of perfect competition in the labour market, there is no way for firms or industries to influence the wage rate. The wage rate is decided by the market. In perfect competition, firms are the price taker. This implies that every firm is obliged to go by the prevailing wage rate. Thus, what a firm can do in such cases is decide how much labour to employ at the existing wage rate, without any capacity to alter the wage determined by the market. Hence, the marginal productivity theory is regarded as the theory of employment at the micro level.

Workers who compete with one another for work will bid lower wages if some of them find themselves unemployed. The DD curve displays the overall labour demand from all employers. It is made by combining each employer's marginal product curve horizontally. DD represents the total demand for labour in an economy. LS represent the total supply of labourer in an economy. The maximum amount of labour is employed in an economy is OL. It is also called as full employment. It does not mean employment is 100 %. Full employment means the unemployment rate is low which is close to 4 to 5%. After OL, whatever might be the demand for labour, the supply of labour is fixed at OL. As more labour units are employed in the economy, the marginal product of labour falls, assuming the quantities of other elements remain constant.

The employer will hire OL' units of labour, if the wage increases to OW'. As a result, LL' units of labour will remain unemployed. This happens because at a higher wage rate, fewer workers are demanded.

Note: Wage rate and demand for labour is inversely related. On the other hand, wage rate and supply of labour is positively related. For instance, if wage rate increases from Rs 1,000 to Rs 1500 many workers will come forward to work.



With increase in wage rate the demand for labour reduces. Unemployed workers will take lower wages in an effort to find employment. This will gradually lower the wage rate. The wage rate will finally reach $OW (=LE)$, where all workers get employment.

At a wage rate of OW'' , employers desire to hire OL'' units of labor. At this level, firms can earn higher profits.

However, only OL units of labour are available in the economy. When wage rate is OW' , the employer demand for labor exceeds the available supply of labour. In this scenario, each employers will compete to hire additional workers. This competition will increase the wage rate till the time wage rate reached to OW .

Note: both demand for and supply of labour intersect with each other at point E where there is no unemployment in the economy. Point E is more stable and any other points excluding point E, is not stable.

Critical Evaluation of Marginal Productivity Theory

The marginal productivity theory significantly influences the traditional theory of distribution. Even today, it remains important in the modern study of factor pricing. However, this theory is not free from criticism.

- ❖ Marginal productivity theory is criticized for relying on unrealistic assumptions. It assumes perfect competition in the market. Factors of production are expected to move freely from one use to another. Buyers and sellers are considered to have equal bargaining power. The theory also assumes that everyone has complete knowledge of market conditions. Competition in the factor market is characterized by significant imperfections rather than perfection.

- ❖ Another important criticism of the marginal productivity theory applies to both the Clarkian and Marshall–Hicks versions. The theory assumes perfect competition. It assumes that both product and factor markets are perfectly competitive. However, such conditions are not always found in the real world.
- ❖ Marginal productivity theory's suggests that trade unions or collective bargaining cannot increase labor wages without resulting in unemployment. Higher wage rates reduce the demand for labor due to the downward-sloping marginal productivity curve. Higher wages achieved by a trade union may result in employers hiring fewer workers. As a result, more workers may become unemployed. Increasing wage rates by unions does not always lead to unemployment; in monopsony labor markets, higher wages may result in increased employment instead.
- ❖ The marginal productivity theory is criticized for ignoring the positive relationship between productivity and factor rewards. Specifically, it does not give enough emphasis on the relationship between labour productivity and wages. Increased wages could boost workers' productivity. It has been noted that increases in wage have a positive impact on workers' productivity and efficiency. Workers who earn more money can afford to live better and be healthier, which will increase their efficiency and output.

Wages and worker productivity are frequently positively correlated, particularly in developing nations like India. Wages in many industries may continue to be extremely low and may drop below what is required for a minimal standard of living. In these circumstances, higher wage rate can boost workers' productivity and efficiency. However, this positive impact of high wages on worker productivity is entirely ignored by marginal productivity theory.

- ❖ Another significant criticism of the marginal productivity theory of distribution is its inability to adequately explain entrepreneurs' earnings. The theory fails to explain how profits are determined.
- ❖ The marginal productivity theory of distribution inadequately explains the determination of all factor prices.

Externality

An economic event that affects a third party is termed an externality in economics. The individual will not be actively involved in the economic process. It does not matter if one chooses to participate; they may experience both a benefit and a cost. An externality will either be a cost or a benefit experienced by a third party as a consequence of the economic activity.

Consumers may generate externality through their consumption activities, such as causing pollution or creating benefits for others. Just like firms, producers can generate externalities through production processes. Governments generate externalities via their policies and

actions. Building roads or parks can provide benefits to individuals who do not pay for them directly.

Projects like industrial expansion or large dams can impose costs, such as pollution or displacement, on specific groups. Government decisions can lead to both positive and negative externalities. These extra costs or benefits are not included in the price of a good or service. As a result, markets are inefficient. Externalities can therefore result in market failure. Externalities, which are prevalent in daily life, can be classified into two types: positive and negative externalities.

Understanding externalities enables MBA students to enhance their business decision-making capabilities. It shows how a firm's actions, like pollution or social benefits, can have an impact on people who are not directly involved. This knowledge is useful for designing responsible strategies, improving brand image, and avoiding legal or regulatory risks. Additionally, it helps managers in coordinating corporate objectives with long-term value creation and sustainability.

Positive Externality

Positive externality: When one party's actions help another. A positive externality is one that is advantageous to society.

For example:

- ❖ **Environmental clean-up-** Positive externalities are produced by environmental cleanup since it benefits a large number of people in addition to those who perform it. Public health is enhanced and medical expenses are decreased by cleaner air and water. Additionally, it promotes tourism, improves local quality of life, and safeguards animals. Even individuals who did not participate in the cleanup can take use of these advantages.
- ❖ **Covid 19 vaccine:** The COVID-19 vaccine creates positive externalities because it protects not only the person who takes it but also others. The virus spreads less when more people receive vaccinations.
- ❖ Government created gardens in multiple places in your State
- ❖ Government set up one hospital which is close to your house
- ❖ Following traffic rules
- ❖ No corruption
- ❖ Planting trees in multiples places in India



Critical thinking: How planting of a plant creates positive externality to the society

Negative Externality

When the action of one economic agent imposes cost rather than benefit to another producer or consumers. In other words, that action of a producer or consumer which affects the society negatively. Negative externality always create cost to the society. In the real-world negative externalities are more common as compared to positive externality. Recently ongoing US-Israel-Iran war has a negative impact to the society. Similarly, the company discharges waste into the river, posing a health risk to the public and create negative externality.

- ❖ Environmental pollution
- ❖ Industrial pollution from factories
- ❖ Chemical use in agriculture harming soil and water
- ❖ Consuming alcohol can lead to abusive behaviors towards family members and others.



Critical thinking question: Does the River Ganga generate both positive and negative externalities, and how can its benefits be increased while minimizing its harms?

Does having a girls' hostel in front of a boys' hostel create positive, negative, or both types of externalities?

Negative consumption externalities



Particulates from vehicle pollution



Household waste



Noise pollution from neighbours



Traffic congestion



Impact of addition on families



Litter from tourists

Private vs. Public Goods

Private Goods: Private goods the focus is on individual. Private goods example: your own laptop, your mobile, your home etc. Economists define private goods as being rival and excludable.

Private Goods



Private goods have two characteristics

- ❖ **Excludability** (if you don't have willingness and ability to pay you will be excluded from consumption)
- ❖ **Rivalry** (one movie ticket but two people are standing in Queue. Both are rival to each other)

Public Goods: Public goods are non-excludable and non-rival resources. They can be used by anyone without diminishing their availability to others. Public goods examples government hospitals, Roads, Garden, Défense, Ocean water, etc. Public goods the focus is on a group.



Public goods have two characteristics

Non rival: It implies that the use of a good by one individual does not restrict the use of the same good by another.

Non-excludable: refers to goods that cannot be withheld from others. Examples include national defense, highways, police services, and fire protection.

Asymmetric Information

The phrase "asymmetric information" describes a situation in which one side to a transaction has more information than the other. Due to asymmetric information, in which the vendor knows more about the good being sold than the buyer, sellers may exploit purchasers in some transactions.

Let's begin with an illustration. You own a vehicle. Nobody knows your car's condition better than you do. Buyers and sellers don't have same information. The seller possesses more information than the buyer. We refer to this as asymmetric information. Consider insurance providers. We don't have all the information, but they do. Likewise think about any emergency treatment.

Doctors have more information's than the patients. If you look at the stock price, everyone has access to the same data—that is, the starting and closing prices. It is called symmetric information which is opposite to asymmetric information. Note: if everyone has access to the same information, there is no chance for arbitrage and the market is efficient.

Asymmetric information complicates managerial roles, as various individuals within an organization possess differing levels of information. Employees might exaggerate their performance, while managers may not fully understand ground-level challenges. This imbalance can lead to poor decisions, inefficiency, and mistrust within the team. This leads to issues such as adverse selection, leading to diminished productivity and effectiveness.



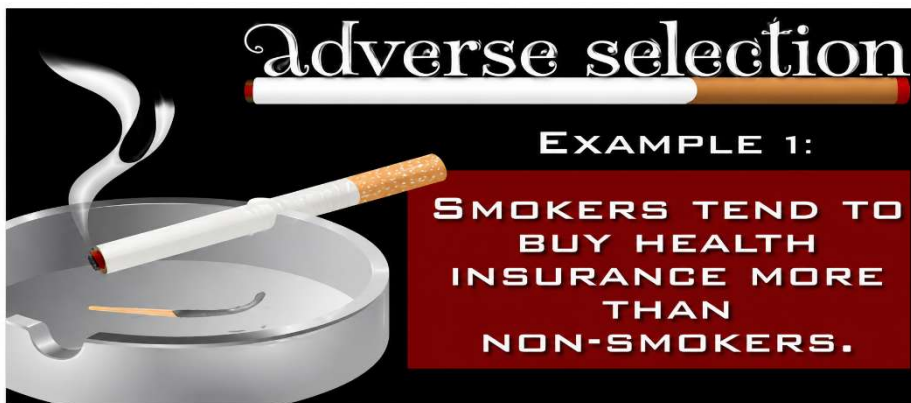
Adverse Selection: Adverse selection means wrong selection. It means choosing a wrong product or people. Because of Asymmetric information, adverse selection occurs and you are buying a wrong product. In the context of car sales, asymmetric information leads to the purchase of low-quality used cars by consumers.

According to the lemon theory, sellers in the used car market typically possess more information about the true condition and value of a vehicle than buyers do. Because of this information gap, buyers hesitate to pay a high price. They often provide a standard price despite the potential high quality of the car.

Think about insurance company they may tell you all diseases are covered. But if someone will go for clam, many are not. This is adverse selection. Suppose you want to buy one laptop. Instead of doing review you only asked your friend and brought it. You faced a lot of problem. In this context adverse selection occurs.

Adverse selection occurs in selection process. For instance, X prepared some questions and interview panellist asked these questions and finally they selected X. Later, it was determined that X is an underperformer, indicating adverse selection.

Adverse selection complicates managerial decision-making due to the lack of complete or accurate information regarding individuals or products. This can result in the selection of low-quality employees, ultimately reducing efficiency and increasing costs.



Moral Hazard

“Moral hazard” describes a situation where a person becomes more willing to take risks. This happens because there is an expectation that an insurer will cover the costs in the event of an issue. Moral hazard is the tendency for individuals to take on higher risks when they believe they are protected from the consequences. This protection may be provided through insurance or a guarantee. Because of this, someone else frequently bears the expense of such dangerous activities. In all kinds of insurance markets, moral hazard is prevalent. For instance, if X does not insure his or her home, X will be responsible for any losses resulting from theft or fire. X is probably going to be more cautious as a result. To secure his or her property, X might hire guards, set security alarms, and take other measures.

Insuring your house for its full value protects you from losses in case of damage or destruction. Hence, there is reduced motivation to safeguard against potential incidents. Hence, there is reduced motivation to safeguard against potential incidents.

- ❖ Person will become careless while driving after taking personal and vehicle insurance.
- ❖ Person will become careless after taking product insurance policy
- ❖ People may be careless with property that is insured
- ❖ Having insurance lowers the cost of an accident, thereby reducing the likelihood of risky behavior.
- ❖ Moral hazard in health insurance primarily leads to an increased utilization of medical services. People often seek more healthcare services due to the financial security provided by insurance.



Market failure

When the market does not operate efficiently, there is a market failure. Market failures do not mean the market will stop working. I mean to say the market will function but not efficiently. Market failures means market provides wrong signal. Because of that, there is loss of output and producer and consumer surplus comes down and there is loss to the society. When there is market failure, there is no proper balance of supply and demand. In other words, supply is greater or smaller than demand. Market failure happens when the marginal benefit does not match the marginal cost.

Market failure talks about the social aspects of the economy such as consumption, Production etc. Market failure happens when resources are inefficiently allocated. Despite individuals pursuing their own rational self-interest, the overall outcome is not optimal for society. When a market fails, society incurs high costs or receives fewer benefits than it ideally should. Market failure occurs when the market is dis-equilibrium. This situation is often caused by market distortions. Market failure occurs when the supply of goods or services does not match the demand for those goods or services. Factors that disrupt the

free market include monopoly power, price controls, minimum wage laws, and various government interventions. Market failures occur due to various reasons.

Market failures occur when the market does not yield the optimal outcomes for society. Resources are not used efficiently, so the goods and services produced do not maximize



overall benefit. As a result, society either gets too little of something valuable. Market failure leads to less welfare creation than what is possible with the available resources. Market failure can occur due to inadequate information, market control, public goods issues, and externalities.

Causes of Market Failure

Market failure can be caused by a lack of information (Asymmetric information), monopoly, public goods, and negative externalities.

❖ Inadequate information (Asymmetric information)

Information gap is one of the causes of market failure. Perfect information is assumed in a market with perfect competition. But in the real-world market often involve cheating and inaccurate information. When information is limited, buyers and sellers don't have same level of knowledge. This imbalance can cause the market to work poorly or inefficiently. A lack of perfect information can result in market failure. When buyers and sellers lack proper information, they may make poor decisions. Consequently, a product can be purchased or sold at a price that does not match its actual cost or value.

❖ **Presence of public goods**-When some people use public goods without paying for them, this may result market failure. These individuals even if they do not contribute, still enjoy the benefits. This imbalance can lead to market failure Examples of such products include national defense, police, and national roadways. These services are accessible to all citizens, regardless of their tax payments.

❖ **Presence of negative externality:** social cost is more than social benefits in the presence of negative externality. Social costs of a factory's pollution exceed social benefits. This leads to inefficient resource utilization and results in market failure.

- ❖ **Market failure also occurs in the presence of positive externality.** In the presence of a positive externality marginal benefit is more than marginal cost. Education benefits both individuals and society by enhancing productivity and promoting better citizenship. Education may be underprovided as individuals primarily focus on their private benefits. This under consumption causes inefficient resource allocation and market failure.
- ❖ **Presence of Monopoly power:** In a monopoly, a single firm dominates the entire market supply of a product. The firm has the ability to restrict output to maintain high prices. This creates artificial scarcity in the market. Resources are not allocated efficiently as a result. Deadweight loss occurs when supply and demand are not in equilibrium, highlighting inefficiencies in market transactions. Some economists contend that market failure is solely the result of government monopolies.
- ❖ **Lack of competition-** Lack of competition can cause market failure because firms gain too much control over the market. They may increase profits by raising prices and reducing output. Without competition, there is less pressure to improve quality or reduce costs. Consumers face reduced choices and potentially higher prices. This leads to inefficient use of resources and creates market failure.



Government policy to correct market failures

To prevent market failure, most countries have anti-trust or anti-monopoly legislation and government practices to combat abuse of power.

- ❖ In the presence of positive externality govt intervene in the market by providing subsidy. On the contrary in the presence of negative externality government in the market by imposing tax.
- ❖ Promote a transparent system
- ❖ Maintain competition through government regulation
- ❖ Excludes self-interest think about the country's interest

- ❖ Increase economic efficiency
- ❖ Economists who don't like government interference point out that information problems are not a problem of the market, it is the problem of government regulation. Market failures should not automatically call for government intervention. Why you know. Because the government may fail too.
- ❖ **Government failure:** Government failure occurs when governmental interventions intended to resolve market issues do not achieve their intended outcomes. These actions exacerbate the situation rather than improve it. Resources are used inefficiently due to intervention.

Reasons for government failure

- ❖ Government don't have information to deal with the problem (for ex. covid 19 scenario)
- ❖ Intervention in the market is almost more complicated then it initially looks
- ❖ Political interest

Merit Goods

Good may be categorized into merits good and demerits goods. Merits goods refer to goods or services which are provided by the government to promote welfare in the community. Remember that the merits good will create a positive externality effect within the society. For this reason, all governments tend to promote merits goods. Examples of merits goods include education services, health care services, COVID-19 vaccination program, and public libraries.

Demerit goods: It is appositive of the merit goods. Demerit goods are those goods or services that are not considered beneficial for the individuals and the society; rather it is considered as harmful to the society. For instance, Gambling, Tobacco, Alcohol, Drugs consumptions. Governments usually not encourages the consumption of demit goods. Government discourages it by imposing taxes, reducing subsidy or by imposing strict regulations.

Role of government in promoting welfare

In order to enhance the overall well-being of the people and the society each and every government plays a major role. Indian government is not an exception. Since the inception of Indian planning process, various policies are given more priorities to enhance the standard of living, reduction of poverty, employment generation, reduction of inequality etc. All the above-mentioned issues are threat to any economy. The progress of any economy depends on how policy authorities are addressing these issues. Through welfare programs the Government of India aims to reduce inequality, improve standard of livings, and promote inclusive development. These initiatives play an important role in achieving sustainable and equitable economic growth.

Key Roles of the Indian Government in Promoting Welfare

❖ Poverty Alleviation

The government introduces schemes to reduce poverty and improve the income of disadvantaged groups.

Deendayal Antyodaya Yojana - National Rural Livelihood Mission (DAY-NRLM):

Deendayal Antyodaya Yojana - National Rural Livelihood Mission (DAY-NRLM) is a flagship program that aims to significantly alleviate poverty in rural regions. It is operated by the Ministry of Rural Development of the Government of India. The programme assists low-income families in securing employment. It supports both self-employment and skilled jobs. The aim is to develop more stable and improved sources of livelihood.

Creating Self-Help Groups (SHGs) is another goal of DAY-NRLM. Poor households are brought together into these groups. They receive regular guidance and assistance. These groups engage in small economic activities. Over time, their income increases, allowing them to move out of deep poverty. The success of DAY-NRLM is evident in its transformative impact on rural lives.

(DAY-NRLM) focuses on building Self-Help Groups (SHGs) for poor people, especially women. They assist members in providing mutual support. The programme also helps these groups get financial services. It helps them to find out alternative sources of income. Members of these groups receive ongoing support. They assist individuals in exploring various livelihood options, thereby decreasing reliance on a single source of income. These factors contribute to increased earnings over time, enabling people to achieve better quality of life. DAY-NRLM aims to alleviate poverty by facilitating access to financial services for poor households, especially women, and promoting livelihood diversification to enhance their quality of life. Financially, the mission supports women by creating community institutions that offer essential financial, technical, and marketing resources.

❖ Employment Generation (MNAREGA)

India's government focuses on employment generation for the impoverished segments of society to improve economic conditions and enhance livelihoods. Rural India largely relies on agriculture, which offers employment for only 5 to 6 months annually. Workers remain unemployed for the rest of the months. Unemployment disrupts the continuous inflow of money into households. In urban areas, there are generally more employment opportunities compared to rural areas. Urban areas have employment opportunities in both the manufacturing and service sectors. In rural areas, employment opportunities are primarily limited to the agriculture sector.

Social welfare in India can be improved by creating employment opportunities for the poorer segments of society. Mahatma Gandhi National Rural Employment Guarantee Act (MNAREGA) is a scheme by the Government of India that guarantees 100 days of

employment for workers in rural areas. The scheme aims to limit rural worker migrations and create employment opportunities in rural areas. This scheme worked very well during the COVID-19 pandemic. During the COVID-19 pandemic, economic activities were halted due to a nationwide lockdown. Most workers are compelled to return to their villages. MNAREGA has been found to positively affect worker welfare by generating income since its inception, according to multiple studies. MNAREGA has the potential to improve the living standards of rural workers, but its effectiveness is limited by funding constraints. MNAREGA guarantees rural workers the right to work and saw a 55% increase in budget allocation from 2019-20 to 2020-21.

❖ **Pradhan Mantri Awas Yojana (Gramin)**

Pradhan Mantri Awas Yojana (Gramin) is India's largest public housing program, launched in 2016. The scheme aim is to eradicating poverty by providing housing to economically weaker section of the society. The scheme aims to enhance the quality of life for rural people. The program's main goal was to furnish "Pucca" with built-in basic amenities, such as LPG connections, water supply, and electricity. By 2024, PMAY-G wants to provide a pucca house with basic amenities to all homeless people and households residing in kutcha and dilapidated houses. In three years, from 2016–17 to 2018–19, the immediate goal is to assist one crore households residing in decrepit or kutcha houses. The beneficiary selection process is one of PMAY-G's most crucial components. To ensure targeted assistance, PMAY-G selects beneficiaries based on housing deprivation parameters from the 2011 Socio Economic and Caste Census (SECC), verified by Gram Sabhas, rather than solely from BPL households. The SECC data identifies specific housing-related deprivation among households. Households that are houseless, living in 0, 1, and 2 kutcha wall and kutcha roof houses, can be segregated and targeted for assistance. During the COVID-19 pandemic, the share of funds allocated to PMAY-G in total expenditure decreased as the government redirected resources to healthcare, COVID-19 vaccine R&D, food security, and rural employment. PMAY-G implementation and monitoring will utilize an end-to-end e-Governance model through Awaas Soft and Awaas App.

The programme's implementation is monitored through multiple methods, not just electronic tracking. Officials from both Central and State Governments, as well as National Level Monitors, are involved in monitoring and reviewing the programme's progress.

❖ **Pradhan Mantri Gram Sadak Yojana (PMGSY)**

Rural Road Connectivity is essential for Rural Development in India. It enhances access to economic and social services, leading to higher agricultural incomes and more productive employment opportunities. It is a key ingredient in ensuring sustainable poverty reduction.

Government launched the Pradhan Mantri Gram Sadak Yojana (PMGSY) on December 25, 2000, to provide all-weather access to unconnected habitations, enhancing connectivity as part of its poverty reduction strategy. The Pradhan Mantri Gram Sadak Yojana (PMGSY) is

a fully Centrally Sponsored Scheme. The scheme aims to improve road connectivity in rural areas by constructing concrete roads. It aims to improve the quality of life for the rural poor by increasing access to essential facilities like hospitals, schools, and markets. As of 2023-24, approximately 93% of the road construction under the scheme has been completed.

❖ **Health and Social Security**

Health and social security are essential sectors through which the Government of India enhances welfare. They assist people in exploring other sources of income. This lessens their reliance on a single source of revenue.

Health sector: The government aims to deliver affordable and accessible healthcare by operating public hospitals and health centers, particularly in rural regions. Programmes such as Ayushman Bharat provide low-cost healthcare to economically disadvantaged families.

In September 2018, the Ayushman Bharat-Pradhan Mantri Jan Arogya Yojana was introduced. It is the largest government-funded health insurance program in the world. With almost 55 crore beneficiaries, the program has revolutionised healthcare in India. It enables cashless and paperless treatment throughout the nation by offering families an annual health coverage of ₹5 lakh. Over 10.30 crore hospital admissions have been approved under the program thus far. In 2022, the number of beneficiaries was increased to 12 crore families. In 2024, Anganwadi workers, ASHA employees, and their families were included. Ayushman Bharat is an effort that aims to provide people with high-quality, reasonably priced healthcare. In addition, Public health is improved by immunization campaigns and disease prevention initiatives. These initiatives lengthen life expectancy and lessen disease.

Social security: People are protected from financial hazards including old age, disability, and income loss by social security programs. For instance, financial services are accessible under the Pradhan Mantri Jan Dhan Yojana. Income after retirement is guaranteed by pension plans like the Atal Pension Yojana. Insurance programs that provide cash assistance in the event of death include Pradhan Mantri Jeevan Jyoti Bima Yojana.

Social security and health programs work together to lessen inequality and poverty. They make society safer and enhance people's quality of life.

❖ **Financial Inclusion**

Financial inclusion is a key strategy used by the Government of India to promote welfare. Financial inclusion ensures that all individuals and households can access affordable and beneficial financial services. These services include banking, savings, credit, insurance, and payment facilities. Financial inclusion aim to bring more people into the formal financial system. On August 28, 2014, the Pradhan Mantri Jan Dhan Yojana (PMJDY) was introduced as the National Mission for Financial Inclusion. The aim is to make financial services

accessible to all, particularly the poor and vulnerable segments of society. Its main goal is to provide basic banking services, such as at least one bank account, to every family. In addition, it seeks to support social security and increase financial literacy. Financial inclusion links individuals, particularly those from low-income and vulnerable backgrounds, to the formal financial system. Having bank accounts allow individuals to safely save money, receive payments, and access credit.

This reduces dependence on informal lenders who charges high interest rates. It aids the government in the effective delivery of welfare benefits. Through schemes like Pradhan Mantri Jan Dhan Yojana, individuals receive various financial supports directly into their bank accounts, including subsidies and pensions. This system, known as Direct Benefit Transfer (DBT), aims to minimize leakages, corruption, and delays. Financial inclusion enhances financial literacy and security. Access to insurance and pension schemes enhances risk management related to illness, accidents, or aging, thereby improving overall well-being.

❖ **Women and Social Welfare**

Women play a crucial role in promoting social welfare. Women play a crucial role in family and community dynamics. When one woman is empowered, benefits spread to the entire family. When women have access to bank accounts, they spend money on children education, healthcare, nutrition, and overall, wellbeing of the family. Consensus believes that when male member have access to bank accounts, they spend money on consumption of Alcohol, tobacco etc. Hence, women are treated as engine of the growth of the economy.

The future of country's progress depends on the hands of women. Government initiatives such as Beti Bachao Beti Padhao promote the education and protection of girls, contributing to a more equitable society. Beti Bachao Beti Padhao enhances social welfare by improving the status and well-being of the girl child within society.

The scheme promotes awareness of the importance of girls and encourages for equal treatment of boys and girls within families. Efforts to promote girls' education enhance school enrolment and decrease dropout rates, fostering independence and improving future opportunities for girls. Overall, it enhances social welfare by promoting gender equality, improving education, and providing better opportunities for girls, benefiting families and society.

❖ **Education and Skill Development**

Education is the modification of behavior. The country's progress depends on quality of education and health care system. The country where there is easy access to quality of education and health care, they are more developed. Quality education along with quality health care brings innovative ideas. Education, health care and innovation are interrelated and feed each other of overall economic growth and development of a country. The

government promotes human capital development through education initiatives and skill training.

Example: Skill India Mission focuses on training youth with industry-relevant skills to enhance employability

Pareto Optimality

Pareto Optimality is a condition which says that it is impossible in an economy to make one person better off without making another person worse off. In other words, we can't make benefit to one person without making worse of another person. According to this, both persons can't be simultaneously better off or worse off. It holds the principle similar to zero sum game where one person win is exactly equal to the other person loss.

The economic efficiency or Pareto optimality can be explained in following three aspects (i) Consumption Efficiency(ii) Production Efficiency(iii) Allocative efficiency or Product mix efficiency.

Consumption Efficiency

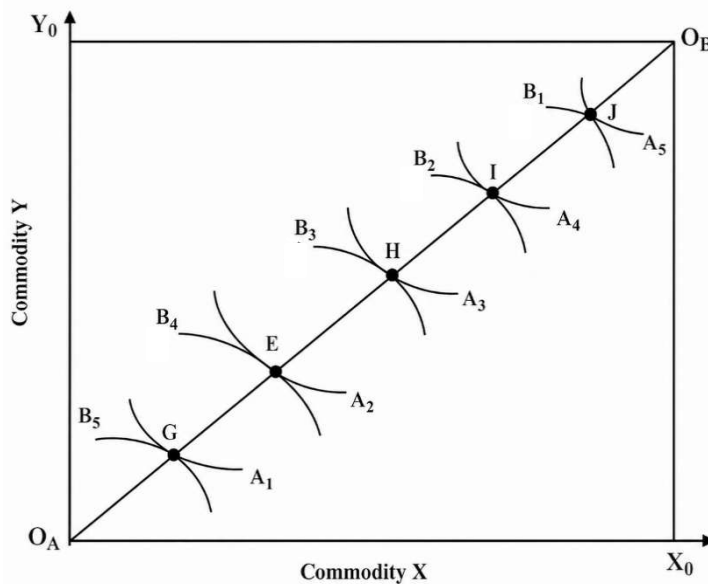
Consumption Efficiency: maximum utility for all consumers. The Consumption efficiency is based on the assumptions of two individuals and two goods. Perfect competition prevails in the market.

Let's assume an economy is operating at full employment level where all the labour force and available resources are fully utilized. The existing producers make profits which attracts new producers to enter the market. New producers need raw material and labour force to produce output. As mentioned above, all the labour force in an economy is fully employed. In this scenario the producers will increase the wage rate to attract existing labour force. New producers take resources from the existing producers. For instance, with increase in wage rate from the current level, new producers are reducing resources from the existing producers. In this scenario, some producers are benefitted by making other producers worse off.

According to this condition, we cannot increase the utility of one consumer without reducing the utility of another consumer. Let's assume that there are two consumers in an economy namely consumer A and B. The reason for taking two consumers is for simplicity though there are N numbers of consumers. The two consumers A and B, who consume both commodities X and Y. The economy produces two commodities namely X and Y. X axis represent X commodity and Y axis represent Y commodity. The total available amount of commodity X is OX_0 and of commodity Y is OY_0 . The available amount of the two commodities, OX_0 and OY_0 determines the dimension of the Edgeworth box. The quantity of X available with the individual A measured from the left to right along the X axis with

bottom left hand corner O_A as the origin. The quantity of commodity Y available with the individual A measured along the Y axis from the bottom upwards with the origin O_A . For individual B, the top right hand corner O_B has been taken as the origin and with the given quantities of X and Y, the quantity of X available for consumption for individual B is measured, right to left, with the origin O_B and the quantity of Y available for B is measured, from the top to bottom, from the origin O_B . Edge worth box has fixed dimensions representing the maximum available quantities of X and Y to be distributed between the two individuals. We further assume that the two individuals between them will entirely consume all the available quantities of two goods. As we move upward from bottom left to top right, the satisfaction of individual A increases and that of B decreases, that is, A moves to successively higher indifference curves and individual B to successively lower indifference curves.

The black line namely $A_1, A_2, \dots, A_5$, represents the indifference curve of consumer A. Similarly, the red line namely $B_1, B_2, \dots, B_5$ represents the indifference curve of consumer B. An indifference curve represents different combinations of two goods that give a consumer an equal level of satisfaction. According to indifference curve properties, higher is the indifference curve, higher is the level of satisfaction. When consumer A moves from A_1 to A_2 ,



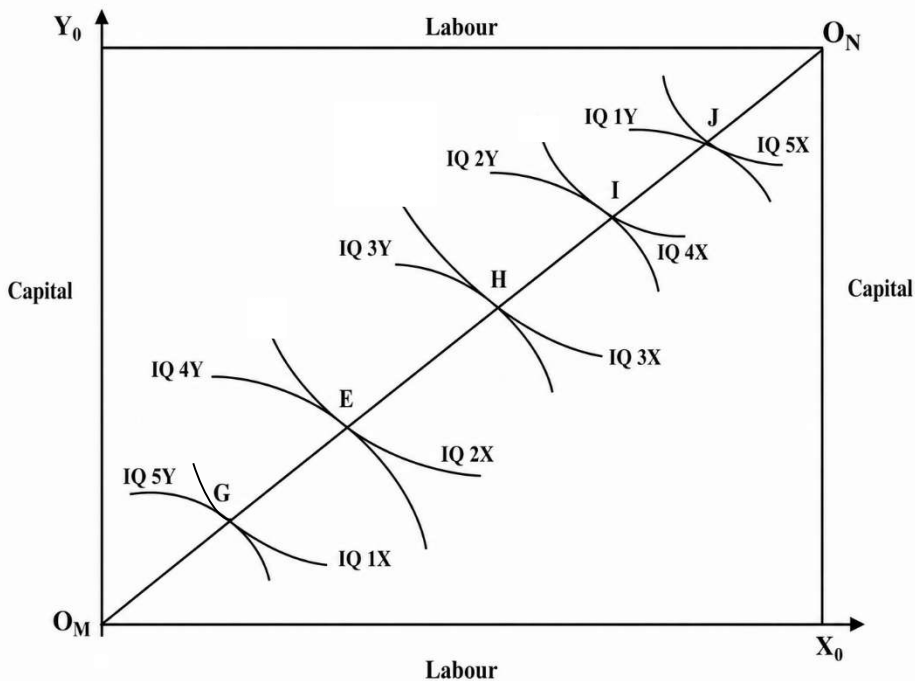
his/her satisfaction level increases. Indifference curve of B starts from B_1 . As consumer B moves from B_1 to $B_2, \dots, B_5$, his/her satisfaction level increases. GJ is the contract curve that connects the tangency points of the indifference curves for two individuals. All points on the contract curve such as points G, E, H, I, J represents Pareto optimality. How we can say all points in the contract curve represent Pareto optimality. As explained earlier, Pareto optimality means improving one person's condition is not possible without reducing the well-being of another person. All points on the contract curve illustrate the principle that improving one individual's utility necessitates a decrease in another's utility. Let's assume

that initially the economy is equilibrium at point G. As we move from point G to point E, the satisfaction of consumer A increases. Consumer A moves from lower indifference curve to higher indifference curve (A_1 vs A_2). On the other hand, the satisfaction of B falls, as B moves from higher indifference curve B_5 to the lower indifference curve of B_4 . As the economy moves from point G to E, the utility of one person increases and utility of another person decreases. As the economy moves from point G to E, the utility of consumer A increases whereas, utility of consumer B decreases. Similarly, as we move from point J to I, then the utility of consumer A reduces (A_5 vs A_4), whereas, the utility of consumer B increases from B_1 to B_2 . Consumer B moves from lower indifference curve to the higher indifference curve (B_1 vs B_2). This is called as Pareto optimality. Pareto optimality is a situation where $MRS_{XYA} = MRS_{XYB}$

Under Pareto optimality, the marginal rate of substitution for consumer A must be the same as that of consumer B. MRS, or marginal rate of substitution, refers to the slope of the indifference curve. The slopes of the indifference curves of consumers A and B are equal. All points on the contract curve, including G, E, H, I, and J, indicate that the slope of consumer A's indifference curve equals the slope of consumer B's indifference curve. Any point beyond the contract curve are not the Pareto optimality points.

(II) Production Efficiency

Production Efficiency tells that maximum possible output with the given resources. Let's assume that two commodities are produced in an economy namely X and Y. There are two



firms namely M and N that produces two commodities. Firm M produces commodity X and firm N produces commodities Y. Factors of production, namely labor and capital, are utilized as inputs in the production process by firms. An allocation of inputs is production efficient if increasing the output of one commodity (X) is not possible without decreasing the output of another commodity (Y).

Now we are discussing the producers' perspective, having previously addressed the consumer viewpoint. In this diagram X axis represents labour and Y axis represents capital. Firm M's ISO Quants include IQ 1X, IQ 2X, IQ 3X, IQ 4X, and IQ 5X. Firm N's ISO Quants consist of IQ 1Y, IQ 2Y, IQ 3Y, IQ 4Y, and IQ 5Y. OM represents the origin point of firm M, while ON indicates the origin point of firm N. Both the firm M and N tries to reach its highest ISO Quant. ISO Quant shows the various combinations of two inputs which gives the produces same output. Higher is the ISO Quant higher is the output.

When firm M moves from IQ1X to IQ2X, IQ3X, IQ4X, and IQ5X, its level of output rises. In the same way, firm N's output rises as it moves from IQ1Y to IQ2Y, IQ3Y, IQ4Y, and IQ5Y. GJ represents the contract curve that connects the tangency points of the isoquants of the two firms. All these points on the contract curve show Pareto optimality. Moving along the contract curve enhances one firm's welfare while reducing the other's. One good's output cannot be increased without decreasing another's output. Thus, all points on the contract curve are "Pareto Optimal" points.

Suppose the producer is equilibrium at point G. When we move from point G to E, we can see the output of firm M increases. Firm M is moving from lower ISO Quant to higher ISO quant (IQ 1X vs IQ 2X). Higher is the ISO Quant, higher is the output and vice versa. On the other hand, firm N's output falls as firm N is moving from point G to E (IQ 5Y vs. IQ 4Y). The output of firm N reduces as it moves from higher ISO Quant IQ 5Y to the lower ISO Quant of IQ 4Y. If firm M produces at the level of IQ5X, firm N will be able to produce only IQ1Y. If firm N produces IQ 5Y, then firm M must produce IQ 1X. At point H, firm M produces IQ 3X units of output and firm N produces IQ 3Y units of output. Is it possible to improve this situation? If firm M and N decided to move from point H to point I, then firm M can increase its output to IQ 4X whereas firm N can reduce its output to IQ 2Y. In this scenario, firm M is better off while firm N is worse off. Similarly, if both the firms moves from point H to point E, then firm M's output falls to IQ 2X and firm N's output increases to IQ 4Y. In this scenario, firm M is worse off and firm N is better off.

At points G, E, H, I, and J, the slopes of the ISO Quant and ISO cost curves are same. Producers achieve equilibrium when the slope of the isoquant is equal to the slope of the isocost line. The slope of ISO quant of firm M and N

$$MRTS^X = MRTS^Y$$

The slope of ISO cost of capital and labour for the two firms= W/i

The marginal rate of technical substitution of labour for capital for firm M equals that of firm N.

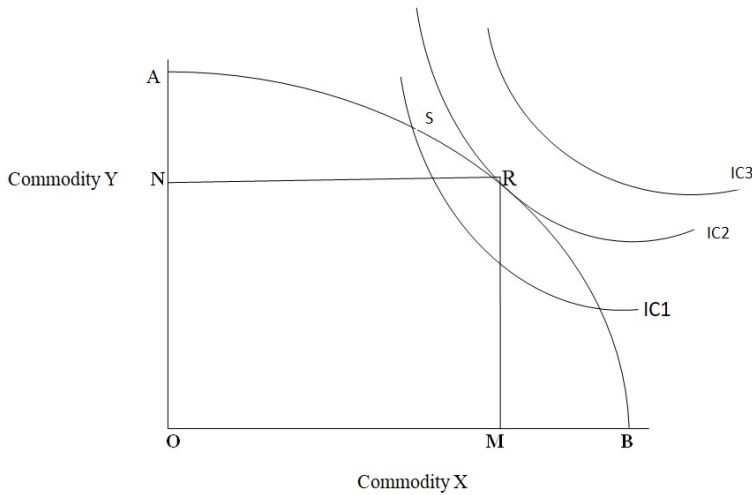
MRTS is nothing but the slope of ISO quant. The slope of the ISO Quant for firm M is equal to that of firm N. All the points such as G, E, H, I, J, the slope of the ISO Quant of firm M is equal to the slope of ISO Quant of firm N. All the points of the contract curve is called Pareto optimality.

At each equilibrium points GEHIJ on the contract curve $MRTSX = MRTSY = W/i$

Product mix efficiency: optimum mix of commodities

Product mix efficiency: linking the production and consumption. Product mix efficiency means producing the best combination of goods in an economy. The economy must produce specific combination of goods that society value the most. Let's assume that farmers in Gobindapur village of Odisha can produce rice and wheat more efficiently. But if the Gobindapur villagers prefer rice, then growing equal amount of both crops would be inefficient, no matter how well farmers cultivate them. Product mix efficiency ensure that production decision align with consumer preferences, creating a harmony between the supply side and demand side of the economy. Product mix efficiency refers to resource allocation for the production of goods and services based on people's preferences. If an economy produces the wrong combination of goods and services, it can increase the production of one good and reduce another. This can improve the well-being of some people without harming others. Therefore, product mix efficiency is about choosing the best point on the production possibility curve based on consumer preferences. Goods should be produced in accordance with the consumer preferences for the attainment of maximum social welfare.

Commodity X is presented in X axis and commodity Y is presented in Y axis. AB represent production possibilities curve between any pairs of goods X and Y. IC_1 , IC_2 and IC_3 are the indifference curve of a consumer. The slope of an indifference curve shows the marginal rate of substitution between two goods for a consumer. Production efficiency necessitates that the economy operates along the production possibilities curve.



At Point R the IC_2 is tangent to the production possibilities curve. At point R the $MRT_{xy} = MRS_{xy}$. Point R indicates the optimal production composition for commodities X and Y, with their respective consumption quantities being OM and ON. This is because point R lies on the highest attainable indifference curve (IC_2) among all the points on the production possibility curve. Therefore, it gives the consumer the maximum satisfaction possible. For example, if goods X and Y are produced and consumed at point S, the consumer will have a lower level of satisfaction. This is because point S lies on a lower indifference curve (IC_1). It intersects the production possibility curve rather than merely touching it at a single point. At point S, the Marginal Rate of Substitution (MRS_{xy}) of the consumer does not equal the Marginal Rate of Transformation (MRT_{xy}) of the community.

Review Questions

1. Elucidate what is meant by Moral Hazard and Adverse selection with the help of suitable example.
2. Analyze the different causes of market failure. Explain how each factor leads to inefficiency in the market system.
3. Illustrate the major causes of market failure with suitable real-life examples. Explain how these factors affect the efficient allocation of resources in an economy.
4. Use a Production Possibility Curve (PPC) to analyze the idea of production efficiency. Describe how various efficiency levels are represented by a point inside, on, and outside the curve.
5. Examine how an economy can achieve product mix efficiency using the Production Possibility Curve (PPC) concept.

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Index

A

Accounting costs 132
Adverse selection 236, 237
Allocative efficiency 174, 246
ARC method 63, 65, 68
ARC price elasticity of demand 65
ARC income elasticity of demand 68
Asymmetric information 174, 236, 239
Average cost (AC) 114, 127, 133, 134, 136, 139
Average total cost (ATC) 133, 134, 136
Average variable cost (AVC) 134, 135, 143
Average fixed cost (AFC) 158, 171,
Average product (AP) 109, 110
Average product for labour (AP_L) 110
Average revenue (AR) 164, 166, 169
Auction pricing 212

B

Barriers to entry 161, 162, 177, 198
Break-even point 182,
Budget line 91, 94
Budget constraint 86, 92, 95
Bundling 212

C

Capitalist economy 22
Capital goods 6

Capital 12
Cardinal utility 77
Causes of oligopoly 108,
Causes of market failure 239
Cartel 205
Cartel success and failure story 207
Centralized cartel 206
Change in quantity demanded 30
Change in quantity supply 41
Change in supply 42
Change in demand 30
Choice 76
Command economy 21, 22
Consumer goods 5
Consumer behaviour 76
Consumer equilibrium 82
Consumer surplus 84
Constant returns to scale 124, 126
Complementary goods 34, 66
Collusive oligopoly 199
Cournot's duopoly model 200
Conditions for price discrimination 185
Conditions for cartel success 207
Cross price elasticity of demand 66

D

Demand schedule 28
Demand curve 29
Demerit goods 241
Dead weight loss 183
Decreasing returns to scale 126
Degrees of price discrimination 186
Diminishing marginal utility 79
Diminishing marginal rate of substitution 119
Diminishing marginal return 216
Diminishing return to factor 114
Diseconomies of scale 154
Disequilibrium 46
Diminishing marginal rate of technical substitution 119
Diminishing marginal utility of money 79
Differentiated products 12, 160
Dominant strategy 203, 204
Durable goods 5
Duopoly 197

E

Economic goods 8
Economic costs 132
Economies of scale 152
Economies of scope 152
Elasticity 56

Elastic demand 60
Elasticity of supply 69
Elastic supply 69
Engel curve 100
Entry and exit 106
Entry barriers 161
Equi marginal utility 82
Explicit costs 132
External economies of scale 154
External diseconomies of scale 155
Externality 231
Expansion path 123
Excess capacity 196

F

Factor of production 24, 107
Fixed cost 130
Fixed input 110, 133
First-degree price discrimination 186
Free goods 7

G

Geometric method 64
Gillette pricing 199
Giffen goods 37
Goods 7, 8, 10
Government failure
Government intervention 241

H

Homogeneous product 203

Heterogeneous product 196

Horizontal summation 41

I

Implicit costs 132

Imperfectly competitive market 160, 189, 196

Income elasticity of demand 68

Income consumption curve (ICC) 98

Income effect 103

Indifference curve 85

Indifference schedule 86

Indifference map 86

Increasing returns to factor 135

Increasing returns to scale 145

Internal economies of scale 152

Internal diseconomies of scale 154

Inferior goods 37

Individual demand curve 29

Individual demand schedule 29

Individual supply curve 41

Individual supply schedule 40

Inelastic demand 63

Inelastic supply 70

Inputs 107

Interest 218

Interest rate 218

Investment 219

Investment demand curve 219

ISO quant 115

ISO quant schedule 115

ISO quant curve 116

ISO quant map 117

ISO-cost line 121

J

K

Keynesian theory of the demand for money 221

Kinked demand curve 208

L

Law of demand 27

Law of supply 39

Law of variable proportions 110

Learning curve 148

Least cost combination of factors 122

Linear demand curve 63

Liquidity preference theory of interest 220

Liquidity trap 225

Limit pricing 199

Long run production function 115

Long run average cost (LAC) 144

Long run marginal cost (LMC) 194

Long run cost curve 144

Long run total cost(LTC) 144

Luxury goods 37

M

Market structure 160

Marginal utility 82

Marginal benefit (MB) 240

Marginal cost (MC) 210

Marginal revenue (MR) 216

Market demand curve 29

Market supply curve 41

Market equilibrium 46

Marginal rate of substitution (MRS) 88

Marginal rate of technical substitution (MRTS) 118

Market failure 238

Market sharing cartel 205

Marginal product (MP) 109

Marginal product of labour (MP_L) 110

Marginal product of capital (MP_K) 120

Merit goods 241

Microeconomics 1

Macroeconomics 13

Mixed economy 21

Minimum wage law 55

Monopolistic competition 189

Monopoly 178

Money income 29

Movement along a demand curve 30

Movement along a supply curve 40

N

Natural monopoly 177

Natural resources 4

Negative income elasticity

Negative marginal utility 80

Necessary goods 38

Negative externality 233

Negative return to factor 114

Nominal interest rate 218

Normal goods 32

Non durable goods 5

Non pricing strategy 197

Non price competition 190

Non-collusive oligopoly 199

O

Oligopoly 196

Opportunity cost 131

Operating cost 132

Optimum plant 149

Optimum output 149

Ordinal utility 84

P

Peak load pricing 211

Perfectly elastic demand 165

Perfectly inelastic demand 63

Perfectly elastic supply 71

Perfectly inelastic supply 71

Perfectly competitive market 173

Perfect competition 161

Perfect substitute 258

Positive marginal utility 80

Point method 65

Positive externality 232

Private goods 6

Production possibilities frontier 16

Preferences 27

Prestige goods 37

Price ceiling 53

Price floor 54

Production function 107

Price elasticity of demand 57

Promotional elasticity of demand 69

Properties of ISO quant 117

Producer's surplus 175

Product differentiation 190

Price discrimination 185

Pricing strategy 199

Prestige pricing 211

Predatory pricing 212

Price matching 212

Price effect 103

Price taker 11

Price maker 160

Price war 197

Price rigidity 198

Price consumption curve (PCC) 96

Price lining 212

Predatory pricing 212

Precautionary demand for money 222

Price leadership model 203

Problems with the cartel 207

Profit maximization 180

Profit 226

Pricing in practice 211

Public goods 235

Public policy 12

Q

Quantity demanded 30

Quantity supplied 39

Quotas 3

R

Return to factor 113

Returns to scale 127

Real income 29

Real interest rate 218

Relatively elastic in demand 60

Relatively inelastic in demand 60

Rent 215

S

Saving 219

Scope of microeconomics 10

Scarcity 4

Services 5

Semi variable cost 131

Second degree of price discrimination 186

Speculative demand for money 222

Short run 109

Shift in demand 30

Shut down point 156

Shut down rule 171

Shortage 46

Surplus 84

Supply of money 223

Short run production 109

Short run production function 109

Short run cost function 139, 140, 141

Short run marginal cost curve (SMC) 194

Skimming 212

Snob effect 37

Social costs 131

Substitution effect 103

Substitute goods 34

Sunk costs 131

Supply 38

Supply schedule 40

Supply curve 40

T

The law of variable proportion 110

The marginal productivity theory of distribution 227

Third degree of price discrimination 187

Total utility (TU) 80

Total cost (TC) 133

Total fixed cost (TFC) 133

Total variable cost (TVC) 133

Total revenue (TR) 155

Total product (TP) 190, 112, 113, 114

Transaction demand for money 221

U

Utility 246

Unitary elastic demand 61

Unitary elastic in supply 70

V

Variable cost 130

Veblen goods 37

W

Wage 23

What to produce 24

How to produce 24

For whom to produce 24

X

Y

Z

Zero Marginal Utility 80

ABOUT THE BOOK

This book presents the fundamental principles of microeconomics in a clear, structured, and application-oriented manner. Covering key topics such as demand and supply, consumer behaviour, production, cost analysis, market structures, and pricing strategies, it connects economic theory with real-world business practices. Rich with illustrations, examples, and practical insights, it equips students with the analytical skills required for informed managerial decision-making. An ideal resource for undergraduate students of Business, Management, Commerce, and Economics.

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